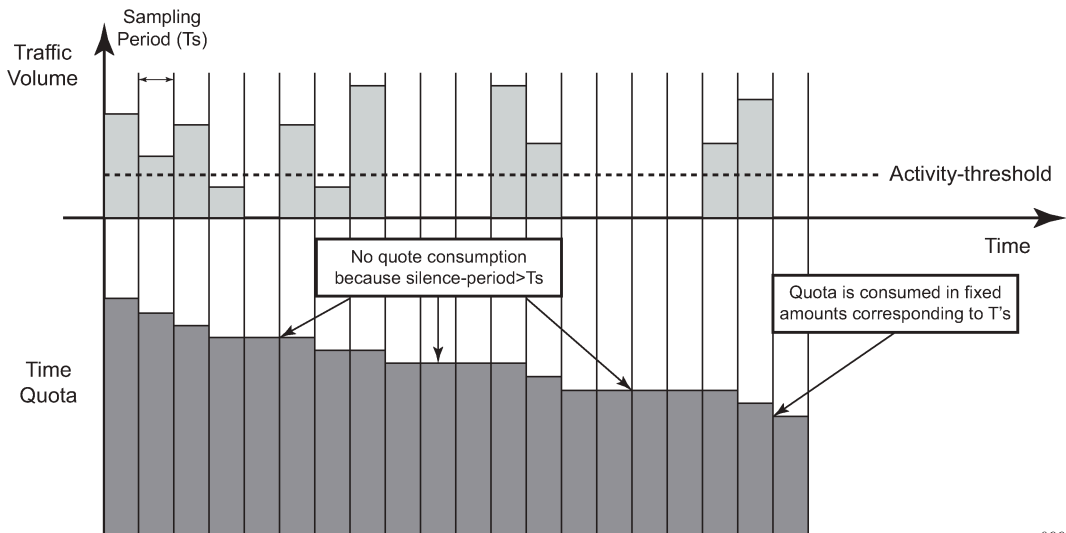


The consequence of sampling method is that the time quota can be consumed only in discrete units which are equal to T_s . The exact algorithm of time quota consumption is shown in [Time quota consumption example](#). As shown, the time quota is consumed as long as period of silence (traffic level is below activity-threshold) does not exceed T_s .

Figure 172: Time quota consumption example

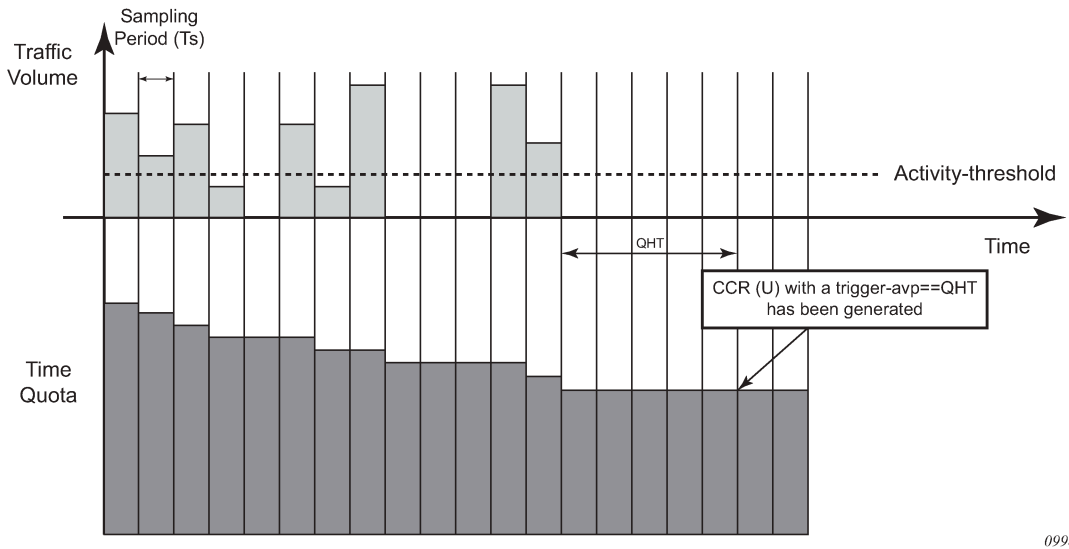


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The algorithm is aligned with Continuous-Time-Period (CTP) mechanism and is a simplification of the generic time-based quota consumption mechanism which is based on restarting time envelope (Quota-Consumption-Time - QCT) every time data packet of the given category is seen by the system. Implementing the generic mechanism is not scalable in the system with high speed interfaces and high capacity.

Quota-holding-time (QHT) is used to prevent an amount of credit being **stranded** with given CC-client while the end-user is not using the service anymore. Per its definition (QHT is reset every time new packet arrives for a given category) this timer also does not map with 7xx0 implementation of metering. To approximate its operation, QHT is considered expired whenever there is no change in usage during its period.

Figure 173: Time quota QHT example



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12.18.5 Support of tariff change

The Tariff-Time-Change AVP is used to determine the tariff switch time as described by RFC 4006, section 8.20. When this event is encountered, all current counts are captured and new counts are started. The counts are sent to the OCS in next CCR.

The client (gateway) reports usage which straddles the change time by sending two instances of the Used-Service-Unit AVP, one with Tariff-Change-Usage = 'unit-before-tariff-change' and one with 'unit-after-tariff-change'.

12.18.6 Termination action and redirection

The OCS can perform termination synchronously (if it has a CCR pending processing) by returning CCA with an appropriate failure code at the command level. Alternatively, the OCS can send an ASR message to terminate the Diameter Session. The gateway follows by sending a CCR (Terminate) to the OCS.

Final-Unit-Indication AVP in a CCA from the OCS at the Multiple Services Credit Control level (MSCC) is supported. Supported Final-Unit-Actions include TERMINATE and REDIRECT. When FUA = REDIRECT, OCS must send the redirect-server URL.

12.18.6.1 HTTP traffic redirection sample configuration

This section provides configuration examples for HTTP redirection with and without AA enabled. The configuration sample in this section is for enabling HTTP redirection using Gy FUI.

Auto-redirect rule

If AA rules are present in the activated policy for the session, an additional 5-tuple rule matching TCP port 80 traffic must be present in the same rulebase with higher precedence (lower value) than other

overlapping filters. The auto-redirect rule is configured with the offline charging method and a rating group different from the other rules.

The auto-redirect rule is only activated if there is at least one other rule in the redirect state. This is required for both the Gy FUI-based redirection as well as the Gx policy rule base redirection with AA rules present.

Sample configuration

```
mobile-gateway
  profile
    dcca "DCCA-Prof1"
    exit
    diameter "diam-prof1"
    exit
    diameter-peer "Gy-Peer-Prof1"
      application-type gy
      destination-realm "nokia.com"
      diameter-profile "diam-prof1"
      interface "gy"
      peer 10.10.18.10 port 3868
      no shutdown
    exit
  exit
  charging
    pgw-charging 1
      on-line
        dcca-profile "DCCA-Prof1"
        enable-online
        pri-ocs-dia-peer "Gy-Peer-Prof1"
      exit
    exit
  exit
exit
pdn 1
  gy
    dcca-profile "DCCA-Prof1"
    interface
      diameter "gy"
    exit
    primary-diameter-peer "Gy-Peer-Prof1"
  exit
exit
```

Redirection configuration (DPI)

```

application-assurance
  group 1
    policy
      app-profile "App_1" create
        divert
      exit
      app-group "AA_chargingGrp_1" create
      exit
      app-group "AA_chargingGrp_2" create
      exit
      application "app_1" create
        app-group "AA_chargingGrp_1"
      exit
      application "app_2" create
        app-group "AA_chargingGrp_2"
      exit
      app-filter
        entry 1 create
          expression 1 http-uri eq "**facebook*"
          application "app_1"
          no shutdown
        exit
        entry 2 create
          expression 1 http-uri eq "**AoC*"
          application "app_2"
          no shutdown
        exit
      exit
    exit
  exit
exit
policy-options
  begin
  policy-rule-unit "TCPPort80"
    flow-description 10
    match
      protocol 6
      remote-port eq 80
    exit
  exit
  flow-gate-status auto-redirect
exit
policy-rule-unit "any/any"
  flow-description 1
  exit
exit
policy-rule-unit "facebook"
  flow-description 10

```



```

        match
            aa-charging-group "AA_chargingGrp_1"
        exit
    exit
exit
policy-rule-unit "AoC"
    flow-description 1
        match
            aa-charging-group "AA_chargingGrp_2"
        exit
    exit
exit
charging-rule-unit "TCPPort80"
    rating-group 101
    charging-method offline
exit
charging-rule-unit "AoC"
    rating-group 1
    charging-method online
exit
charging-rule-unit "facebook"
    rating-group 2
    charging-method online
exit
charging-rule-unit "any/any"
    rating-group 5
    charging-method online
exit
policy-rule "AoC" policy-rule-unit "AoC" charging-rule-
unit "AoC" qci 9 arp 11 precedence 101
policy-rule "facebook" policy-rule-unit "facebook" charging-rule-
unit "facebook" qci 9 arp 11 precedence 102
policy-rule "TCPPort80" policy-rule-unit "TCPPort80" charging-rule-
unit "TCPPort80" qci 9 arp 11 precedence 106
policy-rule "any/any" policy-rule-unit "any/any" charging-rule-
unit "any/any" qci 9 arp 11 precedence 107
policy-rule-base "PRB_1"
    policy-rule "TCPPort80"
    policy-rule "anyany"
    policy-rule "AoC"
    policy-rule "facebook"
exit
exit
commit

```

Redirection configuration (IP 5 Tuple)

```

policy-options
begin

```

```

policy-rule-unit "TCPPort80"
    flow-description 1
        match
            protocol 6
            remote-port eq 80
        exit
    exit
exit
policy-rule-unit "any/any"
    flow-description 1
    exit
exit
policy-rule-unit "facebook"
    flow-description 1
        match
            remote-ip 10.10.10.30/32
        exit
    exit
exit
policy-rule-unit "AoC"
    flow-description 1
        match
            remote-ip 30.30.30.30/32
        exit
    exit
exit
charging-rule-unit "TCPPort80"
    rating-group 99
    charging-method online
exit
charging-rule-unit "AoC"
    rating-group 1
    charging-method online
exit
charging-rule-unit "facebook"
    rating-group 5
    charging-method online
exit
policy-rule "AoC" policy-rule-unit "AoC" charging-rule-
unit "AoC" qci 9 arp 11 precedence 90
policy-rule "facebook" policy-rule-unit "facebook" charging-rule-
unit "facebook" qci 9 arp 11 precedence 91
policy-rule "TCPPort80" policy-rule-unit "TCPPort80" charging-rule-
unit "TCPPort80" qci 9 arp 11 precedence 92
policy-rule "any/any" policy-rule-unit "any/any" charging-rule-
unit "any/any" qci 9 arp 11 precedence 100
policy-rule-base "Online-Charging"
    policy-rule "AoC"
    policy-rule "facebook"
    policy-rule "TCPPort80"

```

```
        policy-rule "any/any"  
    exit  
    commit  
exit
```

12.18.6.2 UE URL append upon redirection

This section describes the functionality where the UE URL is appended to the Redirect-Server-Address AVP value, if the operator has enabled the `url-append` command.

When OCS redirects traffic for an MSCC instance, for example, due to the UE being out of quota, the operator can append the UE URL to the Redirect-Server-Address AVP provided by the OCS.

If the UE requests quota at the CCR-U and receives a CCA with the Url-Append AVP value set to 1 from the OCS and FUI=REDIRECT, and if the Redirect-Server-Address value is set to `http://www.a1.net/landingpage_redirect/kobev.cgi?url=`, then the mobile gateway redirects the flow to the URL `http://www.a1.net/landingpage_redirect/kobev.cgi?url=https://www.google.com`. The initial site that the UE browses is `www.google.com`.

If the URL-Append AVP is not received, the operator can append the UE URL to the Redirect-Server-Address value.

To append the UE URL to the Redirect-Server-Address value, use command in the following context.

```
configure mobile-gateway profile dcca append-url
```

12.18.6.3 Termination action with Reporting Reason Final

When deleting a session (for example after entering flight mode), the mobile gateway reports in the CCR-T message all the active MSCC instances and USU quota, with Reporting-Reason set to Final at the MSCC level.

To configure the maximum number of MSCC instances per CCR message in the Gy interface, use the following command:

```
configure mobile-gateway profile dcca ccr-mscc-limit
```

To enable the report of MSCC instances in the CCR-U and not the CCR-T message, use the following command:

```
configure mobile-gateway profile dcca ccrt-no-mscc
```

The `ccrt-no-mscc` command provides the option to report the Final Units for all the active MSCC instances in the CCR-U message. In this case, when a DCCA session is closed there are no MSCC instances included in the CCR-T message. To configure this, the `num-mscc` parameter must be set to 1 in the `config>mobile>profile>dcca>ccr-mscc-limit` command. For example, if a UE has 10 active MSCCs, at the bearer release the mobile gateway generates 10 CCR-U messages (with one MSCC instance per CCR-U) and a CCR-T message without an MSCC instance.

12.18.6.4 Termination action with non-3GPP triggers

As specified by 3GPP, the OCS terminates an MSCC instance by sending an FUA = TERMINATE message. The mobile gateway supports a customized trigger for terminating an MSCC instance.

When this trigger is enabled, the following OCS behavior triggers the termination of an MSCC instance in the mobile gateway.

1. The mobile gateway reports quota and requests more for an MSCC instance in a CCR-U.
2. The OCS replies with a CCA-U in which:
 - a. the command-level Result-Code AVP is set to DIAMETER_SUCCESS
 - b. the MSCC-level Result-Code AVP is set to DIAMETER_SUCCESS
 - c. the FUI AVP is not present
 - d. the GSU AVP is not present or grants zero quota

The mobile gateway interprets the preceding behavior as equivalent to the OCS sending an FUA = TERMINATE message.

To enable this functionality, use the following command:

```
configure mobile-gateway profile dcca wait-for-first-packet-on-no-gsu
```



NOTE:

If this functionality is disabled, the mobile gateway considers this combination of OCS behavior as an invalid answer, as specified by 3GPP.

12.18.7 Failure handling

When the PGW determines a failure condition with the current OCS, the PGW uses the Credit-Control-Failure-Handling AVP (as defined in RFC 4006, sections 5.7 and 8.14) to determine what to do:

- This AVP may be received from the OCS in the CCA message or may be locally configured.
- The value received from the OCS shall always override any already existing value.

Failure Handling mechanisms according to the behavior described in Annex B of 3GPP TS 32.251 are supported.

12.18.8 Credit Control Failure Handling

Configuration of Credit Control Failure Handling (CCFH) is supported using the CLI command `cc-failure-handling` in the DCCA profile context.

To facilitate interoperability with vendor-specific OCSs, credit control handling can also be configured on a per-error-code basis at the Diameter command level, to override the system CCFH command action if the OCS returns a specific result code.

For example, if the CCFH command is set to **continue**, but the result code 4201 to 4204 is configured to **terminate** at the Diameter command level, then the PGW-GGSN will automatically terminate the session if the OCS returns a result code of 4201 to 4204.

For more information, see the following CLI commands that are configured in the DCCA failure-handling and Gx profile contexts:

- `result-code`
- `result-code-3xxx`
- `result-code-4xxx`
- `result-code-5xxx`

Result codes 4010, 4012, 5003, and 5030 need to be specifically configured as **continue** or **retry-and-continue** in order for Assume Positive mode to be entered for these error codes. For example, even if the 4xxx result code is configured as **continue** or **retry-and-continue**, result codes 4010 and 4012 need to be specifically configured in order for Assume Positive mode to be entered for these error codes. A retry to secondary will be attempted for error codes 3002, 3004, and 3005 regardless of whether the 3xxx result code is configured as **continue**.

Currently CCFH is applied upon Gy transaction (Tx) timer expiration, as per 32.251 Appendix B (Tx expiration; Failure Handling procedure and session failover mechanism description). CCFH is also applied for certain error conditions.

Table 53: Error code handling

Diameter command level result code	CMG behavior	Diameter MSCC level result code	CMG behavior
DIAMETER_SUCCESS (2001)	Session continues	DIAMETER_SUCCESS (2001)	Session continues
DIAMETER-PARTIAL-SUCCESS (2002)	Continue (treated as success)	N/A	—
DIAMETER-COMMAND-UNSUPPORTED (3001)	Follow CCFH	N/A	—
DIAMETER-UNABLE-TO-DELIVER (3002)	(Failover Enabled) Sent to Secondary OCS. If no secondary OCS, CCFH	N/A	—

Table 53: Error code handling (continued)

Diameter command level result code	CMG behavior	Diameter MSCC level result code	CMG behavior
DIAMETER-TOO-BUSY(3004)	(Failover Enabled) Sent to Secondary OCS. If no secondary OCS, CCFH	N/A	—
DIAMETER-LOOP-DETECTED(3005)	(Failover Enabled) Sent to Secondary OCS. If no secondary OCS, CCFH	N/A	—
DIAMETER-OUT-OF-SPACE(4002)	Follow CCFH	N/A	—
DIAMETER_END_USER_SERVICE_DENIED (4010)	Terminate the session. When received in CCA-U, CCR-T is sent	N/A	—
DIAMETER_CREDIT_CONTROL_NOT_APPLICABLE (4011)	User session allowed	N/A	—
DIAMETER_CREDIT_LIMIT_REACHED (4012)	Terminate the session. When received in CCA-U, CCR-T is sent	N/A	—
DIAMETER-AVP-UNSUPPORTED (5001)	Follow CCFH	N/A	—
DIAMETER_UNKNOWN_SESSION_ID (5002)	Follow CCFH	N/A	—
DIAMETER_AUTHORIZATION_REJECTED(5003)	Terminate the session. When received in CCA-U, CCR-T is sent	N/A	—
DIAMETER-INVALID-AVP-VALUE(5004)	Follow CCFH	N/A	—

Table 53: Error code handling (continued)

Diameter command level result code	CMG behavior	Diameter MSCC level result code	CMG behavior
DIAMETER_MISSING_AVP (5005)	Follow CCFH	N/A	—
DIAMETER-AVP-NOT-ALLOWED(5008)	Follow CCFH	N/A	—
DIAMETER_UNABLE_TO_COMPLY (5012)	Follow CCFH	N/A	—
DIAMETER_USER_UNKNOWN (5030)	Terminate the session	N/A	—
DIAMETER-RATING-FAILED(5031)	Follow CCFH	N/A	—
DIAMETER_SUCCESS (2001)	Session continues	DIAMETER_END_USER_SERVICE_DENIED (4010)	Traffic matching the RG is blocked. After the configurable delay timer expires, traffic for the RG will result in CCR-U to OCS asking for quota.
DIAMETER_SUCCESS (2001)	Session continues	DIAMETER_CREDIT_CONTROL_NOT_APPLICABLE (4011)	Allow traffic for this RG
DIAMETER_SUCCESS (2001)	Session continues	DIAMETER_CREDIT_LIMIT_REACHED (4012)	Traffic matching the RG is blocked. After the configurable delay timer expires, traffic for the RG will result in CCR-U to OCS asking for quota.
DIAMETER_SUCCESS (2001)	Session continues	DIAMETER_INVALID_AVP_VALUE (5004)	Block the traffic. CCR-U is sent with FINAL

Table 53: Error code handling (continued)

Diameter command level result code	CMG behavior	Diameter MSCC level result code	CMG behavior
DIAMETER_SUCCESS (2001)	Session continues	DIAMETER_MISSING_AVP (5005)	Block the traffic. CCR-U is sent with FINAL
DIAMETER_SUCCESS (2001)	Session continues	DIAMETER_RATING_FAILED (5031)	Block the traffic. CCR-U is sent with FINAL
DIAMETER_SUCCESS (2001)	Session continues	DIAMETER_UNABLE_TO_COMPLY (5012)	Block the traffic. CCR-U is sent with FINAL

The PGW/GGSN also supports a configurable delay timer as part of failure handling for the OCS error codes at the MSCC level. Service is denied (traffic dropped) when termination is requested, (for example, credit limit reached, end user service denied, and so on) by the network (GGSN/PGW, OCS), until the timer expires. Subsequent to the timer expiry, traffic for an RG will result in a CCR-U sent to the OCS asking for quota.

Handling of result codes on the command and MSCC levels is also configurable.

The CCFH also supports the ability to grant service to the IP-CAN session for a configured volume-limit. The volume limit can be enabled along with the FH session timer, to ensure that the session is terminated upon whichever of these conditions occur first. This feature also provides a configurable option to re-start the Gy session with the OCS (send CCR-I) upon either of the conditions being met. If the OCS is still unavailable, the session is terminated. A configurable retry timer is supported to periodically retry (CCR-I) to restore the Gy session during the grace window.

For CLI configuration information, refer to the DCCA Profile Commands in the *CMG CLI Reference Guide*.

12.18.8.1 Failure Handling Continue options

This feature enhances the functionality of the 3GPP Failure Handling Continue mechanism.

Besides the 3GPP Continue mechanism, this feature allows the operator to select among three proprietary behaviors to be applied during the period that starts when the Continue action starts, until the bearer is released (by normal release or by the expiration of the Continue timer).

The supported options are the following:

- Block (all online-charged flows are blocked and packets are dropped for the duration of the bearer)
- Allow (the user can access all online-charged flows that have not been explicitly blocked or denied by the OCS, for the duration of the bearer)
- Offline (the session is converted to normal postpaid session and CDRs are created)

To configure the failure handling options, use the following command:

```
configure mobile-gateway profile charging ccfh-profile fh-continue-action
```

Flows which were not under OCS control are not affected by those options. For example, if a Rating Group (RG) quota request was replied with Credit Control Not Applicable (the OCS commands that the flow is free of charge), then the trigger of failure handling Continue does not affect that flow.

The `fh-session-continue-timer` and `fh-volume-limit` commands in the `config>mobile>profile>charging>ccfh-profile` context, apply to all the Continue options. At timer expiration (or volume limit reached), the bearer is released regardless of the Continue option (block, allow or offline).

The RGs that are online only, will start being reported to CDRs in the case where failure handling Continue with option offline is triggered.

To configure the failure handling parameters, use the following commands:

```
configure mobile-gateway profile charging ccfh-profile ccfh-fh-action
```

```
config mobile-gateway profile charging ccfh-profile session-restore-retry-timer
```

To link a DCCA profile to a CCFH profile, use the following command:

```
configure mobile-gateway profile dcca ccfh-profile
```

To view failure handling statistics, use the following command:

```
show mobile-gateway pdn ref-point-stats gy
```

12.18.9 Statistics collection for rating groups

The PGW/GGSN supports KCI for the maximum number of simultaneously handled rating groups on the Gy interface (on a per peer basis under the APN context). This is the number of rating groups with active counters in OCS, which is essentially the RGs that have been allocated quota.



NOTE:

The OCS is impacted by the number of counters/service instances in addition to the transactions. This information can be considered as a capacity indicator.

An active rating group instance becomes inactive once quota is returned back to the OCS. So the rating group instance count will be the effective count for the reporting interval.

The number of Rating Group Instances for a this peer on this APN would be indicated as follows:

- KCIReferencePointGy
- NumRatingGroupInstances
- RG1 - x instances
- RG2 - y instances

12.18.10 Gy Assume Positive mode

The 7750 SR MG supports Gy Assume Positive behavior on a PGW to enhance credit control failure handling (CCFH).

For more information about CCFH, see [Credit Control Failure Handling](#). When a PGW is in Gy Assume Positive mode during an OCS outage, the following occurs:

- service to the end user is not interrupted during the OCS outage
- the PGW continues to check if the OCS is back in service
- when Gy connectivity to the OCS is restored, data usage that occurred during the outage is reported to the OCS so it can take appropriate action

While in Gy Assume Positive mode, each subscriber type can be assigned a temporary quota, validity time (VT), and interim quota which is configured locally on the PGW. When the temporary quota is exhausted or the VT is expired, the PGW retries the OCS to determine whether the functionality has been restored. If OCS service has been restored, the quota or VT assignment and tracking proceeds. If service has not been restored, the PGW reallocates the locally configured quota and/or VT to the user. When the OCS returns to service, the PGW reports all unreported quota usage to the OCS in the subsequent CCR-U/CCR-T. If there were multiple retries and interim allocations, the PGW reports quotas used for all allocation intervals.

If the maximum interim quota is reached, the session is terminated. This applies only to sessions with instantiated rating-groups that are exclusively online charged, and only when the maximum interim quotas are reached for all instantiated online charged rating-groups. Until then, any rating-group that has exhausted its interim quota, is blocked. If an offline charged (or non-charged) rating-group is instantiated for a session (that is, traffic matched to this rating-group, has been exchanged), the session is not terminated when maximum interim quota is reached for the last active online-charged rating-group. Therefore, the online charged rating-groups remain blocked.



NOTE:

In case of the special rating-group 0, or a rule with flow-gate-status auto-redirect, the rule/rating-group is by default instantiated.

If a subscriber terminates a session while the OCS is down, the usage data will be lost. To prevent this loss, usage data can be written to a file on the local compact flash so that it can be retrieved and reconciled by the OCS. New log files are written to a primary directory. After the OCS retrieves a log file, it will move the file to the secondary directory so that the file cannot be retrieved multiple times. The PGW periodically clears out old files in both the primary and secondary directory as configured via the

config>mobile>pdn>gy>outage>file-params>obsolete-time command. The OCS retrieves the Gy Assume Positive log files through an SFTP pull (port number 22) from the PGW.

Use the CLI commands in the following contexts to configure Gy Assume Positive functionality:

```
configure mobile-gateway profile charging ccfh-profile assume-positive
  access-type
  message
  rating-group
  credit-pool-id
enable
```

```
configure mobile-gateway profile dcca service-type
```

```
configure mobile-gateway profile charging credit-pool-id
```

```
configure mobile-gateway pdn apn default-cc-group
```

```
configure mobile-gateway pdn gy outage
  file-params
  location
```

When Gy Assume positive mode is enabled, configure the following commands in the config>mobile>profile>dcca context for cross check purposes:

- Set the usu-options-no-quota-mid-session command to **do-not-suppress**.
- Set the usu-options-no-quota-initial command to **cc-total**.

The Gy Assume Positive functionality supports the following KPIs:

- the number of sessions that currently have Gy Assume Positive triggered by a CC group
- the number of Gy Assume Positive sessions triggered over the PGW/HA lifetime per CC group
- the error conditions that triggered Gy Assume Positive over the lifetime of the PGW/HA
- the last session that applied Gy Assume Positive, the reason, and timer

The error conditions that triggered Gy Assume Positive provide the following details:

- the CCR-I, CCR-U, and CCR-T failure
- the specific cause code that triggered Gy Assume Positive
- the timeout for messages sent to the OCS
- the messages that did not leave the PGW due to backpressure
- the availability of the Gy link

The gateway uses the node ID name configured under the `config>mobile>pdn>charging>node-id` context as a prefix to the names of files that are created when storing a CCR, as specified by the Gy Assume Positive command `config>mobile>pdn>gy>outage`. For example, if the `node-id` is configured as `NOKIA-SAEGW01`, the name of the files could be `NOKIA-SAEGW01-AP-AS-RC.date-time`.

When `supported-vendor custom-1` is enabled, the CCFH sent by the OCS will be ignored and the local configuration will be applied as a result. This is required because the PGW honors the CCFH value sent by the OCS in the CCA message. If the OCS sends CCFH = Retry & Terminate, this it is not compatible with the Assume Positive feature (which requires CCFH set to = continue).

Use the following show commands to view information about Gy Assume Positive:

```
show mobile-gateway pdn apn gy-ap-statistics
```

```
show mobile-gateway profile dcca
```

```
show mobile-gateway pdn bearer-context acct gy detail
```

```
show mobile-gateway pdn statistics
```

```
show mobile-gateway profile charging credit-pool-id
```

12.18.10.1 Sample configuration

The following sample configuration shows Gy Assume Positive enabled with credit pools, and with the `service-type`, `access-type`, and `rating-group` parameters defined as wildcards. The storage location parameters are also defined.

```
config>mobile>profile>charging
  pool-id 10
    volume-quota      10000
    validity-time      300
    max-interim-quotas 3
  pool-id 50
    volume-quota      90000
    validity-time      400
    max-interim-quotas 1
  pool-id 60
    volume-quota      30000
    validity-time      0
    max-interim-quotas 2
dcca "abc"
  assume-positive
    enabled
```

```

service-type *
  access-type *
    message ccr-I ccr-u ccr-t
    rating-group *
      pool-id 60
service-type UC
  access-type *
    rating-group 400
      pool-id 60
    rating-group *
      pool-id 50
service-type RTR
  access-type wlan
    message ccr-I ccr-u
    rating-group 100
      pool-id 10
    rating-group 200
      pool-id 50

    rating-group *
      pool-id 60

  access-type *
    rating-group 300
      pool-id 60

    rating-group *
      pool-id 50
config>mobile>pdn>gy#
  outage
    location cf1
      primary
    exit
    location cf2

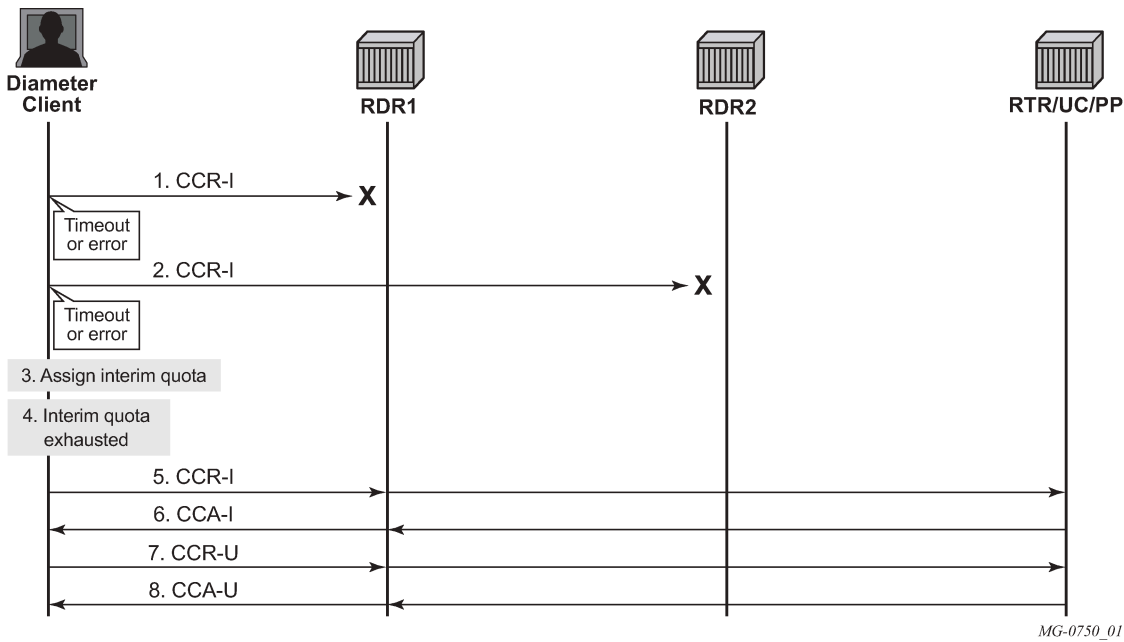
```

12.18.10.2 Gy Assume Positive use cases

The following call flows show various use cases when Gy Assume Positive is enabled.

12.18.10.2.1 CCR-I failure

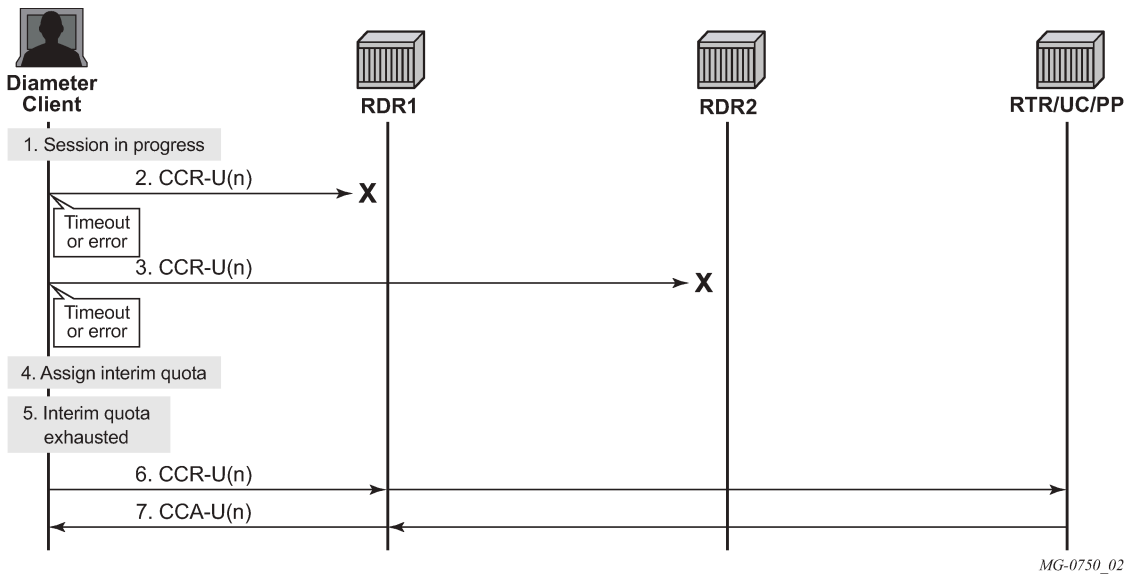
Figure 174: Gy Assume Positive response to CCR-I failure



1. A CCR-I to the primary RDR times out.
2. A Diameter peer retries the CCR-I on the secondary RDR. This also times out.
3. The Diameter peer starts working in Gy Assume Positive mode and assigns an interim quota to the session.
4. The interim quota is exhausted (either the volume is consumed or the time has expired).
5. The Diameter peer attempts the CCR-I to the primary RDR again.
6. The Diameter peer succeeds in getting a successful CCA-I response.
7. The Diameter client sends a CCR-U with the used amount from step 4.
8. The OCS records that usage and responds back with a successful CCA-U.

12.18.10.2.2 CCR-U failure

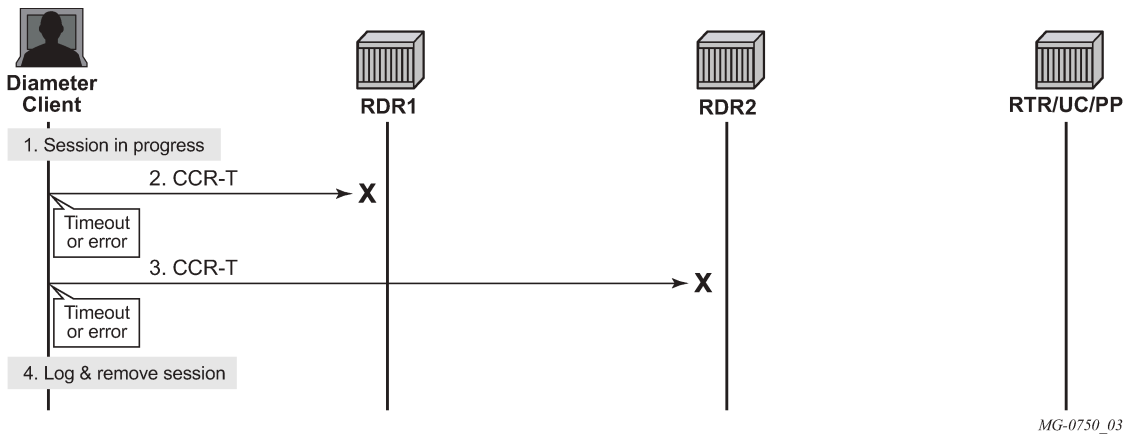
Figure 175: Gy Assume Positive response to CCR-U failure



1. A session is in progress where CCR-Is and CCR-Us are successfully exchanged between the Diameter peer and the OCS.
2. During the session, a CCR-U gets a timeout or error from the primary RDR.
3. The Diameter peer retries the secondary RDR and gets a timeout or error.
4. The Diameter peer starts working in Gy Assume Positive mode and assigns an interim quota to the session.
5. The interim quota is exhausted (either the volume is consumed or the time has expired).
6. The Diameter peer reattempts the CCR-U to the primary RDR. This time it receives a successful response from the OCS. This CCR-U reports the consumed amount from the interim quota.
7. The OCS records the reported usage and responds back with a successful CCA-U.

12.18.10.2.3 CCR-T failure

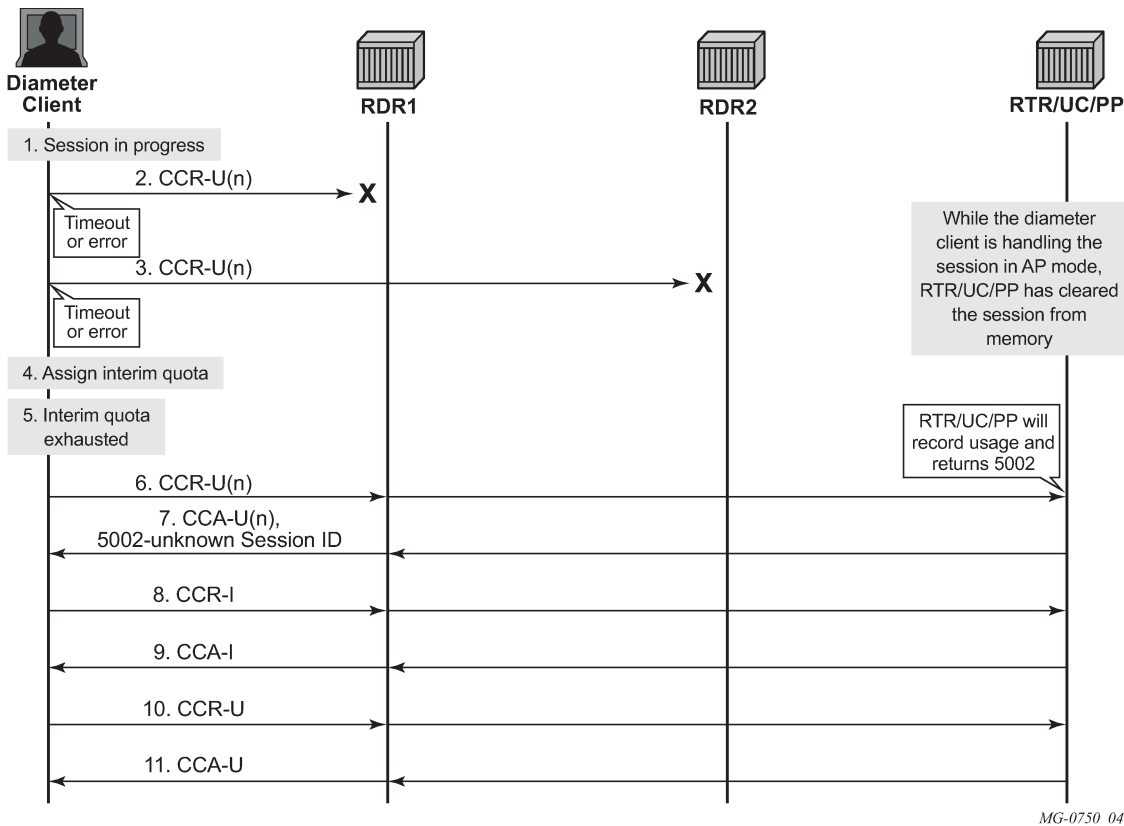
Figure 176: Gy Assume Positive response to CCR-T failure



1. A session is in progress where CCR-Is and CCR-Us are successfully exchanged between the Diameter peer and the OCS.
2. The session is terminated and a CCR-T is issued; however, the Diameter peer receives a timeout or error from the primary RDR.
3. The Diameter peer retries the CCR-T on the secondary RDR and still gets a timeout or error.
4. The Diameter peer logs the event in its local Gy Assume Positive file/directory and clears the session.

12.18.10.2.4 CCR-U unknown session with Gy session restart

Figure 177: CCR-U unknown session with Gy Assume Positive session restart

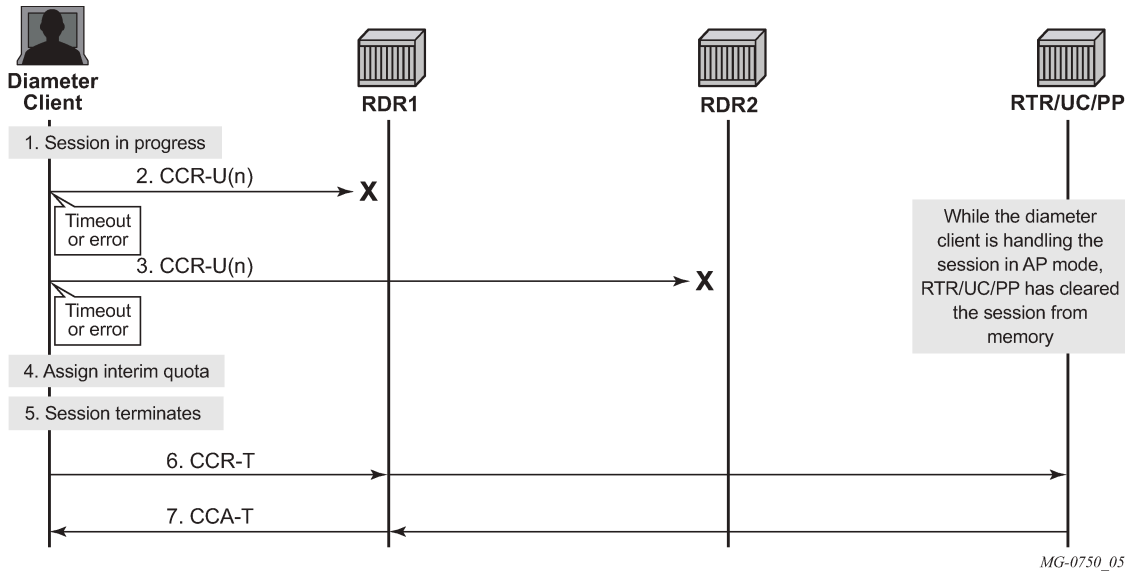


1. A session is in progress where CCR-Is and CCR-Us are successfully exchanged between the Diameter peer and the OCS.
2. During the session, a CCR-U gets a timeout or error from the primary RDR.
3. The Diameter peer retries the secondary RDR and gets a timeout or error.
4. The Diameter peer starts working in Gy Assume Positive mode and assigns an interim quota to the session.
5. The interim quota is exhausted (either the volume is consumed or the time has expired).
6. The Diameter peer reattempts the CCR-U to the primary RDR. It is now able to reach the OCS. This CCR-U reports the consumed amount from the interim quota.
7. The OCS no longer has the session in memory; therefore, it first records the reported usage and responds back with error code 5002 (unknown session ID).
8. Upon receipt of error code 5002, the Diameter peer gracefully clears the Gy session and starts a new Gy session by sending a CCR-I.
9. The OCS responds with a successful CCA-I.
10. Upon quota exhaustion, the Diameter peer sends a CCR-U.

11. The OCS responds with a successful CCA-U.

12.18.10.2.5 CCR-U session disconnect

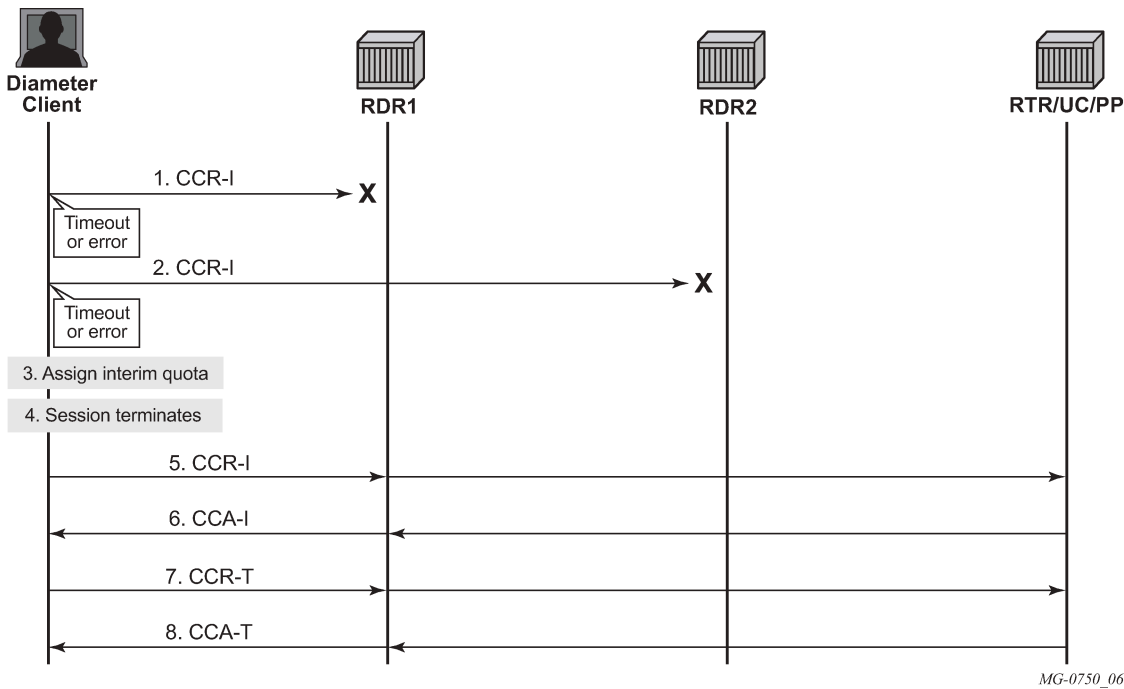
Figure 178: CCR-U session disconnect



1. A session is in progress where CCR-Is and CCR-Us are successfully exchanged between the Diameter peer and the OCS.
2. During the session, a CCR-U gets a timeout or error from the primary RDR.
3. The Diameter peer retries the secondary RDR and gets a timeout or error.
4. The Diameter peer starts working in Gy Assume Positive mode and assigns an interim quota to the session.
5. The session is terminated.
6. The Diameter client successfully sends the CCR-T to the OCS. The CCR-T reports the used amount from the interim quota.
7. The OCS records the reported usage and responds back with a successful CCA-T.

12.18.10.2.6 CCR-I session disconnect

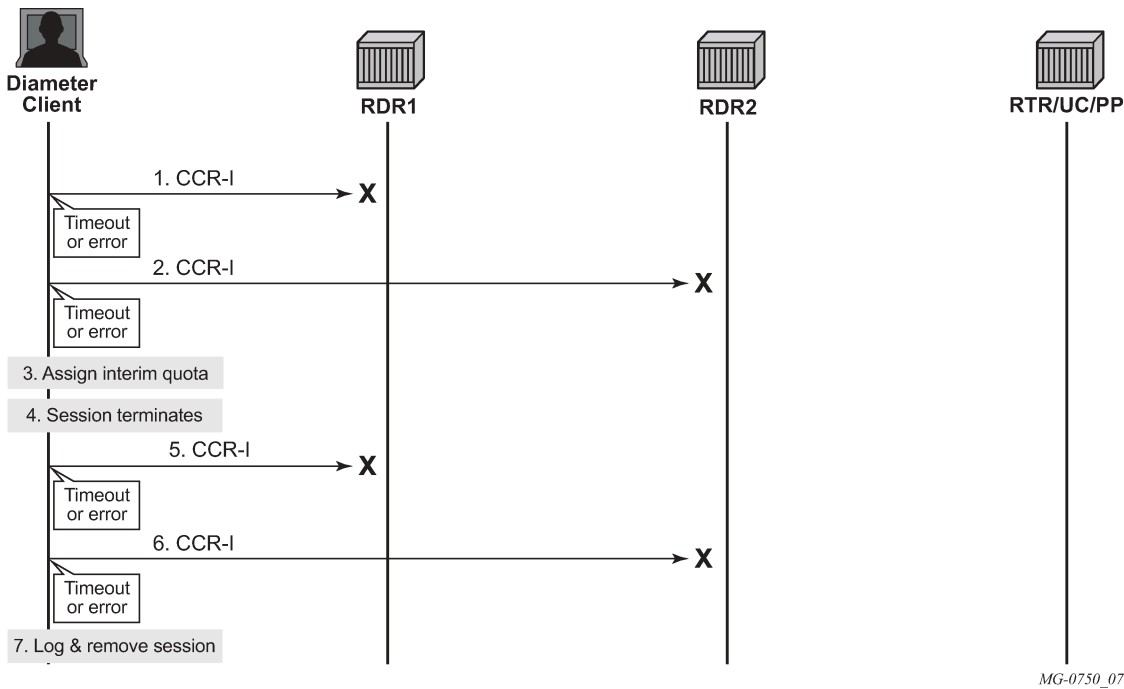
Figure 179: CCR-I session disconnect



1. A CCR-I to the primary RDR receives a timeout or an error.
2. The Diameter peer retries the CCR-I on the secondary RDR. This receives a timeout or an error.
3. The Diameter peer starts working in Gy Assume Positive mode and assigns an interim quota to the session.
4. Before the interim quota is exhausted, the session is terminated.
5. The Diameter peer successfully sends a CCR-I to the OCS.
6. The OCS responds back with a successful CCA-I.
7. The Diameter peer sends a CCR-T reporting usage from the interim quota.
8. The OCS responds back with a successful CCA-T.

12.18.10.2.7 Variation on a CCR-I session disconnect

Figure 180: Variation of a CCR-I session disconnect



1. A CCR-I to the primary RDR receives a timeout or an error.
2. The Diameter peer retries the CCR-I on the secondary RDR. This receives a timeout or an error.
3. The Diameter peer starts working in Gy Assume Positive mode and assigns an interim quota to the session.
4. Before the interim quota is exhausted, the session is terminated.
5. The Diameter peer attempts to send a CCR-I to the OCS. It receives a timeout or error from the primary RDR.
6. The Diameter peer retries the CCR-I on the secondary RDR. This also receives a timeout or an error.
7. The Diameter peer logs the event in its local Gy Assume Positive file/directory and clears the session.

12.18.11 Signaling message rate limiting

Rate limiting can be configured to signal messages over the Gy interface.

This capability accomplishes the following:

- controls the rate of sending messages (via a rate TPS)
- controls the queue size to enable some buffering when the rate exceeds the transactions per second (TPS)
- specifies the maximum rate (TPS), burst size

- specifies the delay tolerance (queue limit)
- specifies the queue high and low watermark threshold
- supports traps to indicate the status of the queue per threshold

The following commands configure rate limiting for messages over the Gy interface:

```
configure mobile-gateway profile sig-rate-limit-profile
  max-burst-size
  delay-tolerance
  max-message-rate
  queue-threshold
```

```
configure mobile-gateway pdn gy sig-rate-limit-profile
```

The following command associates a limit profile within a Diameter peer profile:

```
configure mobile-gateway profile diameter-peer sig-rate-limit-profile
```



NOTE:

The Diameter peer profile supports limit profiles only for Gx and Gy applications. Therefore, the application-type command in the config>mobile>profile>diameter-peer context can only be set to either **gx** or **gy**

The following alarms monitor Diameter message queue utilization per Diameter peer:

- tmnxMobGwDiaMsgQueueFullAlarm
- tmnxMobGwDiaMsgQueHighThrsAlarm
- tmnxMobGwDiaMsgQueLowThrsAlarm

For more information, refer to the *CMG Log Events Guide*.

To view information about signaling rate limit profiles, use the command in the following context:

```
show mobile-gateway profile sig-rate-limit-profile
```

12.18.12 Final Unit Indication

This section provides information about the supported FUI functionality and configuration including FUI support at the MSCC level and command level, filter IDs with FUI Restrict and Redirect and FUA local policy for filter IDs.

FUI is supported at the Multiple Services Credit Control (MSCC) level and at the command level of a CCA message. FUI at the command level in a CCA supports FUA Terminate or FUA Redirect. For FUA Terminate, a CCR-T is sent to the OCS, to terminate the credit control session. The FUA Terminate can be received within

FUI at the command level in a CCA-I or a CCA-U. In both cases, the credit control session with the OCS is terminated and the corresponding PDP context/bearer is terminated.

FUI is present either in the MSCC AVP or at the command level, when independent credit control of multiple services is used. A FUI Redirect at the MSCC level, puts the corresponding Rating Group (RG) in Redirect mode. FUA Redirect at the command level puts the whole session in redirect mode.

For the command-level FUI redirect the following applies:

- The OCS sends the RAR and the gateway sends the CCR-U with RSU for the requested RG or a default RG (if configured), or one of the RGs, when no RG is specified in the RAR. Successful quota allocation by the OCS for the RG results removes the Redirect for the session.
- The VT (specified at command level by OCS) expiration, results in the gateway sending a CCR-U with RSU for a default RG (if configured), or one of the RGs. Successful quota allocation by the OCS for the RG results in the Redirect being removed for the session.

For Gy sessions set up on a per IP-CAN session basis (for example, hybrid LTE/CDMA deployments), Terminate and Redirect apply for the entire session.

To configure command-level FUI, use the following command:

```
configure mobile-gateway profile dcca command-level-fui
```

To view command-level FUI configuration information, use the following show command:

```
show mobile-gateway profile dcca
```

To view command-level FUI redirect information for the bearer-level Gy, use the following command:

```
show mobile-gateway pdn bearer-context imsi bearer acct gy detail
```

To view command-level FUI redirect information for the session-level Gy, use the following command:

```
show mobile-gateway pdn pdn-context imsi acct gy detail
```

To view information about the number of OCS sessions that have one or more RGs in Redirect or Terminate state (MSCC Level FUI (terminate, restrict, redirect) and command-level redirect), use the following command:

```
show mobile-gateway pdn ref-point-stats gy
```

12.18.12.1 Multiple filter IDs with FUI Restrict and Redirect

The OCS can specify multiple filter IDs with FUI Redirect and FUA Restrict for the Gy interface.

A filter ID maps to a predefined Policy Rule Base (PRB) (post-processing) on the PGW. The specified filter IDs are enabled typically to allow traffic to the DNS server, Redirect server, and so on. The HTTP traffic is redirected to the URL specified by the OCS.

**NOTE:**

Filter IDs with FUI Restrict and Redirect are supported at the MSCC level only. They are not supported at the command level of a CCA message.

To view information about FUI Redirect with filter IDs, use the following command:

```
show mobile-gateway pdn bearer-context imsi bearer acct gy detail
```

12.18.12.2 Final Unit Action local policy for filter IDs

This feature provides the ability to configure the Final Unit Action (FUA) rules for filter IDs sent from the OCS within a local policy.

This can be used if the OOC event trigger notification experiences a problem interacting with the PCRF. Normally when the PCRF is available, the OOC is sent and a FUI is propagated along with filter IDs received from the OCS. However, when the PCRF is unavailable, the filter IDs can be enforced locally.

The policy rule selected by a filter ID is activated and de-activated locally by the gateway. It is activated when the OCS refers to the filter ID and de-activated when the FUA condition is removed. The filter ID policy rule is activated prior to reporting an OOC to a PCRF.

Typically IP filters within the policy rule selected by a filter ID allow traffic to specific sites while the FUI action (FUA Restrict and FUA Redirect) is in effect.

For a local policy with FUA Restrict, the following applies:

- the action is applied to already installed rules associated with this RG
- filter IDs are removed for the session locally, when Restrict is no longer applicable

For a local policy with FUA Redirect, the existing behavior for getting out of redirect applies:

- a RAR from the OCS is followed by a CCR-U/CCA-U with a valid quota
- a VT expiration or Event-Trigger results in a CCR-U/CCA-U with a valid quota

To configure filter ID action, use the following command:

```
configure mobile-gateway profile policy-options action-rule-unit gy-filter-id-action
```

To view information about filter ID actions, use the following command:

```
show mobile-gateway pdn bearer-context acct gy detail imsi bearer
```

12.18.13 CCR event reporting

CCR Events (CCR-E) are used to report PLMN, time zone, and location changes to the OCS.

The OCS uses these real-time updates for certain business logic; for example, to apply Usage Control time-of-day restrictions. The Gy CCR-E messages are enabled by the PCRF over Gx via custom operator Trigger AVPs at the command level in a CCA. This feature applies to 4G/LTE subscribers anchored on the PGW.

**NOTE:**

CCR-E messages will not be sent if there is an online charging session established with the OCS (CCR-I) for any bearers within the PDN session. Instead a CCR-U would be sent for any application OCS subscribed triggers.

Table 54: Trigger types for PLMN and time zone reporting

Trigger type	PGW action
Change_in_TimeZone(5)	Time Zone Change Event reporting
Changeinlocation_MCC (30)	MCC Change reporting (first half of PLMN)
Changeinlocation_MNC (31)	MNC Change Reporting (second half of PLMN)
Change_in_Location(3)	Updated ULI for all ULI/PLMN changes (sent by the PGW) Should not be enabled in conjunction with MCC and MNC location change Trigger types

Use the following show commands to view information about PLMN, time zone, and location event reporting:

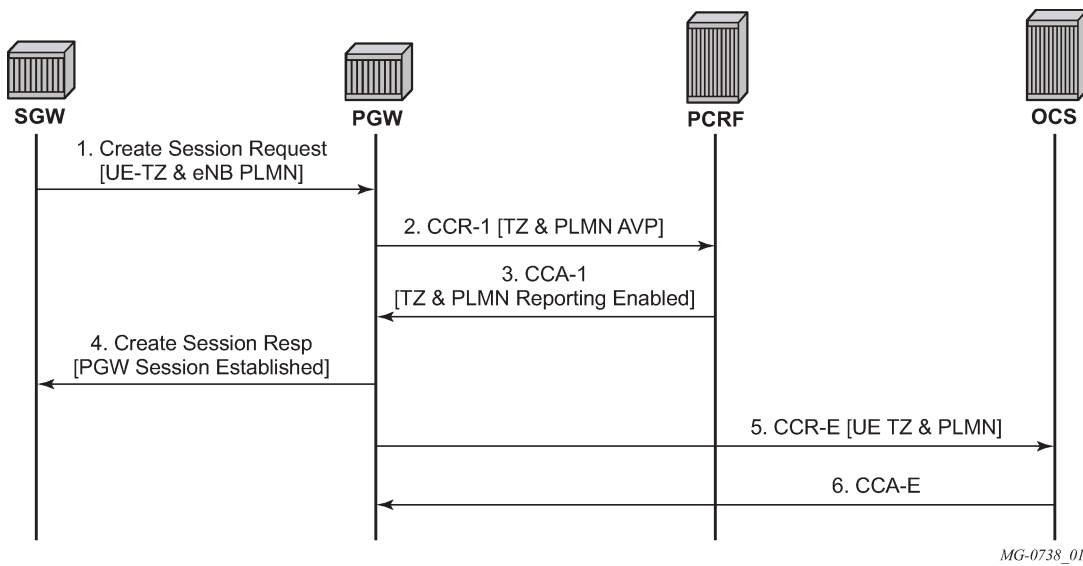
```
show mobile-gateway pdn ref-point-stats gy
```

```
show mobile-gateway pdn pdn-context imsi detail
```

12.18.13.1 Session creation call flow

The PGW reports the Time Zone and PLMN to the UE on initial attach.

Figure 181: Session creation for time zone and PLMN reporting

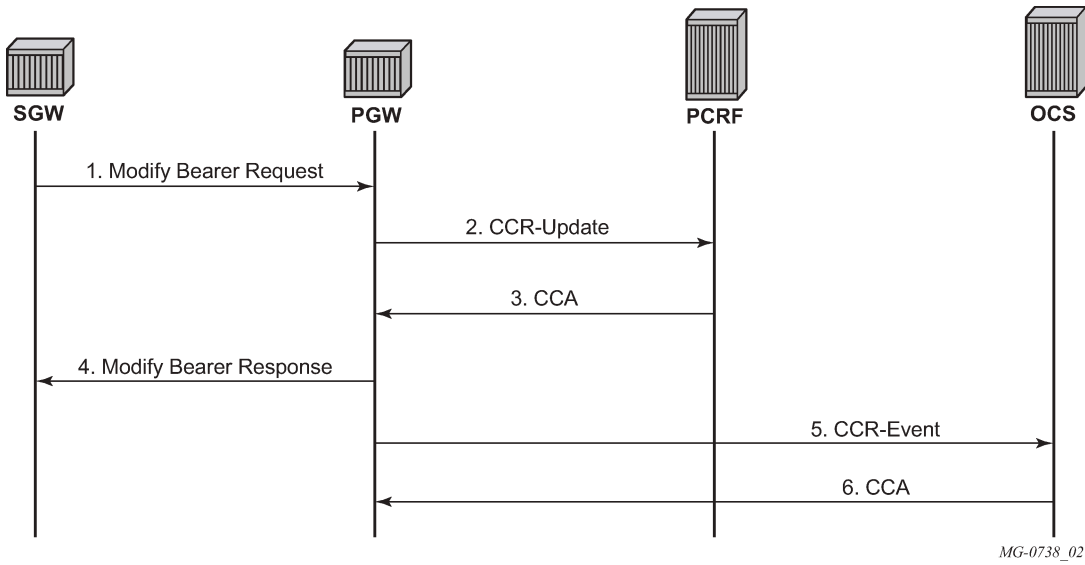


1. The create session request for the IMS PDN contains the UE time zone and serving network eNodeB PLMN.
2. The CCR-I to the PCRF includes the UE time zone and PLMN AVPs.
3. The CCA-I to the PGW includes the event triggers for changes in the time zone and PLMN.
4. The PGW session is established with the Create Session Response to the SGW.
5. The PGW sends the CCR-E to the OCS with the UE time zone and PLMN.
6. The OCS returns the CCA-E.

12.18.13.2 PLMN and time zone update call flow

When the UE attaches to a new eNodeB with a new time zone or PLMN identifier, the serving network (eNodeB, MME, SGW) generates a modify bearer request to the PGW. The MME or SGW may also change during inter-eNodeB handover.

Figure 182: PGW update for changes in UE time zone and PLMN

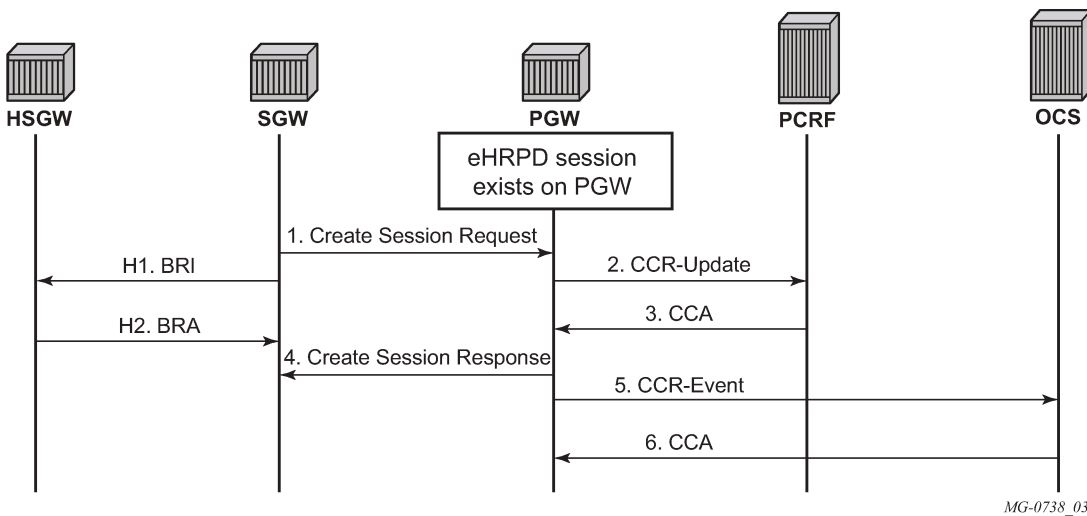


1. The SGW sends the Modify Bearer Request to the PGW for update in the serving network (eNodeB) PLMN.
2. The PGW sends a CCR-U (including the updated UE time zone and/or PLMN AVPs) to the PCRF.
3. The PCRF responds with a CCA that includes the trigger AVP.
4. The PGW sends the Modify Bearer Response to complete the PLMN/time zone update procedure.
5. The PGW sends a CCR-E (including the updated UE time zone and PLMN AVPs) to the OCS.
6. The OCS responds with a CCA.

12.18.13.3 eHRPD to LTE handover call flow

The PGW reports the UE time zone and PLMN during all eHRPD to LTE handovers. Note that the LTE to eHRPD handover does not require Gy event handling.

Figure 183: eHRPD to LTE handover



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1. The SGW sends a Create Session Request for IMS PDN attaches to the PGW. The Create Session Request contains the UE time zone and serving network (eNodeB) PLMN.
2. The PGW sends a CCR-U (including the UE time zone and PLMN AVPs) to the PCRF.
3. The PCRF responds with a CCA that includes the trigger AVP.
4. The PGW sends a Create Session Response to establish the PDN session on LTE and complete the handover.
5. The PGW sends a CCR-E (including the updated UE time zone and PLMN AVPs) to the OCS.
6. The OCS responds with a CCA.

12.18.14 CCR-I on first rating group detection

The ability to configure sending a Credit Control Request Initial (CCR-I) message on first online rating group (RG) detection is provided.

This feature supports enabling and disabling online charging during the life of a bearer, while maintaining the Gy session throughout. This includes inter-RAT handover scenarios where both billing method and rating group can be changed for a specified bearer.

The Gy online charging session is started toward the OCS when a PCC rule with online enabled is installed for the bearer. For example, a PDN session with a default bearer exists without online charging enabled. When a dedicated bearer with online enabled (PCC-rule) is established, a CCR-I is sent to the OCS. Online charging toward the OCS continues as long as the bearer exists. The Gy session is terminated when the corresponding bearer is terminated.

Note that CCR-I message on first online RG detection and CCR-I message on first packet detection (config>mobile>profile>dcca>ccr-initial-on-first-pkt) cannot be enabled at the same time.

Use the following CLI command to configure CCR-I on first online RG detection:

```
configure mobile-gateway profile dcca ccr-initial-on-first-online-rg
```

12.18.15 Multiple Diameter peers with load balancing for Gx and Gy interfaces

This feature provides the ability to create a list of Diameter peer groups and associate that list with Gx and Gy interfaces to facilitate load balancing.

Each group can support a primary and a secondary pair of Diameter peers. In this case, S1 acts as a secondary peer for primary peer P1 and S2 acts as a secondary peer for primary peer P2. The default priority for a group is set to 20, when the priority is not specified. Priority 1 is the highest and 20 is the lowest. The priority can change on the fly. When the priority is not specified or is the same for more than one group, load-balancing (round-robin) applies across all the groups. The traffic is sent through the connection with the highest priority. To support a single cluster, the secondary pair is not specified. If the secondary pair is not specified, failover does not occur to the secondary pair in a single group or cluster and load-balancing applies across all groups.

Use the following CLI commands to create a list of Diameter peer groups:

```
configure mobile-gateway pdn gx pcrf-selection diameter-peer-group-list
```

```
configure mobile-gateway pdn gx pcrf-selection user-range
```

```
configure mobile-gateway pdn apn pcrf-selection diameter-peer-group-list
```

```
configure mobile-gateway pdn gy diameter-peer-group-list
```

```
configure mobile-gateway pdn apn charging diameter-peer-group-list
```

```
configure mobile-gateway profile charging pgw-charging on-line diameter-peer-group-list
```

```
configure mobile-gateway profile charging tdf-charging on-line diameter-peer-group-list
```

```
configure mobile-gateway pdn gy ocs-selection user-range
```

```
configure mobile-gateway profile diameter-peer-group-list
```

12.18.16 OCS selection based on charging characteristics and IMSI/MSISDN range

This feature provides the ability to select an OCS based on a combination of charging characteristics and the IMSI or MSISDN range.

To specify whether the OCS selection criteria is based on an IMSI or MSISDN range and to specify the user range, use the following commands:

```
configure mobile-gateway profile charging pgw-charging on-line selection-type
```

```
configure mobile-gateway profile charging tdf-charging on-line selection-type
```

```
configure mobile-gateway profile charging pgw-charging on-line user-range
```

```
configure mobile-gateway profile charging tdf-charging on-line user-range
```

To view PGW and PDF charging information, use the following commands:

```
show mobile-gateway profile charging pgw-charging
```

```
show mobile-gateway profile charging tdf-charging
```

12.18.17 Configurable event triggers

As per standard behavior, events are enabled by the OCS server in a CCA message. CMG also supports configurable event triggers for Gy online charging. Operators can configure the required event triggers locally.

PGW/GGSN currently supports the following Bearer/Session level Event Triggers:

- QoS change
- RAT change
- Serving node change
- User location

The OCS-provided event triggers will override the locally configured event triggers. For example, if the OCS sends certain event triggers in a CCA, those event triggers are the ones that are enabled (ignoring the locally configured ones).

For configuration information, see the `event - triggers` command and related command descriptions in the `config>mobile>profile>dcca` context.

For show command output, see the `show>mobile>profile>dcca` command.

12.18.18 Gy terminate session on receiving FUA=Terminate for an MSCC failure code

This feature facilitates terminating a PDP context/bearer upon receiving an FUA=Terminate message in the CCA for a configured MSCC result code.

The `fui-terminate-session` command (in the CCFH profile context) can be configured on a per MSCC result code basis, and when enabled, the PDP context/bearer immediately terminates when an FUA=Terminate message in CCA is received for the configured MSCC result code (for example, 4012) and no GSU or GSU with zero units are included. The PDP context/bearer is terminated regardless of whether other RGs have ongoing accounting tasks.

12.18.19 Out-Of-Credit message with FUA=Terminate for Gy result code 4012

This feature enhances the Out-Of-Credit message sent to the PCRF by including a Final-Unit-Indication AVP that has a Final-Unit-Action sub-AVP with a Terminate value.

This happens when the OCS terminates an MSCC instance with Result Code 4012 but does not include the Final-Unit-Indication (FUI) AVP.



NOTE:

This feature is enabled by default when the `rating-group-in-ch-rule-report` command is enabled.

12.18.20 OCS reply on MSCC request for quota

Configure a list of rating groups for which quota is requested at the CCR-I that is sent to the OCS.

The PCEF includes in the CCR-I the Requested-Service-Units AVP within MSCC for the list of preconfigured rating groups according to the active rulebase. Apart from the Requested-Service-Units AVP, the MSCC is empty.

The OCS may grant quota to the requested MSCCs or not. If OCS does not reply to a CCR-I, MSCC is not considered an error. It is treated as Bad-Answer.

When quota for an MSCC at CCR-I is requested, the mobile gateway can be configured to treat the CCA-I, which does not contain the MSCC instance, as if zero quota is allocated. Further quota is requested for that MSCC only when the first UE packet matches the MSCC.

To configure the functionality for CCR-I quota request handling with no MSCC instance in CCA-I, use the following command:

```
configure mobile-gateway profile dcca cca-init-no-mscc-answer-ignore
```

To configure the list of rating groups for which quota is requested at the CCR-I, use the following command:

```
configure mobile-gateway profile dcca default-rg-list rating-group
```

12.18.21 Serializing CCR handling on Gy sessions with OCS

The OCS is unable to process multiple CCR messages. You can configure the gateway to wait for a CCA before sending the next CCR and sends CCR messages per RG in sequential order only if a reply from a previous CCR message has been received.

Only one CCR message can be processed at a time and all the following CCR messages sent before a reply from the previous CCR message is received, are rejected with error code DIAMETER_OUT_OF_SPACE (4002).

To enable the gateway to wait for a CCA for a current CCR before sending the next CCR, use the following command:

```
configure mobile-gateway profile dcca serialize-ccr
```



NOTE:

The `serialize-ccr` command is supported only for bearer-level Gy sessions.



NOTICE:

When the `serialize-ccr` command is enabled, the following commands must not be enabled:

- `config>mobile>profile>dcca>disable-first-pkt-behavior`
- `config>mobile>profile>charging>ccfh-profile>assume-positive`
- `config>mobile>profile>dcca>pre-emptive-quota`
- `config>mobile>profile>dcca>ccr-initial-on-first-pkt`
- `config>mobile>profile>dcca>ccr-initial-on-first-online-rg`
- `config>mobile>profile>charging>chf-profile>default-rg-list`
- `config>mobile>profile>command-level-fui`

It is possible that two or more events can occur for the same RG while the gateway waits for a chance to send a CCR(U). In this case, a single event is generated. USU reported are cumulative and usage reported is accurate.

The order of importance for the event generated is as follows:

- Reporting Reason Final
- Forced re-authorization
- Rating Condition Change

- Validity Time reached
- QHT expiration

Quota exhaustion and quota threshold are reported at the USU level and the other events are reported at the MSCC level. Thus, reporting is made at both levels in the same MSCC which means quota is exhausted and then VT expires. The MSCC has VT as the reporting reason at the MSCC level and quota exhaustion at the USU level.

13 Policy configuration

Information and configuration guidelines for policy control implementation for EPC nodes.

For CUPS-specific policy information, refer to the *CMG CUPS Configuration Guide* and *CMG PFCP (Sx and N4) Interface Description*.

13.1 Enabling release 13 policy and charging control CRs

The Change Requests (CRs) for Release 13 3GPP standards applicable to Policy and Charging Control functionality are supported.

The following specifications are applicable:

3GPP TS 23.203 v13.10.0 (2016-12)	Policy and Charging Control Architecture
3GPP TS 29.212 v13.8.0 (2016-12)	Policy and Charging Control; Reference Points
3GPP TS 29.213 v13.8.0 (2016-12)	Policy and Charging Control; Signaling Flows and QoS Parameter Mapping

The following functionality is supported:

- mission critical QCIs (new QCI values 65, 66, 69, 70 are supported to help operators better serve the needs of public safety services)
- RAN-NAS cause
- time-conditioned APN-AMBR
- PCC enhanced with ADC
- SSG and TDF functionality with Sd interface to PCRF

To enable the Application Based Charging (ABC) feature bit settings for supported features AVP sent to the PCRF, use the following command:

```
configure mobile-gateway profile gx supported-features app-based-charging
```

To view the related feature settings, use the following commands:

```
show mobile-gateway profile gx
```

```
show mobile-gateway profile policy-options gx-avp-options
```

13.2 Policy configuration using CLI batch capability

This section describes the use of the CLI batch capability for configuring policies (policy rule units, policy rules, and policy rule bases) in the following contexts:

```
configure mobile-gateway profile policy-options
```

```
configure mobile-gateway pdn apn static-local-pcc
```

13.2.1 Begin, commit, and abort commands

CLI batch capability is supported when configuring policy definitions (policy rule units, policy rules, policy rule bases, and action rule units) in the following contexts:

```
configure mobile-gateway profile policy-options
```

```
configure mobile-gateway pdn apn static-local-pcc
```

CLI batch commands are used as follows for creating and editing policies.

- The `begin` command enters the mode to create or make modifications to policies. The editing session can continue until the changes are either committed or aborted. The session will not be interrupted by high-availability activity switches or CLI session terminations, such as closing a Telnet session.
- The `commit` command saves and activates the changes made during an editing session. The newly committed policy takes effect immediately for all new flows. Existing flows will transition onto the new policy shortly after the commit.
- The `abort` command discards changes that have been made during an editing session.

To allow flexible order for policy editing, the `commit` function cross-references policy components to verify the update, including the following:

- all ASO characteristics have a default value and are defined in the `app-profile`
- limits checking

13.2.2 Beginning and committing a policy configuration

To enter the mode to create or edit policies, you must enter the `begin` command. The changes you make in edit mode will take effect in the system when you issue the `commit` command.

If you attempt to modify a policy without first entering the `begin` command, the following error message displays.

```
A:ALA-B>config>mobile>profile>policy-options#
```

```
MINOR: AA #1005 Invalid Set - Cannot proceed with changes when in non-edit mode
```

Use the following CLI syntax to begin a policy configuration.

```
configure mobile-gateway profile
policy-options
begin
```

```
configure mobile-gateway pdn apn
static-local-pcc
begin
```

Use the following CLI syntax to commit a policy configuration.

```
configure mobile-gateway profile
policy-options
commit
```

```
configure mobile-gateway pdn apn
static-local-pcc
commit
```

13.2.3 Aborting a policy configuration

The abort command can be used to discard changes that have been made to policies during an edit session.

Use the following CLI syntax to abort a policy configuration.

```
configure mobile-gateway profile
policy-options
abort
```

```
configure mobile-gateway pdn apn
static-local-pcc
abort
```

13.3 Policy modification without session shutdown

When modifying policies (policy rules, policy rule bases, policy units) that are referenced under an APN or installed on a UE (via PCRF or APN configuration), it is not necessary to remove the policy from the APN or

the UE session. By default, modifications to policies are applied to both new and existing sessions, except if there is a conflict with an existing session, in which case the policy will be rolled back.

Policy modifications configured in the following contexts support this functionality:

```
configure mobile-gateway profile policy-options
  policy-rule-unit
  policy-rule
  charging-rule-unit
  policy-rule-base
```

```
configure mobile-gateway pdn apn static-local-pcc
  policy-rule
  policy-rule-base
  dedicated-bearers default-bearer-qci arp
```

The following is the system default behavior for policy updates without a session shutdown:

- policy updates are applied to both new and existing sessions
- if a conflict occurs with an existing session, policies are automatically rolled back

You can change the default system behavior for policy updates. Modified policies can be applied to new sessions only or to new and existing sessions with a `reactivate` option. When the `reactivate` option is selected and a session conflict occurs, the session terminates with a "reactivation-required" cause code and then picks up the updated policy upon reattachment.

You can modify the default behavior for policy updates, using the following command:

```
configure mobile-gateway pdn apn policy-on-the-fly
```

The following options can be selected for applying policy updates:

- Apply policy updates to new sessions only.
- Apply policy updates to new and existing sessions, using policy roll back in the case of a session conflict. This is the default mode.
- Apply policy updates to new and existing sessions using the `re-activate` option. In the case of a conflict, the session will terminate with a "reactivation-required" cause code and pick up the updated policy upon re-attach.

You can view information for policy rules, policy rule units, policy rule bases, and charging rule units, such as policies in use, session references, policy rules, and configuration references, using the applicable options for the following `show` command:

```
show mobile-gateway profile policy-options
```

You can monitor the results of policy updates using the following command:

```
tools dump mobile-gateway policy-on-the-fly-update-stats
```

The following constraints apply for policy updates without session shutdown:

- If any modification fails toward the UE, the PDN session terminates with a reactivation-required cause code.
- References to profiles within a policy rule unit can change, but the contents of those profiles cannot change.

13.4 Gx interface configuration

The Gx interface policy and charging configuration options supported by the CMG.

13.4.1 Gx profile for APN-level timer and error handling

A Gx profile can be configured and applied for the PDN or on a per APN basis. The profile supports timers that are configurable on a per message basis and error handling that is configurable on a per error-code basis.

For deployments with the PCRF, failure handling knobs are supported (no response, fh-3xxx, fh-4xxx) to enable a PDN session to be setup and continue without further PCRF control. Such sessions could end up as long standing sessions until the UE detaches or the session is terminated at the PGW due to idle timers (if used). To provide better control over how long such PDN sessions exist (fall back to static policy without PCRF control), configurable volume and time limits are supported. Upon expiration of the specified volume or time limit (whichever occurs first) the session is terminated.

As part of the Gx profile, CLI commands are provided to enable and disable population of the feature ID bits for optional features within the supported features AVP sent to PCRF.

For information about the CLI configuration, refer to the *Gx Profile Commands* and the `dynamic-pcc` and `gx-profile` commands in the `config>mobile>pdn>apn>pcrf-selection` context in the *CMG CLI Reference Guide*.

13.4.2 Gx synchronization after session or message loss

A stateless session discovery and recovery Diameter application is supported between the PCRF and PGW, for Gx session synchronization and recovery. The application enables the PCRF to query the PGW about a subscriber's IP-CAN session and request the PGW to re-establish/recover the Gx session.

The session recovery only applies to custom IMS APNs and is only relevant to home scenarios.

A set of request/answer commands is supported for this feature:

Session Discovery Request (SDR)

Enables the PCRF to do the following:

- query the PGW for Gx sessions associated with an IMSI and IPv6 address prefix
- request the PGW to re-establish the Gx session, if one exists

Session Discovery Answer (SDA)

Enables the PGW to inform the PCRF if it finds corresponding sessions; the PGW returns a success message if it locates the IMSI.

NOTE:

Following a successful SDA, if the PGW does not find a Gx session corresponding to IMSI and IPv6 prefix, it will not send a subsequent CCR-U to recover the session.

If the PCRF and PGW indicate support for this feature, the following event triggers are enabled by default, without requiring a subscription:

SESSION_RECOVERY

Supports session recovery scenarios

SESSION_SYNC

Supports session synchronization scenarios and is used in responses when the PCRF sends a "Session-Sync-Requested" AVP within a CCA or a RAR over the Gx session

Figure 184: PCRF-initiated session recovery

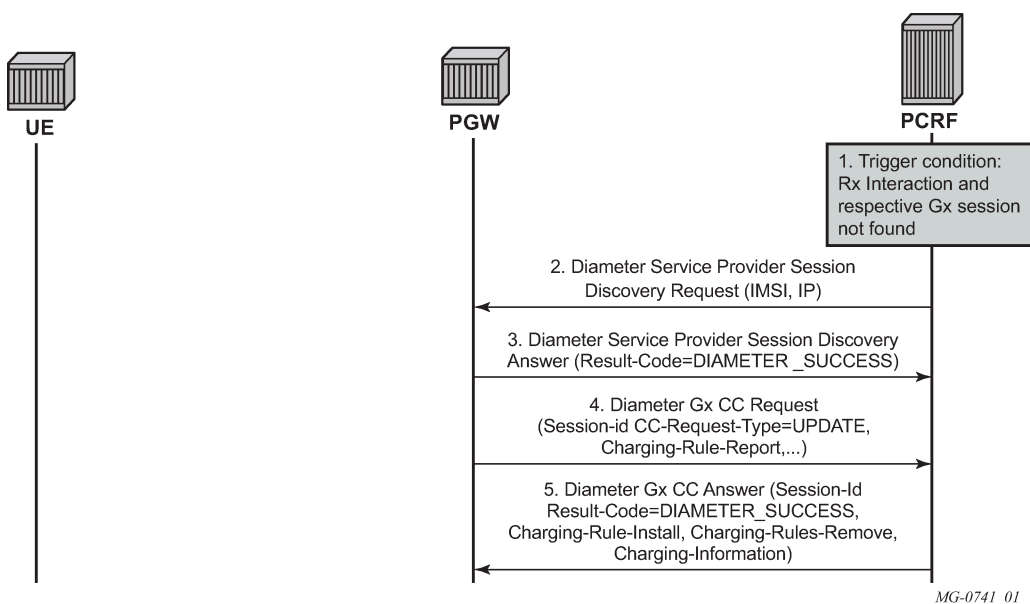
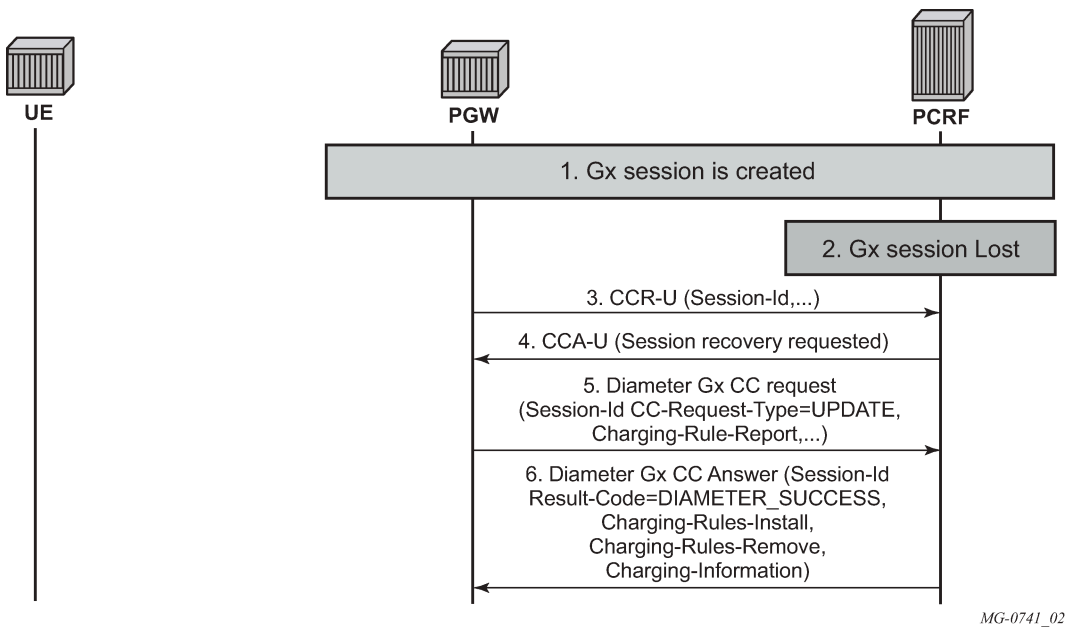


Figure 185: PGW-initiated session recovery



To enable Gx synchronization after session or message loss functionality, use the following commands:

```
configure mobile-gateway pdn apn enable-gx-session-recovery
```

```
configure mobile-gateway pdn gx supported-vendor
```

13.4.3 Gx profile based failure handling

For Gx profile based failure handling, *Gx profile CCFH action per call flow type* shows expected behaviors for "Continue" and "AP-Continue", when the system receives a CCA-U with a failure code from the PCRF. For more information about Gx-profile configuration, see the Gx Profile Commands in the *CMG CLI Reference Guide*.

Table 55: Gx profile CCFH action per call flow type

–	Gx profile CCFH setting	
	Continue	AP-Continue
Handover	Terminate Session	Activate Gx AP and Continue
ULI Change		

Table 55: Gx profile CCFH action per call flow type (continued)

–	Gx profile CCFH setting	
	Continue	AP-Continue
SGW Relocation		
Failure code 5002 (DIAMETER_UNKNOWN_SESSION_ID)	Terminate Session	Terminate Session

The mobile gateway supports terminating the next attempt as a failure handling action. When configured, if the SCTP connection is down (or a new OSI has been received, indicating that PCRF has restarted), the mobile gateway does not take any action. The session is terminated when the gateway attempts to send a CCR.

To configure the terminate next attempt failure handling action, use the following commands:

```
configure mobile-gateway profile gx profile-name fh-session-no-response
terminate-next-attempt
```

```
configure mobile-gateway profile npcf profile-name fh-session-no-response
terminate-next-attempt
```

For more information about configuring the fh-session-no-response command, refer to the *CMG CLI Reference Guide*.

13.4.4 VoLTE UE ULI support for billing

This operator custom feature supports the propagation of the ULI to the IMS domain at the time of a VoLTE call or dedicated bearer setup. The PCRF subscribes to the SUCCESSFUL_RESOURCE_ALLOCATION Event Trigger. The PGW sends the ULI received in a create bearer response to the PCRF via a CCR-U, with the SUCCESSFUL_RESOURCE_ALLOCATION Event Trigger, and the PCRF propagates it to the P-CSCF.



NOTE:

The ULI is also sent to the PCRF when the Event Trigger is IP-CAN-CHANGE or RAT-CHANGE [to LTE].

Use the following command to facilitate the described behavior:

```
configure mobile-gateway pdn gx supported-vendor
```


13.4.5 Override control for predefined and static rules

The Override-Control AVP applies to predefined and static rules that are instantiated subsequently. It may be used, for example, to change the RG of predefined or static rules when a session is being throttled (at a lower APN-AMBR).

On the Gx interface, Override-Control is a command-level AVP that can be present in a CCA-I, CCA-U, or RAR. This grouped AVP is used to override configured charging parameters for predefined and static rules. This limits the need to configure multiple rules on the gateway.

The following override control functionality is supported:

- Applies to predefined and static rules.
- Static rules apply to the default bearer only.
- When requested by the PCRF in a CCA/RAR, applies to the specific subscriber PDN session only (predefined rules can be referenced by multiple sessions).
- Does not apply to post processing rules (for example, filter IDs received from the OCS).
- Takes precedence over the inheritance feature and replaces it.
- Does not support QCI/ARP changes within the current definition of the Override Control AVP.
- Supports wildcards for the Override-Rule-Name and Override-Charging-Action-Exclude-Rule AVPs within the Override-Control AVP, for sending of partial rule names and partial exclude-rule names respectively.

Rules or exclude-rule names matching the partial string can be overridden or excluded. The delimiter, `<*>`, is used to send partial names over the Gx interface as follows:

- Pattern `<*>` indicates the names start with partial string and end with anything.
 - `<*>` Pattern indicates that the names start with anything and end with a partial string.
 - `<*>` Pattern`<*>` indicates that the names contain a partial string.
- Supports Override-QoS-Information and Override-Max-Requested-Bandwidth-UL/DL AVPs within the Override-Charging-Action-Parameters.

Predefined rules within the PGW are defined with QoS values that can be changed based on the subscriber entitlement and/or other relevant service usages. This can be done by creating new equivalent rules in the PGW, and/or changing the predefined rule QoS values via a PGW MOP. However, since these options may be time-consuming, alternatively it is possible to pass the new QoS values via a policy, using Override-Control QoS AVPs to change the predefined rules that are applied to a subscriber.

The following command must be enabled to activate Override Control for predefined and static rules (the default is no supported-vendor):

```
configure mobile-gateway pdn gx supported-vendor custom-1
```

Supported sub-AVPs of the Override-Control AVP describes the supported sub-AVPs of the Override-Control AVP. These AVPs can be sent in CCA-I, CCA-U, and/or RAR.

Table 56: Supported sub-AVPs of the Override-Control AVP

AVP		Category	Description	Type
Override-Rule-Name		O	Specifies the rule or rule base name (predefined or static) for which override values are sent Supports sending of partial rule names May be included more than once, if the override values apply for multiple rules	Octet String
Override-Charging-Action-Parameters		O	Overrides the charging parameters configured at the PGW for a static/predefined rule The overridden parameters apply to all static/predefined rules except those specified by Override-Charging-Action-Exclude-Rule	Grouped
	Execution Time	O	As defined in 3GPP TS 29.212 The Vendor ID bit of this sub AVP is 3GPP	Time
	Override-Charging-Action-Exclude-Rule	O	Defines the rule or rule base name for which override-control shall not be applied Rules may be excluded from the PCRF override action by specifying them within the Override-Charging-Action-Exclude-Rule AVP	Octet String
	Override-Charging-Parameters	O	Overrides the charging parameters configured at the PGW for a static/predefined rule	Grouped
	Override-Rating-Group	O	Overrides the value of the rating group configured for a static/predefined rule	Unsigned32
	Override-Online	O	Overrides the Online value configured for a static/predefined rule	Enumerated
	Override-Offline	O	Overrides the offline flag configured in the charging action specified by the Charging-Action-Name AVP	Enumerated

Table 56: Supported sub-AVPs of the Override-Control AVP (continued)

AVP		Category	Description	Type
	Override-Online ResponseRequired	O	Indicates WAIT or DONT_WAIT of data and session setup when the online AVP is enabled	Enumerated
	Override-Policy-Parameters	O	Overrides QoS parameters configured at the PGW for a static/predefined rule or rule base	Grouped
	Override-QoS-Information • Override-Max-Requested-Bandwidth-UL • Override-Max-Requested-Bandwidth-DL	O	Overrides QoS options configured at PGW for a static/predefined rule • Overrides the maximum bitrate for the uplink direction configured for a static/predefined rule or rule base • Overrides the maximum bitrate for the downlink direction configured for a static/predefined rule or rule base	Unsigned 32

13.4.6 Called Station ID AVP

It is possible to specify whether the real, public APN name, or virtual APN (if applicable) is sent as the value for the Called-Station-ID AVP of Diameter and RADIUS Accounting/Authentication/Pre-authentication messages, for the Service-Selection AVP of S6b Diameter messages, or the APN Network Identifier (APN NI) for Ga interfaces.

Some operators for example, require that only the Access Point Name Network Identifier (APN-NI) be sent in the Called-Station-ID AVP. This can be configured using the **apn-ni-only** option for the APN network ID.

When the APN network ID is not configured, the default value sent is the APN to which the user is attached.

To configure an APN option that will be sent in the Called Station ID AVP, use the following command:

```
configure mobile-gateway pdn apn apn-network-id
```

13.4.7 Diameter race condition handling for Gx and Sd interfaces

Diameter race condition handling for the Gx and Sd on SSG interfaces is supported, as specified in the 3GPP Policy and Charging Specifications.

The Gx and Sd interfaces allow the server (PCRF) to update a session in two ways; unsolicited and solicited. The PCRF pushes policy decisions and provision event triggers in an unsolicited fashion using a RAR. It can also install policy decisions in a solicited manner by responding to a CCR sent by the client (gateway). The client and the server can initiate transactions that modify the state of the session independently (for

example, CCR from the client and RAR from the server) and potentially concurrently. Additionally, there may be Diameter agents in between the client and the server (for example, DRA or in general Diameter relays/proxies) which could cause messages to be delivered out of order. This can lead to race conditions that may result in the wrong information maintained by the client and/or server for a session.

On receipt of a Diameter request for an existing session, if there is an ongoing transaction for the session, and the incoming request cannot be safely handled without creating a state mismatch, the incoming request is rejected by the gateway, with a Diameter experimental result code of `DIAMETER_PENDING_TRANSACTION`. An exception to this is a request for session release (RAR with Session-Release-Cause) which is handled immediately. Alternately, if the PCRF sends a `DIAMETER_PENDING_TRANSACTION` result code, the gateway retries the request after the expiration of the configurable pending transaction timer. The number of times the same request is re-attempted due to the `DIAMETER_PENDING_TRANSACTION` code is also configurable.

**NOTE:**

Pending transaction on the Gx interface for the SSG is currently not supported.

To enable and configure this functionality, use the following commands.

```
configure mobile-gateway profile gx supported-features pending-transaction
```

```
configure mobile-gateway profile gx pending-transaction-timer
```

```
configure mobile-gateway profile gx pending-transaction-retry-count
```

```
configure mobile-gateway profile sd supported-features pending-transaction
```

```
configure mobile-gateway profile sd pending-transaction-timer
```

```
configure mobile-gateway profile sd pending-transaction-retry-count
```

13.4.8 Gx Assume Positive

Gx Assume Positive enhances the existing PGW failure handling behavior by applying local pre-defined policies on the PGW to maintain and continue subscriber sessions in case failure result codes are generated or there is no response from the PCRF.

A PDN session that falls into Assume Positive mode continues without any further interaction with the PCRF until the user is detached or the PDN session is disconnected.

The existing failure handling timer (`fh-time-limit` under the Gx profile) can be leveraged to control the length of time a session in Assume Positive mode continues before it is terminated.

The default QoS policy can be applied per APN and RAT type. Configurable failure handling per message type and error code is supported.

No event triggers are enabled for sessions that fall into Assume Positive mode at the start of a session (CCR-I). The PGW applies a local pre-defined policy and the session continues. When a session falls into Assume Positive mode at mid-session (CCR-U failure or no response), all PCRF enabled event triggers are disabled.

The following summarizes the Gx Assume Positive behavior for specific events:

AN-GW-Change	Ignore a CCR-U failure and continue the session.
Out-Of-Credit	The PGW acts on the filters enabled via Gy.
Reallocation-Of-Credit	The PGW acts on the filters enabled via Gy.
RAT-Change	The PGW applies a local pre-defined policy for the new RAT type (similar to CCR-I).
ULI-Change	Ignore a CCR-U failure and continue the session.
PLMN-Change	Ignore a CCR-U failure and continue the session.
UE-Time-Zone-Change	Ignore a CCR-U failure and continue the session.
Successful-Resource-Allocation	Ignore a CCR-U failure and continue the session.

To define a QoS profile, use the following commands:

```
configure mobile-gateway profile qos-profile
```

```
configure mobile-gateway profile qos-profile arp
```

```
configure mobile-gateway profile qos-profile dl-ambr
```

```
configure mobile-gateway profile qos-profile qci
```

```
configure mobile-gateway profile qos-profile ul-ambr
```

To associate a QoS profile with an access type, use the following commands:

```
configure mobile-gateway pdn apn access-type
```

```
configure mobile-gateway pdn apn access-type default-qos-profile
```

To define a trigger rule unit and trigger rule unit access types, use the following commands:

```
configure mobile-gateway profile policy-options trigger-rule-unit
```

```
configure mobile-gateway profile policy-options trigger-rule-unit access-type
```

To associate a trigger rule unit with a policy rule, use the following command:

```
config mobile-gateway profile policy-options policy-rule
```

To define an Assume Positive action profile and policy action, use the following commands:

```
configure mobile-gateway profile policy-options policy-rule ap-action-profile
```

```
configure mobile-gateway profile policy-options policy-rule ap-action-profile  
policy-action
```

To specify result codes for entering Assume Positive mode, use the following commands:

```
configure mobile-gateway profile gx msg result-code
```

```
configure mobile-gateway profile gx msg result-code-3xxx
```

```
configure mobile-gateway profile gx msg result-code-4xxx
```

```
configure mobile-gateway profile gx msg result-code-5xxx
```

To specify a failure handling time limit and failure handling options for Assume Positive mode, use the following commands:

```
configure mobile-gateway profile gx fh-session-no-response
```

```
configure mobile-gateway profile gx fh-time-limit
```

To view Assume Positive content and related statistics, use the following commands:

```
show mobile-gateway pdn apn access type
```

```
show mobile-gateway pdn apn gx-ap-statistics
```

```
show mobile-gateway profile gx
```

```
show mobile-gateway profile policy-options ap-action-profile
```

```
show mobile-gateway profile policy-options trigger-rule-unit
```

```
show mobile-gateway profile qos-profile
```

13.4.8.1 Gx Assume Positive configuration example

This section provides an example of the Gx Assume Positive configuration.

Procedure

1. Define a QoS Profile.

Step example

```
# configure mobile-gateway profile qos-profile QoS-Profile-1 qci 9
# configure mobile-gateway profile qos-profile QoS-Profile-1 arp 11
# configure mobile-gateway profile qos-profile QoS-Profile-1 dl-ambr 5000
# configure mobile-gateway profile qos-profile QoS-Profile-1 ul-ambr 5000
```

2. Associate the QoS profile with an access point and access type.

Step example

```
# configure mobile-gateway pdn 1 apn "ipd.nokia.1.1.com" access-type eutran
default-qos-profile "QoS-Profile-1"
```

3. Define a trigger rule unit.

Step example

```
# configure mobile-gateway profile policy-options
>config>mobile>profile>policy-options# begin
>config>mobile>profile>policy-options# trigger-rule-unit Trigger-rule-unit-1
access-type eutran
```

```
>config>mobile>profile>policy-options# commit
```

4. Associate the trigger rule unit with a policy rule.

Step example

```
# configure mobile-gateway profile policy-options
>config>mobile>profile>policy-options
# begin
>config>mobile>profile>policy-options
# policy-rule "pr1" policy-rule-unit "pru1" charging-rule-unit "cru1" qci 9
  arp 11 precedence 101 trigger-rule-unit "Trigger-rule-unit-1"
```

5. Assign the policy rule to a policy rule base and add the rule base to the access point.

Step example

```
>config>mobile>profile>policy-options# begin
>config>mobile>profile>policy-options# policy-rule-base "prb1" policy-rule
  "pr1"
>config>mobile>profile>policy-options# commit
# configure mobile-gateway pdn 1 apn "ipd.nokia.1.1.com" static-predefined-
  pcc default-policy-rule-base "prb1"
```

6. Define an Assume Positive action profile.

Step example

```
# configure mobile-gateway profile policy-options ap-action-profile ap-
  action-profile-1 policy-action pre-defined-policy
```

7. Specify result codes for entering Assume Positive mode in Gx profile and associate the result codes with the configured Assume Positive action profile.

Step example

```
# configure mobile-gateway profile gx Gx-profile-1 message ccr-u result-
  code-5xxx retry-and-ap-continue ap-action-profile "ap-action-profile-1"
```

8. Specify a failure handling time limit and failure handling options in Gx profile for Assume Positive mode and associate with the configured Assume Positive action profile.

Step example

```
# configure mobile-gateway profile gx "Gx-profile-1" fh-session-no-response
  ap-continue ap-action-profile "ap-action-profile-1"
```



```
# configure mobile-gateway profile gx "Gx-profile-1" fh-time-limit 10
```

9. Associate the Gx profile with the access point.

Step example

```
# configure mobile-gateway pdn 1 apn "ipd.nokia.1.1.com"
>config>mobile>pdn>apn# shutdown
>config>mobile>pdn>apn# pcrf-selection dynamic-pcc gx-profile "Gx-profile-1"
>config>mobile>pdn>apn# no shutdown
```

10. Display Assume Positive statistics and related content.

To view Assume Positive statistics, use the following commands:

```
show mobile-gateway profile qos-profile
```

```
show mobile-gateway profile policy-options trigger-rule-unit
```

```
show mobile-gateway profile policy-options ap-action-profile
```

```
show mobile-gateway profile gx
```

```
show mobile-gateway pdn apn gx-ap-statistics
```

13.4.9 PCRF migration

PCRF migration from one segment (primary or secondary) to another segment (primary or secondary) is facilitated by leveraging the Diameter peer group list.

Each peer group within the peer group list is associated with a priority. By modifying the priority, the migration is initiated and new session traffic is routed to the peer group with the highest priority.

Priority 1 is the highest and 20 is the lowest and can be changed at any time for the specified peer group. When the priority is not specified, the default priority is set to 20. A priority value of 0 for a specific peer group within a Diameter peer group list will not be considered for load-balancing.

For example, peer-group 1's priority is changed from 1 to 20 while peer group 2's priority is changed to 1. Existing session traffic continues to be routed to the supported hosts within segment one (peer group 1) while new session traffic is routed to the peer group with the highest priority (peer group 2).

Use the following command to apply a configured Diameter peer group list for PCRF selection to support PCRF migration for Gx:

```
configure mobile-gateway pdn gx pcrf-selection diameter-peer-group-list
```

**NOTE:**

The Diameter peer group list must first be configured in the `config mobile profile diameter-peer-group-list` context.

The priority of a peer group can be modified using the following command:

```
configure mobile-gateway profile diameter-peer-group-list peer-group
```

Validation of the received origin-host from the server (PCRF) with the supported hosts that are locally configured within the Diameter peer profile can be enabled using the following command:

```
configure mobile-gateway profile diameter-peer validate-supported-host
```

The following command can be used to control the population of the destination host in the first message sent to an alternate server, after a switchover to a secondary server after the primary failure:

```
configure mobile-gateway profile diameter-peer exclude-dest-host-after-switchover
```

13.4.10 Supported event triggers for the Gx interface

From a Gx perspective, the PCRF subscribes to the event triggers it is interested in using. Some of the event-triggers are enabled by default and do not require an explicit subscription from the PCRF. Operators who are not interested in using these event triggers can disable them using CLI commands.

The Gx interface also supports custom tethering event triggers, for which the PCRF must subscribe. The reporting value associated with these event triggers is configurable, to meet the requirements of different operators.

[Gx interface event triggers](#) lists the event triggers that are supported for the Gx interface. For more detailed command descriptions, see the Gx Profile Commands in the *CMG CLI Reference Guide*.

Table 57: Gx interface event triggers

Context	Command	Description
config>mobile>profile>gx> event-triggers	tethering-start <i>tethering-start-value</i> no tethering-start	Enables reporting to the PCRF and assigns a value that is used to notify the PCRF of a tethering-start event
	tethering-stop <i>tethering-stop-value</i>	Enables reporting to the PCRF and assigns a value that is used to notify the PCRF of a tethering-stop event

Table 57: Gx interface event triggers (continued)

Context	Command	Description
	no tethering-stop	
config>mobile>profile> gx> gx-forced-triggers	[no] disable-ambr-modification-failure	Disables reporting to the PCRF in case of an AMBR modification failure event
	[no] disable-default-eps-bearer-qos-change	Disables reporting to the PCRF in case of a default EPS bearer QoS change event
	[no] disable-default-eps-bearer-qos-modification-failure	Disables reporting to the PCRF in case of a default EPS bearer QoS modification failure event
	[no] disable-qos-change	Disables reporting to the PCRF in case of a Qos change event

13.4.11 Gx access network information event trigger

The PCRF is able to obtain access network information, such as the user location (TAI, ECGI), from the PGW (PCEF) and provide it to the P-CSCF, when an IMS call is initiated, modified, or released.

The PCRF subscribes to the access network information report event trigger and includes the required access information AVP within the charging rule definition of the affected rules. The gateway provides the access network information to the PCRF, when the PCC rules that contain the required access information AVP are installed, modified or removed, or upon bearer deactivation or session termination. This capability may be used during an IMS call to allow a P-CSCF to gather current access network information for a UE.

13.4.12 PCC based on charging characteristics for Gx interface

PCC can be enabled on the Gx interface for a subscriber's IP-CAN session, based on the Charging Characteristics received from HSS.

Charging Characteristics are used to select a locally configured Charging Profile on the gateway. The charging profile is enhanced to support configuration that can be used to enable or disable PCC (Gx).

Charging Characteristics-based PCC selection is used to selectively enable PCRF control for a subscriber's session. This allows the operator to control loading on Gx interface toward the PCRF. Subscribers belonging to specific service tiers may be allowed to be dynamically controlled by a PCRF.

Charging characteristics-based PCC selection can be configured using CLI commands in the following context:

```
configure mobile-gateway profile charging pgw-charging pcrf-selection
```

13.4.13 Usage monitoring over Gx

Usage Monitoring over Gx is described in detail in 3GPP standard 29.212. Usage monitoring can be applicable to a single SDF, a group of SDFs, or to all the traffic in an IP-CAN session. Usage monitoring can be applicable to SDFs associated with dynamic PCC rules and pre-defined PCC rules.

A monitoring key is used to group services that share a common allowed usage and identifies a usage monitoring control instance. Many PCC rules may share the same monitoring key. The PCRF controls usage monitoring by providing the required usage monitoring information to the gateway, which includes the monitoring key, associated volume threshold, and the monitoring level (session-level or PCC rule-level).

The Usage-Monitoring-Level AVP indicates if the monitoring is applicable at the session-level or PCC-rule level. If the Usage-Monitoring-Level AVP is not provided, its absence indicates that monitoring is applicable at the PCC-rule level.

For session-level monitoring, all traffic that gets successfully forwarded for the IP-CAN session (including any flows associated with pre-defined static policy) is accounted toward the threshold associated with session-level monitoring. Service-level monitoring is achieved by associating the SDFs belonging to a particular service with a specific monitoring key.

For PCC rule-level monitoring and a PCRF provided rule, the monitoring key is provided within the charging rule definition. For a pre-defined rule, the monitoring key is included as part of the rule definition (static policy) at the gateway. To specify the monitoring key as part of the charging rule unit of a PCC rule, use the following command:

```
configure mobile-gateway profile policy-options charging-rule-unit monitoring-key
```

When usage monitoring is enabled, the gateway measures the volume of the IP-CAN session or the volume of the applicable service data flows and reports the accumulated usage to the PCRF in the following conditions:

1. when a usage threshold is reached
2. when all the PCC rules associated with a monitoring key are removed or deactivated
3. when usage monitoring is explicitly disabled by the PCRF
4. when an IP-CAN session is terminated
5. when requested by the PCRF (on demand)

Accumulated volume reporting can be done by the gateway for the total volume, the uplink volume, or the downlink volume as requested by the PCRF, and set in CC-Total-Octets, CC-Input-Octets, or CC-Output-Octets AVPs of Used-Service-Unit AVP, respectively.

Time-based usage monitoring applies to session-level as well as PCC rule-level usage monitoring. The only difference, with respect to the operation details, is how the time threshold is specified by the PCRF and how the time usage is reported by the gateway. Granted Service Units (GSU) and Used Service Units (USU) grouped AVPs include the CC-Time AVP that will be used for time threshold and time usage reporting.

Time-based usage monitoring is set by the TimeBasedUM feature bit within the supported features AVP sent to the PCRF, and is enabled using the `config>mobile>profile>gx>supported-features>time-based-usage-monitoring` command.

Simultaneous volume and time-based usage monitoring is supported for session-level usage monitoring and rule-level usage monitoring.

- When both time and volume thresholds are provided for usage monitoring, the accumulated usage, since the last report, is reported when either the time or the volume threshold is reached. Accumulated usage is reported for both measurements.
- If a new time or volume threshold is provided while usage is being reported for both time and volume measurements, then the measurements will continue only for that provided type and any accumulated usage for the non-provided type will be discarded.
- For time-based usage monitoring, active time usage is supported; however when an inactivity detection time (Quota-Consumption AVP) is specified within the usage monitoring Information sent from the PCRF, the QCT received from PCRF is ignored and locally configured activity-threshold is used to account active usage. The activity threshold can be configured on a per monitoring-key basis using the following command: `config>mobile>profile>gx>usage-monitoring>monitoring-key>activity-threshold`.

To summarize:

- CC-Time within GSU is used to specify the time threshold by the PCRF
- CC-Time within USU is used to report the time usage by the gateway
- Simultaneous volume and time-based usage monitoring is supported for session-level usage monitoring and rule-level usage monitoring

This feature also supports consuming time for active usage, when enabled on the gateway. An activity threshold can be configured on the gateway. The activity threshold defines an average data rate under which the subscriber host is considered silent. As long as the effective rate of the application usage (for a monitoring key instance) will not exceed the rate defined by the activity threshold, the specified subscriber host will be considered silent and its corresponding quota will not be used. As long as the application usage (for a monitoring key instance) will exceed the rate, the corresponding time quota will be consumed (in terms of time). When the `active-usage` command in the `config>mobile>pdn>gx` context is not enabled, time is consumed linearly. When the **activity-threshold** keyword of the `active-usage` command is enabled, it applies to all time-based usage monitoring keys (that is, all of them can be active usage-based, when enabled or elapsed time-based, when disabled).

Table 58: Presence of usage monitoring AVPs in Gx messages

AVP		CCR-I	CCA-I	CCR-U	CCA-U	CCR-T	CCA-T	RAR	RAA
Usage-Monitoring-Information			√	√	√	√		√	
	Monitoring-Key		√	√	√	√		√	
	GSU		√		√			√	
	USU			√		√			
	Usage-Monitoring-Level		√		√			√	
	Usage-Monitoring-Report				√			√	
	Usage-Monitoring-Support				√			√	

The USAGE_REPORT (33) Event-Trigger is supported.

According to 3GPP 29.212, this value shall be used in CCA and RAR commands by the PCRF when requesting usage monitoring at the PCEF. The PCRF shall also provide in the CCA or RAR command, the Usage-Monitoring-Information AVPs including the Monitoring-Key AVP and the Granted-Service-Unit AVP.

**NOTE:**

Gx usage monitoring is not intended as a charging solution. The standardized solution does not cover all the aspects that are defined for offline and online charging solutions.

For improved granularity of usage reporting, an accounting priority profile can be optionally created and assigned to an APN.

To create and configure an accounting priority profile, use the commands in the following context:

```
configure mobile-gateway profile accounting-priority-profile
```

To specify an accounting priority profile for an APN, use the following command:

```
configure mobile-gateway pdn apn accounting-priority
```

13.4.13.1 Usage monitoring reporting accuracy enhancement

When enabling Gx usage monitoring, follow the guidelines in this section.

- Configure the threshold for triggering the usage reporting toward the PCRF already when this threshold is reached. The threshold is configured as a percentage of the GSU, for example at 95%.

To configure the threshold, use the following command:

```
configure mobile-gateway pdn gx usu-options-reporting-threshold
```

- The 3GPP specification does not allow the PCRF to communicate to the PCEF what to do when the quota runs out. You can specify whether to allow or drop traffic while the CCR-U/CCA-U transaction is in progress. The gating starts to take effect when reaching the real GSU threshold. Gating is stopped when new quota is granted by the PCRF, or usage monitoring is stopped.

To configure if traffic will be allowed or dropped, use the following command:

```
configure mobile-gateway pdn gx usu-options-over-quota-traffic-handling
```

- It is possible that for a high bandwidth session, the next polling would result in an USU being larger than a GSU. This could also happen if the threshold setting is too low. In this case, the gateway will report the same for the USU and the GSU, and save the excess quota for later reporting.

This means that the excess quota is reported when monitoring is stopped:

- explicitly by the PCRF
- when an IP-CAN session is terminated
- when all PCC rules associated with a monitoring key are removed or deactivated

Typically, the unreported excess quota is added to the current USU immediately when the new GSU is granted by the PCRF, and therefore it is no longer considered as excess quota.

To specify if the last excess usage report in those cases is suppressed, use the following command:

```
configure mobile-gateway pdn gx usu-options-suppress-excess-quota
```

To enhance the reporting accuracy:

- configure the system with a smaller number of stats (RG and service ID mapped rules) which results to a tighter polling frequency for the relevant stats due to the decreased total number of stat objects to be polled
- configure a small granularity for grants from the PCRF instead of granting all available quota in one go (the downside is significantly increased Gx signaling)
- if the PCRF implements a threshold by informing the PCEF of a GSU which is slightly lower than the actual usage limit for that subscriber, this could assist in mitigating the reporting inaccuracy
- according to 3GPP 29.212, the PCRF is also allowed to query the PCEF to report accumulated usage for one or more enabled monitoring keys regardless if a usage threshold has been reached by sending to the PCEF within the Usage-Monitoring-Information AVP the Usage-Monitoring-Report AVP set to the value USAGE_MONITORING_REPORT_REQUIRED (this will increase the signaling rate between gateway and the PCRF)

To view Gx usage statistics, use the following command:

```
show mobile-gateway pdn pdn-context imsi gx-usage-monitoring monitoring-key
```

13.4.13.2 Monitoring key usage report event trigger override configuration

An override for the default usage report event trigger value can be configured using the `config>mobile>profile>gx>usage-monitoring>override-usage-report-value` command. When enabled, the USAGE_REPORT event trigger value will be 26 instead of 33. In 3GPP Release 9, the value of the USAGE_REPORT event trigger changed from 26 to 33. Enabling the override option helps interoperability with PCRF vendors that continue to use the USAGE_REPORT event trigger value of 26 for usage monitoring.

Enabling this command prevents gateway support for the TAI_CHANGE event trigger, which uses a value of 26. If both TAI_CHANGE and usage monitoring are deployed, the PCRF should be capable of supporting both TAI_CHANGE (value 26) and USAGE_REPORT (value 33) event triggers accordingly.

13.4.13.3 Service/application usage exclusion from IP-CAN/TDF session usage

This feature allows excluding usage of a service or application from IP-CAN session-level usage monitoring, by adding an indication in the Gx interface when the PCRF issues PCC rules for services and applications whose usage should be excluded from IP-CAN session-level usage monitoring.

Prior to this feature, when session-level monitoring and PCC-rule level monitoring were simultaneously active, the session level counts included all traffic for that IP-CAN session, even if certain traffic was being monitored at the PCC-rule level via a different monitoring key.

The PGW/GGSN supports excluding service or application usage information from the IP-CAN session-level usage monitoring by setting the ExUsage feature bit within the supported features AVP sent to the PCRF.

If the PCC rule is associated with an indication of exclusion from the session-level monitoring, the PCEF will not consider the corresponding service data flow for the volume and time measurement on the IP-CAN session. If the Monitoring-Flags AVP is included within the Charging-Rule-Install AVP, it is used to indicate the monitoring action related to the corresponding service data flow. It applies to all the PCC rules within the Charging-Rule-Install AVP. If the Monitoring-Flags AVP is provided, the value will be valid for the PCC rules until the new value is provided.

This feature is configured using the `config>mobile>profile>gx>supported-features>exclude-usage` command.

13.4.14 PCC rule-level redirection configuration for the Gx interface

This functionality supports the PCC rule-level redirection for pre-defined rules with an IP address filter as a classification criterion.

Specifically, the following enhancements apply:

- only the pre-defined/static rules are supported

- either the session-level or the rule-level redirection can be supported for a Gx session; not both at the same time
- for an IP-CAN session, only Gx or the Gy redirection is supported; that is, a single entity (PCRF or OCS) controls the redirection
- if a redirect rule is associated with an action rule unit (ARU) that contains a rule level redirection, then the mobile gateway performs a rule-level redirection for that rule (only rule-level redirection for this session)
- any other PCC rules are treated according to their precedence and flow-gate status; for example, forwarding is performed on the traffic matching rules when flow-gate is enabled

To support rule-level redirection, a redirect rule must be associated with an ARU that has the `rule-level-redirect` command enabled. Therefore, any traffic that matches this rule classification criterion is redirected to the specified URL.

To configure the PCC rule-level redirection, enable the rule-level redirection flag in the ARU using the following command:

```
configure mobile-gateway profile policy-options action-rule-unit rule-level-redirect
```

13.5 Filtering criteria configuration in policy rules

A policy rule unit is the fundamental unit of service awareness. With policy rule units the operator can define which parts of the ingress packet are used for matching to a certain rule.

13.5.1 Configuring source address in policy rules filtering

Configure a source IP address prefix/subnet as the match criteria for flow descriptions in policy rules. The source IP address (IPv4 address/subnet or IPv6 prefix/subnet) is the address of the UE. If the command is not configured, the source IP address is not used as match criteria for traffic classification.

Procedure

1. Configure the IP address of the UE as match criteria in policy rules.

```
configure mobile-gateway profile policy-options policy-rule-unit flow-description match local-ip ip-prefix/ip-prefix-length
```

2. Show the filters under a policy rule unit.

To show all the filters under a policy rule unit, use the following command:

```
show mobile-gateway profile policy-options policy-rule-unit policy-name
```

To show only the filters that are installed for a specific UE under a policy rule unit, use the following command:

```
show mobile-gateway pdn bearer-context imsi detail
```

**NOTE:**

Upon a create session request, where the UE IP address is known, the mobile gateway parses all the local-ip filters and decides which ones are installed for the specific session. This means that different sessions that have the same policy rules assigned will most likely get a different set of filters based on local-ip installed.

13.5.2 Configuring port lists in policy rules filtering

Configure the same policy rules for the same protocols and IP addresses but for different ports.

Purpose

You can configure the same filtering criteria for a list of non-contiguous local/remote ports by using the `local-port-list` and `remote-port-list` commands respectively within the flow description. The list of ports must be configured within the context of `config>mobile>profile>list`. Those ports might be scattered and non-contiguous. In this case, configuration in the flow description, where local or remote port/port range parameters are used, results in multiple same configuration sets. Only the values of those parameters are different.

**NOTE:**

The maximum number of port lists that can be configured is 100, with a maximum number of 32 entries per list (either port numbers or ranges).

Procedure

1. Configure the name of the list of ports that will be used for PCC rule filtering criteria.

```
configure mobile-gateway profile list port-list
```

2. Specify the TCP or UDP port numbers to use in the port list definition configured above.

Use the following commands:

```
configure mobile-gateway profile list port-list port
```

```
configure mobile-gateway profile list port-list port range
```

3. Configure the previously defined list of local/remote ports in PCC rule filtering criteria.

Use the following commands:

```
configure mobile-gateway profile policy-options policy-rule-unit flow-  
description match local-port-list
```

```
configure mobile-gateway profile policy-options policy-rule-unit flow-  
description match remote-port-list
```

4. Show the configured port list profiles.

```
show mobile-gateway profile list port-list
```

13.5.3 Configuring IP address lists in policy rules filtering

There are cases where the operator needs to configure the same flow description for similar services which are provided by servers with different IP addresses. The same flow description must be configured several times for different IP addresses (IPv4 or IPv6).

Purpose

The operator can configure only the same filtering criteria for a list of IP addresses, thus reducing configuration. The list of IP addresses must be configured in the context of `config>mobile>profile>addr-list`.



NOTE:

- A single address list must only contain either IPv4 or IPv6 addresses. If both types of addresses are required in the same PCC rule, an IPv4 address list must be configured within one flow description and an IPv6 address list within another.
- The maximum number of IP address lists that can be configured is 100, with a maximum number of 100 entries per list (either single IP addresses or ranges). When an IP address list is configured under a policy rule unit, a list larger than 100 entries will be rejected.

Procedure

1. Configure the name of the list of IPv4 or IPv6 addresses that will be used for PCC rule filtering criteria.

```
configure mobile-gateway profile addr-list
```

2. Configure the possibility to use the previously defined list of IPv4/IPv6 addresses in PCC rule filtering criteria.

```
configure mobile-gateway profile policy-options policy-rule-unit flow-
description match remote-ip-list
```

13.6 Gateway-derived flow descriptions using DNS resolution

The EPC gateway supports the ability to dynamically derive flow descriptions using DNS resolution. A fully qualified domain name (FQDN) group can be configured with an FQDN list. The list is resolved (periodically refreshed), to obtain the remote IP addresses and derive the flow descriptions.

FQDN groups are configured within the flow description part of a policy rule unit. They are referenced under an APN for DNS resolution using the DNS servers configured under `config>mobile>pdn>apn>pco-option`. The resolved flow descriptions can be associated with the needed rating group and service IDs within a charging rule unit, to facilitate zero-rating, differentiated charging, and so on. When the optional `aa-advanced-functions` command is enabled within a policy rule unit, the derived flow descriptions are used to selectively divert traffic to AA, to facilitate the enhancement of the maximum throughput of an ISM-MG card.

For FQDN group command descriptions, refer to the following commands in the *CMG CLI Reference Guide*.

- Profile Commands
- APN PDN Commands
- Tools and OAM Commands

```
configure
  mobile-gateway
    profile
      fqdn-group "zero-rate-http"
        dns-refresh-interval 10
        ip-expiry-refresh-count 3
        fqdn 1 internal1.nokia.com
        fqdn 2 internal2.nokia.com
      exit
      fqdn-group " zero-rate-dns"
        dns-refresh-interval 10
        ip-expiry-refresh-count 3
        fqdn 1 internal3.nokia.com
        fqdn 2 internal4.nokia.com
      exit
    exit
  exit

  policy-rule-unit "pru_zeroRate_HTTP"
    flow-description 1
```

```

        fqdn-group "zero-rate-http"
        protocol 6
        remote-port eq 80
    exit
exit

```

```

policy-rule-unit "pru_zeroRate_DNS"
    flow-description 1
        fqdn-group "zero-rate-dns"
        protocol 17
        remote-port eq 53
    exit
exit

```

```

charging-rule-unit "rg10-svc-id-100"
    charging-method offline
    rating-group 10
    reporting-level rating-group
    service-id 100
exit

```

```

charging-rule-unit "rg10-svc-id-200"
    charging-method offline
    rating-group 10
    reporting-level rating-group
    service-id 200
exit

```

```

config>mobile-gateway>profile>policy-options /*create PCC rules */
begin
    policy-rule ZR-group1
    policy-rule-unit "pru_zeroRate_HTTP" charging-rule-unit "rg10-svc-id-
100" qci 9
    precedence 10
    policy-rule ZR-group2
    policy-rule-unit "pru_zeroRate_DNS" charging-rule-unit "rg10-svc-id-200"
qci 9
    precedence 20
config>mobile-gateway>profile>policy-options /*create PRB */
    policy-rule-base ZeroRate-PRB
        policy-rule ZR-group1
        policy-rule ZR-group2
    commit

config>mobile-gateway>

```

```

    pdn 1
      apn "NOK_internal"
      pco-option
        dns-server-ipv4
          primary 22.22.22.2
          secondary 33.33.33.2
        exit
      exit
      router "vprn1"
        ip-local-pool "pool1.1.0"
      exit
      fqdn-group zero-rate-http
      fqdn-group zero-rate-dns
    exit
    apn "NOK_corp"
      pco-option
        dns-server-ipv4
          primary 88.88.88.8
          secondary 99.99.99.9
        exit
      exit
      router "vprn2"
        ip-local-pool "pool2.2.0"
      exit
      fqdn-group zero-rate-http
      fqdn-group zero-rate-dns
    exit
  exit
exit

```

```

configure
  mobile-gateway
    profile
      fqdn-group "NOK_enrichment"
        dns-refresh-interval 10
        ip-expiry-refresh-count 3
        fqdn 1 host1.nokia.com
        fqdn 2 host2.nokia.com
      exit
      fqdn-group "NOK_shallow"
        dns-refresh-interval 10
        ip-expiry-refresh-count 3
        fqdn 1 doNotDivert1.nokia.com
        fqdn 2 doNotDivert2.nokia.com
    exit
  exit

```

```

        exit
    exit

    policy-rule-unit "pru_NOK_httpEnrichment"
        flow-description 1
            fqdn-group "NOK_enrichment"
        exit
        aa-advanced-functions
    exit

    policy-rule-unit "pru_NOK_doNotDivert"
        flow-description 1
            fqdn-group "NOK_shallow"
        exit
        shallow-inspection-only
    exit
    pdn 1
        apn "NOK_internal"
        pco-option
            dns-server-ipv4
                primary 22.22.22.2
                secondary 33.33.33.2
            exit
        exit
        router "vprn1"
            ip-local-pool "pool1.1.0"
        exit
        fqdn-group NOK_enrichment
        fqdn-group NOK_shallow
    exit
    apn "NOK_corp"
        pco-option
            dns-server-ipv4
                primary 88.88.88.8

```

13.7 Policy rules precedence

This feature allows UE traffic to be matched to policy rules according to the policy rule precedence, regardless of whether the policy rule is diverted to AA or not.

If this functionality is not enabled (the default), all policy rules that are handled by AA are given higher precedence (matched first) over non-AA rules.

To configure policy rule precedence at the APN level, use the following command:

```
configure mobile-gateway pdn apn enable-global-pcc-precedence
```

13.8 Installing different rules with the same charging group

This feature adds support for installing different rules/rule bases with the same AA charging group and a different flow-gate status (allow or drop).

This feature can be used, for example, when a UE attaches and the PCRF installs rule-1 with flow-gate-status configured to drop, and rule-2 with flow-gate-status configured to allow. Both rule-1 and rule-2 use the same aa-charging-group.

If rule-2 has a higher precedence, matching traffic is forwarded by default (allow). If the UE runs out of credit, the PCRF removes rule-2, the traffic starts matching rule-1, and then the traffic is dropped.



NOTE:

In the current release, only the values allow and drop are supported for the flow-gate status.

13.9 Sponsored data connectivity

This feature provides support for sponsored data connectivity procedures as specified in 3GPP Policy and Charging specifications.

With sponsored data connectivity, the sponsor has a business relationship with the operator and the sponsor reimburses the operator for user data connectivity, to allow the user access to an associated Application Service Provider's (ASP) services.

According to TS 23.203 two deployment scenarios exist for sponsored data connectivity. The sponsor identity and ASP identity are provided for sponsored services to the PCRF from the AF over the Rx interface.

In the first scenario, the PCRF assigns a service specific charging key for a sponsored IP flow. The charging key is used by the PCEF to generate separate accounting records for offline charging and/or usage data records for online charging for the sponsored flows. Correlation of accounting records and usage data records from multiple users per sponsor and/or application service provider is then performed using the charging key.

In the second scenario, the sponsor identity and ASP identity is included in PCC-rules from the PCRF to the PCEF. For this scenario, the same charging key may be used both for IP flows that are sponsored and for flows that are not sponsored. Accounting records generated by the PCEF for offline charging include the sponsor identity and the ASP Identity. Correlation of accounting records from multiple users per sponsor and/or application service provider can then be based on sponsor identity and ASP identity, instead of the charging key. Usage reporting for online charging including sponsor identity and ASP identity has not been

specified in this release of the specification. PCC rules that include a sponsor identity and an ASP identity include a charging method that indicates offline charging.

This feature supports the following for the Gx interface, the PGW Rf interface, and PGW-CDRs.

- The PCRF can include sponsor identity and ASP identity as part of the PCC rule.
- The PCRF can also set the reporting level to a new value for sponsored connectivity, which causes the PGW to report usage for traffic on the combination of rating group and sponsor identity.
- The PGW includes the sponsor identity and ASP identity in PGW-CDRs as well as part of the PS-Information in Rf charging records.

The following commands are provided to view and configure a PCC rule to support Sponsored Data Connectivity including reporting level, sponsor and application service provider ID:

```
configure mobile-gateway profile policy-options charging-rule-unit
```

```
show mobile-gateway profile policy-options charging-rule-unit
```

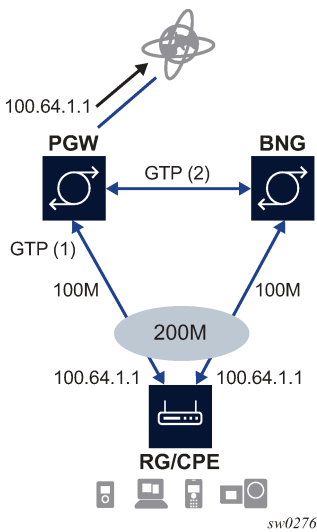
13.10 Hybrid access bonding

This feature enables the PGW to "bond" a pair of PDN connections using two different access types (typically a fixed access such as DSL and a wireless access such as LTE).

The PGW will assign the same IP address to both PDN connections and use a hashing mechanism, similar to that used with ECMP, to load-balance flows for downlink traffic to both PDN connections. Similarly, the UE can load-balance uplink flows allowing the operator to provide the sum of the bandwidth of both access types to the UE.

No change is required to the rest of the network infrastructure and handheld devices. A single Gx session toward the PCRF, Charging (Gy, Ga/Rf), and LI interfaces provides operational efficiency. Bonding can be controlled per APN.

Figure 186: Hybrid access bonding



13.10.1 Fixed access fill-out

Support for policy control and enforcement on fixed access (for example, DSL) filling out first is supported for hybrid access bonding.

When traffic levels exceed the capacity of the fixed access line, the traffic distribution load is adjusted to allow the overflow traffic to use the mobile access link. Similarly, when the traffic levels drop, the distribution load is switched back to the original policy. The operator can configure the thresholds (high and low watermark) to facilitate change in traffic load distribution.

The PGW/GGSN supports low and high thresholds (watermark) for the fixed access to facilitate change in the traffic distribution load upon hitting these thresholds. The solution is catered toward change in policy for sustained rate increase or decrease rather than an instantaneous rate change. The low threshold (for example, 80% of the capacity) is used to monitor the sustained increase or decrease in rate. The rates are monitored for a particular duration (for example, 3 polling intervals with a polling frequency of 60 seconds) before effecting the change in traffic load distribution.

To configure the hybrid access bonding policy options for fixed access (for example, DSL) filling out first, use the following CLI commands:

```
configure mobile-gateway profile policy-options usage-monitoring-actions
```

```
configure mobile-gateway profile policy-options usage-monitoring-actions
usage-monitoring-action
```

```
configure mobile-gateway profile policy-options usage-monitoring-actions
usage-monitoring-action after-fill-delta-percent
```

```
configure mobile-gateway profile policy-options usage-monitoring-info
```

```
configure mobile-gateway profile policy-options usage-monitoring-info
monitoring-key
```

To configure the bonding support and monitoring key profile for the APN, use the following CLI commands:

```
configure mobile-gateway pdn apn hybrid-access
bonding-support
monitoring-key
```

13.11 Static predefined and local policies on PGW/GGSN

The predefined policy rules that are configured on the gateway can be activated and deactivated for a PDN session by the PCRF referencing the corresponding rule name and/or rule base name.

In addition, there are two alternative local solutions that can be used.

Static predefined policies

They can be configured under an APN using the command `config>mobile>pdn>apn>static-predefined-pcc`. These policies are applied to new PDN sessions that have no PCRF control. A PDN session may be without PCRF control, for example, if the PCRF is not deployed or the PCRF is unreachable and a fall-back to the local policy occurs for new sessions.

Static local policies

They can be configured under an APN using the command `config>mobile>pdn>apn>static-local-pcc`. The PCEF (gateway) applies these policies are applied to PDN sessions locally. Static local policies are applicable to the default bearer only.

These local policies instantiated by the PCEF (gateway) are never communicated to the PCRF (for example, in a charging rule report).

The Override Control feature applies to both predefined and static rules.

For additional configuration information, see the following commands:

```
configure mobile-gateway pdn apn static-predefined-pcc
```

```
configure mobile-gateway pdn apn static-local-pcc
```

**NOTE:**

The CLI batch commands `begin`, `commit`, and `abort` must be used when configuring policy rules and policy rule bases.

Policy modifications without session shutdown are also supported for policies (policy rules, policy rule bases, policy units) configured in the following contexts:

```
configure mobile-gateway profile policy-options
```

```
configure mobile-gateway pdn apn static-local-pcc
```

Configuring static local policies can trigger dedicated bearers.

Related information

Policy configuration using CLI batch capability

Policy modification without session shutdown

ADC extended to static PCC and VoIP optimization

This feature allows the gateway to create dedicated bearers for rules and rule bases that are locally configured and are not under PCRF control. This is an enhancement to the dedicated bearer creation for deployments under PCRF control, where static rules are installed only on the default bearer.

13.12 Dedicated bearer configuration options

The following dedicated bearer configuration options are supported:

- *Dedicated bearer idle timer per QCI*
- *Support of multiple bearers diversion to AA*
- *Dedicated bearer creation upon application detection*
- *ADC extended to static PCC and VoIP optimization*

**NOTE:**

Different dedicated bearer rule types (immediate, data-driven, or ADC) must not be combined in the same rule base.

13.12.1 Dedicated bearer idle timer per QCI

This feature provides operator support for configuring an idle timer for an established dedicated bearer per APN and QCI level.

Idle timers allow the gateway to clean up sessions that otherwise may not have been cleaned up by existing signaling mechanisms. A unique dedicated bearer idle timer may be configured per APN, depending on the QCI of the bearer. This feature may be used for example to configure shorter idle times for high priority QCIs associated with dedicated bearers on the SGW and PGW.

To configure a dedicated bearer idle timeout profile, use the following command:

```
configure mobile-gateway profile dedicated-bearer-idle-timeout-profile
```

To specify the QCI value for which the idle timer is specified and the amount of time a dedicated bearer must be idle before it is released, use the following command:

```
configure mobile-gateway profile dedicated-bearer-idle-timeout-profile qci
```

To specify the dedicated bearer idle timeout profile for an APN, use the following command:

```
configure mobile-gateway pdn apn dedicated-bearer-idle-timeout-profile
```

To enable the deletion of the ADC dedicated bearer on idle timeout, use the following command:

```
configure mobile-gateway pdn apn delete-adc-dedicated-bearer-on-idle-timeout-only
```

13.12.2 Support of multiple bearers diversion to AA

In a VoLTE call, the default bearer is used for passing IMS signaling (non-GBR) while a dedicated bearer is created for voice traffic which requires a specific QoS. The dedicated bearer can then be diverted to AA for further analysis.

This feature enables the diversion of more than one bearers per subscriber for AA analysis, by specifying bearer diversion at the APN level. The bearers are diverted if their QCI value matches to a QCI list that includes QCI values that require AA diversion.

**NOTE:**

When more than one bearer per session are diverted to AA, only one bearer can have AA rules installed. The other bearers will use 5-tuple rules.

To configure a QCI list, use the following command:

```
configure mobile-gateway profile list qci-list
```

To enable bearer traffic diversion to AA based on the bearer's QCI, use the following command:

```
configure mobile-gateway pdn apn aa-divert-per-bearer-qci
```

**NOTE:**

Renaming the QCI list only affects new bearers and sessions. The QCI values which are configured in a QCI list cannot be modified if that QCI list is referenced under an APN.

13.12.3 Dedicated bearer creation upon application detection

Support for standards defined PCC rules enhanced with Application Detection Control (ADC) functionality for a PCEF with an integrated TDF.

The PGW can initiate a dedicated bearer creation upon application detection without further PCRF interaction, if the QCI/ARP of the PCC rule is enhanced with ADC functionality that is different than that of the default bearer, and has mute notification enabled.

When the PCRF activates a pre-defined AA rule with a QCI/ARP that is different from that of the default bearer (or a PRB containing such rules), upon application detection the PGW/GGSN initiates dedicated bearer creation without any further interaction with the PCRF (mute-notification enabled). After the eNodeB or MME rejects the dedicated bearer creation upon initial flow detection of an application, the gateway will continue to attempt to initiate the dedicated bearer upon detection of subsequent flows of the same application. This results in a significant amount of unnecessary messages.

To stop the gateway from initiating the dedicated bearer of the same application, a retry wait time can be configured. Additional controls are provided to manage the dedicated bearer creation, including a retry wait timer that ensures the gateway will wait the specified amount of time before retrying.

A configurable dedicated bearer idle timeout is also supported. The dedicated bearer is torn down when the idle timeout timer expires or the PGW detects that the last application flow associated with the dedicated bearer has stopped.

Either service data flow filters or an application identifier can exist in a PCC rule. The application identifier is used to reference an application detection filter, which is predefined in the PCEF. The same application identifier value can occur in more than one PCC rule. If so, the PCRF ensures that there is at most one PCC rule active per application identifier value and IP CAN session at any time. The same application identifier value could be used for a dynamic PCC rule and a predefined PCC rule or for multiple predefined PCC rules.

QoS information includes the QoS class identifier (authorized QoS class for the service data flow), the Allocation and Retention Priority (ARP) and authorized bitrates for uplink and downlink. The redirect indicates whether the uplink part of the detected application traffic is redirected to a controlled address. The target redirect address may also be included.

**NOTE:**

The redirect is applicable when the application identifier exists in the PCC rule. This is applicable to PCC rules with DPI (ADC rules), except those that have the capability for gateway initiated dedicated Bearer creation.

For a dynamic PCC rule, the redirect address may be provided as part of the dynamic PCC rule or may be preconfigured in the PCEF. A redirect destination provided within the Redirect-Server-Address AVP in a dynamic PCC Rule shall override the redirect destination preconfigured in the PCEF for this PCC rule.

If the PCC rule contains the application identifier, it describes a gate for the corresponding detected application traffic. The Flow-Status AVP of the PCC rule specifies if the possible uplink and possible downlink gate is opened or closed.

For the solicited application reporting, the PCRF may instruct the PCEF to detect applications by providing the Charging-Rule-Install AVP with the corresponding parameters as follows:

- The application to be detected is identified by the TDF-Application-Identifier AVP, which is either provided under Charging-Rule-Definition AVP for dynamic PCC Rules or pre-provisioned for the corresponding predefined PCC Rule, and in such a case only Charging-Rule-Name/Charging-Rule-Base-Name is provided.
- When starting or stopping of the application's traffic (identified by TDF-Application-Identifier) is detected, the PCEF reports the information regarding the detected application's traffic in the Application-Detection-Information AVP in the CCR command.
- The PCRF may then make the policy decision based on the information received and send the updated PCC and PCC rules to the PCEF.
- The Mute-Notification AVP value cannot be changed during an active IP-CAN session. Absence of this AVP means that application start/stop notifications shall be sent for the detected application.

13.12.3.1 Dedicated bearer creation for ADC configuration

To configure dedicated bearer creation for ADC, including the dedicated bearer idle timeout, use the following commands. The dedicated bearer idle timeout profile that is referenced by the APN must be pre-configured with the permitted idle time before an established dedicated bearer is removed by the PGW (QCI).

```
configure mobile-gateway pdn gw-id apn apn-name  
    dedicated-bearer-create-retry-count count  
    dedicated-bearer-create-retry-wait-timer wait-timer  
    dedicated-bearer-idle-timeout-profile profile-name
```

Upon application detection, the PGW/GGSN initiates dedicated bearer creation without further interaction with the PCRF, if mute notification is enabled for the PRU. To enable mute notification for a TDF application, (AA charging group) for a predefined PCC rule, use the following command:

```
configure mobile-gateway profile policy-options policy-rule-unit mute-
notification
```

To enable dedicated bearer creation for a PRB that contains a PCRF-activated pre-defined AA rule with a QCI/ARP rule that is different from that of the default bearer, use the following command:

```
configure mobile-gateway profile policy-options policy-rule-base ded-bearer-
on-app-start
```

13.12.4 ADC extended to static PCC and VoIP optimization

This feature allows the gateway to create dedicated bearers for rules and rule bases that are locally configured and are not under PCRF control. This is an enhancement to the dedicated bearer creation for deployments under PCRF control, where static rules are installed only on the default bearer.

The creation of dedicated bearers for rules and rule bases that are locally configured is supported for the following types of static rules:

5-tuple rules	Mappings to the rating group are handled locally by the ISM-MG blades.
Application rules	Diverted to the AA where Application-Start and Application-Stop create or delete the dedicated bearers.

Whether the rule is handled by the ISM-MG or diverted to AA, the ISM-MG performs the dedicated bearer creation. Application-Start and Application-Stop messages or traffic matching the 5-tuple filters will trigger the creation of dedicated bearers.

Both application and 5-tuple rules can be active for the same dedicated bearer at the same time, but only according to the QCI/ARP. The gateway keeps track of all the rules (5-tuples and AA rules) that are using the dedicated bearer and deletes the dedicated bearer when no rule needs it.

In addition, dedicated bearers can be of the following two types:

- created at rule installation with a different QCI/ARP than the default bearer (UE attach)
- data-driven, that is, created only when traffic matches the configured rule

The gateway activates all policy rules which share the same rating group at the same time when traffic matches any policy rule for that rating group. In the current implementation, charging (offline and/or online) is enabled for data-driven dedicated bearers to be installed so that traffic that matches the rating group triggers the data driven dedicated bearer.

In the case of data-driven dedicated bearers, the rules are first activated in the default bearer until the traffic is matched to the rule, which will trigger the dedicated bearer creation. Subsequent traffic for the

data driven rule will be passed through the dedicated bearer. Data-driven functionality via the action rule unit applies to both PCRF-installed rules and static local PCC dedicated-bearers installed rules.

**NOTE:**

CMG allows a maximum configuration of 32 charging groups with application start/stop per AA group partition.

There can be a maximum of 40 static policy items per session, so the number of dedicated bearers triggered based on feature *ADC extended to static PCC and VoIP optimization* is limited. One L3/L4 data-driven rule is considered as a rule with two static policy items when data-driven dedicated bearer is active due to that rule. Local immediate dedicated bearer triggering rule is considered as a rule with one static policy item.

The following functionality that is available for dedicated bearers is reused in the creation of dedicated bearers by static PCC rules feature.

- Signaling flows toward the access network for creating, updating, and deleting dedicated bearers are the same as what is already supported.
- MBR, GBR, and TFT are re-calculated when new flows are created or terminated; this may trigger a dedicated bearer update.
- L4 filters (5-tuples) and L7 filters (handled by AA) can both trigger dedicated bearers.
- PCC rules enhanced with ADC are supported, that is, an Application-Start and an Application-Stop trigger dedicated bearer creation, update, or deletion, and the PCRF is not informed (similar to a Mute Notification case). If the creation of a dedicated bearer fails, then the logic for retrying the creation is used. Rules for creation of dedicated bearers upon application stop and start at the PRB level are also supported.

**NOTE:**

The ADC enhancements to this feature do not apply to the `config>mobile>pdn>apn>static-predefined-pcc` context. The `static-predefined-pcc` command supports configuration of a rule base that is used when the PCRF is not available (for example, when the PCRF fails) and those static rules are deactivated when PCRF control of the PDN session is restored. Static predefined PCC rules are active only in the default bearer and do not create dedicated bearers. This functionality is not modified by this feature.

Related information

Dedicated bearer creation for ADC configuration

13.12.4.1 Configuring static PCC rules for ADC

Purpose

If a rule has the `data-driven-dedicated-bearer` command enabled, the dedicated bearer is created at traffic detection. If the command is disabled, the dedicated bearer is created when the rule is installed (that is, when the UE initially attaches).

PCC rules enhanced with ADC are also supported; for information about configuring dedicated bearer creation for ADC at the APN and PRB levels, including creation, deletion, and retry logic, see [Dedicated bearer creation for ADC configuration](#).

Procedure

1. Enable creation of dedicated bearers by static rules.

```
configure mobile-gateway pdn apn apn-name static-local-pcc dedicated-bearers
  default-bearer-qci qci-value arp arp-value
  policy-rule policy-rule-name
  policy-rule-base base-policy-name
```

If the configured rules and rule bases under static local PCC require a different QCI/ARP than the default bearer, then the PCEF will install the rules.

Step example

The following configurations examples show the differences between previous and current implementations.

From:

```
static-local-pcc
  policy-rule <policy-rule-name>
  policy-rule-base <policy-rule-base-name>
exit
```

To:

```
static-local-pcc
  dedicated-bearers
    default-bearer-qci <qci> arp <arp>
    policy-rule <rule>
    policy-rule-base <rulebase>
  exit
  default-bearer-qci <qci> arp <arp>
  policy-rule <rule>
  policy-rule-base <rulebase>
exit
```

```

exit
policy-rule <rule>          <-
rules listed here will be installed on default bearer if matching default-
bearer qci/arp
policy-rule-base <rulebase> <-
rule bases listed here will be installed on default bearer if matching
default-bearer qci/arp
exit

```

2. Configure the type of dedicated bearer.

```

configure mobile-gateway profile policy-options
action-rule-unit action-rule-unit-name
data-driven-dedicated-bearer

```

Step example

```

action-rule-unit aru1
no data-driven-dedicated-bearer
exit

```

13.13 Protocol Configuration Options (PCO)

The PCO Information Elements (IE) are used to transfer information between the UE and the network (PGW).

The PCO message types and values used by the UE and network are described in 3GPP TS 24.008. These values may be sent in the PCO or the Additional Protocol Configuration Options (APCO). The PCO is a type 4 IE with a minimum length of 3 octets and a maximum length of 253 octets. Each unit is of variable length and consists of the following:

- protocol identifier (2 octets)
- length of the protocol identifier contents of the unit (1 octet)
- protocol identifier contents itself (n octets)

To view the supported PCO IEs, see *Protocol identifiers in configuration protocol options list of PCO IE* in *Appendix A: Messages and information element reference*.

To configure the PCO using, use the commands in the following context:

```
config>mobile>pdn>apn>pco-option
```

13.14 AES encryption for HTTP redirection configuration

CMG supports AES-CBC-128 encryption for HTTP redirection. This feature facilitates encrypting sensitive fields such as the IMSI, MSISDN, IMEI, and so on, that are appended to a URL that is sent to the UE for HTTP redirection. Macros within the URL indicate which fields are encrypted.

AES-CBC-128 encryption is supported in non-CUPS and CUPS deployment architectures.

Table 59: URL encryption for HTTP traffic redirection in CUPS and non-CUPS deployments

Gy/Gx redirection	URL Encryption Configuration		Non-CUPS	CUPS
	Command/Option	Context		
Gy redirection with response FUI Action Redirect	url-encryption	config>mobile>profile>dcca	✓	
	url-encryption	config>mobile>pdn		✓
Gy redirection with response FUI Action Restrict and a filter ID PRB	http-redirect and encryption	config>mobile>profile>policy-options>action-rule-unit>gy-filter-id-action	✓	✓
Gx policy-rule based redirection	http-redirect and encryption	config>mobile>profile>policy-options>action-rule-unit>flow-gate-status	✓	✓

To configure URL encryption for Gy charging response FUI Action Redirect for non-CUPS deployments, use the following commands.

```
configure mobile-gateway profile dcca url-encryption {aes-cbc-128}
[passphrase string]
```

To configure URL encryption for Gy charging response FUI Action for CUPS deployments, use the following command at the PDN level.

```
configure mobile-gateway pdn url-encryption {aes-cbc-128}[passphrase string]
```

To configure URL encryption for Gy charging response with FUI Action Restrict and a filter ID PRB, for non-CUPS and CUPS deployments, use the following commands.

```
configure mobile-gateway profile policy-options
action-rule-unit aru_name
```

```
gy-filter-id-action http-redirect url-string [encryption {aes-cbc-128}]
[passphrase string]
```

To configure URL encryption for Gx policy-rule based redirection for non-CUPS and CUPS deployments, use the following commands.

```
configure mobile-gateway profile policy-options
  policy-rule-unit pru_name
    flow-gate-status http-redirect url-string [encryption {aes-cbc-128}]
    [passphrase string]
```

13.14.1 AES encryption using salt

This feature enhances the AES encryption feature by adding support for using salt to enhance the security. A salt is a random data that is used as additional input to a one-way function that hashes a password or passphrase.

While encrypting with salt, the following is prepended to the encrypted data:

1. Salted__
2. 8 byte salt value used for encryption

The cipher text is as follows:

```
53616c7465645f5f34dcffff00000000bdc1f82b4202aa74c3ea825ba7417c2a
```

where:

- 53616c7465645f5f is the Salted__ (prefix),
- 34dcffff00000000 is the Salt
- bdc1f82b4202aa74c3ea825ba7417c2a is the encrypted text

The decode side based on the Salted__ marker starts decrypting using the salt.

```
echo 53616c7465645f5f34dcffff00000000bdc1f82b4202aa74c3ea825ba7417c2a | xxd -p -r | openssl aes-128-cbc -d -md sha1 -pass pass:tigris
```

where 12345098762 is the decrypted value.

13.15 Out of Credit and Reallocation of Credit event triggers

The Out of Credit (OOC) and Reallocation of Credit (ROC) events trigger notifications toward the PCRF. Credit control interactions with the OCS are normally on a per RG basis. However the OOC and ROC event triggers support interactions with the PCRF for rules associated with the corresponding RG.

As defined in 3GPP TS 32.240 and 32.299, the OOC value is used in CCA and RAR commands by the PCRF to indicate that the PCEF should inform the PCRF about the PCC rules for which credit is no longer available, together with the applied termination action. When used in a CCR command, this value indicates that the

PCEF generated the request because the PCC rules indicated by the corresponding Charging-Rule-Report AVP ran out of credit. The termination action indicated by the corresponding Final-Unit-Indication AVP applies.

The ROC value is used in CCA and RAR commands by the PCRF to indicate that the PCEF should inform the PCRF about the PCC rules for which credit has been reallocated, after an OOC indication. When used in a CCR command, this value indicates that the PCEF generated the request because the PCC rules indicated by the corresponding Charging-Rule-Report AVP have been reallocated credit, after an OOC indication.

By default, reporting of the OOC and ROC triggers is done according to 3GPP TS 29.212, where the Charging-Rule-Report AVP includes the Charging-Rule-Name, PCC-Rule-Status, and Final-Unit-Indication.

It is also possible to enable a customized report layout that includes the rating group instead of the policy rule. In this case, the Charging-Rule-Report AVP includes the Rating-Group AVP and the Final-Unit-Indication AVP (includes the Final-Unit-Action AVP).

To enable the customized report layout option, use the following command:

```
configure mobile-gateway profile dcca rating-group-in-ch-rule-report
```

13.16 Locally-triggered default bearer modification with QoS update

This feature supports locally triggered default bearer modification with QoS update upon application start, without PCF interaction.

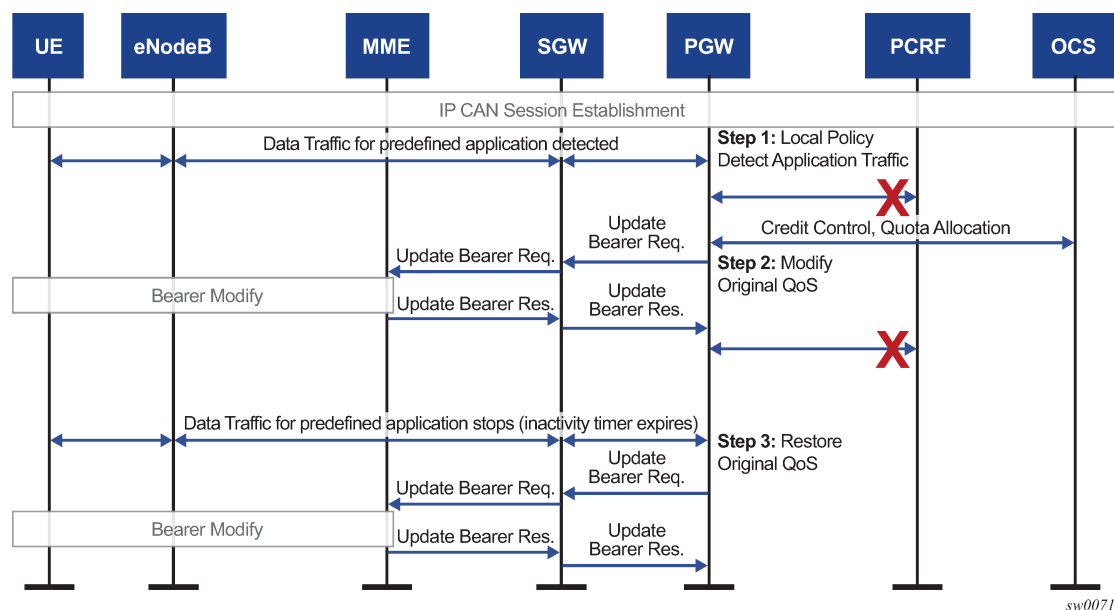
It enhances the functionality described in 3GPP 23.203, section 7.4.1 "IP-CAN Session Modification; GW (PCEF) initiated", by reducing signaling toward the PCRF in case there is need for QoS update when certain application traffic is detected in the default bearer.

If this feature is enabled, the PCEF enhanced with Application Detection and Control (ADC) can locally decide to modify the QoS of the default bearer when start or stop is detected for the configured applications.

The operator can configure an enhanced QoS profile (qci, arp, apn-ambr uplink/downlink) per APN, and configure PCC rules in the default bearer that will trigger the update to the configured QoS parameters.

When such application traffic is detected, the PCEF will trigger an Update Bearer Request for the default bearer without involvement of the PCRF, thus reducing Gx signaling. When application traffic stops and there are no application traffic detected which require the enhanced QoS, a new Update Bearer Request is triggered (without PCRF involvement) to update the default bearer's QoS to the default value.

Figure 187: Locally trigger default bearer modification with QoS update



To configure an enhanced QoS profile, use the commands in the following contexts:

```
configure mobile-gateway profile policy-options action-rule-unit modify-qos-
default-bearer
```

```
configure mobile-gateway pdn apn default-qos-application-modify
```

```
configure mobile-gateway pdn apn default-qos-application-modify apn-ambr-attr
```

```
configure mobile-gateway pdn apn default-qos-application-modify apn-ambr-attr
aggr-rate
```

```
configure mobile-gateway pdn apn default-qos-application-modify arp
```

```
configure mobile-gateway pdn apn default-qos-application-modify qci
```

```
configure mobile-gateway pdn apn default-qos-application-modify qos-holding-
time
```

```
configure mobile-gateway pdn apn default-qos-application-modify retry-wait-
timer
```

13.17 QCI/ARP modification of default bearer

QCI/ARP modification of a default bearer instantiated with wildcard QCI/ARP PRB is supported.

This applies for QCI/ARP modification as a result of the following:

- handovers (3G/4G, Wi-Fi/LTE and so on)
- initiated by the PCRF
- triggered as result of a handover with an MB command
- initiated by the HSS

**NOTE:**

QCI/ARP change of default bearer for non-PCRF deployments that use fixed QCI/ARP instead of wildcard is not supported.

If any dynamic rule ends up with a previous QCI/ARP as a result of the QCI/ARP change, the PCRF is notified of the rules that have been removed

To enable for the APN the dedicated bearer creation for dangling rules that end up with a previous QCI/ARP as a result of default bearer QCI/ARP change, use the following command:

```
configure mobile-gateway pdn apn allow-dangling-rule-ded-bearer
```

13.18 DSCP-based bearer binding

The packet filter attributes DSCP Type of Service (TOS) (IPv4) and Traffic class (IPv6) are supported for SDF classification and bearer binding.

These attributes can be specified by the UE as part of packet filter information in a TFT / TAD and as part of the flow-information AVP from the PCRF. Valid and allowed flow description packet filter attribute combinations are as specified in section 15.3.2 of 23.060.

These attributes are used as part of the match criteria for SDF classification and bearer binding. DSCP is also supported as part of match criteria for predefined PCC rules on the gateway.

To configure DSCP-based bearer binding, use the following commands:

```
configure mobile-gateway profile policy-options policy-rule-unit flow-  
description match tos-tc
```

```
configure mobile-gateway profile policy-options policy-rule-base disable-  
bearer-bind-verification
```

```
configure mobile-gateway pdn apn disable-bearer-bind-verification
```


**NOTE:**

- The policy-rule-base level command `disable-bearer-bind-verification` disables uplink bearer binding verification for the bearer on which that policy-rule-base is installed. The APN level command `disable-bearer-bind-verification` applies to all sessions on that APN. The APN level command therefore overrides the policy-rule-base level command, if both are configured and apply for the same session.
- The `disable-bearer-bind-verification` command is not restricted to DSCP-based bearer binding. The command will work if the DSCP value has not been set (in the PCC rule, flow-information, or TFT), and DSCP as a match criterion is supported regardless of whether this command is enabled.

13.19 No QoS negotiation indication

Configure the PGW to authorize the QoS locally without sending a CCR to the PCRF.

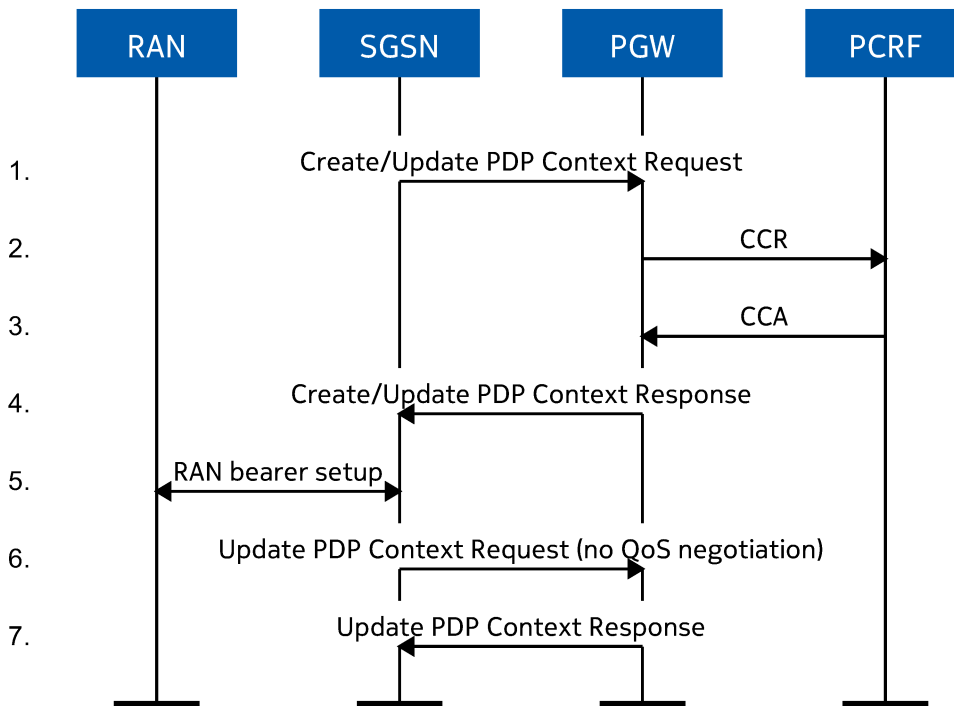
When a PDP context is created or updated, the SGSN passes the requested QoS to the PGW, which passes it to PCRF for authorization. The authorized QoS is returned to the SGSN, which passes the QoS to the RAN. The RAN can downgrade the authorized QoS, if it does not have enough resources for the QoS.

If the authorized QoS is downgraded, the SGSN will inform the PGW about the QoS downgrade. The request also contains indication that no further QoS negotiation is allowed. The PGW will authorize the QoS locally without sending a CCR to the PCRF if the following command has been set:

```
configure mobile-gateway pdn apn pcrr-selection suppress-ccr-with-no-qos-negotiation
```

No QoS negotiation indication shows how No QoS negotiation indication is handled during PDP context activation or modification if `suppress-ccr-with-no-qos-negotiation` is set:

Figure 188: No QoS negotiation indication



QoS authorization is done locally in the PGW according to *B.3.3.3 QoS handling for interoperation with Gn/Gp SGSN* section of the 3GPP TS 29.212. The Update PDP Context Request is rejected in the following cases:

- Derived QCI or ARP has been changed
- GBR is changed
- MBR or derived APN-AMBR is upgraded

The PGW may still send a CCR to PCRF, if the Update PDP Context Request contains any other event that needs to be reported to PCRF, but the CCR does not contain a QoS update. If the CCA contains a QoS update in this scenario, the QoS update is done with the Update PDP Context Request, after the Update PDP Context Response has been sent.

If the `suppress-ccr-with-no-qos-negotiation` is not set, the PGW ignores the no QoS negotiation indication and sends the CCR to the PCRF. This may lead to an infinite QoS negotiation loop if the PCRF upgrades the QoS back to the original authorized QoS, which the RAN is not able to support.

13.20 RAN/NAS cause

A new RAN/NAS Cause IE has been specified for GTPv2. It assists operator call performance analysis, user QoE analysis and proper billing reconciliation.

The MME, ePDG, or TWAG sends the RAN and/or NAS cause to the combined SGW/PGW, it will be passed over the Ga (SGW-CDR and PGW-CDR), Gx, Gy, and Rf interfaces.

The MME may include the RAN and/or NAS failure code when it rejects the network initiated bearer creation, modification or deletion. The MME may also indicate the RAN and/or NAS failure code when it deletes the dedicated bearer. The RAN and/or NAS code may be passed when the MME, ePDG or TWAG deletes the PDN connection.

The MME may pass S1-AP, EMM or ESM failure codes in the RAN/NAS Cause IE. The ePDG may pass IKEv2 or Diameter error codes in the RAN/NAS Cause IE. The TWAG may pass WLCP or Diameter error code in the RAN/NAS Cause IE.

For IMS sessions, the operator may need to pass the RAN/NAS cause value to the P-CSCF via the PCRF. If there is a radio error, the other party may notice the error and terminate the call. The P-CSCF would see this as a normal termination, unless it receives the RAN/NAS code value indicating that there was actually a radio error.

The RAN/NAS cause value forwarded in Ga, Gx, Gy, and Rf interfaces is conditional. In Gx, the RAN/NAS cause is passed to the PCRF only if the PCRF has advertised support for the Gx feature RAN NAS Cause. The local configuration defines whether the PGW advertises support for this Gx feature. The local configuration will also define whether the RAN/NAS cause is visible in SGW-CDR, PGW-CDR, Gy, or Rf.

To configure the RAN/NAS cause, use the following commands:

```
configure mobile-gateway profile gx supported-features ran-nas-cause
```

```
configure mobile-gateway profile gtp gtpv2-options exclude-ran-nas-cause
```

```
configure mobile-gateway profile charging gy-avp-options-profile include-enhanced-diagnostics
```

```
configure mobile-gateway profile charging cdr-avp-options enhanced-diagnostics
```

13.21 AA application profile and ASO characteristics as part of pre-defined PCC rule

The functionality for PCRF signaling AA-profile override and/or ASO override (using pre-defined PCC rule bases) has been enhanced to support the AA application profile and ASO characteristics as part of a pre-defined PCC rule.

The individual PCC rule facilitates taking differentiated AQP action per application per subscriber. For example, when accumulated usage of specific applications exceeds the daily threshold, a rate limit can be applied to the application for a basic subscriber.

To specify an existing application profile name and an existing characteristic, use the following commands:

```
configure mobile-gateway profile policy-options action-rule-unit app-profile-name
```

```
configure mobile-gateway profile policy-options action-rule-unit characteristic
```

Notes

- Provided that the AA application profile's scope is per an IP-can session, at any point in time, there can be only one AA application profile that applies to the subscriber's IP-CAN session. The instantiated rules and rule bases (if there is more than one rule base activated) must reference the same AA application profile.
- Provided that an ASO characteristic is global, at any point in time, it can be associated with a single value for an IP-CAN session. This implies that the activated rules and rule bases for an IP-CAN session cannot have an ASO characteristic with different values.
- The AA application profile and ASO characteristics can be part of a pre-defined PCC rule configuration. If a rule includes an AA application profile and one or more AA attributes, the specified attributes override the values in the referenced profile.
- The AA application profile configured as part of the pre-defined PCC rule or rule base overrides any default AA application profile configured for an APN, when the rule base or the rule is activated for the IP-CAN session.
- In the event of a detected conflict, the gateway sends a charging rule report to indicate that the conflicting rule base or rule was not activated (the PCC rule status is inactive and the Rule Failure Code is Resource-Allocation-Failure).
- It is possible to have a PRB with an ASO characteristic and an AA application profile along with some pre-defined rule or rules within the PRB that also have an ASO characteristic, as long as there is no conflict.
- This feature is applicable to the Gx and Sd interfaces.
- If the policy rule unit does not have any flow descriptions and the action rule unit has an application profile or ASO, then the QCI/ARP must be a wildcard. Otherwise, the policy rule configuration will be rejected at the configuration time.

13.22 QoS policing per RG/SID

This feature provides an option to apply QoS policy at the RG (Rating Group) or RG/SID (Service ID) level without the need to divert traffic to the AA. Each MG-VM node implements eight million policers.

Policers can be assigned in a downlink, uplink, or both directions. Each policer can be allocated to police at any of the following levels:

- bearer level
- RG level
- a combination of RG and SID level

The gateway will aggregate the QoS requirements of the installed PCC rules according to the above levels. For example, all PCC rules that share the same RG/SID will be assigned one policer.

This functionality is independent from the PCRF control. If policy rules are activated by the PCEF, policing at the RG/SID level is also available.

Use case 1

Application A is reported as RG-1 SID-1 and application B as RG-1 SID-2. Both are active in the same bearer (same QCI/ARP). Application A can be policed at 5 Mb/s and application B at 2 Mb/s. Each RG/SID is allocated one policer and is then policed independently according to the PCC rule configuration.

If application C is installed and uses RG-1 SID-1 (same as application A) and requires policing at 3 Mb/s, then application A and application C traffic is policed at the aggregated rate of 8 Mb/s (5 Mb/s + 3 Mb/s).

For deployments with AA, this feature will not limit the SDF-level policing applied to flows diverted to the AA.

Use case 2

An operator can use RG/SID policing at the MG-ISM node and divert the flows which require SDF policing to the AA. In this case, a policer is assigned to the RG/SID level, which polices the aggregated flows for the same RG/SID (AA flows and non-AA flows). Another policer is handled by the AA for the flow which was sent to the AA.

To enable policing by the local MG-ISM node without diverting traffic to the AA, use the following command:

```
configure mobile-gateway profile policy-options action-rule-unit sdf-level-  
policing
```

13.23 QoS policing per SDF

This feature provides an option to apply QoS policy at SDF level without the need to divert traffic to the AA.

Traffic policing per SDF enables policing of application traffic per flow. If enabled, traffic is not policed at bearer level. The policer values of multiple SDFs that share the same RG/SID are not aggregated.

Enabling sdf-level-policing for an action rule unit that relates to a policy rule or policy rule base, creates SDF level policers with the traffic matching the policy rule being policed at the gbr and mbr configured under the policy rule unit.

To enable SDF level policing for non-CUPS on uplink, downlink, or both directions at the same time, use the following commands:

```
configure mobile-gateway system policer-distribution dl-only dl-percent
```

```
configure mobile-gateway profile policy-options policy-rule-unit qos down-link  
gbr gbr mbr mbr
```

```
configure mobile-gateway profile policy-options policy-rule-unit qos up-link  
gbr gbr mbr mbr
```

```
configure mobile-gateway profile policy-options action-rule-unit sdf-level-  
policing direction direction group group
```

Uplink and downlink SDF level policing

```
configure mobile-gateway system policer-distribution dl-only 50  
configure mobile-gateway profile policy-options policy-rule-unit qos down-link  
gbr 100 mbr 100  
configure mobile-gateway profile policy-options policy-rule-unit qos up-link  
gbr 100 mbr 100  
configure mobile-gateway profile policy-options action-rule-unit sdf-level-  
policing direction bi-direction group sdf
```

Configuration for downlink SDF level policing

```
Policy-Rules:  
Policy-Rule 1:  
Remote IP Address 15.1.1.1  
Remote Port 10000  
Protocol 17  
GBR 100 MBR 100
```

```
Policy-Rule 2:  
Remote IP Address 15.1.1.2  
Remote Port 2000  
Protocol 17  
GBR 200 MBR 200
```

Traffic matching policy rule 1 is policed at 100 kbps downlink. Traffic matching policy rule 2 is policed at 200 kbps downlink.

If sdf-level-policing is disabled:

- If the policy rules are both installed on a GBR bearer, the policer for that bearer will be 300 kbps.

- If the policy rules are installed on a non-GBR bearer, the policer will be based on the APN-AMBR values.

CUPS/5G support

SDF level traffic policing for 5G is supported in downlink or both uplink and downlink direction with the gbr and mbr configured under a QoS rule unit without diverting the traffic to the AA. Uplink-only direction is not supported.

Enabling sdf-level-policing for a QoS rule unit that relates to a policy rule or policy rule base, creates SDF level policers with the traffic matching the policy rule being policed at the gbr and mbr configured under the QoS rule unit.



NOTE:

QRU must be added to all the policy rules in the same policy rule base.

To enable SDF level policing for 5G/CUPS, use the following commands:

```
configure mobile-gateway profile policy-options qos-rule-unit qru-name
```

```
configure mobile-gateway profile policy-options qos-rule-unit qos down-link  
gbr gbr mbr mbr
```

```
configure mobile-gateway profile policy-options qos-rule-unit qos up-link  
gbr gbr mbr mbr
```

```
configure mobile-gateway profile policy-options qos-rule-unit sdf-policing  
direction direction
```

Uplink and downlink SDF level policing

```
configure mobile-gateway profile policy-options qos-rule-unit QRU1  
configure mobile-gateway profile policy-options qos-rule-unit qos down-link  
gbr 100 mbr 100  
configure mobile-gateway profile policy-options qos-rule-unit qos up-link gbr  
100 mbr 100  
configure mobile-gateway profile policy-options qos-rule-unit sdf-policing  
direction both
```

Configuration for both uplink and downlink SDF level policing

```
Policy-Rules:  
Policy-Rule 1:  
qos-rule-unit QRU1 :
```

```
sdf-policing direction both
Downlink GBR 200 MBR 200
Uplink GBR 200 MBR 200
policy-rule-unit PRU1 :
Remote IP Address 15.1.1.1
GBR 100 MBR 100
```

```
Policy-Rule 2:
qos-rule-unit QRU2 :
sdf-policing direction both
Downlink GBR 300 MBR 300
Uplink GBR 300 MBR 300
policy-rule-unit PRU2 :
Remote IP Address 15.1.1.2
GBR 100 MBR 100
```

Traffic matching policy rule 1 is policed at 200 kbps downlink configured in the QRU. The policing is done in both uplink and downlink direction since sdf level policing is enabled in both directions.

Traffic matching policy rule 2 is policed at 300 kbps downlink configured in the QRU. The policing is done in both uplink and downlink direction since sdf level policing is enabled in both directions.

If sdf-level-policing is disabled:

- If the policy rules are both installed on a GBR bearer, the policer for that bearer will be 200 kbps (MBR/GBR from the policy-rule-unit).
- If the policy rules are installed on a non-GBR bearer, the policer will be based on the APN-AMBR values.

13.24 Bandwidth roundup for GBR and MBR

This feature allows operators to configure the PGW to round upwards the MBR/GBR APN-AMBR received from PCRF, into a valid kb/s value as specified in 24.301 (NAS specification).

This feature solves a problem detected in some VoLTE networks which do not round the value received from the PGW into a valid NAS value. In such cases a VoLTE call failure might be experienced by some VoLTE UEs due to the invalid bitrate.

Notice that the round upwards is used only to inform the LTE access. The PGW will internally remain aligned to Gx specifications, thus internally it will continue enforcing the PCRF provided bitrates. All other interfaces (Gx, charging, non LTE accesses, and so on.), will continue reporting the PCRF provided values.

If this feature is disabled, the behavior is the standard (backward compatible with previous mobile gateway releases) which follows Gx interface specifications.

The following CLI command can be used to configure this functionality:

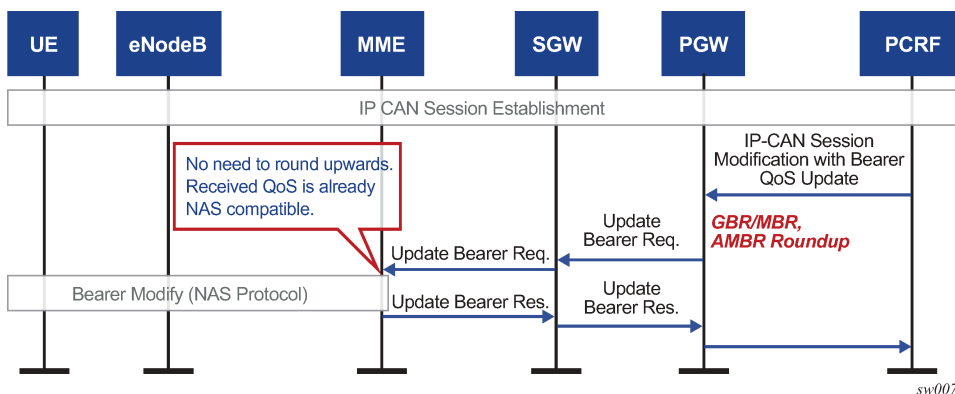
```
configure mobile-gateway pdn apn bitrate-roundup
```


Table 60: Examples for rounding upwards

PCRF	Gx specs (29.274)	Non Access Stratum (NAS) specs (24.301)
101850 b/s	102 kb/s	104 kb/s (8 kb/s steps in this range)
8670456 b/s	8671 kb/s	8700 kb/s (100 kb/s steps in this range)

From the example in *Examples for rounding upwards*, if the PCRF sends value 101850 b/s, if the feature is disabled, the value sent to the network is 102 kb/s (rounded to the next kb/s value using 1000 b/s = 1 kb/s). If the feature is enabled, the value sent to the LTE network is 104 kb/s, as this is the next step allowed in NAS protocol.

Figure 189: GBR/MBR AMBR roundup process flow



13.25 Enable/disable dynamic PCC per RAT type

This feature allows operators to define the RAT types that are allowed to use PCRF resources and the RAT types that are not controlled by the PCRF.

UE mobility across different RAT types indicates that during an IP-CAN session, there may be multiple handovers between RAT types with and without the PCRF's control. Dedicated bearers and services provided to the UE will not be affected if the operator decides to allow such services to remain active in the RAT types with no PCRF control.

When a PDN connection is established, the gateway will check the configuration to determine whether the UE (IP-CAN session) is under the PCRF's control. If the UE is under PCRF control, a new parameter indicates the RAT types that are allowed to be under the PCRF's control.

The functionality is as follows:

1. If an IP-CAN session is created under a RAT type which is not under the PCRF's control, a CCR-I is not sent. The functionality is the same for sessions that are not under the PCRF's control.
 - a. If during the life of the session, there is a handover to a RAT type which is allowed to be under the PCRF's control, the Gx session is started (a CCR-I is sent toward the PCRF). Rule bases and rules may be changed according to the PCRF indication. This might trigger the closure of service data flows.
 - b. If there is a second handover, back to a RAT type without the PCRF's control, a new configurable timer starts running.

If at timer expiration the RAT has not changed (not allowed for PCRF control), a CCR-T is triggered to the PCRF to terminate the Gx session. Rules and rule bases will be reverted back to their initial values allowed for the UE without the PCRF's control. This might trigger the closure of service data flows.
2. If an IP-CAN SESSION is created under a RAT type which is under the PCRF's control, the CCR-I is sent as usual. Functionality is the same for sessions under the PCRF's control.
 - a. If during the life of the session, there is a handover to a RAT type which is not allowed to be under the PCRF's control, the behavior is identical to that described in 1b.
 - b. If there is a second handover, back to a RAT type under the PCRF's control, the behavior is identical to that described in 1a.

To configure dynamic PCC and to set the RAT types not allowed under the PCRF's control at the charging profile level, use the following commands:

```
configure mobile-gateway profile charging pgw-charging pcrf-selection dynamic-  
pcc enable
```

```
configure mobile-gateway profile charging pgw-charging pcrf-selection pcc-  
disable-per-rat
```

```
configure mobile-gateway profile charging pgw-charging pcrf-selection pcc-  
disable-per-rat-timer
```

(optional)

To set the dynamic PCC and RAT types not allowed under PCRF control at the APN level, use the following commands:

```
configure mobile-gateway profile charging pgw-charging pcrf-selection dynamic-pcc inherit
```

```
configure mobile-gateway pdn apn pcrf-selection dynamic-pcc
```

```
configure mobile-gateway pdn apn pcrf-selection pcc-disable-per-rat
```

```
configure mobile-gateway pdn apn pcrf-selection pcc-disable-per-rat-timer
```

(optional)

To configure a combination of PGW charging and APN-level, use the following commands:

```
configure mobile-gateway profile charging pgw-charging pcrf-selection dynamic-pcc enable
```

```
configure mobile-gateway profile charging pgw-charging pcrf-selection pcc-disable-per-rat
```

```
configure mobile-gateway pdn apn pcrf-selection pcc-disable-per-rat
```

For example, if at the PGW charging profile level the RAT type is set to WLAN, then the WLAN RAT type will be excluded from the PCRF's control. At the APN-level, if the RAT type is set to UTRAN and GERAN, these RAT types will be excluded from the PCRF's control. In this example, subscribers under the PGW charging profile will not be under the PCRF's control when WLAN, GERAN, or UTRAN are used. However, GERAN and UTRAN will be excluded from the PCRF's control for subscribers using the same APN but a different charging profile.

To view the PGW charging profile configuration, use the following command:

```
show mobile-gateway gateway profile charging pgw-charging
```

13.26 Dynamic modification of AA start/stop notifications

The system can be configured to provide start/stop notifications when a user generates a flow which matches a particular AA charging group.

This is possible when PCRF requests ADC start/stop notifications or when a predefined PCC rule triggers the activation of a dedicated bearer. The charging group for which the AA must provide notifications and the reporting granularity must be configured.

To configure the charging group, use the following command:

```
configure application-assurance group policy charging-group charging-group-name create notify-start-stop [flow-based]
```

If the **flow-based** option is configured, the AA provides notifications for each flow that matches the configured charging group. If not, the AA provides a start notification when the first flow that matches the charging group is detected, and a stop notification when the last flow that matches the charging group terminates. This is a static configuration and applies to all sessions.

To configure the granularity level, at which the AA provides start and stop notifications dynamically per session, use the following command:

```
configure mobile-gateway profile policy-options action-rule-unit aa-notify-action {charging-group | flow-level | none}
```

The ARU contains the requested granularity level, which can be configured with the following actions:

- request notifications on charging group level
- request notifications on flow level
- suppress notifications

The operator can activate a PCC rule that contains the above ARU and modify the notifications for that user session. The operator must always activate start-stop notifications for a charging group (in AA), to be able to dynamically modify or suppress notifications.

13.27 High data rate support for 5G New Radio (option 3x)

High bit rates for New Radio (NR) are supported: 20 Gb/s for downlink and 10 Gb/s for uplink, per UE according to the requirements in TR38.913 in Rel-15.

Currently QoS-specific data over PCC interfaces are defined as Unsigned32 type values, indicating the bit rate in bits per second. According to this definition, the maximum supported bandwidth value is about 4,3 Gb/s ($2^{32} - 1 = 4.294.967.295$ bits per second). The GTP-C can signal data rates up to 4 terabits per second (Tb/s).

New AVPs have been introduced to support bit rates higher than 4 Gb/s. These AVPs are also of type Unsigned32 but indicate kilobits per second. The maximum supported bandwidth value is therefore approximately 4,3 Tb/s.

In Release 15 of the protocol, the UE and the network interpret the following:

- any maximum bit rate value and guaranteed bit rate value for uplink larger than 10 Gb/s as 10 Gb/s
- any maximum bit rate value and guaranteed bit rate value for the downlink larger than 20 Gb/s as 20 Gb/s

When the Extended-BW-NR feature is supported, extended bandwidth AVPs representing bit rates in kb/s will be used to support bandwidth values higher than $2^{32}-1$ bits per second (b/s) instead of the bandwidth AVPs representing bit rates in b/s. That is:

- Extended-Max-Requested-BW-DL/UL AVPs will be used instead of Max-Requested-Bandwidth-DL/UL AVPs.
- Extended-GBR-DL/UL AVPs will be used instead of Guaranteed-Bitrate-DL/UL AVPs.
- Extended-APN AMBR DL/UL AVPs will be used instead of APN-Aggregate-Max-Bitrate-UL/DL AVPs.
 - For values lower or equal to $2^{32}-1$ b/s AVPs representing bit rates in b/s will be used.
 - When the IP-CAN session is established and the PCEF supports the Extended-BW-NR feature and bandwidth values are higher than $2^{32}-1$ b/s, AVPs representing bit rates in b/s (that is, APN-Aggregate-Max-Bitrate-UL/DL AVPs) will be provided with a value set to $2^{32}-1$ b/s and bandwidth AVPs representing bit rate in kb/s (that is, Extended-APN-AMBR-UL/DL AVPs) will be provided with the actual required bandwidth.

**NOTE:**

When the Diameter session is being established, the originator node does not yet know the features supported by the peer node.

13.27.1 AVPs and information elements

The gateway supports sending the Extended-BW-NR feature bit in the Supported-Features AVP sent to the PCRF. Population of this feature bit is controlled by a CLI command.

Table 61: Features of Feature-List-ID 2 used in Gx

Feature bit	Feature	M/O	Description
7	Extended-BW-NR	O	This feature indicates the support of extended bandwidth values for NR.

For more information about extended bandwidth AVPs, refer to [29.212 section 5.3.16] QoS-Information AVP.

Table 62: Gx-specific Diameter AVPs

				AVP flag rules						
Attribute name	AVP code	Clause defined (29. 212)	Value type	Must	May	Should not	Must not	May Encr.	Acc. type	Applicable to
Extended-APN-AMBR-DL	2848	5.3.134	Unsigned32	V	–	–	M	Y	All	Extended-BW-NR
Extended-APN-AMBR-UL	2849	5.3.135	Unsigned32	V	–	–	M	Y	All	Extended-BW-NR
Extended-GBR-DL	2850	5.3.136	Unsigned32	V	P	–	M	Y	All	Extended-BW-NR
Extended-GBR-UL	2851	5.3.137	Unsigned32	V	P	–	M	Y	All	Extended-BW-NR

Table 63: Gx re-used Diameter AVPs

Attribute name	Reference	Description	Acc. type	Applicable to
Extended-Max-Requested-BW-DL	3GPP TS 29. 214	Defines the maximum authorized bandwidth in kilobits per second for downlink.	All	PC Extended-BW-NR
Extended-Max-Requested-BW-UL	3GPP TS 29. 214	Defines the maximum authorized bandwidth in kilobits per second for uplink.	All	PC Extended-BW-NR
Max-Requested-Bandwidth-UL	3GPP TS 29. 214	Defines the maximum authorized bandwidth in bits per second for uplink.	All	–
Max-Requested-Bandwidth-DL	3GPP TS 29. 214	Defines the maximum authorized bandwidth in bits per second for downlink.	All	–

The gateway supports the extended QoS parameters to support high data rates for NR, for online charging over the Gy interface, and for offline charging over the Ga interface.

The following AVPs defined in 29.212 section 5.3.16 QoS-Information AVP are reused over Gy and Rf Diameter interfaces.

- Extended-Max-Requested-BW-UL
- Extended-Max-Requested-BW-DL
- Extended-GBR-UL
- Extended-GBR-DL
- Extended-APN-AMBR-UL
- Extended-APN-AMBR-DL

This feature includes support for the following IEs (within EPC QoS Information) on the Ga interface:

- extendedMaxRequestedBWUL
- extendedMaxRequestedBWDL
- extendedGBRUL
- extendedGBRDL
- extendedAPNAMBRUL
- extendedAPNAMBRDL

13.27.2 CLI commands

To configure High Data Rates for 5G NR (Option 3x), use the following commands:

```
configure mobile-gateway profile gx supported-features extended-bw-nr
```

```
configure mobile-gateway profile charging cdr-avp-options extended-bw-nr
```

```
configure mobile-gateway profile charging gy-avp-options include-extended-bw-nr
```

13.27.3 QCI values

The mobile gateway supports the following QCI values (for example, for low latency with normal reliability, URLL support for new QCI values C) as part of the EPC enhancements for 5G NR.

- QCI = 67 (GBR)
- QCI = 75 (GBR)

- QCI = 79 (non-GBR)
- QCI = 80 (non-GBR)
- QCI = 82 (GBR)
- QCI = 83 (GBR)
- QCI = 84 (GBR)
- QCI = 85 (GBR)

Existing CLI and show commands support these new QCI values. For more information, refer to 3GPP 23.203, Table 6.1.7-A: Standardized QCI characteristics.

13.28 DNS snooping

This functionality provides the option to configure L3/L4 filters using FQDNs instead of IP addresses.

The mobile gateway allows the configuration of FQDN lists as L3/L4 filters with each FQDN list including a set of FQDNs (regular expressions). The L3/L4 domain name functionality dynamically creates internal destination IP address filters, according to the FQDN list, by snooping the DNS responses (following the DNS requests by the end user). If a PCC rule filter for the specific session is configured with a set of domain names (FQDN List), the mobile gateway detects the domain name in the DNS response. Matching with the configured domain name is performed by searching if the string is a sub string in the domain name.

The L3/L4 domain name configuration eliminates the requirement for (static) configuration of multiple IP addresses for that domain name, but does not prohibit the configuration of a host name and/or URI on L7 in the same filter. Domain names on L3/L4 are preferred to L7 host names in cases where improved performance is needed by skipping the AA. However, in cases where the AA is needed (for example, HTTP header enrichment, and so on), the DNS snooping functionality does not prevent other traffic to be analyzed by the AA. The DNS snooping functionality is available in the default bearer.

13.28.1 Wildcard matching

Wildcards (*) and dots (.) can be used when configuring regular expressions of domain names (FQDNs). The wildcard is a quantifier which matches to zero or more preceding characters, and the dot matches to any single character.

When using a wildcard, a detected domain name can match to more than one configured domain names. In this case, the mobile gateway does not only matches to the one with the greatest number of rightmost characters of the detected domain name, but to all matching domain names.

For example, the regular expression `^*\my-domain\com$` matches `wap.my-domain.com`, `www.my-domain.com`, and so on.



NOTE:

- A dot is used to match a single character. If a dot needs to be matched, use "\.". In the preceding example, the "\" characters after the wildcard are optional and prevent the matching of, for example, fakemy-domain.com.
- If the first character needs to be matched, use the "^" character. For example, to match only www.my-domain.com, the configured domain name must be ^www\my-domain\com\$.

Table 64: Examples of configured and DNS-matched domain names

Configured domain name	Matching DNS response
^www\my-domain\com	www.my-domain.com
www\my-domain\com	www.my-domain.com abcwww.my-domain.com
www.my-domain.com	www.my-domain.com abcwww-my-domain.com
my-domain.com	academy.trainings.my-domain.com
www.*domain.com	www.domain.com www.domain3.domain2.domain.com
www.my.*domain.com	www.mysubdomain.com www.my-secondlevel.domain.com
www.domain.*.com	www.domain-host.com www.domain2.host.com
www.my-domain.*	www.my-domain.ph
my-domain.*	demo-stats.statistics.my-domain.iq

13.28.2 Dynamic storage of returned IP addresses from DNS replies

When DNS snooping is enabled for the UE-generated DNS traffic, the mobile gateway snoops into the DNS responses and dynamically creates internal tables.

If a single DNS response contains more than 20 TYPE A (resolved IPv4 address) and/or TYPE AAAA (resolved IPv6 address) answer fields, snooping is performed only for the first 20 IP addresses and the rest are ignored. This does not affect the maximum number of IP addresses that can be cached under any domain name.

**NOTE:**

The mobile gateway snoops:

- UDP DNS responses from port 53 (DNS over TCP is ignored)
- DNS traffic from APN PCO, trusted DNS servers, or from all DNS responses (configurable)

The mobile gateway allows the configuration of up to 2000 FQDN lists, with each FQDN list including a maximum of 25 FQDN (regular expressions) entries. A maximum of 60000 resolved IP addresses can be stored, and each FQDN list can store up to 1000 IP addresses. The mobile gateway limits the learned IP addresses per session to 50 FQDN lists.

When the number of stored resolved IP addresses reaches 45000, a caution alarm is raised. This alarm indicates that the IP address list of the filters may not be updated with newly resolved IP addresses.

The DNS table is synchronized between the active and the standby nodes to ensure that no data is lost during switchovers. In geo-redundant deployments, the DNS database is also synchronized every 5 minutes between the master and the slave mobile gateways.

**NOTE:**

The snooped DNS responses are matched with an internally created rule for downlink traffic to UDP port 53 and charged to RG 0. The packets can be charged to another RG only if a specific rule to catch UDP port 53 has been configured.

13.28.3 Refresh of stored entries

The mobile gateway deletes and refreshes any outdated IP addresses based on their TTL and a hard-coded time threshold (two minutes). The time threshold serves as a safety margin for the deletion of an IP address after its TTL has expired.

Every IP address acquires an estimated time-stamp (ETS) which serves as its deletion criterion. The ETS is defined as the sum of the update time (the time when the outdated IP addresses are updated), the TTL, and the time threshold ($ETS = \text{update time} + \text{TTL} + \text{time threshold}$). There is a periodic scan of the ETS values of all stored (resolved) IP addresses (two minutes + the lowest TTL of all IP addresses in the specific FQDN group). During this scan, if the scan time is greater than the sum of the ETS and the time threshold ($\text{scan time} > ETS + \text{time threshold}$), the IP address is deleted.

**NOTE:**

The mobile gateway limits the minimum and maximum TTL values for the stored entries to 5 minutes and 12 hours respectively.

13.28.4 Configuring DNS snooping

Purpose



NOTE:

For more information about the commands used in this section, refer to the *CMG CLI Reference Guide*.

Procedure

1. Create an FQDN list for DNS snooping.

The FQDN list must include the domain names for DNS snooping used for storing and matching the resolved IP addresses.

```
configure mobile-gateway profile list
fqdn-list fqdn-list-name
fqdn fqdn-name-id fqdn-name-string
```

Step example

```
configure mobile-gateway profile list
fqdn-list list-dns-snoop
fqdn 10 .*\.my\.domain\.com
```

2. Enable DNS snooping for the FQDN list.

```
configure mobile-gateway profile policy-options policy-rule-unit policy-name
flow-description entry-id
match
dns-snooping fqdn-list-name
```

Step example

Use the FQDN list created in step 1.

```
configure mobile-gateway profile policy-options policy-rule-unit policy1
flow-description 10
match
dns-snooping list-dns-snoop
```

3. Create an IP address list of the DNS servers.

The IP address list must include the IP addresses of the DNS servers for DNS snooping.

```
configure mobile-gateway profile addr-list addr-list-name ip-address ip-address/prefix-length
```

```
configure mobile-gateway profile addr-list ip-list-1 ip-address 10.10.10.10
```

4. Enable DNS snooping for the APN.

```
configure mobile-gateway pdn gw-id apn apn-name dns-snooping [dns-server-list addr-list] [snoop-all]
```

Sub-steps

a) To snoop DNS responses from the DNS servers defined in APN PCO, use the following command:

```
configure mobile-gateway pdn gw-id apn apn-name dns-snooping
```

Step example

```
configure mobile-gateway pdn 1 apn APN_1 dns-snooping
```

b) To snoop DNS responses from APN PCO and specific DNS servers, use the following command and specify the IP address list created in step 3.

```
configure mobile-gateway pdn gw-id apn apn-name dns-snooping dns-server-list addr-list
```

Step example

```
configure mobile-gateway pdn 1 apn APN_1 dns-snooping dns-server-list ip-list-1
```

c) To snoop all DNS responses, use the following command:

```
configure mobile-gateway pdn gw-id apn apn-name dns-snooping snoop-all
```

Step example

```
configure mobile-gateway pdn 1 apn APN_1 dns-snooping snoop-all
```

13.28.5 Forwarding DNS traffic to AA

To enable the forwarding of DNS traffic to AA for deployments that require DNS snooping and AA functionality related to DNS analysis, use the following command:

```
configure mobile-gateway pdn gw-id apn apn-name  
[no] dns-snooping-forward-to-aa
```

The following is an example of `dns-snooping-forward-to-aa`:

```
configure mobile-gateway pdn 1 apn APN_1  
dns-snooping-forward-to-aa
```

13.29 Non-terminating policy rules

By default, traffic is matched to a PCC rule according to the rule precedence. When a UE packet matches traffic filters of multiple installed policy rules, traffic is matched to the policy rule with the lowest precedence value.

Non-Terminating Policy Rules is a Nokia proprietary functionality that allows you to match a UE packet to more than one policy rules.

CMG allows policy rules to be defined as “Non-Terminating”. When the functionality is enabled, when the UE traffic matches a non-terminating rule, the traffic matching process continues. CMG first attempts to match traffic to the installed non-terminating rules, and then it runs a second pass to match traffic to one Terminating Policy Rule (the matching terminating policy rule with the lowest precedence value).

This allows you, for example, to configure steering policy independently of charging policy, by installing non-terminating policy rules with steering rules, and a different policy rule (Terminating) for defining the charging method and, or rating group.

The following actions can be configured in non-terminating policy rules:

- `svc-steering` [steering-profile *profile-name*]
- `flow-gate-status` *flow-gate-status*
flow-gate-status values: allow/drop/allowUL/allowDL/auto-redirect



NOTE:

The Drop action can be selected from the highest priority non-terminating or terminating rule.

The steering action must be selected from the non-terminating rule. If the non-terminating rule does not have a steering action, then it is selected from the terminating rule, provided a steering action is configured under it.

On-the-fly changes: In the current Steering implementation, CMG does not update existing sessions when a rule, or a rule base, is changed from steering to non-steering, and vice-versa.

Traffic matching when functionality is enabled

When the functionality is enabled, traffic matching is first applied against all non-terminating policy rules installed in the session in precedence order. Traffic matching is then applied against all terminating policy rules. For terminating policy rules, only the highest priority terminating rule is applied.

For a certain action type (for example, flow gate status or SVC steering), the first matched rule (according to precedence) overwrites the others.

Packets matching only non-terminating rules are also valid.

For example, when a packet matches the traffic filters for the following rules:

- pcc-rule-1, precedence value: 5000, non-terminating # Port 80, 8080
- pcc-rule-2, precedence value: 4000, terminating
- pcc-rule-3, precedence value: 3000, non-terminating # Port 25
- pcc-rule-4, precedence value: 2000, terminating
- pcc-rule-5, precedence value: 1000, non-terminating # Port 44
- pcc-rule-6, precedence value: 10, terminating # Youtube L7 rule

Where:

- pcc-rule-5 (highest priority) configures flow-gate-status and does not configure svc-steering
- pcc-rule-3 configures flow-gate-status and svc-steering

In this case, first matching is applied to non-terminating rules pcc-rule-1, 3, 5.

Steering configuration applied is according to pcc-rule-3.

Flow gate status applied is according to pcc-rule-5.

14 GPRS Tunneling Protocol

This section describes information related to supported GPRS Tunneling Protocol (GTP) functionality.

14.1 GTP configuration

To configure GTP profiles, use the GTP Profile commands in the `config>mobile>profile>gtp` context. This includes commands for enabling or disabling IEs to be included in GTPv1 and GTPv2 messages, such as change and CSG reporting action IEs, APN restriction IEs, charging gateway address information, and so on.

To configure GTP-C and GTP-U for supported interfaces and signaling (for example, S11, S12, S2b, S5, S8, Gn, Gp and so on), use the `gtp-c` and `gtp-u` commands in the `config>mobile>pdn` context.

For more information, refer to the Profile and PDN Commands sections in the *CMG CLI Reference Guide*.

14.2 GTP error indication messages

GTP Error Indication is a method for the data path entity in the mobile core to inform its peers of inconsistent forwarding states and recovery from them.

If a GTP packet is received with an unknown TEID, the receiving node informs its sending peer via Error Indication messages of the inconsistent mappings between the sender and receiver. The sender includes the received TEID in the Error Indication message to provide the bearer context that is in the wrong state.

Support is provided for the following GTP Error Indication behavior:

- processing GTP Error Indication messages received on N3, N9, S11-U, S1u, S4u, S5u, S8, S2b, S2a, Gn, and Gp interfaces
- sending GTP Error Indication messages to the peer node upon receiving a GTP-U packet with an unknown TEID, for GTP-U interfaces (N3, N9, S11-U, S1u, S4u, S5, S8, S2b, S2a, Gn, and Gp)

**NOTE:**

GTP Error Indication is generated for GTP-U interfaces independent of the interface type on which the GTP-U traffic is received.

With support for receiving and processing GTP Error Indication messages, corrective action is taken as per the 3GPP defined specifications. The default or dedicated bearer context is deleted or put in idle mode upon receiving the GTP Error Indication message from the peering node with the TEID value of associated bearer context.

Corrective action is taken for the bearer associated with the received TEID value on SGW (S1/S4/S5/S8), PGW (S5/S8,S2a,S2b), ePDG (S2b), and TWAG (S2a). The action taken depends on the gateway type:

- On PGW, if the TEID value of a default bearer is received in an error indication message from an S5/S8/S2a/S2b peer, the default bearer and all associated dedicated bearers are deleted. If the TEID value of a dedicated bearer is received in an error indication message from S5/S8/S2a/S2b peer, only the associated dedicated bearer is deleted.
- On SGW, if the TEID value of a default bearer is received in an error indication message from an S5/S8 peer, the default bearer and all associated dedicated bearers are deleted. If the TEID value of a dedicated bearer is received in an error indication message from an S5/S8 peer, only the associated dedicated bearer is deleted. If the TEID value of a default or dedicated bearer is received in an error indication message from the S1u or S11-U peer, the UE is put in idle mode and DDN is initiated toward MME with a specific cause code. If the TEID value of the default or the dedicated bearer is received in an error indication message from the S4 peer, the following action is configurable:
 - The UE is put in idle mode and the DDN is initiated toward the SGSN with a specific cause code.
 - The default or dedicated bearer identified by the TEID value in an error indication message is deleted. If the default bearer is deleted, all the associated dedicated bearers are also deleted.
- On ePDG, if the TEID value of a default bearer is received in an error indication message from an S2b peer, the default bearer and all associated dedicated bearers are deleted. If the TEID value of a dedicated bearer is received in an error indication message from an S2b peer, only the associated dedicated bearer is deleted.
- On TWAG, if the TEID value of a default bearer is received in an error indication message from an S2a peer, the default bearer and all associated dedicated bearers are deleted. If the TEID value of a dedicated bearer is received in an error indication message from an S2a peer, only the associated dedicated bearer is deleted.
- On GGSN, if the TEID value of a bearer context is received in an error indication message from a Gn/Gp peer, the primary or secondary PDP contexts are deleted. If there is a direct tunnel to the RNC for the PDP context that maps to the TEID received in the error indication on the Gn/Gp interface, then the GGSN does not delete the PDP context but marks it as invalid (release downlink TEID) and updates the SGSN that it has received the error indication from the RNC.

The following command can be used to enable sending of GTP Error Indication messages on GTP-U interfaces:

```
configure mobile-gateway system error-indication-tx
```

The following show command displays the number of GTP Error Indication messages generated per peer IP address:

```
show mobile-gateway pdn error-indication
```

The following show command displays the number of bearers or sessions deactivated due to Error Indication messages received on the S5/S8, S2a, S2b, S4 and Gn/Gp interfaces and the number of S1 release procedures due to Error Indication messages received on the S1-U interface:

```
show mobile-gateway pdn call-flow-stats
```


The following show command has been enhanced to display the number of Update PDP Context Request transmitted toward SGSN on receiving Error Indication for PDP for which Direct Tunnel exists with RNC:

```
show>mobile>pdn>ref-point-stats>gn
```

The following KPI counters are supported:

- total number of default bearer deactivations due to Error Indication message received on the S5/S8, S2a, S2b, S4 and Gn/Gp interfaces
- total number of dedicated bearer deactivations due to Error Indication message received on the S5/S8, S2a, S2b, S4 and Gn/Gp interfaces
- number of S1 release procedures due to Error Indication message received on the S1-U interface

For information about KPI-KCI counters, refer to the *CMG KPI-KCI Counters Guide* and the *CMG KPI-KCI Counters Spreadsheet*.

14.3 GTP retransmit timer configuration

The GTP profile provides message retransmit configuration options.

The signaling procedure on most network elements for an interface has the same no answer timer. This can hinder the timer optimization effort for different call flow scenarios, where network elements that precede others are timing out before the network elements later in the sequence. To overcome this limitation, each element later in the sequence requires a reduced timeout period than the previous node. For example, the MME must wait less than 9 seconds because the eNB timeout is configured at 9 seconds.

The SGW and various other EPC elements support configuration of timeout values in multiples of seconds. This restricts fine tuning of end-to-end (for example, call setup) values. For example, the MME can normally only fine tune its timer at 8 seconds to be less than 9 seconds. Or the SGW timer value can be 6 seconds, which is less than 8 seconds.

A granular timer value of 100 milliseconds (100-ms) is required for Timeout and Retry per GTP message type, to support better tuning of the network. The GTP retransmission timer can also be set at 100-ms granularity. By allowing this finer timeout granularity, an operator can cascade the GTP timeouts so that upstream elements will timeout more quickly. For example, the SGW S5 reference points may be set to timeout before the MME's S11 reference points. Therefore the MME will be able to differentiate between a case where the SGW is simply not responding and the case where the SGW has not yet timed out on its respective S5 interfaces.

For configuration information, see the `config>mobile>profile>gtp>message-retransmit` command.

14.4 SMS paging distinction

The GTP profile can be configured for SMS paging distinction. This feature enhances the VoLTE call experience for end users by reducing the time it takes to set up a voice call.

Currently, a single IMS PDN session is used for both voice calls and SMS delivery. Because the mobility management entity can not differentiate between a voice call and SMS over the same PDN session, it treats both applications with the same priority during paging. However, it is possible to assign a relative priority to IMS applications based on the DSCP marking. For example, SMS delivery is handled at a lower priority if the P-CSCF marks SMS traffic with a lower priority DSCP marking in the IP header compared to the DSCP marking for a voice call. The SGW receives the DSCP value in the user packet header during paging for either voice call or SMS, and forwards it to the MME in the DDN message. The MME then applies distinction paging priority based on the application that is being activated.

For configuration information, see the `config>mobile>profile>gtp>gtpv2-options>exclude-paging-priority` command.

14.5 Pause of charging

PGW pause charging

Pause charging is supported on the PGW so that charging records on the SGW and PGW are not mismatched.

This feature protects subscribers from overcharging when excessive downlink traffic is dropped due to loss of radio coverage while the UE is active, or because of extended paging while the PGW continues to forward user traffic to the SGW. The SGW indicates support for pause charging using the PPSI bit set in the Indication IE of a Create Session request for a new session or a Modify Bearer request for a handover. The PGW responds with the PPSI bit set and the PPEI bit set in the Indication IE to enable the SGW to use pause charging. If the SGW detects an abnormal radio link release or experiences an extended period of paging, the SGW sends the PGW a Modify Bearer request with a pause charging indicator set to TRUE. The SGW can clear this condition by sending a Modify Bearer request with the pause charging indicator set to FALSE.

The PGW receives a Modify Bearer request with pause charging indicator set to FALSE or TRUE depending on the active state or unreachable state of the UE. When pause charging is enforced by the SGW, the PGW stops counting traffic that is being forwarded to the SGW even though the PGW continues to generate charging records.

Pause charging is supported on PGW sessions only and applies to Rf, Gy, and Ga reference points.

For configuration information, see the `pgw-pause-charging` command in the following contexts:

```
configure mobile-gateway pdn s5 signaling gtp-c pause-charging
```

```
configure mobile-gateway pdn s8 signaling gtp-c pause-charging
```



NOTE:

The PDN pause charging feature must not be enabled with high latency communication devices.

SGW pause charging with packet loss trigger

Pause charging is designed to solve the mismatching of charging records on the SGW and PGW and to avoid overcharging a subscriber if excessive downlink traffic is dropped when the UE is unreachable.

If the SGW detects that a UE is unreachable, it can send a pause charging indication up to the PGW to tell the PGW to stop charging for the applicable PDN connection. The SGW will send the pause charging indication when it receives a certain error code in a DDN Acknowledgment from the MME. The SGW can also trigger the pause charging indication based on a configurable per UE buffer threshold.

The gateway supports configurable triggering of the pause charging to the PGW after the 9th packet. 12 packets is the total buffer size on the MSCP (Mobile Subscriber Control Plane) per UE. Triggering of the pause charging to the PGW requires signaling between the SGW and the PGW. If the 12 packet limit is reached quickly, it may lead to excessive signaling, which may not be desirable. To suppress excessive signaling between the SGW and the PGW, a configurable hold-down timer (per UE) is also supported, to allow extra time for paging to be successful before pause charging is triggered toward the PGW.

The pause charging and the hold timer can be configured for GTP-C signaling for the s5 and s8 interfaces, using the `data-threshold` and `data-threshold-hold-time` CLI commands in the following contexts:

```
configure mobile-gateway pdn s5 signaling gtp-c pause-charging
```

```
configure mobile-gateway pdn s8 signaling gtp-c pause-charging
```

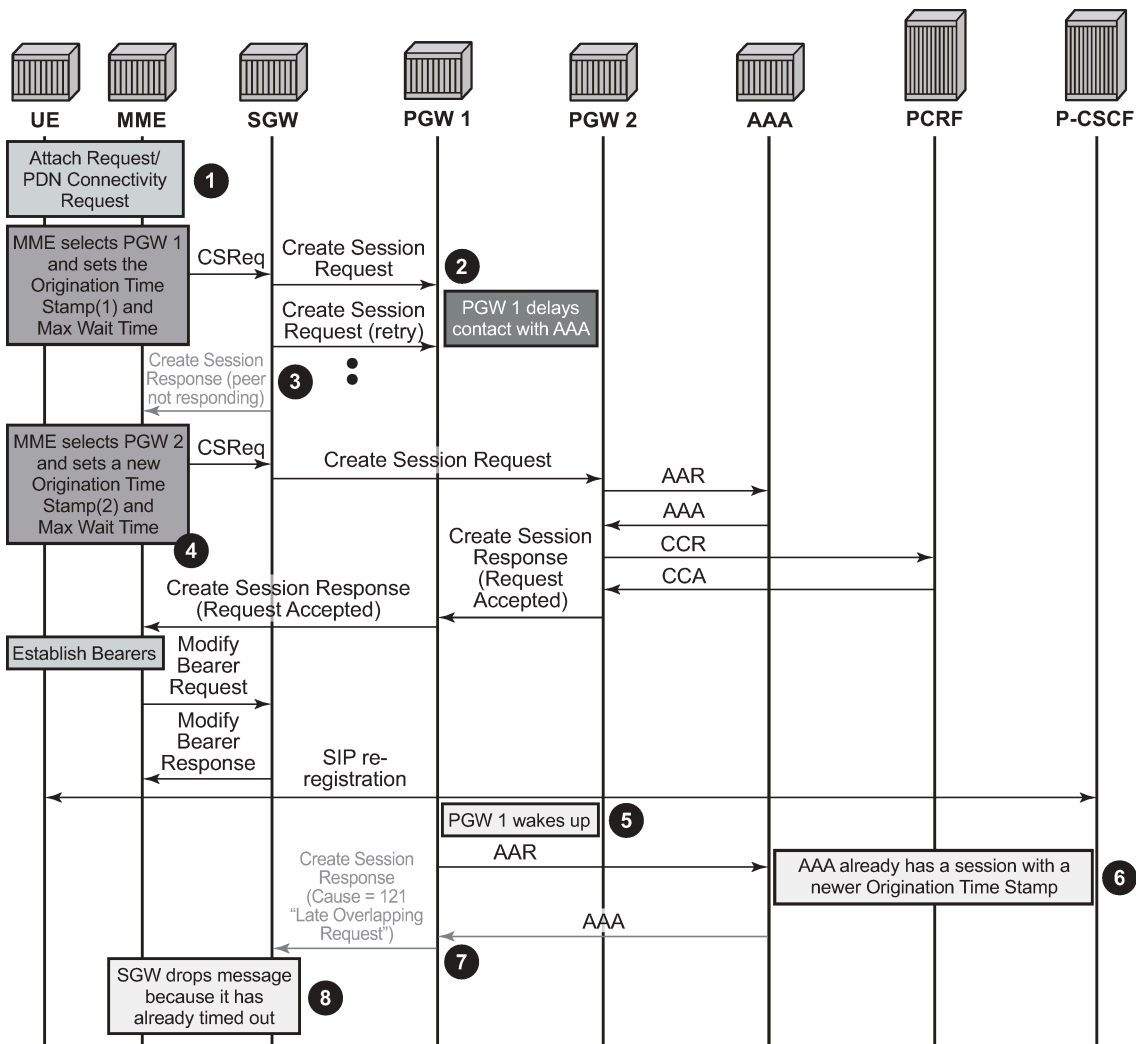
14.6 GTP Diameter timestamp

Certain processing delays in EPC elements can cause the MME to select a different set of SGW and PGW elements after giving up on previous selections. This can lead to differing views amongst 3GPP elements regarding which set of EPC elements (primarily SGW and PGW) provides the PDN connection for a UE. This inconsistency can then result in VoLTE call failures.

To address these issues, the SGW and PGW support, the GTP IEs and the Diameter AVPs (Origination TimeStamp and Maximum Wait Time).

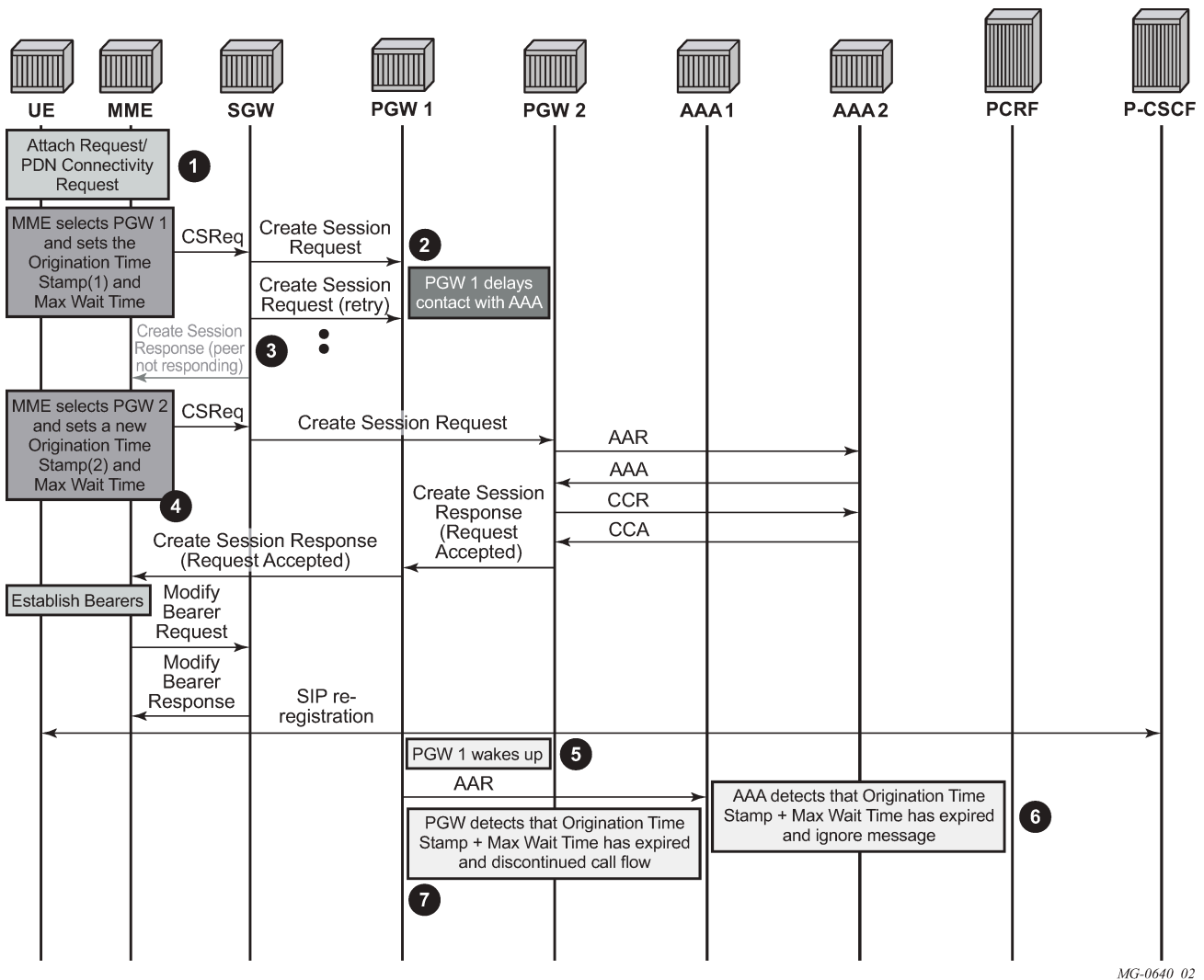
The call flows shown in *AAA detects failure based on origination time stamp*, *AAA and PGW clean up after Maximum Wait Time*, and *PGW detects SGW delay failure based on origination time stamp* provide examples of how the combination of GTP Time Stamp and Max Wait Time can prevent a failure.

Figure 190: AAA detects failure based on origination time stamp



MG-0640_01

Figure 191: AAA and PGW clean up after Maximum Wait Time



MG-0640_02

The following describes the steps in the call flows shown in *AAA detects failure based on origination time stamp* and *AAA and PGW clean up after Maximum Wait Time*. The first 5 steps are identical in both call flows.

1. The UE attempts to establish a new PDN connection. The MME chooses an SGW/PGW pair, sets the Origination Time Stamp to the current time, and sends the Maximum Wait Time.
2. PGW 1 delays further processing of the PDN connection request.
3. The SGW gives up and sends a Create Session Response with Cause = "Peer Not Responding".
4. The MME selects PGW 2 and uses the same SGW. The MME creates a new Origination Time Stamp. The call progresses successfully through PGW 2, which establishes sessions with the AAA (S6b) and the PCRF, and the UE registers with the P-CSCF. All nodes store the Origination Time Stamp for the PDN connection.
5. PGW 1 continues with the original PDN connection request and tries to establish the S6b session with the AAA.

6. The following occurs:

- In *AAA detects failure based on origination time stamp*, the AAA notices that it already has a session for this PDN connection, compares the Origination Time Stamp received from PGW 1 with the value stored for the session, and finds that PGW 1 is using an older timestamp. The AAA rejects the new session attempt and sends an AA-Answer with Experimental-Result-Code = "5453 DIAMETER ERROR LATE OVERLAPPING REQUEST".
- In *AAA and PGW clean up after Maximum Wait Time*, the AAA does not have an existing session, but notices that the Origination Time Stamp and Maximum Wait Time have expired. The AAA 1 ignores the message or rejects the new session with "5454 DIAMETER ERROR TIMED OUT REQUEST".

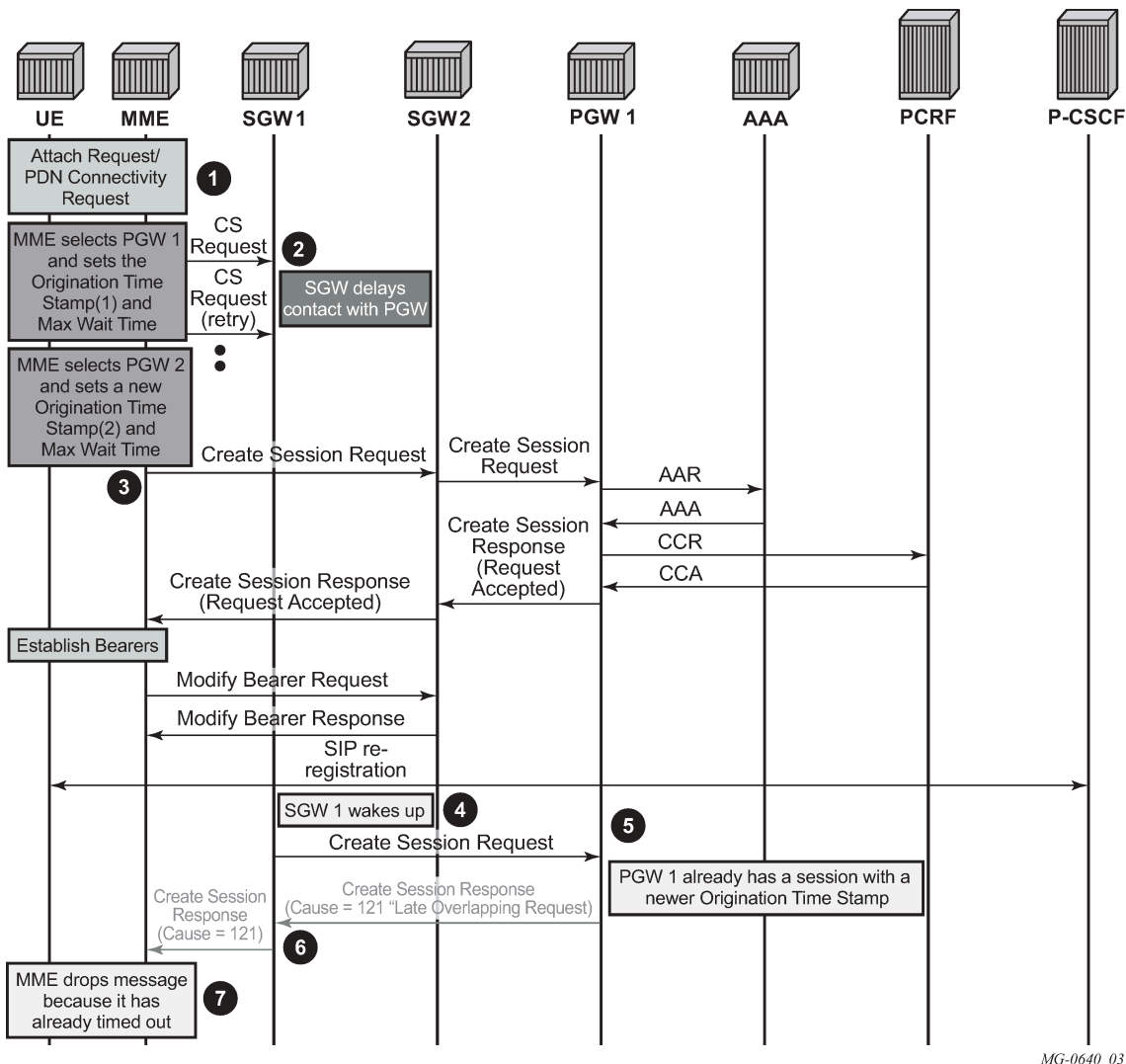
7. The following occurs:

- In *AAA detects failure based on origination time stamp*, the PGW 1 sends a Create Session Response with Cause = 121 "Late Overlapping Request".
- In *AAA and PGW clean up after Maximum Wait Time*, the PGW 1 notices that the Origination Time Stamp and Maximum Wait Time have expired and discontinues the call flow.

8. In *AAA detects failure based on origination time stamp*, the SGW discards the Create Session Response because it has timed out for PGW 1.

At the end of both call flows, the PGW 2 will have the PDN connection and the PGW 1 will have cleaned up any state for this PDN connection.

Figure 192: PGW detects SGW delay failure based on origination time stamp



The following describes the steps in the call flow shown in *PGW detects SGW delay failure based on origination time stamp*.

1. The UE attempts to establish a new PDN connection. The MME chooses an SGW/PGW pair, sets the Origination Time Stamp to the current time, and sends the Maximum Wait Time.
2. SGW 1 delays further processing.
3. The MME selects SGW 2 and the same PGW. The MME creates a new Origination Time Stamp. The call progresses successfully through SGW 2 and PGW 1. PGW 1 establishes sessions with AAA 2 (S6b) and the PCRF and the UE registers with the P-CSCF. All nodes store the Origination Time Stamp for the PDN connection.
4. SGW 1 wakes up and continues with the original PDN connection request and sends a Create Session Request to PGW 1.

5. PGW 1 notices that it already has a session for this PDN connection, compares the received Origination Time Stamp with the value stored for the session, and finds that SGW 1 is using an older timestamp. PGW 1 sends a Create Session Response with Cause = 121 "Late Overlapping Request".
6. SGW 1 sends a Create Session Response with Cause = 121 "Late Overlapping Request".
7. The SGW discards the Create Session Response because it already timed out for that PGW.

14.7 GTP-C load and overload control

The GTP-C load control and overload control are two distinct but complementary concepts.

Load control is an overload prevention concept. It enables an "originator" node (such as an SGW or PGW) to send its load information to a "consumer" node (SGW, PGW, MME or ePDG). The consumer node can use this load information for node selection to load balance the session across entities supporting the same function (a PGW cluster) according to their effective load.

Overload control is an overload mitigation concept. It enables an "originator" node (an SGW, a PGW, or an MME) that is overloaded or at risk of becoming overloaded to gracefully reduce incoming signaling by instructing its "consumer" node (an MME, an ePDG, or a PGW) to reduce sending traffic. A GTP-C originator node is in an overload state when it operates over its signaling capacity, resulting in diminished performance including impacts to the handling of incoming and outgoing traffic.

When load control reporting is enabled, GTP-C load control information can be configured to be sent in every n th GTP-C message; it can also be configured to be sent in every x percentage load change.

When GTP-C load control reporting is enabled on the combined SGW/PGW, it handles load control information in the GTP-C messages being sent toward the MME and the ePDG as listed below.

- For PDN sessions where the combined SGW/PGW is both an SGW and a PGW, load control information for both the SGW and PGW is sent in the GTP-C messages to the MME.
- For PDN sessions where the combined SGW/PGW is only an SGW, the SGW load control information is added in the GTP-C messages sent to the MME and any PGW load control information received in GTP-C S5/S8 messages is relayed to the MME.
- For PDN sessions where the combined SGW/PGW is only a PGW, the PGW load control information is added in the S5/S8 GTP-C messages sent to the SGW.
- For untrusted WiFi PDN sessions where the combined SGW/PGW is a PGW, the PGW load control information is added in the S2b GTP-C messages sent to the ePDG.

A combined SGW/PGW or an MME are considered to be in overload when they are operating over their nominal capacity, resulting in diminished performance including impacts to handling of incoming and outgoing traffic. When the combined SGW/PGW or the MME are in overload, GTP-C overload control information is sent to the consumer node to reduce and/or throttle the amount of GTP-C signaling traffic being sent toward the originating node. The combined SGW/PGW overload information can be configured with one of three levels of traffic reduction.

When overload control is enabled, the mobile gateway (acting as SGW, PGW, combined SGW/PGW, or ePDG) relays Overload Control Information (OCI) to the MME, SGW, PGW, and ePDG as follows:

- For PDN sessions where the mobile gateway is both an SGW and a PGW and operating over its nominal capacity, the mobile gateway will include both the SGW OCI IE and PGW OCI IE in the GTP-C message to the MME.
- For PDN sessions where the mobile gateway is the SGW and operating over its nominal capacity, the mobile gateway will include the SGW OCI IE (and include any PGW OCI IE received from the PGW) in the GTP-C message to the MME. It will also include the SGW OCI IE in the GTP-C messages to the PGW.
- For PDN sessions where the mobile gateway is the PGW and is operating over its nominal capacity, the mobile gateway will include the PGW OCI in the GTP-C message to the SGW.
- For untrusted WiFi PDN sessions where the mobile gateway is the PGW and is operating over its nominal capacity, the mobile gateway will include the PGW OCI IE in the GTP-C message responds to the ePDG.
- For untrusted WiFi PDN sessions where the mobile gateway is the ePDG and is operating over its nominal capacity, the mobile gateway will include the ePDG OCI IE in the GTP-C message to the PGW.

When GTP-C overload control is enabled, the PGW takes action when receiving GTP-C OCI IE from the MME, SGW and/or the ePDG. Thus, based on the traffic reduction received in the OCI IE from the MME, SGW and/or the ePDG, the PGW takes action on any network-initiated call flows to the MME, SGW, or ePDG based on the message prioritization as shown in *Message priority of network-initiated call flows*.

Message priority of network-initiated call flows lists the order from the highest priority (dropped last) to the lowest priority messages (dropped first).

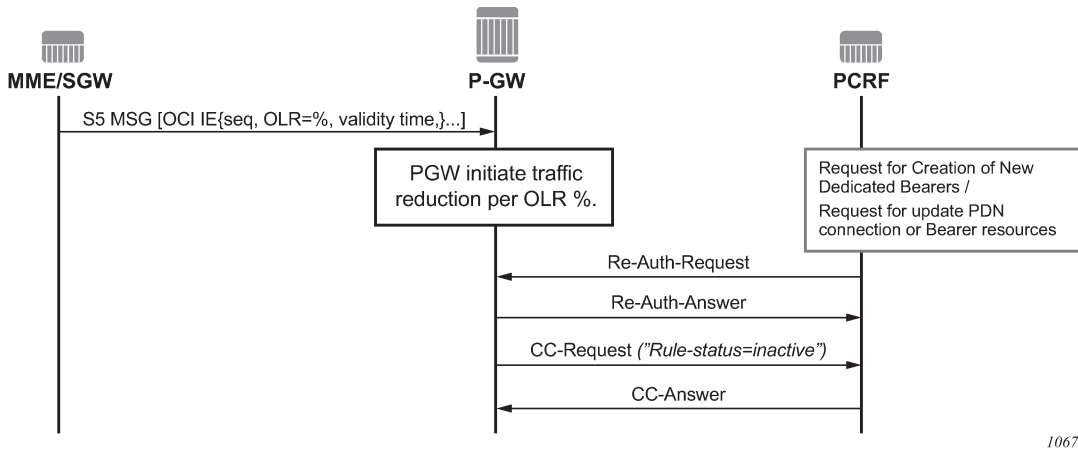
Table 65: Message priority of network-initiated call flows

Prioritization group	Message prioritization
1. Release of PDN Connection or Bearer Resources	1. Delete Bearer Request for Dedicated Bearer 2. Delete Bearer Request for Default Bearer
2. Update of PDN Connection or Bearer Resources	1. Update Bearer Request for Dedicated Bearers 2. Update Bearer Requests for QoS Change - Default Bearer
3. Creation of New Dedicated Bearers	1. Create Bearer Request

In the case where the PCRF is in use, the PGW informs the PCRF about messages that are marked to be dropped, as shown in *Message priority of network-initiated call flows*.

The PGW accepts RARs for Bearer Creation or modification, but sends a CCR-U with Rule-Status = INACTIVE to the PCRF based on the Overload Indication.

Figure 193: PGW informing PCRF for dropped messages



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14.7.1 GTP load and overload configuration

To configure GTP load control, use the following commands:

```
configure mobile-gateway pdn gw-id gtp-load-control
enable-gtp-load-control
load-change-trigger
nth-message
```

To configure GTP overload control, use the following commands:

```
configure mobile-gateway pdn gw-id gtp-overload-control
enable-gtp-overload-control
overload-reduction
critical
major
minor
exit
s8
validity-time
```

The thresholds that define when the mobile gateway reaches a minor, major, or critical GTP-C overload level, can be configured in the range 50-97%. The gateway comes out of overload when the GTP-C overload is 2% below the configured (or default) minor value.

To configure the minor, major, and critical GTP-C overload thresholds, use the following commands:

```
configure mobile-gateway pdn gw-id overload-control
```

```

thresholds
  minor
  major
  critical
exit

```

The CPU sampling period can be configured on the fly when the state of the mobile gateway changes from:

- no overload to overload
- overload to different overload conditions (minor, major, and critical state)
- overload to no overload

To configure the CPU sampling period, use the following command:

```

configure mobile-gateway pdn gw-id overload-control cpu-sampling-interval
[overload overload-interval] [overload-state-change ovrld-state-change-
intrvl] [no-overload no-overload-interval]

```

To exclude LCI and OCI IEs that are sent in the GTP-C messages to GTP peer nodes, use the following commands:

```

configure mobile-gateway profile gtp
  gtpv2-options
    [no] exclude-lci
    [no] exclude-oci
  exit
exit

```

To view load and overload control information, use the following command:

```

show mobile-gateway pdn gtp-load-overload-control

```

To view local load and overload statistics of a node, use the following commands:

```

show mobile-gateway pdn
  gtp-local-load-control stats
  gtp-local-overload-control stats

```

14.7.2 PGW GTP-C node-level overload throttling

In addition to the GTP-C overload control, throttling action is supported when the PGW enters node-level GTP-C overload.

To enable throttling actions on the PGW, use the following command:

```
configure mobile-gateway pdn gtp-overload-control local-throttling local-overload-throttling
```

When enabling throttling actions on the PGW, GTP-C overload control must also be enabled using the following command:

```
configure mobile-gateway pdn gtp-overload-control enable-gtp-overload-control
```

Table 66: GTP-C overload condition and PGW throttling actions

GTP-C overload	Description	Configuration
Minor The PGW reports the minor overload condition to the MME (or ePDG) informing it to reduce traffic by the minor overload of 30% (default).	The PGW does not throttle any GTP-C traffic. Thus, at this overload level it allows the MME (or ePDG) to reduce traffic with its own priority procedures.	To configure the percentage of traffic reduction requested for minor overload conditions, use the following command: <pre>config>mobile>pdn>gtp-overload-control>minor</pre>
Major The GTP-C overload continues to increase and the traffic increases to the next level (major). The PGW node reports the major overload to the MME (or ePDG), informing it to reduce traffic by the major overload of 60% (default).	The PGW does not start throttling traffic at the same traffic reduction percentage it informed the MME (or ePDG) of. That is, if the PGW has reached the major GTP-C overload level and traffic reduction for this level is set to 60%, then the PGW throttles GTP-C traffic from the MME (or ePDG) as follows: - Reject 6 out of 10 received GTP-C Create Session Requests, Bearer Resource Command or Modify Bearer Command call flows for new or modified UE attaches with cause code GTP-C Entity Congestion and include the OCI IE with traffic reduction of 60% to the MME. - Reject 6 out of 10 received PCRF RARs that would trigger	To configure the percentage of traffic reduction requested for major overload conditions, use the following command: <pre>config>mobile>pdn>gtp-overload-control>major</pre>

Table 66: GTP-C overload condition and PGW throttling actions (continued)

GTP-C overload	Description	Configuration
	a GTP-C bearer update or new bearer create.	
Critical The GTP-C traffic still increases and the traffic increases to the next level (major). The PGW node reports the critical overload to the MME (or ePDG), informing it to reduce traffic by the critical overload of 90% (default).	The PGW starts throttling traffic at the same traffic reduction percentage it informed the MME (or ePDG) of. That is, if the PGW has reached the critical GTP-C overload level and traffic reduction for this level is set to 90%, then the PGW throttles GTP-C traffic from the MME (or ePDG) as follows: - Reject 9 out of 10 received GTP-C Create Session Requests, Bearer Resource Command or Modify Bearer Command call flows for new or modified UE attaches with cause code GTP-C Entity Congestion and include the OCI IE with traffic reduction of 90% to the MME. - Reject 9 out of 10 received PCRF RARs that would trigger a GTP-C bearer update or new bearer create.	To configure the percentage of traffic reduction requested for critical overload conditions, use the following command: <pre>config>mobile>pdn>gtp-overload-control>critical</pre>

The following are not throttled during GTP-C overload conditions.

Major GTP-C overload

- emergency APN and delete command call flows
- eMPS IMS call flows activated via Gx
- handover, unless it is based on CSReq; then, these handovers fall into 6 of 10 CSReqs throttled (when major overload is set to 60%)

Critical GTP-C overload

- delete command call flows
- eMPS IMS call flows activated via Gx

- handover, unless it is based on CSReq; then, these handovers fall into 9 of 10 CSReq throttled (when critical overload is set to 90%)

To configure the percentage of the traffic to be throttled when the PGW reaches the minor, major, or critical GTP-C overload level, use the following command:

```
configure mobile-gateway pdn gtp-overload-control local-throttling local-
overload-throttling
  minor
    throttling-percent percent-value
  exit
  major
    throttling-percent percent-value
  exit
  critical
    throttling-percent percent-value
  exit
exit
```

The type of messages that can be throttled when the gateway reaches the GTP-C minor, major, or critical overload level, can be controlled using the drop-only command with the **csreq** option.

When the **csreq** option is specified in the drop-only command, the configured (by the throttling-percent command) percentage of the incoming Create Session Requests is throttled.

When the **csreq** option is not specified in the drop-only command, the configured (by the throttling-percent command) percentage of the GTP-C traffic is throttled as described in [GTP-C overload condition and PGW throttling actions](#).

To exclude the incoming Create Session Requests from throttling based on the ARP values, use the following command:

```
configure mobile-gateway pdn gtp-overload-control local-throttling major
exclude-arp-list list-name
```

To create the ARP list, and configure the ARP values under this list, use the arp-list and arp commands.

The exclude-arp-list command defines which Create Session Requests must be excluded from throttling based on the ARP values that are listed in the list name, configured under the context:



NOTE:

The exclude-arp-list command can be configured only by enabling the drop-only command with the **csreq** option in the list name.

The following configuration example shows the behavior of the `exclude-arp-list` and the `drop-only` commands:

```
configure mobile-gateway profile list
  arp-list "priority-users-arp-list"
    arp 6
    arp 7
  exit
exit
configure mobile-gateway pdn gtp-overload-control local-throttling local-
overload-throttling
  minor
    throttling-percent 0
  exit
  major
    throttling-percent 70
    drop-only csreq
    exclude-arp-list "priority-users-arp-list"
  exit
  critical
    throttling-percent 95
  exit
exit
```

Based on this configuration example, the following can be stated:

- When the gateway reaches the minor GTP-C overload level, zero percentage of the traffic is throttled.
- When the gateway reaches the major GTP-C overload level, 70% of the incoming Create Session Requests are throttled, except those with the ARP values listed in the ARP list that is configured with `arp-list` command.
- When the gateway reaches the critical GTP-C overload level, 95% of the traffic is throttled as described in *GTP-C overload condition and PGW throttling actions*, that is:
 - 90% of the received GTP-C Create Session Requests, Bearer Resource Command or Modify Bearer Command call flows for new, or modified UE attaches
 - 90% of the received PCRF RARs that would trigger a GTP-C bearer update, or new bearer create

To view overload throttling information, use the following command:

```
show mobile-gateway pdn gtp-load-overload-control
```

To view the incoming Create Session Requests that are excluded from throttling in the ARP list, use the following command:

```
show mobile-gateway profile arp-list list-name
```

To view whether the LCI IE and OCI IEs are included or excluded in GTP-C messages, use the following command:

```
show mobile-gateway profile profile-gtp profile-name
```

14.7.3 GTPv2-C APN load and overload control

The GTPv2-C APN-level load control advertises the load information at the APN level granularity toward the consumer nodes (for example, MME). This can assist the consumer node to achieve an evenly load balanced network at the APN level by using the dynamic load information provided within the load control Information.

The GTP-C APN-level overload control advertises the overload information at the APN level granularity to the consumer nodes (for example, MME) and applies the mitigation policies based on this information to the signaling traffic related only to this APN.

Note that this APN is a real APN and not virtual.

The GTPv2-C APN-level load and overload control is supported to be advertised on the S5/S8, S2b, S4, and S11 interfaces.

To enable the APN load and overload control, the node level load and overload must also be enabled using the following commands:

```
configure mobile-gateway pdn gtp-load-control
```

```
configure mobile-gateway pdn gtp-overload-control
```

In addition, the maximum number of PDN connections or maximum rate of PDN connection activations must be configured using the following commands:

```
configure mobile-gateway pdn apn gtp-pdn-threshold max-activations
```

```
configure mobile-gateway pdn apn gtp-pdn-threshold max-connections
```

14.7.3.1 APN load control

When the APN load control is activated, the PGW advertises the APN load information based on the maximum number of PDN connections or maximum rate of PDN activations configured for this APN. The APN Load Control Information (LCI) IE is then piggy-backed in the existing GTPv2-C response messages and can be included in every *n*th response message or due to a change in load triggered by a load change of *x* %.

To configure the message interval for sending load control information, use the following command:

```
configure mobile-gateway pdn gtp-load-control nth-message
```


To configure the load change value which will trigger a new load control update, use the following command:

```
configure mobile-gateway pdn gtp-load-control load-change-trigger
```

Table 67: APN LCI IE

Information	Description
Load Control Sequence Number	A sequence number that is incremented every time the load metric changes.
Load-Metric	Includes the APN-Load-Metric that indicates the current resource utilization, as a percentage based on the following: • If only load control is enabled, the Load-Metric is based on the configured values of gtp-pdn-threshold max-connections or gtp-pdn-threshold max-activations • If both load and overload control are enabled, then the APN-Load-Metric will be based on the configured values of gtp-pdn-threshold max-connections or gtp-pdn-threshold max-activations and overload-threshold.
List of APN and Relative Capacity tuple	• APN name: The full APN name including the APN Network Identifier and APN Operator Identifier as specified in 3GPP TS 23.003. • Relative Capacity: The relative capacity for the APN based on the resources configured for the APN compared to the total resources on the node as a percentage.

A maximum of 10 LCI IEs with a single instance of an APN-list (single APN per LCI IE) is supported.

To configure the APN load control, use the following command:

```
configure mobile-gateway pdn apn apn-load-control
```

14.7.3.2 APN overload control

When the APN overload control is activated, the PGW advertises the APN overload information based on the maximum number of PDN connections or maximum rate of PDN activations configured per APN. The APN Overload Control Information (OCI) IE is then piggy-backed in the existing GTPv2-C response messages when the configured APN overload threshold is reached.

Table 68: APN OCI IE

Information	Description
Overload Control Sequence Number	A sequence number that is only incremented when information in the OCI IE changes.
Overload Reduction Metric	Includes the traffic reduction percentage as configured for the APN.
Period of Validity	The validity time configured for the APN.
List of APN	The full APN name including the APN Network Identifier and APN Operator Identifier as specified in 3GPP TS 23.003.

A maximum of 10 OCI IEs with a single instance of an APN-list (single APN per LCI IE) is supported.

When the APN overload threshold is reached, the PGW will reject the PDN connections from being attached by responding with Cause Code APN Congestion (113) in the Create Session Response.

To configure the APN overload control, use the following commands:

```
configure mobile-gateway pdn apn apn-overload-control
  overload-reduction
  overload-threshold
  validity-time
```

14.8 Signaling priority indication

Support for machine-type communication requires the Signaling Priority Indication IE to be included in the GTP-C signaling from MME toward the combined SGW/PGW. The PGW updates the charging domain by including indication of signaling priority using the Low Priority Indicator AVP.

Machine-type communication devices tend to stay dormant for an extended period of time, with some devices remaining dormant for over 30 days.

To configure the time that a PDP context is allowed to be idle before it is removed by the PGW, use the following command:

```
configure mobile-gateway pdn apn idle-timeout
```

To exclude sending the Signaling Priority Indication IE, use the following command:

```
configure mobile-gateway profile gtp gtpv2-options exclude-signaling-priority
```

14.9 GTP-C collision handling

GTP-C collision handling enables the PGW to recover and retry the call flow procedures, if a parallel overlapping procedure is initiated that will lead to a collision of messages.

The following cases are supported:

- Create Bearer Request initiated from the PGW and Modify Bearer Request is initiated from the MME during handover with Inter- SGW relocation.
Create Bearer Response is received on the PGW from the SGW/MME with cause code 110 (Temporary reject due to handover in progress). Create Bearer Request is re-initiated after Modify Bearer Response is sent to the SGW from the PGW.
- Create Bearer Request is initiated from the PGW and Modify Bearer Request is initiated from the MME during handover without change in the SGW.
Create Bearer Response is received on the PGW from the SGW/MME with cause code 110 (Temporary reject due to handover in progress). Create Bearer Request is re-initiated after Modify Bearer Response is sent to the SGW/MME from PGW.
- Update Bearer Request is initiated from the PGW and Modify Bearer Request is initiated from the MME during handover with inter-SGW relocation.
Update Bearer Response is received on the PGW from the SGW/MME with cause code 110 (Temporary reject due to handover in progress). Update Bearer Request is re-initiated after Modify Bearer Response is sent to the SGW/MME from the PGW.
- Update Bearer Request is initiated from the PGW and Modify Bearer Request is initiated from the MME during handover with inter MME and inter SGW relocation.
Update Bearer Response is received on the PGW from the SGW/MME with cause code 110 (Temporary reject due to handover in progress). Update Bearer Request is re-initiated after Modify Bearer Response is sent to the SGW/MME from the PGW.
- Delete Bearer Response is received on the PGW from the SGW/MME with cause code 110 (Temporary reject due to handover in progress). Delete Bearer Request is re-initiated after the Modify Bearer Response is sent to the SGW/MME from the PGW.



NOTE:

- Applies only if the Delete Bearer Response (CC:110) is received for a dedicated bearer. The PGW will start the call-procedure-retry timer and retry Delete Bearer Request after the timer expires.
- If the Delete Bearer Response (CC:110) is received for a default bearer, the PGW will delete the PDN.

The retry of the call flow procedure that failed due to cause code 110 takes place based on the configured retry time on the PGW. The retry takes place immediately after the PGW responds with a Modify Bearer Response.

To configure the retry time on the PGW, use the following command.

```
configure mobile-gateway pdn gw-id call-procedure-retry retry-time seconds  
retry-attempts value [retry-delay retry-delay-msec]
```

In case the Create Bearer Request uses different retry settings from the Update/Delete Bearer Request, use the following command.

```
configure mobile-gateway pdn gw-id create-procedure-retry retry-time seconds  
retry-attempts attempts [retry-delay retry-attempt-msec]
```

15 IP fragmentation and reassembly

This section provides information about IP fragmentation and reassembly.

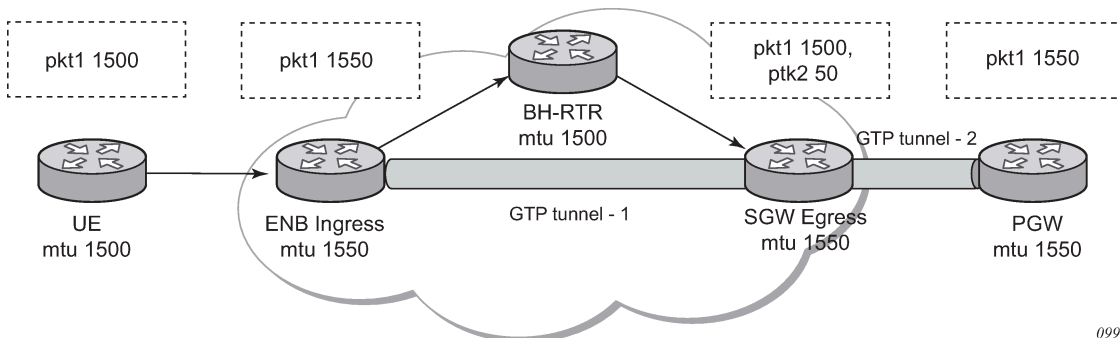
15.1 IPv4 reassembly for GTP-U tunnel ingress packets

Tunneling protocols provide a method of forwarding packets of a particular protocol through a network of a different protocol type. For example, GRE-IP, IPSec, GTP, and L2TP can transport a protocol, such as inner IP payload, through a routed IP network.

This capability requires a pair of devices that define the endpoints of the tunnel. Packets entering the tunnel are processed and encapsulated at the ingress endpoint, and packets exiting the tunnel are processed and de-encapsulated at the egress endpoint.

When packets are tunneled through an IP network, an encapsulation header is added to the original packet. As a result, they may become too large to transport through the network. In this case, they are fragmented to meet the transport network capabilities. Fragmented packets must be reassembled at the tunnel endpoint, otherwise they will be dropped.

Figure 194: Tunnel ingress packet reassembly example



In a GTP tunnel between ENodeB and the gateway as GTP-endpoints, in the uplink direction, ENodeB is the ingress of the GTP tunnel and the gateway acts as the egress for the GTP tunnel. The user IP payload is encapsulated using a GTP UDP header and is routed through the RAN network toward the gateway.

When the GTP tunnel header is added to the inner IP payload by the GTP endpoint and the resulting IP payload exceeds the outgoing link MTU size, the IP packet is fragmented. This is assuming that the MTU size of the outgoing link is set to 1500.

For example:

1. Uplink packets from ENodeB (GTP-ingress) to the gateway (GTP-egress).
2. Downlink packets from the gateway (GTP-ingress) to ENodeB (GTP-egress).

If the user payload is 1500 bytes and MTU of the outgoing link in ENB is set to 1500, the addition of the GTP header results in 1536 bytes, which exceeds the MTU of the outgoing link. In that case, ENB fragments the IP payload packet into two IP packets: pkt-1 is 1500 bytes and pkt-2 is 36 bytes.

The first packet (initial fragment with M bit set 1, OFF=0) contains the GTP header (Layer 4, 3 header). The second packet contains the last fragment (with Layer 3 header) (with non-initial fragment with M bit set 0, OFF=1480) which is sent toward the gateway, which is the tunnel egress.

Without reassembly support, the gateway, which is the tunnel egress, cannot handle the non-initial fragments as they do not contain the TEID-U. Only the initial packet contains the GTP-U header with TEID-U information. Therefore, the non-initial packets are dropped at the gateway egress endpoint, and there will be packet drops that are recovered by the UDP application performing path MTU or TCP applications adjusting the maximum segment size of TCP.

From an End-to-End point of view, this can be defined as follows:

1. The egress GTP endpoint can implement IP reassembly of fragmented GTP-U tunneled packets.
2. The egress GTP endpoint can send an ICMP MTU exceeded message including the MTU of the outgoing link toward the ingress tunnel, so that the ingress GTP endpoint can take action, such as fragmenting the inner IP payload, then adding the GTP payload.

15.2 PGW support for link MTU in PCO

The PGW/GGSN can provide an MTU to the UE for each PDN connection, which the UE can use when sending IP PDUs on the PDN connection. If supported by the UE, this capability can help to avoid IP fragmentation in the packet core network.

The PGW is able to provide an MTU to the UE when the PDN connection is established via the PCO. The PDP PDUs are routed and transferred between the MS and the GGSN or PGW as N PDUs. To avoid IP layer fragmentation between the MS and the GGSN or PGW, the link MTU size in the MS is set to a value provided by the network as a part of the IP configuration.

The link MTU size can be locally configured on the PGW/GGSN at the APN level. The link MTU size for IPv4 is sent to the MS by including it in the PCO (as indicated in TS 24.008). The link MTU size for IPv6 is sent to the MS by including it in the IPv6 router advertisement message (as indicated by RFC 4861).



NOTE:

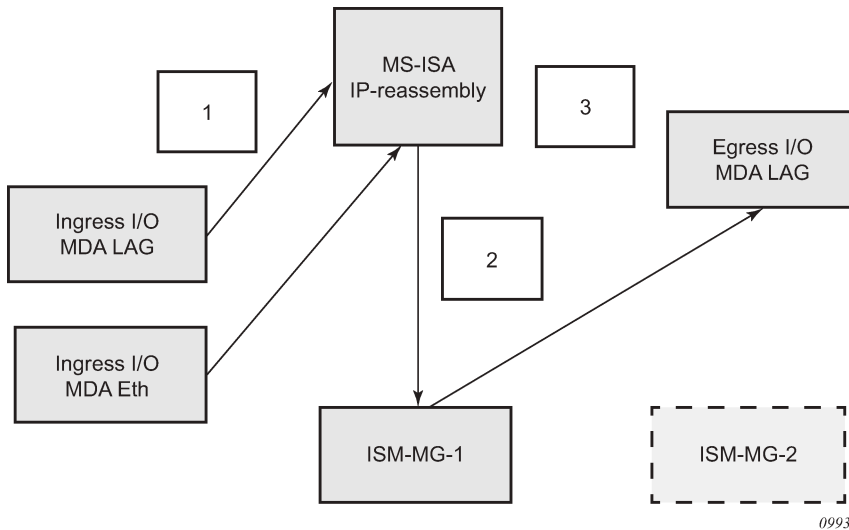
An MTU of 1358 will avoid fragmentation for the majority of cases. This is based on the worst case of GTP-U over IPv6 further encapsulated in IPSec over IPv6. For more information, see 3GPP TS 23.060, Annex C Link MTU considerations.

The sending of a link MTU is disabled by default. To enable the sending of a link MTU, use the following command:

```
configure mobile-gateway pdn apn pco-option link-mtu octets
```

15.3 IP reassembly MS-ISA

Figure 195: Gateway IP reassembly flow example



All the IP packets (fragmented and non-fragmented) from the ingress IOMs are sent to the MS-ISA for IP reassembly to keep the same delivery order when IP reassembly is enabled on an interface. Fragmented packets are reassembled and then redirected to the correct ISM-MG based on the TEID of the packet.

After GTP-U processing, the packet is sent to the egress IOM.

Use case 1: IP reassembly functionality in a dedicated MS-ISA

IP reassembly can be enabled/configured on any Layer 3 IP interface or IP LAG interface. When enabled, all the packets, whether they are fragmented or non-fragmented, are sent to the MS-ISA to maintain the order of delivery of application packets.

In this case, the initial fragment (M=1, OF=0) and non-initial fragments (M=0, OF=x) are sent to the centralized MS-ISA for reassembly. After the packet is reassembled, it is sent to the appropriate anchored ISM-MG through the switch fabric for regular GTP processing.

Each IP reassembly MS-ISA supports full duplex mode for IP fragmented traffic or non-fragmented IMIX traffic. The throughput depends on the module (ISA1 or ISA2), and the buffering and processing demands of the application.

Use case 2: Load balancing/redundancy support achieved by configuring multiple MS-ISAs to support LAG interfaces

If the S1-U or S5-U traffic exceeds levels supported by the MS-ISA module, it's possible to add additional MS-ISA modules to form a logical group to handle the additional traffic. Packets are diverted to the MS-ISAs based on (IP dst) hashing to respective MS-ISAs.

15.4 IP reassembly configuration

IP reassembly and IPSec can be configured to use separate ISA1 or ISA2 modules, each with its own ISA mode.

For ISA2 modules, the system can support IP reassembly on the same ISA2 that is used for IPSec. Sharing a single ISA2 for both functions reduces the number of ISA2s required for the system.



NOTE:

An IP reassembly MDA can be either the second slot of the ISM-MG card or on any complex of any non-MG IOM.

15.4.1 IP reassembly and IPSec on separate MS-ISA configuration

The following example shows IP reassembly configured for a separate ISA or ISA2 module.

- IP reassembly: mda-type = isa2-ip-reas
- IPSec (and GRE): mda-type = isa-tunnel

```
>config# info
-----
config card 1
  card-type ism-mg-2pac-fp3
  mda 2
    mda-type isa2-ip-reas
  exit
```

15.4.2 IP interface for IP reassembly configuration

The following example shows IP interface configuration for IP reassembly. This configuration is the same when IP reassembly and IPSec are configured on separate ISA or ISA2 modules or the same ISA2 module.

```
config>router>if# ip-reassembly port-name port-name
<port-name>      : <port-id>[:context-val]
context-val      - [1..31]
<port-id>        : slot/mda/port
lag-id           - lag-<id>
lag              - keyword
id               - [1..200]
```

```
config>router># info
#-----
```



```

echo "IP Configuration"
#-----
interface "s5"
  address 10.112.24.1/29
exit
interface "system"
  address 10.20.1.1/32
  ip-reassembly 1/2/1:1
exit

```

15.4.3 IP reassembly and IPSec on the same MS-ISA configuration

The system can support IP reassembly on the same ISA2 hardware module that is used for IPSec. Sharing a single ISA2 module for both functions reduces the number of ISA modules required for the system.

For this configuration, there is no new MDA type. Instead, the ISA2 is configured as an ISA Tunnel, using ports 3 or 4 for the tunnel MDA and specifying **isa2-tunnel** for the mda - type. The rest of the IP reassembly configuration remains the same. For example, packets received on an IP interface must be directed to the reassembly ISA, using the `config>router>interface>ip-reassembly` command to point to the slot that contains the tunnel ISA.

This feature includes support for GRE combined with the reassembly function.

The following example shows the IP reassembly function combined with the tunnel ISA function on the tunnel ISA2.

```

config# info
-----
    config card 1
      card-type ism-mg-2pac-fp3
      mda 2
        mda-type isa2-tunnel
      exit
    -----

```

The receiving IP interface configuration is the same as when IP reassembly and IPSec are configured on separate ISA modules.

```

config>router# info
-----
    interface "system"
      address 10.20.1.1/32
      ip-reassembly 1/2/3:1
    exit
    -----

```

15.4.4 IP reassembly ports, VLANs, and LAGs configuration

When an IP-reassembly MDA is configured the system automatically creates two virtual ports. Virtual port 1 is a network port used to enable IP reassembly for network interfaces (interfaces in the base routing instance). Virtual port 2 is used to enable IP reassembly for service interfaces (interfaces in VPRNs).

IP reassembly interface in base routing instance

In this example, card 3/2 is configured as an IP-reassembly MDA with port 1 for the base routing instance.

```
SGW# config router
      interface "S5loopback"
        address 10.33.1.1/32
        loopback
        ip-reassembly 3/2/1:1
      exit
```

IP reassembly in VPRN service interface

In this example, card 3/2 is configured as an IP-reassembly MDA with port 2 for the VPRN service interface.

```
SGW# config services
      vprn 1 customer 1 create
      interface "S11loopback" create
        address 6.7.100.1/32
        loopback
        ip-reassembly 3/2/2:1
      exit
```

IP reassembly LAG

Multiple IP-reassembly MDAs can be grouped onto an IP-reassembly LAG which can be used for handling higher throughput as well as added redundancy. In this example, cards 3/2, 4/2, and 7/2 are IP-reassembly MDAs. Virtual port 2 is used because this LAG will be applied to a GTP interface defined in a VPRN.

```
lag 4
  description "Access LAG for redundant ip-reassembly ISA MDAs"
  mode access
  encap-type dot1q
  port 3/2/2
  port 4/2/2
  port 7/2/2
  no shutdown
```

```
exit
```

```
vprn 1 name "1" customer 1 create
  interface "S1u-loopback" create
    address 100.2.3.4/32
    loopback
    ip-reassembly lag-4:3
  exit
```

If multiple interfaces are using the same IP-reassembly port or LAG, the operator may use a VLAN tag to differentiate between them (as shown in this example). These VLAN tags are internal to the system and are unknown to remote GTP peers.

By default, the load-balancing of traffic across a physical port LAG, IP-reassembly LAG, and ECMP is based on source/destination IP address. To enable TEID-based hashing of incoming GTP traffic, the interface level command `teid-load-balancing` has to be enabled on the physical interfaces for all the routing instances containing GTP protocol interfaces.

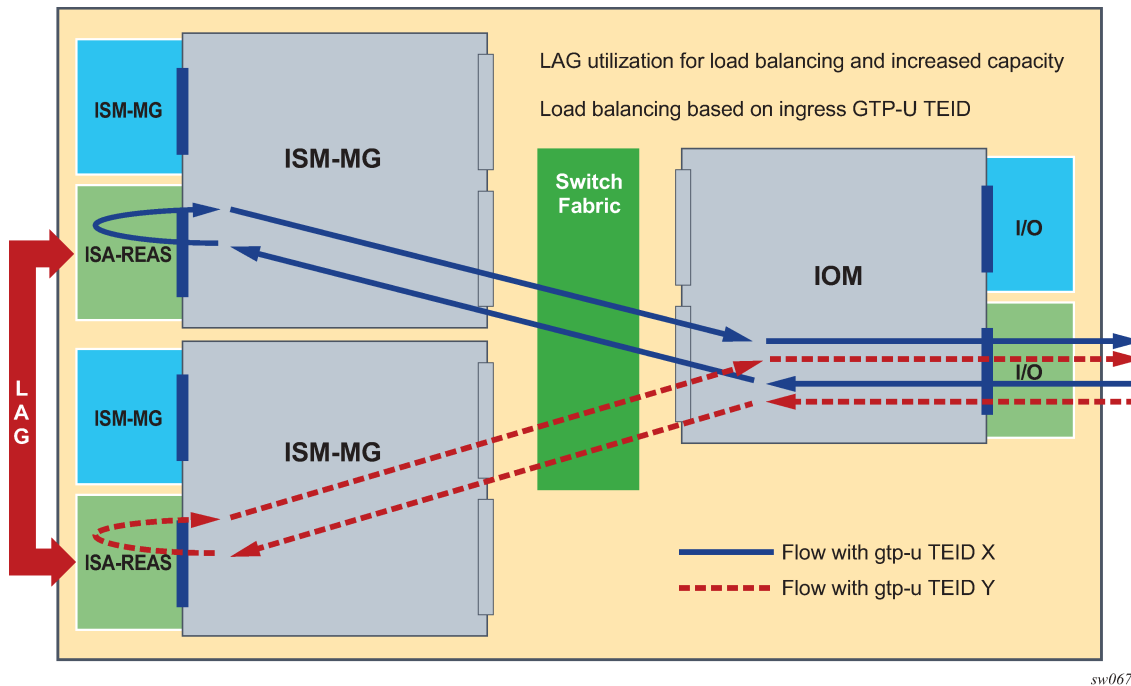
TEID-based hashing improves internal load-balancing of traffic when multiple IP-reassembly MDAs coexist in a LAG. The following applies when TEID-based hashing is enabled:

- Non-fragmented GTP packets are hashed based on GTP TEID, source/destination IP address, and optionally source/destination UDP ports (I4-load balancing)
- Fragmented packets are hashed based on source/destination IP address

For TEID-based hashing to work over MC redirect, the same CLI configuration has to be enabled over the physical interface in front of the MC mobile loopback interface.

Load-balancing of GTP traffic generated by the mobile gateway is always based on TEID.

Figure 196: IP reassembly configuration with LAG utilization and load-balancing based on ingress GTP-U TEID



15.4.5 IP reassembly in CMG

IP reassembly is supported in CMG systems as mda-type "isa2-ip-reas-v" which can be configured in any LB slot (iom-v card). The use of ports, LAGs, and Tags for IP reassembly is supported in CMG the same as in 7750 SR MG.

Example: IP reassembly in CMG

In this example, card 2/2 is configured as IP reassembly in CMG.

```
>config# info
-----
card 2
  card-type iom-v
  mda 1
    mmda-type m20-v
    no shutdown
  exit
  mda 2
    mda-type isa2-ip-reas-v
    no shutdown
  exit
  no shutdowna
exit
```

Configuration notes

IPv4 or IPv6 reassembly is currently not supported with an Internal-VM L3 network. It is therefore expected that the end-to-end MTU of the L3 network must be consistent with the configured fabric MTU.

15.4.6 Show command output

```
show isa ip-reas-stats port-name
<port-name>      : <port-id>[:context-val]
context-val      - [1..31]
<port-id>        : slot/mda/port
lag-id           - lag-<id>
lag              - keyword
id               - [1..200]
```

Example: Outputs

```
show isa ip-reas-stats 3/2/1 ipv4
```

```
=====
IP Reassembly statistics
=====
```

```
-----
Interface           : 1-lag-63:1 (Reassembly)
Selective Reassembly : disabled
Port                : 3/2/1
-----
```

```
IPv4 Reassembly statistics
-----
```

Total Pkts Rx	: 120	Total Bytes Rx	: 65660
Total Pkts Tx	: 60	Total Bytes Tx	: 62980
Fragmented Pkts Rcvd	: 100	Fragmented Bytes Rcvd	: 49400
Packets Reassembled	: 40	Bytes Reassembled	: 48200
Fragments Timed Out	: 0	Fragments Discarded	: 0

```
-----
Number of entries : 1
=====
```

```
show isa ip-reas-stats 4/2/1 ipv4
```

```
=====
IP Reassembly statistics
=====
```

```
-----
Interface           : 1-lag-63:1 (Reassembly)
Selective Reassembly : disabled
```

Port	: 4/2/1		

IPv4 Reassembly statistics			

Total Pkts Rx	: 121	Total Bytes Rx	: 65720
Total Pkts Tx	: 61	Total Bytes Tx	: 63040
Fragmented Pkts Rcvd	: 100	Fragmented Bytes Rcvd	: 49400
Packets Reassembled	: 40	Bytes Reassembled	: 48200
Fragments Timed Out	: 0	Fragments Discarded	: 0

Number of entries : 1			
=====			

15.5 Diversion of fragmented traffic to IP reassembly MS-ISA

If the gateway receives fragmented IP PDUs, the system must reassemble them to perform forwarding based on the GTP tunnel. This is necessary because the GTP TEID will not appear in anything but the first fragment.

If fragmentation is expected, you must configure an ISA to be used for IP reassembly (see [IP reassembly configuration](#)). The IP reassembly ISAs are bound to the GTP reference point (which is typically a loopback). By default, the system sends all packets destined for the GTP reference point to the reassembly ISA first. If there are multiple reassembly ISAs, the system will load-balance to ensure that upper layer flows go through a single ISA.

Also, by default the system will send all PDUs to the reassembly ISAs, regardless of whether or not they are IP fragments.

The following two mechanisms can be used to configure the system to send fragmented PDUs to the reassembly ISA and non-fragmented PDUs directly to the appropriate ISM-MG:

- apply a filter to the receiving physical interfaces
- configure selective reassembly for the system

Related information

[Applying filters to external interfaces](#)

One option to direct only fragmented packets to the reassembly ISAs is to apply an IP filter to the receiving physical interface.

[Configuring selective reassembly](#)

It is possible to configure the system to reassemble fragmented tunneled packets before forwarding them to the appropriate ISM-MG group for processing.

15.5.1 Applying filters to external interfaces

One option to direct only fragmented packets to the reassembly ISAs is to apply an IP filter to the receiving physical interface.

The filter redirects non-fragmented PDUs to a dummy interface from where they are forwarded to the appropriate ISM-MG. Anything not matching the filter is forwarded toward the IP reassembly ISAs first. This option requires you to apply the filter on every external interface that may receive fragments and also requires redirection to a dummy IP interface.

This example shows an IP filter configuration that will send only fragmented PDUs to the reassembly ISA. For more information about the commands, refer to the *7750 SR OS Services Guide*. For information about configuring selective reassembly for the system, see [Configuring selective reassembly](#).

```

echo "Filter Configuration"
#-----
filter
  ip-filter 100 create
    default-action forward
    entry 1 create
      match protocol udp
        dst-port eq 2152
        fragment false
      exit
    action forward next-hop 5.5.5.5
  exit
exit
exit
vprn 105 customer 1 create
  description "S5 VPRN"
  vrf-import "s5-ipv4-bgpvpn-import"
  vrf-export "s5-ipv4-bgpvpn-export"
  autonomous-system 66662
  route-distinguisher 66662:105
  interface "S5_1LoopBack" create
    address 16.101.14.1/32
    loopback
    ip-reassembly lag-63:5
  exit
  interface "Sgw_Router" create
    address 10.71.1.2/24
    sap lag-1:105 create
      ingress
        filter ip 100
      exit
    exit
  exit
  interface "Dummy-S5" create
    address 5.5.5.5/32
    loopback

```

```
exit
```

15.5.2 Configuring selective reassembly

It is possible to configure the system to reassemble fragmented tunneled packets before forwarding them to the appropriate ISM-MG group for processing.

The following describes the supported behavior for the selective reassembly feature:

- When selective reassembly is enabled, fragmented packets will be sent to the reassembly ISAs, if configured. Non-fragmented packets will not be sent to the reassembly ISAs.
- When selective reassembly is disabled, both fragmented and non-fragmented packets will be sent to the reassembly ISAs, if configured.

To enable selective reassembly, use the command:

```
configure router selective-reassembly
```

To determine if selective reassembly is enabled, use the command:

```
show isa ip-reas-stats
```



NOTE:

The selective reassembly feature is not supported for the Home Agent. Both fragmented and non-fragmented packets destined for the Home Agent will be sent to the reassembly ISAs, if configured to do so.

Related information

[IP reassembly configuration](#)

IP reassembly and IPSec can be configured to use separate ISA1 or ISA2 modules, each with its own ISA mode.

15.6 TCP MSS configuration

This section describes TCP Maximum Segment Size (MSS) configuration options. MSS configured traffic is diverted to AA-ISA.

15.6.1 TCP MSS static configuration

TCP MSS represents the maximum amount of data in a TCP message, not including the IP and TCP header lengths.

The ability to statically set the TCP MSS value sent in the SYN packet for TCP connections associated with the IP interface is supported for the IP and IPv6, base and IES/VRPN service interfaces.

The TCP MSS value must be smaller than the calculated value based on the interface MTU. The configured TCP-MSS value will not affect TCP connections that are already established at the time the `tcp-mss` command is issued.

Only TCP connections established after the configuration statement is processed will be affected. The TCP MSS static setting only applies to traffic originating from or terminating at the node, and does not affect transit traffic.

For more information and command descriptions, refer to the *7750 SR OS Services Guide*.

15.6.2 TCP MSS configuration for NAT

The ability to change the TCP MSS for packets that transit the PGW/GGSN is supported when NAT is configured. This can be used to avoid reassembly of GTP-U PDUs.

By setting the TCP MSS to a low enough value, you can control the size of the inner IP payload and therefore avoid fragmentation that may occur when the GTP encapsulation overhead is added. The TCP MSS field for NAT can be adjusted with a preconfigured value. This triggers the interception of transit TCP packets with the SYN bit set and consequently rewrites the MSS field of the intercepted packet. The MSS field will only be rewritten in cases where the configured TCP MSS adjust value is smaller than the one inside the intercepted packet. If the given TCP option field is not present in the TCP SYN packet, it will be inserted with the preconfigured value.

15.6.3 TCP MSS adjust configuration

The TCP MSS field can be adjusted with a preconfigured value without diverting traffic via AA or using the NAT function.

TCP MSS is configured per action rule unit and triggers the interception of transit TCP packets with the SYN bit set and consequently rewrites the MSS field of the intercepted packet. The TCP MSS field is rewritten when the configured TCP MSS adjust value is smaller than the one inside the intercepted packet. If the given TCP option field is not present in the TCP SYN packet, it is inserted with the preconfigured value.

To configure TCP MSS adjust in an ARU in the policy-options profile, use the following command:

```
configure mobile-gateway profile policy-options action-rule-unit mss-adjust
```

To configure the TCP MSS adjust value, use the following command:

```
configure mobile-gateway pdn bearer-traffic mss-adjust
```

For TCP MSS adjust under bearer traffic to work without diverting traffic to the AA, the PRU must match protocol 6 (TCP), as shown in the following example:

```
policy-rule-unit "tcpmatch"  
    flow-description 1  
        match  
            protocol 6
```

```
exit
exit
```

For command descriptions, see the *CMG CLI Reference Guide*.

16 IP tunnels

Overview of IP Security (IPSec) software features for the IPSec ISA.

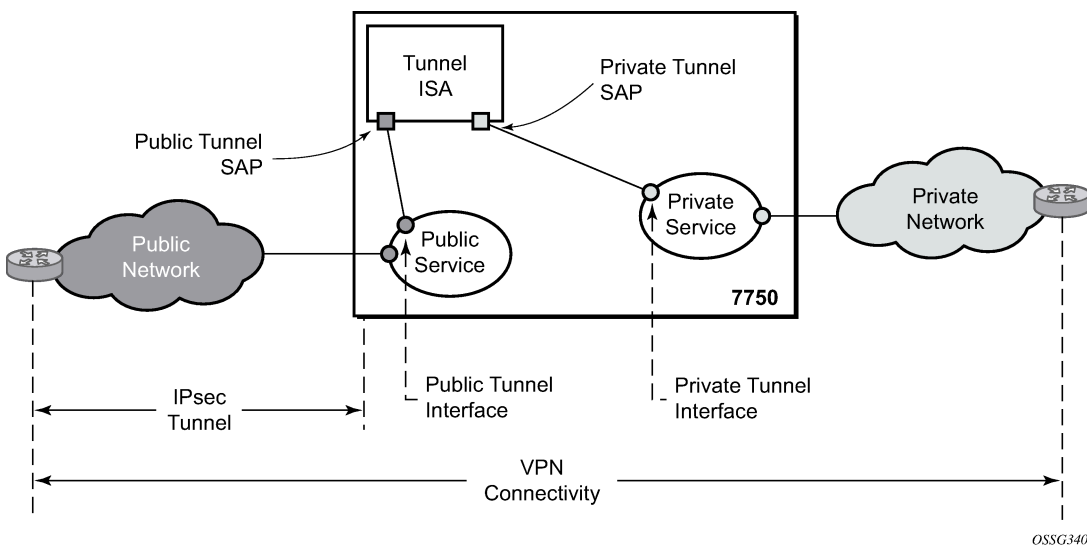
16.1 IP tunnels overview

The IP Security (IPSec), GRE tunneling, and IP-IP tunneling features supported by the MS-ISA.

In these applications, the MS-ISA functions as a resource module for the system, providing encapsulation and (for IPSec) encryption functions. The IPSec encryption functions provided by the MS-ISA are applicable for many applications including: encrypted SDPs, video wholesale, site-to-site encrypted tunnel, and remote access VPN concentration.

7750 IPSec implementation architecture shows an example of an IPSec deployment, and the way this would be supported inside a 7750. GRE and IP-IP tunnel deployments are very similar. IP tunnels have two flavors GRE/IP-IP, in all but a few areas the information for IP Tunnels applies to both types.

Figure 197: 7750 IPSec implementation architecture



7750 IPSec implementation architecture, the public network is typically an "insecure network" (for example, the public Internet) over which packets belonging to the private network in the diagram cannot be transmitted natively. Inside the 7750, a public service instance (IES or VPRN) connects to the public network and a private service instance (typically a VPRN) connects to the private network.

The public and private services are typically two different services, and the MS-ISA is the only "bridge" between the two. Traffic from the public network may need to be authenticated and encrypted inside an IPSec tunnel to reach the private network. In this way, the authenticity/confidentiality/integrity of accessing the private network can be enforced. If authentication and confidentiality are not important then access to the private network may alternatively be provided through GRE or IP-IP tunnels.

The MS-ISA provides a variety of encryption features required to establish bidirectional IPSec tunnels including:

Control Plane

- Manual Keying
- Dynamic Keying: IKEv1/v2
- IKEv1 Mode: Main and Aggressive
- Authentication: Pre-Shared-Key /xauth with RADIUS support/X.509v3 Certificate/EAP
- Perfect Forward Secrecy (PFS)
- DPD
- NAT-Traversal
- Security Policy

Data Plane

- ESP (with authentication) Tunnel mode
- Authentication Algorithm: MD5/SHA1/SHA256/SHA384/SHA512/AES-XCBC
- Encryption Algorithm: DES/3DES/AES128/AES192/AES256
- DH-Group: 1/2/5/14/15
- Anti-Replay Protection
- N:M IPSec ISA card redundancy

**NOTE:**

SR OS uses a configured authentication algorithm in an ike-policy for Pseudorandom Function (PRF).

There are two types of tunnel interfaces and SAPs:

Public tunnel interface

Configured in the public service; outgoing tunnel packets have a source IP address in this subnet.

Public tunnel SAP

Associated with the public tunnel interface; a logical access point to the MS-ISA card in the public service.

Private tunnel interface

Configured in the private service; can be used to define the subnet for remote access IPSec clients.

Private tunnel SAP

Associated with the private tunnel interface, a logical access point to the MS-ISA card in the private service.

Traffic flows to and through the MS-ISA card as follows:

- In the upstream direction, the encapsulated (and possibly encrypted) traffic is forwarded to a public tunnel interface if its destination address matches the local or gateway address of an IPSec tunnel or the source address of a GRE or IP-IP tunnel. Inside the MS-ISA card, encrypted traffic is decrypted, the tunnel header is removed, the payload IP packet is delivered to the private service, and from there, the traffic is forwarded again based on the destination address of the payload IP packet.
- In the downstream direction, unencapsulated/clear traffic belonging to the private service is forwarded into the tunnel by matching a route with the IPSec/GRE/IP-IP tunnel as next-hop. The route can be configured statically, learned by running OSPF on the private tunnel interface (GRE tunnels only), learned by running BGP over the tunnel (IPSec and GRE tunnels only), or learned dynamically during IKE negotiation (IPSec only). After clear traffic is forwarded to the MS-ISA card, it is encrypted if required, encapsulated per the tunnel type, delivered to the public service, and from there, the traffic is forwarded again based on the destination address of the tunnel header.

16.1.1 Tunnel ISAs

A **tunnel-group** is a collection of MS-ISAs (each having `mda - type` set to `isa-tunnel`) configured to handle the termination of one or more IPSec, GRE and/or IP-IP tunnels.

Example: Tunnel-group configurations

```
config isa
    tunnel-group 1 create
        primary 1/1
        backup 2/1
        no shutdown
    exit

config isa
    tunnel-group 2 create
        multi-active
        mda 3/1
        mda 3/2
        no shutdown
```

A GRE, IP-IP, or IPSec tunnel belongs to only one tunnel group. There are two types of tunnel groups:

- A single-active tunnel-group can have one tunnel-ISA designated as primary and optionally one other tunnel-ISA designated as backup. If the primary ISA fails the affected failed tunnels are re-established on the backup (which is effectively a cold standby) if it is not already in use as a backup for another tunnel-group.

- A multi-active tunnel-group can have multiple tunnel-ISAs designated as primary. This is only supported on 7750 SR7/SR12/SR12E with chassis mode D or 7450 mixed mode with IOM3.

The `show isa tunnel-group` allows the operator to view information about all configured tunnelgroups. This command displays the following information for each tunnel-group: group ID, primary tunnel-ISAs, backup tunnel-ISAs, active tunnel-ISAs, admin state and oper state.

16.1.1.1 Public tunnel SAPs

A VPRN or IES service (the delivery service) must have at least one IP interface associated with a public tunnel SAP to receive and process the following types of packets associated with GRE, IP-IP and IPSec tunnels:

- GRE (IP protocol 47)
- IP-IP (IP protocol 4)
- IPSec ESP (IP protocol 50)
- IKE (UDP)

The public tunnel SAP type has the format `tunnel-tunnel-group.public:index`, as shown in the following CLI example.

```
#config>service# info
-----
customer 1 create
  description "Default customer"
exit
ies 1 customer 1 create
  interface "public" create
    address 64.251.12.1/24
    tos-marking-state untrusted
    sap tunnel-1.public:200 create
  exit
exit
no shutdown
exit
vprn 2 customer 1 create
  route-distinguisher 1.1.1.1:65007
  interface "greTunnel" tunnel create
    address 10.0.0.1/24
    dhcp
    no shutdown
  exit
  sap tunnel-1.private:210 create
    ip-tunnel "toCel" create
    dest-ip 10.0.0.2
```

```

gre-header
source 64.251.12.88
remote-ip 64.251.12.2
backup-remote-ip 64.251.12.22
delivery-service 1
no shutdown
exit
exit
exit
no shutdown
exit
-----

```

16.1.1.2 Private tunnel SAPs

The private service must have an IP interface to a GRE, IP-IP, or IPSec tunnel to forward IP packets into the tunnel, causing them to be encapsulated (and possibly encrypted) per the tunnel configuration and to receive IP packets from the tunnel after the encapsulation has been removed (and decryption). That IP interface is associated with a private tunnel SAP.

The private tunnel SAP has the format `tunnel-tunnel-group.private:index`, as shown in the following CLI example where a GRE tunnel is configured under the SAP.

```

# show ip tunnel
=====
IP Tunnels
=====
TunnelName          SapId              SvcId
Admn
Local Address              DlvrySvcId
Oper
OperRemoteAddress
-----
tun-1-gre-tunnel      tunnel-1.private:1  201      Up
141.1.1.2              1201              Up
41.1.1.2
-----
IP Tunnels: 1
=====

```

16.1.1.3 IP interface configuration

In the configuration example of the previous section the IP address 10.0.0.1 is the address of the GRE tunnel endpoint from the perspective of payload IP packets. This address belongs to the address space of the VPRN 1 service and will not be exposed to the public IP network carrying the GRE encapsulated packets. An IP interface associated with a private tunnel SAP does not support unnumbered operation.

It is possible to configure the IP MTU (M) of a private tunnel SAP interface. This sets the maximum payload IP packet size (including IP header) that can be sent into the tunnel - for example, it applies to the packet size before the tunnel encapsulation is added. When a payload IPv4 packet that needs to be forwarded to the tunnel is larger than M bytes:

- If the DF bit is clear, the payload packet is IP fragmented to the MTU size prior to tunnel encapsulation.
- If the DF bit is set, the payload packet is discarded and (if allowed by the ICMP setting of the sending interface) an ICMP type 3/code 4 is returned to the sender (with MTU of the private tunnel SAP interface in the payload).

16.1.1.4 GRE and IP-IP tunnel configuration

To bind an IP/GRE or IP-IP tunnel to a private tunnel SAP, the `ip-tunnel` command must be added under the SAP. To configure the tunnel as an IP/GRE tunnel, the `gre-header` command must be present in the configuration of the `ip-tunnel`.

To configure the tunnel as an IP-IP tunnel, the `ip-tunnel` configuration should have the `no gre-header` command.

When configuring a GRE or IP-IP tunnel, the `dest-ip` command is mandatory as this specifies the private IP address of the remote tunnel endpoint. If the `dest-ip` address is not within the subnet of the local private endpoint then the tunnel will not come up. In the CLI sub-tree under `ip-tunnel`, there are commands to configure the following:

- The source address of the GRE or IP-IP tunnel. This is the source IPv4 address of GRE or IP-IP encapsulated packets sent by the delivery service. It must be an address in the subnet of the associated public tunnel SAP interface.
- The remote IP address - If this address is reachable in the delivery service (there is a route) then this is the destination IPv4 address of GRE or IP-IP encapsulated packets sent by the delivery service.
- The backup remote IP address- If the remote IP address of the tunnel is not reachable then this is the destination IPv4 address of GRE or IP-IP encapsulated packets sent by the delivery service.
- The delivery service- This is the id or name of the IES or VPRN service where GRE or IP-IP encapsulated packets are injected and terminated. The delivery service can be the same service where the private tunnel SAP interface resides.
- The DSCP marking in the outer IP header of GRE encapsulated packets- If this is not configured then the default is to copy the DSCP from the inner IP header to the outer IP header.

A private tunnel SAP can have only one `ip-tunnel` sub-object (one GRE or IP-IP tunnel per SAP).

The `show ip tunnel` displays information about a specific IP tunnel or all configured IP tunnels. The following information is provided for each tunnel:

- service ID that owns the tunnel
- private tunnel SAP that owns the tunnel
- tunnel name, source address
- remote IP address
- backup remote IP address
- local (private) address
- destination (private) address
- delivery service
- DSCP
- admin state
- oper state
- type (GRE or IP-IP)

16.1.1.5 Tunnel configuration for ePDG

IPSec tunnels are used for ePDG (3GPP specifications, Rel 11, Dec. 2013) including:

- Tunnel authentication and authorization (termination of IKEv2 signaling and relay via AAA messages)
- UE authentication using IKEv2 and EAP-AKA over Diameter. The ePDG acts as an EAP passthrough agent, relaying EAP messages between the UE and the 3GPP AAA server via IKE. The particular EAP method (EAP-AKA, EAP-SIM, EAP-TLS, and so on) used between the UE and the AAA server is transparent to the ePDG.
- SWm reference point to carry EAP Request and Diameter EAP Response messages
- De-encapsulation and encapsulation of packets for the following:
 - IPSec
 - GTP, if S2b is used
- IPv4/IPv6 transport for IPSec
- IKEv2 redirection call flows, re-keying parameters, and re-authentication

For information and configuration guidelines for ePDG, including IKE and ESP security associations, ePDG certificates, SWu Reference Points for interfaces, and Diameter Reference Points using SWm, refer to the *7750 SR Mobile Gateway Configuration Guide*.

Related information

[*Tunnel group configuration for ePDG*](#)

16.1.1.6 IP fragmentation and reassembly for IP tunnels

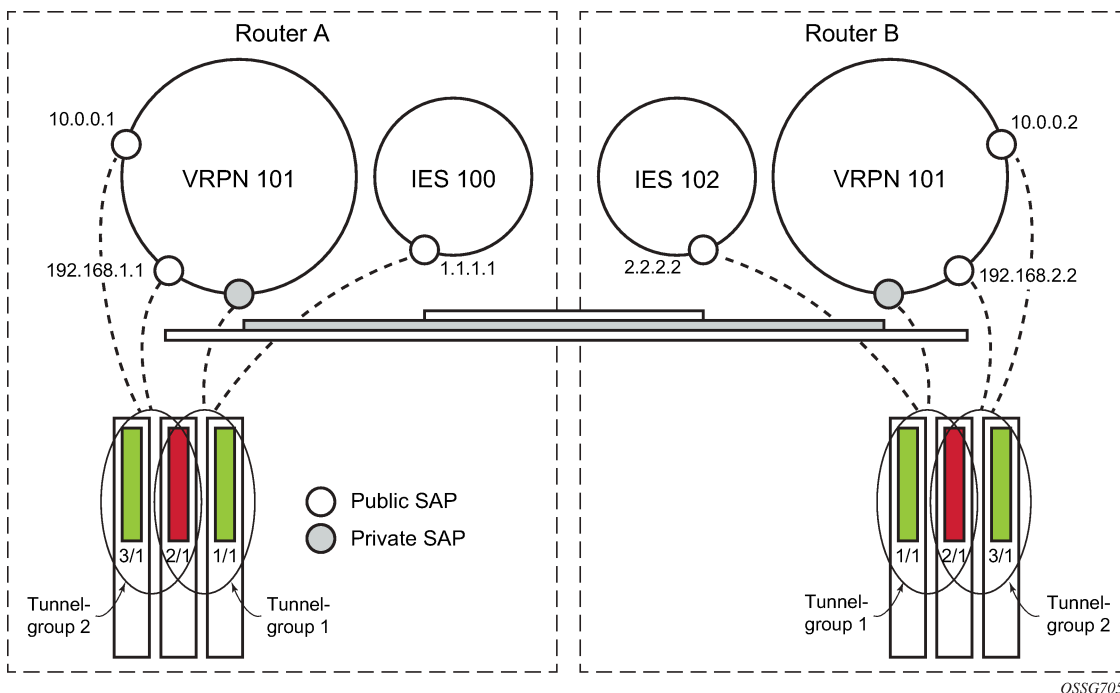
An IPSec, GRE or IP-IP tunnel packet that is larger than the IP MTU of some interface in the public network must either be discarded (if the Do Not Fragment bit is set in the outer IP header) or fragmented. If the tunnel packet is fragmented, it is then up to the destination tunnel endpoint to reassembly the tunnel packet from its fragments. Starting in R10, IP reassembly can be enabled for all the IPSec, GRE, and IP-IP tunnels belonging to a tunnel-group. For IP-IP and GRE tunnels, the reassembly option is also configurable on a per-tunnel basis so that some tunnels in the tunnel-group can have reassembly enabled, and others can have the extra processing disabled. When reassembly is disabled for a tunnel, all received fragments belonging to the tunnel are dropped.

To avoid public network fragmentation of IPSec, GRE, or IP-IP packets belonging to a particular tunnel, one possible strategy is to fragment IPv4 payload packets larger than a specified size M at entry into the tunnel (before encapsulation and encryption if applicable). The size M is configurable using the `ip-mtu` command under the `ip-tunnel`, `ipsec-tunnel` or `tunnel-template` configuration.

If the payload IPv4 packets are all M bytes or less in length then it is guaranteed that all resulting tunnel packets will be less than M+N bytes in length, if N is the maximum overhead added by the tunneling protocol. If M+N is less than the smallest interface, IP MTU in the public network fragmentation will be avoided. In some cases, some of the IPv4 payload packets entering a tunnel may have their Don't Fragment (DF) bit set. And if required, SR OS supports the option (also configurable on a per-tunnel basis) to clear the DF bit in these packets so that they can be fragmented.

The system allows users to configure an encapsulated-ip-mtu for a specified tunnel. This represents the maximum size of the encapsulated tunnel packet. If the v4/v6 tunnel packet size exceeds the configured encapsulated-ip-mtu, then the system will fragment the packet against the encapsulated-ip-mtu.

Figure 198: Example GRE over IPSec tunnel configuration



Example: Router A configuration

```
config isa
    tunnel-group 1 create
        primary 1/1
        backup 2/1
        no shutdown
        exit
    tunnel-group 2 create
        primary 3/1
        backup 2/1
        no shutdown
        exit
    exit all

config ipsec
    ike-policy 1 create
        auth-algorithm sha1
        dh-group 5
        encryption-algorithm aes128
        ike-mode main
        ike-version 1
        exit
    ipsec-transform 1
        esp-auth-algorithm sha1
        esp-encryption-algorithm aes256
        exit
    exit all

config service ies 100 customer 1 create
    interface "public-ipsec-1" create
        address 1.1.1.1/24
        sap tunnel-1.public:200 create
        exit all

config service vprn 101 customer 1 create
    ipsec
        security-policy 1 create
            entry 1 create
                local-ip 192.168.1.0/24
                remote-ip 192.168.2.0/24
                exit
            exit
        exit
    interface "private-ipsec-1" tunnel create
```

```

    sap tunnel-1.private:201 create
        ipsec-tunnel "ipsec-tunnel-for-n-gre-tunnels" create
            security-policy 1
            local-gateway-address 1.1.1.2 peer 2.2.2.2 delivery-
service 100
            dynamic-keying
                ike-policy 1
                pre-shared-key "secret"
                transform 1
            exit
        exit
    exit
interface "public-gre-1" create
    address 192.168.1.1/24
    sap tunnel-2.public:200 create
    exit
interface "private-gre-1" tunnel create
    address 10.0.0.1/30
    sap tunnel-2.private:201 create
        ip-tunnel "protected-gre-tunnel" create
            gre-header
            dest-ip 10.0.0.2
            source 192.168.1.2
            remote-ip 192.168.2.2
            delivery-service 101
            no shutdown
        exit
    exit
exit
static-route 192.168.2.0/24 next-hop ipsec-tunnel "ipsec-tunnel-for-n-gre-
tunnels"

```

16.1.2 Operational conditions

A tunnel group that is in use cannot be deleted. In single-active mode, changes to the primary ISA are allowed only in when the tunnel group is in a shutdown state. Change to the backup ISA (or the addition of a backup ISA) is allowed at any time unless the ISA is currently active for this tunnel group. When the backup module is active, changing the primary module is allowed without shutting down the tunnel group. If it is part of a multi-chassis configuration, you cannot change the mode until it is removed from this configuration as well.

A shutdown of tunnel-group is required to change the following:

- the mode between multi-active and single-active

- the primary-isa in single-active mode
- the active-mda-number in multi-active mode

In multi-active mode, if the active member ISA goes down, system will replace it with backup ISA; however, if there is no backup ISA, the tunnel-group will be oper-down. A multi-active tunnel-group with MC-IPSec enabled cannot be changed into single active-mode unless it is removed from MC-IPSec configuration.

Changes to the ipsec-transform/ike-policy in-use are not allowed.

The public interface address can be changed at any time. However, if changed, tunnels that were configured to use it will require a configuration change. If the public subnet changed is still using an old subnet, the tunnels will be in an operationally down state until their configuration is corrected. The public service cannot be deleted while tunnels are configured to use it. A public service is the IES or VPRN service that hold the regular interface that connects the node to the public network. A private service connects to the private protected service.

A tunnel group ID or tag cannot be changed. To remove an tunnel group instance, it must be in a shutdown state (both front-door and back-door).

A change to the security policy is not allowed while a tunnel is active and using the policy.

The tunnel local-gateway-address, peer address, or delivery router parameters cannot be changed while the tunnel is operationally up (shutdown will make it both admin down and operationally down).

A tunnel security policy cannot be changed while the tunnel is operationally up. An IPSec transform policy or ike-policy assignments to a tunnel requires the tunnel to be shutdown.

16.1.3 QoS interactions

The MS-ISA can interact with the queuing functions on the IOM through the ingress/egress QoS provisioning in the IES or IP VPN service where the IPSec session is bound.

Multiple IPSec sessions can be assigned into a single IES or VPRN service. In this case, QoS defined at the IES or VPRN service level, is applied to the aggregate traffic coming out of or going into the set of sessions assigned to that service.

To keep marking relevant in the overall networking design, the ability to translate DSCP bit marking on packets into DSCP bit markings on the IPSec tunneled packets coming out of the tunnel is supported.

16.1.4 OAM interactions

The MS-ISA is IP-addressed by an operator-controlled IP on the public side. That IP address can be used in Ping and Traceroute commands and the ISA can either respond or forward the packets to the CPM.

For static LAN-to-LAN tunnel, in multi-active mode, a ping requests to public tunnel address would not be answered if the source address is different from the remote address of the static tunnel.

The private side IP address is visible. The status of the interfaces and the tunnels can be viewed using show commands.

Traffic that ingresses or egresses an IES or VPRN service associated with specific IPsec tunnels can be mirrored like other traffic.

Mirroring is allowed per interface (public) or IPsec interface (private) side. A filter mirror is allowed for more specific mirroring.

16.1.5 Redundancy

In single-active mode, every tunnel group can be configured with primary and backup ISAs. An ISA can be used as a backup for multiple IPsec groups. The ISAs are cold standby such that upon failure of the primary the standby resumes operation after the tunnels re-negotiate state. While the backup ISA can be shared by multiple tunnel groups only one tunnel group can fail to a single ISA at one time (no double failure support).

In multi-active mode, the `active-mda-number` value determines the number of ISA MDAs that will be active for this tunnel group, and tunnels are spread across all active ISA MDAs. Additional ISA MDA in this tunnel group will be in cold standby.

IPsec also supports dead peer detection (DPD).



NOTE:

BFD can be configured on the private tunnel interfaces associated with GRE tunnels and used by the OSPF, BGP or static routing that is configured inside the tunnel.

SR OS also supports multi-chassis IPsec redundancy, which provides 1:1 stateful protection against MS-ISA failure or chassis failure

16.1.6 Statistics collection

Input and output octets and packets per service queue are used for billing end customers who are on a metered service plan. Since multiple tunnels can be configured per interface the statistics can include multiple tunnels. These can be viewed in the CLI and SNMP.

Reporting (syslog, traps) for authentication failures and other IPsec errors are supported, including errors during IKE processing for session setup and errors during encryption or decryption.

A session log indicates the sort of SA setup when there is a possible negotiation. This includes the setup time, teardown time, and negotiated parameters (such as encryption algorithm) as well as identifying the service a particular session is mapped to, and the user associated with the session.

16.1.7 Security

The MS-ISA module provides security utilities for IPsec-related service entities that are assigned to interfaces and SAPs.

These entities (such as card, isa-tunnel module, and IES or VPRN services) must be enabled in order for the security services to process. The module only listens to requests for security services from configured

remote endpoints. In the case of a VPN concentrator application, these remote endpoints could come from anywhere on the Internet. In the cases where a point-to-point tunnel is configured, the module listens only to messages from that endpoint.

16.1.7.1 GRE tunnel IPv4 multicast support

GRE tunnels only support unicast and multicast IPv4 packets as payload. From a multicast prospective, GRE tunnel IP interface (associated with a private tunnel SAP) can be configured as an IGMP interface and/or as a PIM interface.

The following multicast features are supported:

- IGMP versions 1, 2 and 3
- IGMP import policies
- IGMP host tracking
- Static IGMP membership
- Configurable IGMP timers
- IGMP SSM translation
- Multicast CAC
- Per-interface, per-protocol (IGMP/PIM) multicast group limits
- MVPN support (draft-rosen)
- MVPN support (BGP-MPLS)
- PIM-SM and SSM operation
- PIM BFD support
- Configurable PIM timers
- Configurable PIM priority
- PIM tracking support
- PIM ECMP (bandwidth or hash-based)
- Static multicast route

16.1.8 IKEv2

IKEv2, defined in RFC 4306, *Internet Key Exchange (IKEv2) Protocol*, is the second version of the Internet Key Exchange Protocol. The main driver of IKEv2 was to simplify and optimize IKEv1. An IKE_SA and a CHILD_SA can be created with only four IKEv2 messages exchanges.

The 7750 SR MG supports IKEv2 with following features:

- static lan-to-lan tunnel

- dynamic lan-to-lan tunnel
- remote-access tunnel
- pre-shared-key authentication, certificate authentication, EAP (remote-access tunnel only)
- liveness check
- IKE_SA rekey
- Child_SA rekey

16.1.9 IKEv2 TS-List

Configure a ts-list per ipsec-gw, to apply to both IKEv2 remote-access tunnels and LAN-to-LAN tunnels.

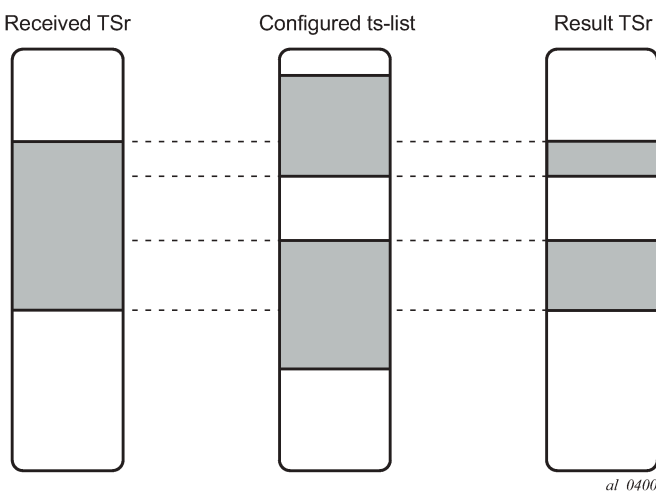
Each ts-list contains up to 32 entries. Each entry contains a local address range/subnet. In case of ipsec-gw, the ts-list represents the TSr payload. Each entry represents one TS inside TSr.

The address range/subnets between entries in the same ts-list are NOT allowed to overlap.

The system will perform address range narrowing for the received TSr as follows:

- The system will compute the intersection list between the received TSr address ranges and the address ranges in the ts-list.
- The system will send back the resulting TSr to the client.
- If there is no intersection, the system will fail the tunnel setup and return a TS_UNACCEPTABLE notification.

Figure 199: IKEv2 TS-List



16.1.10 SHA2 support

According to RFC 4868, *Using HMAC-SHA-256, HMAC-SHA-384, and HMAC-SHA-512 with IPSec*, the following SHA2 variants are supported for authentication or pseudo-random functions:

Use HMAC-SHA-256+ algorithms for data origin authentication and integrity verification in IKEv1/2, ESP:

- AUTH_HMAC_SHA2_256_128
- AUTH_HMAC_SHA2_384_192
- AUTH_HMAC_SHA2_512_256

For use of HMAC-SHA-256+ as a PRF in IKEv1/2:

- PRF_HMAC_SHA2_256
- PRF_HMAC_SHA2_384
- PRF_HMAC_SHA2_512

16.2 Using certificates for IPSec tunnel authentication

The SR OS supports X.509v3 certificate authentication for IKEv2 tunnel (LAN-to-LAN tunnel and remote-access tunnel) and asymmetric authentication. This means the SR OS and the IKEv2 peer can use different methods to authenticate. For example, one side could use pre-shared-key and the other side could use certificate.

The SR OS supports certificate chain verification. For a static LAN-to-LAN tunnel or ipsec-gw, there will be a configurable trust-anchor-profile which specifies the expecting CA that should be present in the certificate chain before reaching the root CA (self-signed CA) configured in the system.

The SR OS own key and certificate are also configurable per tunnel or ipsec-gw.



NOTE:

When using certificate authentication, the SR OS will use the subject of the configured certificate as its ID by default.

16.3 Trust-Anchor-Profile

Since R12.0R1, the SR OS supports multiple trust-anchors per ipsec-tunnel/ipsec-gw. Users can configure a trust-anchor-profile that includes up to eight CAs. The system will build a certificate chain by using the certificate in the first certificate payload in the received IKEv2 message. If any of configured trust-anchor CAs in the trust-anchor-profile appears in the chain, then authentication is successful. Otherwise authentication is failed.

**NOTE:**

The SR OS only supports processing of up to 16 hashes for the trust-anchor list from other products. If the remote end is sending more than 16, and a certificate match is in the > 16 range the tunnel remains down with authentication failure.

The trust-anchor command under the ipsec-tunnel and ipsec-gw contexts will be deprecated in a future release.

16.4 Cert-profile

Since R12.0R1, the SR OS supports sending different certificate/chain according to the received IKEv2 certificate-request payload. This is achieved by configuring a cert-profile which allows up to eight entries. Each entry includes a certificate and a key and optionally also a chain of CA certificates.

The system will load cert/key in cert-profile into memory and build a chain: compare-chain for the certificate configured in each entry of cert-profile upon no shutdown of the cert-profile. These chains will be used in IKEv2 certificate authentication. If a chain computation cannot be completed for a configured certificate, then the corresponding compare-chain will be empty, or only partially computed.

Because there could be multiple entries configured in the cert-profile, the system needs to pick the cert/key in the correct entry that the other side expects to receive. This is achieved by a lookup of the CAs within the received cert-request payload in the compare-chain and then picking the first entry that there is a cert-request CA appearing in its chain. If there is no such cert, the system picks the first entry in the cert-profile. Note that the first entry is the 1st configured entry in cert-profile. The entry-id of first entry does not have to be 1. For example, there are three CA listed in certificate-request payload: CA-1, CA-2 and CA-3, and there are two entries configured in the cert-profile as shown in the following example.

```
cert-profile "cert-profile-1"
entry 1
cert "cert-1"
key "key-1"
entry 2
cert "cert-2"
key "key-2"
send-chain
ca-profile "CA-1"
ca-profile "CA-2"
```

The system will build two compare-chains: chain-1 for cert-1 and chain-2 for cert-2. Assume CA-2 appears in chain-2, but CA-1 and CA-3 do not appear in either chain-1 or chain-2. Then the system will pick entry 2.

After a cert-profile entry is selected, the system generates the AUTH payload by using the configured key in the selected entry. The system will also send the cert in the selected entry as "certificate" payload to the peer.

If a chain is configured in the selected entry, then one certificate payload is needed for each certificate in the configured chain. The first certificate payload in the IKEv2 message will be the signing certificate, which is configured by the `cert t` command in the chosen `cert-profile` entry. With the above example, the system will send three certificate payloads: `cert-2`, `CA-1`, `CA-2`.

The following CA chain-related enhancements are supported

- The no-shut of a `ca-profile` will trigger a re-computation of `compute-chain` in related `cert-profiles`. The system will also generate a new `log-1` to indicate a new `compute-chain` has been generated; the log includes the `ca-profile` names on the new chain. Another `log-2` will be generated if the `send-chain` in a `cert-profile` entry is not in `compute-chain` due to this `ca-profile` change. Another log is generated if the hash calculation for a certificate under a `ca-profile` has changed.
- When no-shutting a `cert-profile`, the system now allows the CAs in the `send-chain`, not in the `compute-chain`. The system will also generate `log-2` as above.
- The system now allows changes of the configuration of `send-chain` without shutdown of `cert-profile`.

16.4.1 Cert-profile/Trust-anchor-profile versus Cert/Trust-anchor

Since R12.0R1, `cert-profile/trust-anchor-profile` provides a superset of function of current `cert t` and `trust-anchor` commands. The current `cert` and `trust-anchor` commands will be deprecated in a future release.

To facilitate transition and also to update the certificate trust-anchor, the following is a list of user configuration actions and corresponding system behavior while the tunnel or `ipsec-gw` is enabled (no shutdown):

- `trust-anchor-profile X` → `trust-anchor-profile Y`: allowed
- `trust-anchor Z` → `trust-anchor-profile Y`: allowed
- `trust-anchor-profile X` → no `trust-anchor-profile`: disallowed
- `trust anchor W` → `trust-anchor Z`: disallowed
- `cert-profile X` → `cert-profile Y`: allowed
- `cert A + key B` → `cert-profile Y`: allowed
- `cert-profile X` → no `cert-profile`: disallowed
- `cert A` → `cert B`: disallowed
- `key C` → `key D`: disallowed



NOTE:

The new configuration will only be used in subsequent tunnel authentication. Existing tunnels will not be affected.

16.5 Certificate Management Protocol version 2 (CMPv2)

CMPv2, RFC 4210, Internet X.509 Public Key Infrastructure Certificate Management Protocol (CMP) is a protocol between a Certificate Authority (CA) and an end entity. It provides multiple certificate management functions like certificate enrollment, certificate update, and so on.

The SR OS supports following CMPv2 operations:

Initial Registration

The process the SR OS uses to enroll a certificate with a certain CA for the first time.

- Public/Private key pair must be pre-provisioned before enrollment by means of local generation or other methods.
- Users can optionally include a certificate or certificate chain in the extraCerts field of the initial registration request.

Key Pair Update

A process for SR OS to update an existing certificate due to reason like refreshes key/cert before it expires or any other reason.

Certificate Update

A process where an initialized SR OS system obtains additional certificates.

Polling

In some cases, the CA may not return the certificate immediately for reasons such as **request processing need manual intervention**. In such cases, the SR OS supports polling requests and responds as described in Section 5.3.22, Polling Request and Response, in RFC 4210, *Internet X.509 Public Key Infrastructure Certificate Management Protocol (CMP)*.

The following lists some implementation details:

- HTTP is the only supported transport protocol for CMPv2. HTTP 1.1 and 1.0 are supported and configurable.
- All CMPv2 messages sent by SR OS consist of only one PKI Message. The size of the sequence for PKI Messages are 1 in all cases.
- Both the password-based MAC and the public key-based signature CMPv2 message protection are supported.
- SR OS only allows one outstanding ir/cr/kur request for each CMPv2 server. The means that no new requests are allowed if a pending request is present.

16.6 OCSP

Unlike CRL, which relies on checking against an offline file, OCSP provides timely, online information regarding the revocation status of a certificate.

Online Certificate Status Protocol (OCSP) (RFC 2560, X.509 Internet Public Key Infrastructure Online Certificate Status Protocol - OCSP) is used by SR OS applications to determine the (revocation) state of an identified certificate.

IPSec is the only supported application to use OCSP. With introduction of OCSP, the system supports both CRL and OCSP as the certificate revocation status checking method. For a given ipsec-tunnel or ipsec-gw, the user could configure a primary method, a secondary method and a default result to achieve a hierarchical fallback mechanism. If the primary method fails to return a result, the system will fall back to the secondary method. If the secondary method fails, the fall back proceeds to a default result.

The following lists implementation details:

- Only an OCSP client function is supported.
- HTTP is the only supported transport protocol.
- OCSP server access via management routing instance is not supported.
- SR OS does not sign an OCSP Request.
- The OCSP response must be signed. The system will verify the response by using the signer's certificate included in the response. If there is no such certificate, the CA certificate in the ca-profile will be used.
- If a nextUpdate exists in the OCSP response, the system will check the current time $\leq$ nextUpdate. If yes, then the response is valid, otherwise the response is considered unreliable. The system will move to next revocation checking method.
- The revocation status result from a valid OCSP response will be cached in the system.
- OCSP can only be used to verify the revocation status of the end-entity certificate. CRL is still needed for CA certificate's status verification.

16.7 Using CRL

By default, the system uses CRL to check the revocation status of a certificate, whether it is an end entity certificate or a CA certificate. This makes CRL a mandatory configuration in the ca-profile.

The revocation-check command in the ca-profile can change this behavior, with revocation-check `crl-optional` configured:

When a user enables the ca-profile (no shutdown), the system will try to load the configured CRL (specified by the `crl-file` command). But, if the system fails to load it for following reasons, then the system will still be ca-profile oper-up, but leave the CRL as non-exist.

- CRL file does not exist
- CRL is not properly encoded, possibly due to an interrupted file transfer

- CRL is not signed by CA certificate
- Wrong CRL version
- CRL expired

If the system needs to use the CRL of a specific ca-profile to check revocation status of an end-entity cert and CRL is non-existent due to the above reasons, then the system will treat it as unable to get an answer from CRL and fall back to the next status-verify method or default-result configured under the ipsec-gw/ipsec-tunnel.

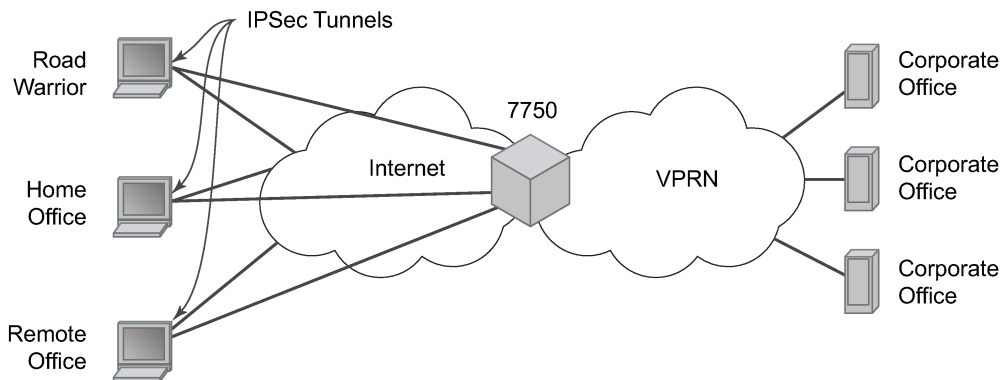
If the system needs to check the revocation of a CA cert in cert chain, and if CRL is non-existent due to the above reasons, then the system will skip checking the revocation status of the CA cert. For example, the CA1 is issued by CA2, if CA2's revocation-check is **crl-optional** and CA2's CRL is non-existent, then the system will not check CA1 cert's revocation status and consider it as good.

**NOTE:**

You must disable the ca-profile to change the revocation-check configuration.

16.8 Video wholesale example

Figure 200: Video wholesale configuration example



OSSG182

As satellite headend locations can be costly, many municipal and second tier operators cannot justify the investment in their own ground station to offer triple play features. However, it is possible for a larger provider or a cooperative of smaller providers to unite and provide a video headend. Each retail subscriber can purchase content from this single station, and receive it over IP. However, encryption is required so the signal cannot be understood if intercepted. A high speed encrypted tunnel is preferred over running two layers of double video protection which is cumbersome and computationally intensive.

16.9 IKEv2 remote-access tunnel

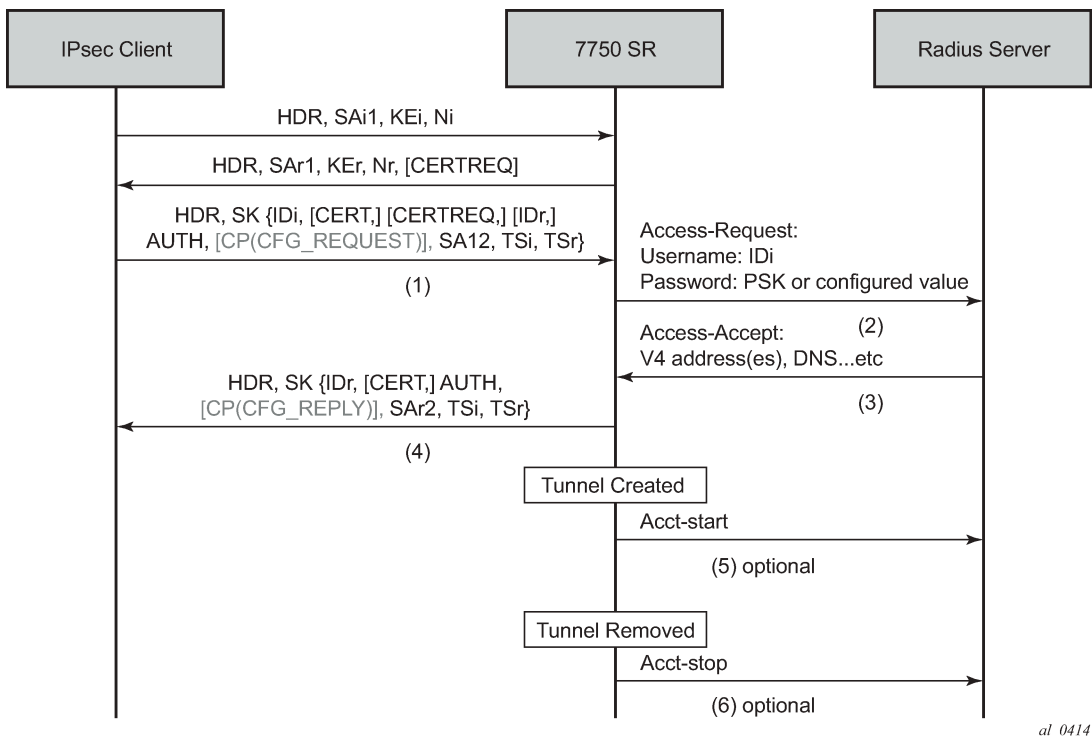
Since R11.0R6, SR OS supports IKEv2 remote-access tunnel. The difference between a remote-access tunnel and a LAN-to-LAN tunnel is that the remote-access tunnel allows the client to require an internal address via the IKEv2 configuration payload. The SR OS supports IKEv2 remote-access tunnel with following features:

- Authentication Methods:
 - Pre-Shared-Key RADIUS (psk-radius)
 - Certificate RADIUS (cert-radius)
 - EAP/EAP-Only
- Internal address assignment via IKEv2 configuration payload
- RADIUS based address assignment
- RADIUS accounting to report address usage
- NAT-Traversal support
- Support MC-IPSec

16.9.1 IKEv2 remote-access tunnel - Authentication

If the auth-method in the `ike-policy` context is configured as auth-method **psk-radius** or auth-method **cert-radius**, then the system will authenticate the client via PSK or certificate accordingly as like a LAN-to-LAN tunnel. The difference being that if **psk-radius/cert-radius** are configured, the system will also perform a RADIUS authentication/authorization and optionally send RADIUS accounting messages.

Figure 201: Call Flow for psk-radius/cert-radius



The Access-Request includes following attributes:

- Username: IDi
- User-Password: One of following value hash according to section 5.2 of RFC 2865, *Remote Authentication Dial In User Service (RADIUS)*.
 - Client's PSK if the **psk-radius** is configured (refer to the CLI).
 - Otherwise, a CLI configured key via the password command in the radius-authentication-policy context; if the password is not configured in this case, the system will not include User-Password attribute in the access-request.
- Acct-Session-Id: Represents the IPsec tunnel session, format is: local_gw_ip-remote_ip:remote_port-time_stamp.
An example: 172.16.100.1-192.168.5.100:500-1365016423.
- Other RADIUS attributes (dependent on the config>ipsec>radius-auth-policy> include-radius-attribute configuration).
 - Called-Station-Id: local tunnel address
 - Calling-station-Id: remote tunnel address: port number
 - Nas-Identifier: the system name
 - Nas-Ip-Address: the system IP
 - Nas-port-id: the public tunnel SAP ID

If the RADIUS authentication is successful, the RADIUS sever will send access-accept back; else, it will send access-reject back. Expected attributes in access-accept:

- Alc-IPsec-Serv-Id
- Alc-IPsec-Interface
- Framed-IP-Address
- Framed-IP-Netmask
- Alc-Primary-Dns
- Alc-IPsec-Tunnel-Template-Id
- Alc-IPsec-SA-Lifetime
- Alc-IPsec-SA-PFS-Group
- Alc-IPsec-SA-Encr-Algorithm
- Alc-IPsec-SA-Auth-Algorithm
- Alc-IPsec-SA-Replay-Window

Only Framed-IP-Address and Framed-IP-Netmask are mandatory, other attributes are optional.

Once a tunnel is successfully created, the system could optionally (depending on the configuration of radius-accounting-policy under the ipsec-gw context) send accounting-start packet to the RADIUS server, and also send accounting-stop when a tunnel is removed. The following attributes are expected in acct-start/stop:

- Acct-status-type
- Acct-session-id: same as it in access-request
- Username
- The following attributes are dependent on the radius-acct-policy>include-radius-attribute configuration:
 - Frame-ip-address: the assigned internal address
 - Calling-station-id
 - Called-station-id
 - Nas-Port-Id
 - Nas-Ip-Addr
 - Nas-Identifier
 - Acct-Session-Time: tunnel session time, only in acct-stop packet.

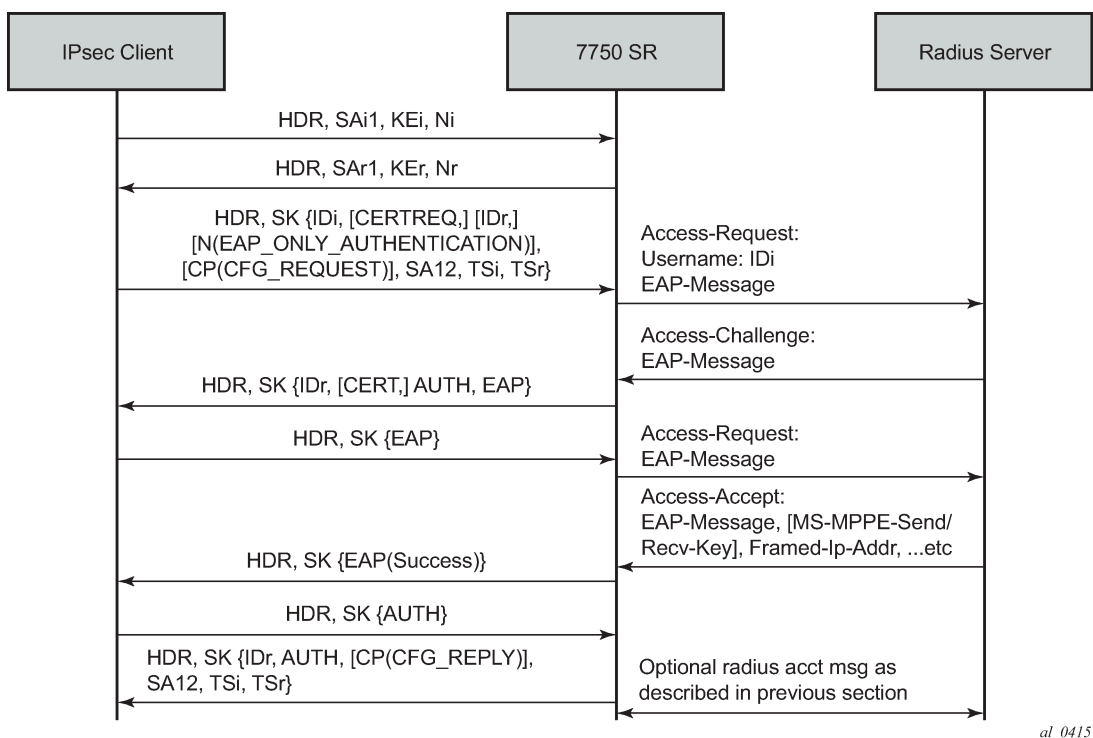
By default, the system will only return what the client has requested in the CFG_REQUEST payload. However, this behavior can be overridden by configuring the relay-unsolicited-cfg-attribute in the ike-policy. With this configuration, the configured attributes returned from the source (such as

the RADIUS server) will be returned to client regardless if the client has requested it in the CFG_REQUEST payload.

16.9.2 IKEv2 remote-access tunnel - EAP authentication

SR OS supports EAP authentication for IKEv2 remote-access tunnel, in which case, the system acts as an authenticator between the IPSec client and the RADIUS server. It transparently forwards an EAP message between the IKEv2 session and the RADIUS session, so the actual EAP authentication happens between the client and RADIUS server.

Figure 202: Typical call flow of EAP authentication



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EAP authentication is enabled by configuring authentication eap. Once enabled, after receiving an IKE_AUTH request from the client, the system sends an EAP-Response/ID with IDi as the value in access-request to AAA. AAA will return a method request and the system starts pass through between client and AAA (*Typical call flow of EAP authentication*).

The generation of AUTH payload in IKE_AUTH response sent by the SR OS (message 4 in the above flow) is dependent on the configuration of the own-auth-method command parameters:

psk

AUTH payload is present and generated by using PSK.

cert

AUTH payload is present and generated by configured public/private key pairs as it does in certificate authentication, any needed certificates will be also sent.

eap-only

Neither AUTH nor CERT payload is present.

The RADIUS attributes in authentication/accounting packets are very similar to psk-radius/cert-radius with following differences:

- support EAP-Message/Message-Authenticator /State attributes
- support Access-Challenge packet
- support MS-MPPE-Send-Key/ MS-MPPE-Recv-Key in access-accept; these two attributes are required for all EAP methods that generate MSK

The system provides a method to support EAP and other authentication methods on the same ipsec-gw. This is enabled by configuring **auto-eap-radius** as the auth-method in the ike-policy.

With **auto-eap-radius**:

- If there is no AUTH payload in IKE_AUTH request, then the system uses EAP to authenticate the client and will also use own-auth-method command to generate AUTH payload.
- If there is an AUTH payload in the IKE_AUTH request:
 - If auto-eap-method is psk, then the system proceeds as auth-method: **psk-radius**.
 - If auto-eap-method is cert, then the system proceeds as auth-method: **cert-radius**.
 - The system uses auto-eap-own-method command to generate the AUTH payload.
 - If auto-eap-method is **psk-or-cert**, then:
 - If the Auth Method field of the AUTH payload is PSK, then the system proceeds as **auth-method:psk-radius**.
 - If the Auth Method field of the AUTH payload is RSA or DSS, then the system proceeds as **auth-method:cert-radius**.

16.9.3 IPv6 over IPv4 IPSec tunnel

SROS IPv6 over IPv4 IPSec allows IPv6 packets to be forwarded within an IPv4 IPSec tunnel, current support include following:

- IKEv2 tunnel only
- Tunnel type support:
 - Static LAN-to-LAN tunnel
 - Dynamic LAN-to-LAN tunnel
 - Remote-access tunnel
- 7750 SR C12 is not supported
- The prefix length of the IPv6 address on private interface must be /96 or longer

16.9.3.1 IPv6 over IPv4: Static LAN-to-LAN tunnel

There are three methods to forward IPv6 traffic into static tunnel on the private side:

The destination address is a configured dest-ip under the tunnel context:

```
configure service vprn if sap ipsec-tun
[no] dest-ip v4/v6 addr
```

- dest-ip could be either an IPv6 address or an IPv4 address.
- In case of IPv6, it must be either an IPv6 global unicast address or an IPv6 link-local address.
- In case of IPv4, it could be used to forward IPv4 traffic into the tunnel.
- In case of unicast address, dest-ip must be within the prefix configured on a private interface.
- Up to 16 dest-ips can be configured per ipsec-tunnel.

A v6 route with configured dest-ip as the next-hop. This route can be either static or dynamic learn from routing protocol like BGP.

A IPv6 static route with ipsec-tunnel as next-hop.

Security-policy supports either an IPv4 entry or an IPv6 entry; however IPv4 and IPv6 cannot coexist in the same entry.

16.9.3.2 IPv6 over IPv4: Dynamic LAN-to-LAN tunnel

With Dynamic LAN-to-LAN tunnel, system will automatically create v6 reverse route in private VPRN based on the received TSi payload with the tunnel as the nexthop.

16.9.3.3 IPv6 over IPv4: Remote-access tunnel

The system supports the following IKEv2 IPv6 configuration attributes:

- INTERNAL_IP6_ADDRESS
- INTERNAL_IP6_DNS

The system supports only one internal IPv6 address per tunnel.

The following IPv6 related RADIUS attributes are also supported in access-accept:

- Framed-IPv6-Prefix — This will be translated into INTERNAL_IP6_ADDRESS in configuration payload, which includes two parts. A 16 bytes v6 prefix and one byte prefix length.
- Alc-Ipv6-Primary-Dns
- Alc-Ipv6-Secondary-Dns

If an internal v6 address has been assigned to the remote-access client, then Framed-IPv6-Prefix will also be included in RADIUS accounting-request packet.

The assigned internal v6 address must be within the prefix configured on the corresponding private interface.

16.10 Configuring IPsec with CLI

Information about configuring IPsec using CLI.

16.10.1 Tunnel ISA provisioning

An IPsec ISA can only be provisioned on an IOM2. The following output displays a card and ISA configuration.

```
*A:ALA-49>config# info
-----
...
  card 1
    card-type iom2-20g
    mda 1
      mda-type m10-1gb-sfp
    exit
    mda 2
      mda-type isa-tunnel
    exit
  exit
...
-----
```

16.10.2 Tunnel group configuration

The following output displays a tunnel group configuration in the ISA context. The primary command identifies the card/slot number where the IPsec ISA is the primary module for the IPsec group.

```
*A:ALA-49>config# info
-----
...
  isa
    tunnel-group 1 create
      primary 1/2
      no shutdown
    exit
  exit
...
-----
```

16.10.2.1 Tunnel group configuration for ePDG

To configure ePDG, the second ISA in an ISM-MG must be used for IPSec. The MDA type must be declared to be **isa2-ipsmg**. A group configuration is used, similar to the Application Assurance groups, where all **isa2-ipsmg** MDA types will then be assigned to a single group. However the ipsmg redundancy will be inherited by the ISM-MG group on which the ISA is operating.

The first step is to declare the second ISA as an IPSMG MDA type. In this example, card slot 2 is being configured.

```
config card slot 2 card-type ism-mg-2pac-fp3
config card mda 1 mda-type isa-mg
config card mda 2 mda-type isa2-ipsmg
```

The next step is to add MDA 2 to an IPSMG group. An IPSMG group refers to an Integrated Services Adapter (ISA) Internet Protocol Security (IPSec) Mobile Gateway (MG) group.

```
config isa ipsmg-group create 1
config isa ipsmg-group 1 mda 2/2
```

Finally you can associate the ipsmg-group with the PDN gateway ID:

```
config mobile-gateway pdn 5 ipsmg-group 1
config isa ipsmg-group 1 mda 2/2
```

For more information and guidelines for configuring ePDG, including IKE and ESP security associations, ePDG certificates, SWu Reference Points for interfaces, and Diameter Reference Points using SWm, refer to the *7750 SR Mobile Gateway Configuration Guide*.

16.10.3 Router interface configuration for IPSec

The following output displays an Internet interface configured using the network port (1/1/1).

```
*A:ALA-49>config# info
-----
...
router
  interface "internet"
    address 10.10.7.118/24
    port 1/1/1
  exit
  interface "system"
```

```

        address 10.20.1.118/32
    exit
    autonomous-system 123
exit
...
-----

```

16.10.4 IPSec parameter configuration

The following output displays an IPSec configuration example.

```

*A:ALA-49>config# info
-----
...
    ipsec
        ike-policy 1 create
            ipsec-lifetime 300
            isakmp-lifetime 600
            pfs
            auth-algorithm md5
            dpd interval 10 max-retries 5
        exit
        ipsec-transform 1 create
            esp-auth-algorithm sha1
            esp-encryption-algorithm aes128
        exit
    exit
...
-----

```

16.10.5 IPSec in services configuration

The following output displays an IES and VPRN service with IPSec parameters configured.

```

*A:ALA-49>config# info
-----
...
    service
        ies 100 customer 1 create
            interface "ipsec-public" create
                address 10.10.10.1/24
                sap tunnel-1.public:1 create
            exit
    exit

```

```

        exit
        no shutdown
    exit
    vprn 200 customer 1 create
        ipsec
            security-policy 1 create
                entry 1 create
                    local-ip 172.17.118.0/24
                    remote-ip 172.16.91.0/24
                exit
            exit
        exit
    exit
route-distinguisher 1:1
    ipsec-interface "ipsec-private" tunnel create
        sap tunnel-1.private:1 create
        ipsec-tunnel "remote-office" create
            security-policy 1
            local-gateway-address 10.10.10.118 peer 10.10.7.91
delivery-service 100
    dynamic-keying
        ike-policy 1
        pre-shared-key "humptydumpty"
        transform 1
    exit
    no shutdown
    exit
    exit
    interface "corporate-network" create
        address 172.17.118.118/24
        sap 1/1/2 create
    exit
    exit
static-route 172.16.91.0/24 ipsec-tunnel "remote-office"
    no shutdown
    exit
    exit
...
-----

```

16.10.6 X.509v3 certificate parameters configuration

The X.509v3 certificate parameters configuration comprises the following:

1. *Configuring the certificate enrollment*
2. *Configuring CA certificate/CRL import*

16.10.6.1 Configuring the certificate enrollment

Procedure

1. Generate a key.

```
admin certificate gen-keypair cf3:/key_plain_rsa2048 size 2048 type rsa
```

2. Generate a certificate request.

```
admin certificate gen-local-cert-req keypair cf3:/key_plain_rsa2048 subject-  
dn "C=US,ST=CA,CN=7750" file 7750_req.csr
```

note: since 12.0R1, the system encodes the common name field as UTF8 instead of a printable string format. If a printable string is required for compatibility add the option "use-printable" to the request for legacy behavior.

3. Send the certificate request to CA-1 to sign and get the signed certificate.
4. Import the key.

```
admin certificate import type key input cf3:/key_plain_rsa2048 output  
key1_rsa2048 format der
```

5. Import the signed certificate.

```
admin certificate import type cert input cf3:/7750_cert.pem output 7750cert  
format pem
```

16.10.6.2 Configuring CA certificate/CRL import

Procedure

1. Import the CA certificate.

```
admin certificate import type cert input cf3:/CA_1_cert.pem output ca_cert  
format pem
```

2. Import the CA CRL.

```
admin certificate import type crl input cf3:/CA_1_crl.pem output ca_crl
format pem
```

Example

The following displays a certificate authentication for IKEv2 static LAN-to-LAN tunnel configuration.

```
config>system>security>pki# info
-----
        ca-profile "nokia-root" create
            cert-file "nokia_root.cert"
            crl-file "nokia_root.crl"
            no shutdown
        exit
-----
config>ipsec# info
-----
        ike-policy 1 create
            ike-version 2
            auth-method cert-auth
        exit
        ipsec-transform 1 create
        exit
        cert-profile "segw" create
            entry 1 create
                cert segw.cert
                key segw.key
            exit
            no shutdown
        exit
        trust-anchor-profile "nokia" create
            trust-anchor "nokia-root"
        exit

config>service>vpn>if>sap
-----
        ipsec-tunnel "t50" create
            security-policy 1
            local-gateway-
address 192.168.55.30 peer 192.168.33.100 delivery-service 300
            dynamic-keying
                ike-policy 1
                transform 1
```

```

        cert
        trust-anchor-profile "nokia"
        cert-profile "segw"
    exit
exit
no shutdown
exit

```

The following displays an example of the syntax to import a certificate from the pem format.

```

admin certificate import type cert input cf3:/pre-import/R1-0cert.pem output
R1-0cert.der format pem

```

The following displays an example of the syntax to export a certificate to the pem format.

```

admin certificate export type cert input R1-0cert.der output cf3:/R1-0cert.pem
format pem

```

16.10.7 MC-IPSec configuration

The MC-IPSec configuration can support the following:

- *MIMP configuration*
- *Multi-chassis synchronization configuration*
- *Routing configuration for MC-IPSec*

16.10.7.1 MIMP configuration

The following is an MIMP configuration example.

```

config>redundancy>multi-chassis
-----
    peer 2.2.2.2 create
        mc-ipsec
        bfd-enable
        tunnel-group 1 create
            peer-group 2
            priority 120
            no shutdown
        exit
    exit
no shutdown
exit

```

The peer tunnel-group id is not necessarily the same as the local tunnel-group id. With `bfd-enable`, the BFD parameters are specified under the interface that the MIMP source address resides on, which must be a loopback interface in the base routing instance. The default source address of MIMP is the system address.

The `keep-alive-interval` and `hold-on-neighbor-failure` define the MIMP alive parameter, however, BFD could be used for faster chassis failure detection.

The SR OS also provides the `tools` command to manually trigger the switchover such as:

```
tools perform redundancy multi-chassis mc-ipsec force-switchover tunnel-group
1
```

16.10.7.2 Multi-chassis synchronization configuration

The following displays an MCS for MC-IPSec configuration.

```
config>redundancy>multi-chassis>
-----
      peer 2.2.2.2 create
        sync
        ipsec
        tunnel-group 1 sync-tag "sync_tag_1" create
        no shutdown
      exit
```

The sync-tag must match on both chassis for the corresponding tunnel-groups.

16.10.7.3 Routing configuration for MC-IPSec

The following configuration is an example using a route policy to export /32 local tunnel address route.

```
config>router>policy-options>
-----
      policy-statement "exportOSPF"
        entry 10
          from
            protocol ipsec
            state ipsec-master-with-peer
          exit
          action accept
            metric set 500
          exit
        exit
      entry 20
```

```

        from
            protocol ipsec
            state ipsec-non-master
        exit
        action accept
            metric set 1000
        exit
    exit
    entry 30
        from
            protocol ipsec
            state ipsec-master-without-peer
        exit
        action accept
            metric set 1000
        exit
    exit
exit

```

The following configuration shows shunting in public and private service.

Example: Shunting in public service

```

config>service>ies>
    interface "ipsec-pub" create
        address 172.16.100.254/24
        sap tunnel-1.public:100 create
    exit
    static-tunnel-redundant-next-hop 1.1.1.1
exit

```

Example: Shunting in private service

```

config>service>vprn>
    cmpv2
        static-tunnel-redundant-next-hop 7.7.7.1
    exit

```

Shunting is enabled by configuring redundant next-hop on a public or private IPSec interface:

static-tunnel-redundant-next-hop Shunting nexthop for a static tunnel

dynamic-tunnel-redundant-next-hop Shunting next-hop for a dynamic tunnel

16.10.8 CMPv2 configuration and usage

CMPv2 server information is configured under the corresponding `ca-profile` by using the following commands:

```
configure system security pki ca-profile
    cmpv2
        url url-string [service-id service-id]
        response-signing-cert filename
        key-list
            key password reference reference-number
```

The `url` command specifies the HTTP URL of the CMPv2 server, the `service` specifies the routing instance that the system used to access the CMPv2 server (if omitted, then system will use base routing instance).

Also note that the service ID is only needed for inband connections to the server via VPRN services. IES services are not to be referenced by the service ID as any of those are considered base routing instance.

The `response-signing-cert` command specifies a imported certificate that is used to verify the CMP response message if they are protected by signature. If this command is not configured, then CA's certificate will be used.

The `key-list` command specifies a list of pre-shared-key used for CMPv2 initial registration message protection.

```
config>system>security>pki>ca-profile
    cmpv2
        url "http://cmp.example.com/request" service-id 100
        key-list
            key passwordToBeUsed reference "1"
```

All CMPv2 operations are invoked by using the `admin certificate cmpv2` command.

If there is no `key-list` defined under the `cmpv2` configuration, the system will default to the `cmpv2` transaction input for the command line in regards to authenticating a message without a sender ID. Also, if there is no sender ID in the response message, and there IS a key-list defined, it will choose the lexicographical first entry only, if that fails, it will have a fail result for the transaction.

The system supports optional commands (such as, `always-set-sender-id`) to support inter-op with CMPv2 servers. For more information about the syntax and usage, refer to the section *CMPv2 Commands* in the *CMG CLI Reference Guide*.

16.10.9 OCSP configuration

OCSP server information is configured under the corresponding ca-profile:

```
configure system security pki ca-profile
    oscp
        responder-url url-string
        service service-id
```

The `responder-url` command specifies the HTTP URL of the OCSP responder. The `service` command specifies the routing instance that system used to access the OCSP responder.

```
configure system security pki ca-profile
    oscp
        responder-url "http://ocsp.example.com/request"
        service 100
```

For a specified `ipsec-tunnel` or `ipsec-gw`, the user can configure a primary method, a secondary method and a default result.

```
configure service ies if sap ipsec-gw
configure service vprn if sap ipsec-gw
configure service vprn if sap ipsec-tun
    cert
        status-verify
        primary {ocsp | crl}
        secondary {ocsp | crl}
        default-result {revoked | good}
```

```
configure service ies if sap ipsec-gw
    cert
        status-verify
        primary ocsp
        secondary crl
```

16.10.10 IKEv2 remote-access tunnel configuration

The following are configuration tasks for an IKEv2 remote-access tunnel:

- Create an `ike-policy` with one of the `auth-methods` that enabled the remote-access tunnel.
- Configure a `tunnel-template/ipsec-transform`. This is the same as configuring a dynamic LAN-to-LAN tunnel.
- Create a `radius-authentication-policy` and optionally, a `radius-accounting-policy` (a `radius-server-policy` and a `radius-server` must be preconfigured)

- Configure a private VPRN service and private tunnel interface with an address on the interface. The internal address assigned to the client must come from the subnet on the private interface.
- Configure a public IES/VPRN service and an ipsec-gw under the public tunnel SAP.
- Configure the radius-authentication-policy and radius-accounting-policy (optional) under the ipsec-gw.
- Certificate the related configuration if cert-radius is used.

Example: cert-radius

```

config>system>security>pki# info
-----
        ca-profile "NOKIA-ROOT" create
        cert-file "NOKIA-ROOT.cert"
        crl-file "NOKIA-ROOT.crl"
        no shutdown
        exit
-----
A:SeGW>config>aaa# info
-----
        radius-server-policy "femto-aaa" create
        servers
            router "management"
            server 1 name "svr-1"
        exit
    exit
-----
A:SeGW>config>router# info
-----
        radius-server
            server "svr-1" address 10.10.10.1 secret "KR35xB3W4aUXtL8o3WzPD."
hash2 create
        exit
    exit
-----

config>ipsec# info
-----
        ike-policy 1 create
            ike-version 2
            auth-method cert-radius
        exit
        ipsec-transform 1 create
        exit
        tunnel-template 1 create

```



```

        transform 1
    exit
    cert-profile "c1" create
        entry 1 create
            cert SeGW2.cert
            key SeGW2.key
        exit
    no shutdown
exit
trust-anchor-profile "tap-1" create
    trust-anchor "NOKIA-ROOT"
exit
radius-authentication-policy "femto-auth" create
    include-radius-attribute
        calling-station-id
        called-station-id
    exit
    password "DJzlyYKCefyhmnFcFSBuLZovSemMKde" hash2
    radius-server-policy "femto-aaa"
exit
radius-accounting-policy "femto-acct" create
    include-radius-attribute
        calling-station-id
        framed-ip-addr
    exit
    radius-server-policy "femto-aaa"
exit

-----
config>service>ies# info
-----

    interface "pub" create
        address 172.16.100.0/31
        tos-marking-state untrusted
        sap tunnel-1.public:100 create
            ipsec-gw "rw"
                cert
                    trust-anchor-profile "tap-1"
                    cert-profile "c1"
                exit
            default-secure-service 400 interface "priv"
            default-tunnel-template 1
            ike-policy 1
            local-gateway-address 172.16.100.1
            radius-accounting-policy "femto-acct"

```

```
        radius-authentication-policy "femto-auth"
        no shutdown
    exit
exit
exit
no shutdown
-----
A:SeGW>config>service>vprn# info
-----
    route-distinguisher 400:11
    interface "priv" tunnel create
        address 20.20.20.1/24
        sap tunnel-1.private:200 create
    exit
exit
interface "l1" create
    address 9.9.9.9/32
    loopback
exit
no shutdown
-----
```

17 Network Address Translation

Information about Network Address Translation (NAT) and implementation notes.

17.1 Terminology

BNG Subscriber

A broader term than the ESM Subscriber, independent of the platform on which the subscriber is instantiated. It includes ESM subscribers on CMG, as well as subscribers instantiated on third party BNGs. Some of the NAT functions, such as Subscriber Aware Large Scale NAT44 utilizing standard RADIUS attribute work with subscribers independently of the platform on which they are instantiated.

Deterministic NAT

A mode of operation where mappings between the NAT subscriber and the outside IP address and port range are allocated at the time of configuration. Each subscriber is permanently mapped to an outside IP and a dedicated port block. This dedicated port block is referred to as deterministic port block. Logging is not needed as the reverse mapping can be obtained using a known formula. The subscriber ports can be expanded by allocating a dynamic port block in case that all ports in deterministic port block are exhausted. In such case logging for the dynamic port block allocation/de-allocation is required.

Enhanced Subscriber Management (ESM) subscriber

A host or a collection of hosts instantiated in CMG Broadband Network Gateway (BNG). The ESM subscriber represents a household or a business entity for which various services with committed Service Level Agreements (SLA) can be delivered. NAT function is not part of basic ESM functionality.

Large Scale NAT (LSN)

Refers to a collection of network address translation techniques used in service provider network implemented on a highly scalable, high performance hardware that facilitates various intra and inter-node redundancy mechanisms. The purpose of LSN semantics is to make delineation between high scale and high performance NAT functions found in service provider networks and enterprise NAT that is usually serving much smaller customer base at smaller speeds. The following NAT techniques can be grouped under the LSN name and are supported by CMG:

- Large Scale NAT44 or Carrier Grade NAT (CGN)
- NAT64

Each distinct NAT technique is referred to by its corresponding name (Large Scale NAT44 [or CGN], and NAT64) with the

understanding that in the context of CMG, they are both part of LSN (and not enterprise based NAT).

Large Scale NAT44 term can be interchangeably used with the term Carrier Grade NAT (CGN) which in its name implies high reliability, high scale and high performance. These are again typical requirements found in service provider (carrier) network.

NAT RADIUS accounting

Reporting (or logging) of address translation related events (port-block allocation/de-allocation) via RADIUS accounting facility. NAT RADIUS accounting is facilitated via regular RADIUS accounting messages (start/interim-update/stop) as defined in RFC 2866, *RADIUS Accounting*, with NAT specific VSAs.

NAT RADIUS accounting

Can be interchangeably used with the term NAT RADIUS logging.

NAT subscriber

In NAT terminology a NAT subscriber is an inside entity whose true identity is hidden from the outside. There are a few types of NAT implementation in 7750 and subscribers for each implementation are defined as follows:

- Large Scale NAT44 (or CGN) — The subscriber is an inside IPv4 address.
- NAT64 — The subscriber is an IPv6 prefix.

Non-deterministic NAT

A mode of operation where all outside IP address and port block allocations are made dynamically at the time of subscriber instantiation. Logging in such case is required.

Port block

A collection of ports that is assigned to a subscriber. A deterministic LSN subscriber can have only one deterministic port block that can be extended by multiple dynamic port blocks. Non-deterministic LSN subscriber can be assigned only dynamic port blocks. All port blocks for a LSN subscriber must be allocated from a single outside IP address.

Port range

A collection of ports that can spawn multiple port blocks of the same type. For example, deterministic port range includes all ports that are reserved for deterministic consumption. Similarly dynamic port range is a total collection of ports that can be allocated in the form of dynamic port blocks. Other types of port ranges are well-known ports and static port forwards.

17.2 NAT overview

CMG supports Network Address (and port) Translation (NAPT) to provide continuity of legacy IPv4 services during the migration to native IPv6. By equipping the Multiservice ISA (MS ISA) in an IOM3-XP, the system can operate in LSN mode.

Source address and port translation is performed as commonly deployed for shared Internet access. A CMG with NAT is used to provide consumer broadband or business Internet customers access to IPv4 internet resources with a shared pool of IPv4 addresses. Native IPv6 services are still growing and a significant amount of Internet content remains IPv4.

17.2.1 Principles of NAT

NAT devices modify the IP headers of packets between a host and server, changing some or all of the source address, destination address, source port (TCP/UDP), destination port (TCP/UDP), or ICMP query ID (for ping). CMG in both NAT modes performs S-NAPT.

S-NAPT devices are commonly deployed in residential gateways and enterprise firewalls to allow multiple hosts to share one or more public IPv4 addresses to access the Internet. The common terms of inside and outside in the context of NAT refer to devices inside the NAT (that is behind or masqueraded by the NAT) and outside the NAT, on the public Internet.

TCP/UDP connections use ports for multiplexing, with 65536 ports available for every IP address. Whenever many hosts are trying to share a single public IP address there is a chance of port collision where two different hosts may use the same source port for a connection. The resultant collision is avoided in S-NAPT devices by translating the source port and tracking this in a stateful manner. All S-NAPT devices are stateful in nature and must monitor connection establishment and traffic to maintain translation mappings. The CMG NAT implementation does not use the well-known port range (1...1023).

In most circumstances, S-NAPT requires the inside host to establish a connection to the public Internet host or server before a mapping and translation will occur. With the initial outbound IP packet, the S-NAPT knows the inside IP, inside port, remote IP, remote port and protocol. With this information the S-NAPT device can select an IP and port combination (referred to as outside IP and outside port) from its pool of addresses and create a unique mapping for this flow of data.

Any traffic returned from the server will use the outside IP and outside port in the destination IP/port fields - matching the unique NAT mapping. The mapping then provides the inside IP and inside port for translation.

The requirement to create a mapping with inside port and IP, outside port and IP and protocol will generally prevent new connections to be established from the outside to the inside as may occur when an inside host wishes to be a server.

17.2.2 Application compatibility

Applications which operate as servers (such as HTTP, SMTP, and so on) or peer-to-peer applications can have difficulty when operating behind an S-NAPT because traffic from the Internet can reach the NAT without a mapping in place.

Different methods can be employed to overcome this, including:

- Port Forwarding
- STUN support

CMG supports both methods following the best-practice RFC for TCP (RFC 5382, *NAT Behavioral Requirements for TCP*) and UDP (RFC 4787, *Network Address Translation (NAT) Behavioral Requirements for Unicast UDP*). Port Forwarding is supported on CMG to allow servers which operate on well-known ports <1024 (such as HTTP and SMTP) to request the appropriate outside port for permanent allocation.

STUN is facilitated by the support of Endpoint-Independent Filtering and Endpoint-Independent Mapping (RFC 4787) in the NAT device, allowing STUN-capable applications to detect the NAT and allow inbound P2P connections for that specific application. Many new SIP clients and IM chat applications are STUN capable.

Even with these three mechanisms some applications will still experience difficulty operating behind a NAT. As an industry-wide issue, forums like UPnP the IETF, operator and vendor communities are seeking technical alternatives for application developers to traverse NAT (including STUN support). In many cases the alternative of an IPv6-capable application will give better long-term support without the cost or complexity associated with NAT.

17.3 Large Scale NAT

A Large Scale NAT is typically deployed in a central network location with two interfaces, the inside toward the customers, and the outside toward the Internet. A Large Scale NAT functions as an IP router and is located between two routed network segments (the ISP network and the Internet).

Large Scale NAT represents the most common deployment of S-NAPT in carrier networks today, it is already employed by mobile operators around the world for handset access to the Internet.

Traffic can be sent to the Large Scale NAT function on CMG using IP filters (ACL) applied to SAPs or by installing static routes with a next-hop of the NAT application. These two methods allow for increased flexibility in deploying the Large Scale NAT, especially those environments where IP MPLS VPN are being used in which case the NAT function can be deployed on a single PE and perform NAT for any number of other PE by simply exporting the default route.

The CMG NAT implementation supports NAT in the base routing instance and VPRN, and through NAT traffic may originate in one VPRN (the inside) and leave through another VPRN or the base routing instance (the outside). This technique can be employed to provide customer's of IP MPLS VPN with Internet access by introducing a default static route in the customer VPRN, and NATing it into the Internet routing instance.

As Large Scale NAT is deployed between two routed segments, the IP addresses allocated to hosts on the inside must be unique to each host within the VPRN. While RFC 1918 private addresses have typically been used for this in enterprise or mobile environments, challenges can occur in fixed residential environments where a subscriber has existing S-NAPT in their residential gateway. In these cases the RFC 1918 private address in the home network may conflict with the address space assigned to the residential gateway WAN interface. Some of these issues are documented in *draft-shirasaki-nat444-isp-shared-addr-02*. Should a conflict occur, many residential gateways will fail to forward IP traffic.

17.3.1 Port range blocks

Port range blocks are configurable as part of outside pool configuration, allowing you to specify the number of ports allocated to each subscriber when a mapping is created.

When a range is allocated to the subscriber, these ports are used for all outbound dynamic mappings and are assigned in a random manner to minimize the predictability of port allocations (draft-ietf-tsvwg-port-randomization-05).

The S-NAPT service on the 7750 BNG incorporates a port range block feature to address scalability of a NAT mapping solution. With a single BNG capable of hundreds of thousands of NAT mappings every second, logging each mapping as it is created and destroyed logs for later retrieval (as may be required by law enforcement) could quickly overwhelm the fastest of databases and messaging protocols. Port range blocks address the issue of logging and customer location functions by allocating a block of contiguous outside ports to a single subscriber. Rather than log each NAT mapping, a single log entry is created when the first mapping is created for a subscriber and a final log entry when the last mapping is destroyed. This can reduce the number of log entries by 5000x or more. An added benefit is that as the range is allocated on the first mapping, external applications or customer location functions may be populated with this data to make real-time subscriber identification, rather than having to query the NAT as to the subscriber identity in real-time and possibly delay applications.

Port range blocks also serve another useful function in a Large Scale NAT environment, and that is to manage the fair allocation of the shared IP resources among different subscribers.

When a subscriber exhausts all ports in their block, further mappings will be prohibited. As with any enforcement system, some exceptions are allowed and the NAT application can be configured for reserved ports to allow high-priority applications access to outside port resources while exhausted by low priority applications.

17.3.1.1 Reserved ports and priority sessions

Reserved ports allow you to configure a small number of ports to be reserved for designated applications should a port range block be exhausted. Such a scenario may occur when a subscriber is unwittingly subjected to a virus or engaged in extreme cases of P2P file transfers.

In these situations, rather than block all new mappings indiscriminately the 7750 NAT application allows operators to nominate a number of reserved ports and then assign a 7750 forwarding class as containing high priority traffic for the NAT application. Whenever traffic reaches the NAT application which matches a priority session forwarding class, reserved ports will be consumed to improve the chances of success. Priority sessions could be used by the operator for services such as DNS, web portal, e-mail, VoIP, and so on to permit these applications even when a subscriber exhausted their ports.

17.3.1.2 Prevent port block starvation

The following are supported to prevent port block starvation:

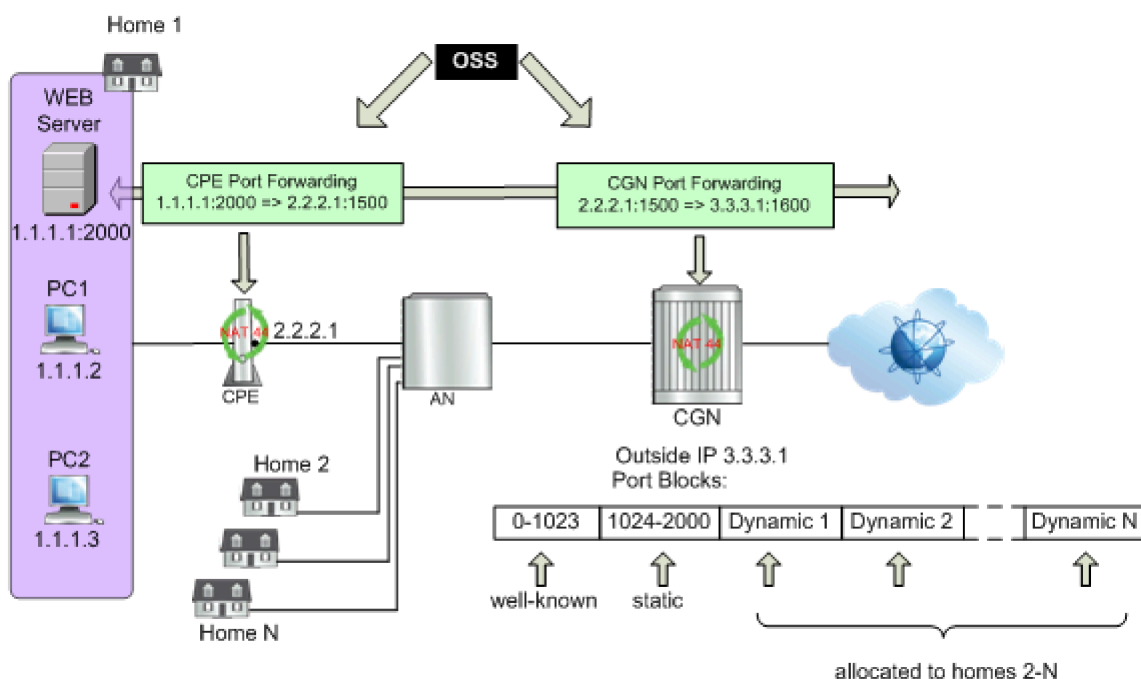
- *Dynamic port block starvation in LSN*
- *Dynamic port block reservation*

17.3.1.2.1 Dynamic port block starvation in LSN

The outside IP address is always shared for the subscriber with a static port forward and the dynamically allocated port block, insofar as the port from the port forward is in the range >1023. This behavior can lead to starvation of dynamic port blocks for the subscriber. An example for this scenario is shown in *Dynamic port block starvation in LSN*.

- A static port forward for the WEB server in Home 1 is allocated in the CPE and the CGN. At the time of static port forward creation, no other dynamic port blocks for Home 1 exist (PCs are powered off).
- Assume that the outside IP address for the newly created static port forward in the CGN is 3.3.3.1.
- Over time dynamic port blocks are allocated for a number of other homes that share the same outside IP address, 3.3.3.1. Eventually those dynamic port block allocations will exhaust all dynamic port block range for the address 3.3.3.1.
- Once the dynamic port blocks are exhausted for outside IP address 3.3.3.1, a new outside IP address (for example, 3.3.3.2) will be allocated for additional homes.
- Eventually the PCs in Home 1 come to life and they try to connect to the Internet. Due to the dynamic port block exhaustion for the IP address 3.3.3.1 (that is mandated by static port forward - Web Server), the dynamic port block allocation will fail and consequently the PCs will not be able to access the Internet. There will be no additional attempt within CGN to allocate another outside IP address. Note that in the CGN there is no distinction between the PCs in Home 1 and the Web Server when it comes to source IP address. They both share the same source IP address 2.2.2.1 on the CPE.
- The solution for this is to reserve a port block (or blocks) during the static port forward creation for the specified subscriber.

Figure 203: Dynamic port block starvation in LSN



17.3.1.2.2 Dynamic port block reservation

To prevent starvation of dynamic port blocks for the subscribers that use port forwards, a dynamic port block (or blocks) will be reserved during the lifetime of the port forward. Those reserved dynamic port blocks will be associated with the same subscriber that created the port forward. However, a log would not be generated until the dynamic port block is actually used and mapping within that block are created.

At the time of the port forward creation, the dynamic port block will be reserved in the following way:

- If the dynamic port block for the subscriber does not exist, then a dynamic port block for the subscriber will be reserved. No log for the reserved dynamic port block is generated until the dynamic port block starts being utilized (mapping created due to the traffic flow).
- If the corresponding dynamic port block already exists, then it will be reserved even after the last mapping within the last port block had expired.

The reserved dynamic port block (even without any mapping) will continue to be associated with the subscriber as long as the port forward for the subscriber is present. The log (syslog or RADIUS) will be generated only when there is not active mapping within the dynamic port block AND all port forwards for the subscriber are deleted.

Additional considerations with dynamic port block reservation:

- The port block reservation should be triggered only by the first port forward for the subscriber. The subsequent port forwards will not trigger additional dynamic port block reservation.
- Only a single dynamic port block for the subscriber is reserved (i.e no multiple port-block reservations for the subscriber are possible).
- This feature is enabled with the configuration command `port-forwarding-dyn-block-reservation` under the `configure>service>vprn>nat>outside>pool` and the `configure>router>nat>outside>pool` CLI hierarchy. This command can be enabled only if the maximum number of configured port blocks per outside IP is greater or equal then the maximum configured number of subscribers per outside IP address. This will guarantee that all subscribers (up the maximum number per outside IP address) configured with port forwards will be able to reserve a dynamic port block.
- In case that the port-reservation is enabled while the outside pool is operational and subscribers traffic is already present, the following two cases will have to be considered:
 - The configured number of subscribers per outside IP is less or equal than the configured number of port blocks per outside IP address (this is permitted) but all dynamic port blocks per outside IP address are occupied at the moment when port reservation is enabled. This will leave existing subscribers with port forwards that do not have any dynamic port blocks allocated (orphaned subscribers), unable to reserve dynamic port blocks. In this case the orphaned subscribers will have to wait until dynamic port blocks allocated to the subscribers without port forwards are freed.
 - The configured number of subscribers per outside IP is greater than the configured number of port blocks per outside IP address. In addition, all dynamic port blocks per outside IP address are

allocated. Before the port reservation is even enabled, the subscriber-limit per outside IP address will have to be lowered (by configuration) so that it is equal or less than the configured number of port blocks per outside IP address. This action will cause random deletion of subscribers that do not have any port forwards. Such subscribers will be deleted until the number of subscriber falls below the newly configured subscriber limit. Note that subscribers with static port forwards will not be deleted, regardless of the configured subscriber-limit number. Once the number of subscriber is within the newly configured subscriber-limit, the port-reservation can take place under the condition that the dynamic port blocks are available. If certain subscribers with port forwards have more than one dynamic port block allocated, the orphaned subscribers will have to wait for those additional dynamic port blocks to expire and consequently be released.

- This feature is supported on the following applications: CGN, LSN NAT44 and NAT64.

17.3.2 Timeouts

Creating a NAT mapping is only one half of the problem - removing a NAT mapping at the appropriate time maximizes the shared port resource. Having ports mapped when an application is no longer active reduces solution scale and may impact the customer experience should they exhaust their port range block. The NAT application provides timeout configuration for TCP, UDP and ICMP.

TCP state is tracked for all TCP connections, supporting both three-way handshake and simultaneous TCP SYN connections. Separate and configurable timeouts exist for TCP SYN, TCP transition (between SYN and Open), established and time-wait state. Time-wait assassination is supported and enabled by default to quickly remove TCP mappings in the TIME WAIT state.

UDP does not have the concept of connection state and is subject to a simple inactivity timer. Nokia-sponsored research into applications and NAT behavior suggested some applications, like the Bittorrent Distributed Hash Protocol (DHT) can make a large number of outbound UDP connections that are unsuccessful. Rather than wait the default five (5) minutes to time these out, the 7750 NAT application supports an udp-initial timeout which defaults to 15 seconds. When the first outbound UDP packet is sent, the 15 second time starts - it is only after subsequent packets (inbound or outbound) that the default UDP timer will become active, greatly reducing the number of UDP mappings.

17.3.3 Watermarks

It is possible to define watermarks to monitor the actual usage of sessions and/or ports.

For each watermark, a high and a low value has to be set. Once the high value is reached, a notification will be send. As soon as the usage drops below the low watermark, another notification will be send.

Watermarks can be defined on nat-group, pool and policy level.

Nat-group

Watermarks can be placed to monitor the total number of sessions on an MDA.

Pool

Watermarks can be placed to monitor the total number of blocks in use in a pool.

Policy

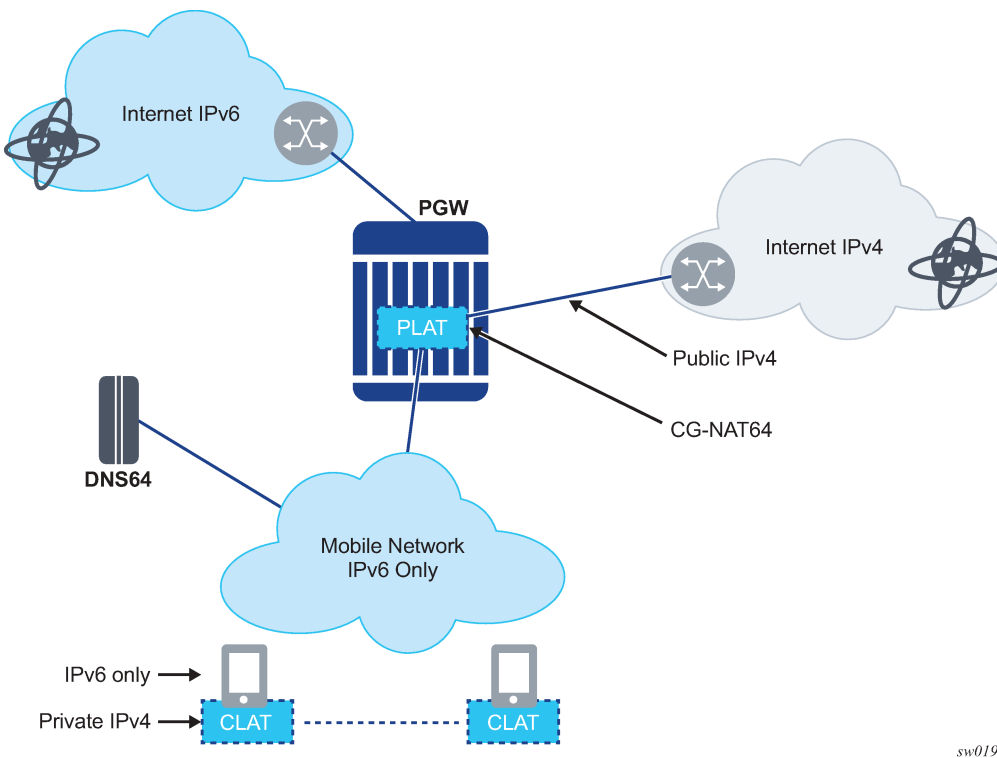
In the policy it is possible to define watermarks on session and port usage. In both cases, it is the usage per host (for large-scale NAT) that will be monitored for the 7750 SR MG.

17.4 464XLAT support

This feature provides 464XLAT support on the PGW with the PLAT (Provider-side Translator) function running on the PGW CG-NAT64 function.

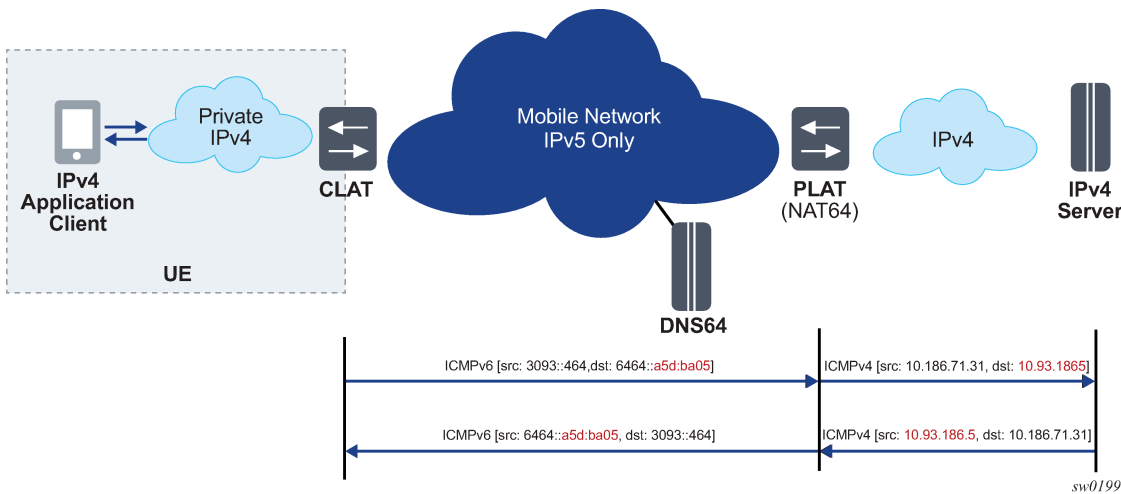
The 464XLAT architecture is based on RFC 6146 and consists of the UE with the Customer-side Translator (CLAT) function, the DNS64, and the PGW with the PLAT function. This allows IPv4 applications on the UE to successfully communicate over IPv6 mobile networks. The IPv6 applications on the UE, are natively routed over the PGW and the 464XLAT function is not used in this case.

Figure 204: IPv4 applications routed over IPv6 mobile networks



The CLAT runs inside the UE and works as a stateless NAT64 (bidirectional one to one mapping). It offers a private IPv4 address and an IPv4 default route to IPv4 applications. Traffic with an IPv4 destination address is translated to an IPv6 address by the CLAT component and uses a dedicated /96, out of the IPv6 /64 prefix assigned to the UE, as the source IP address. The destination IPv6 address is created using the Pref64/n and the embedded destination IPv4 address toward the PLAT.

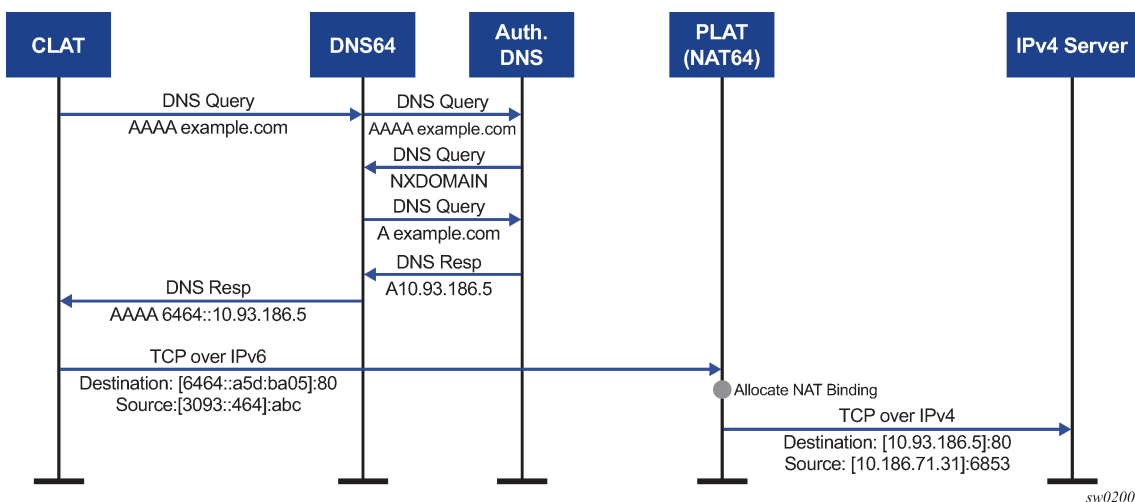
Figure 205: Ping transmission from the IPv4 application to the IPv4 server example



The PGW PLAT function acts as a stateful NAT64. It translates N:1 global IPv6 addresses to public IPv4 addresses and the other way around. The PLAT holds the Pref64/n prefix and all traffic toward this IPv6 prefix must be routed to the PLAT. The PLAT derives the destination IPv4 address from the destination IPv6 address and forwards this to the IPv4 network. The PLAT also implements a scalable logging mechanism, to log all CG-NAT64 bindings for legal purposes. Examples of logging mechanisms include Syslog, SNMP, RADIUS, or IPFIX.

DNS64 is the mechanism for synthesizing IPv6 address (AAAA) resource records from IPv4 address (A) records. If the AAAA query results in one or more AAAA records in the answer section, the result is returned to the requesting client, as per normal DNS semantics. If only an A record is available for the AAAA query, the DNS64 component embeds the IPv4 addresses from the A record in an IPv6 address to be used for responding to the AAAA query. The IPv4 address is embedded in the IPv6 address which always begins with an IPv6 prefix that belongs to the CG-NAT64. This prefix is referred to as Pref64/n. The IPv6 packets that begin with the Pref64/n prefix, are always routed toward the CG-NAT64.

Figure 206: PLAT and DNS64 functionality



The CLAT function may also use the DNS64 for generally discovering the NAT64 Pref64/n prefix using the draft-ietf-behave-nat64-discovery-heuristic (RFC 7050), to discover the Pref64/n prefix. The CLAT component sends an AAAA query to the DNS64 for the known IPv4-only name `ipv4only.arpa`. The Pref64/n prefix is then derived from the received result and the AAAA response is synthesized.

17.5 NAT64 fragmentation

In general, fragmentation functionality is invoked when the size of a fragmentation eligible packet exceeds the size of the MTU of the egress interface/tunnel. Packets eligible for fragmentation are:

- IPv4 packets/fragments with the DF bit in the IPv4 header cleared. Fragmentation can be performed on any routing node between the source and the destination of the packet.
- IPv6 packets on the source node. Fragmentation of IPv6 packet on the transient routing nodes is not allowed.

The best practice is to avoid fragmentation in the network by ensuring adequate MTU size on the transient/source nodes. Drawbacks of the fragmentation are:

- Increased processing and memory demands to the network nodes (especially during reassembly process)
- Increased byte overhead
- Increased latency.

Fragmentation can be particularly deceiving in a tunneled environment whereby the tunnel encapsulation adds extra overhead to the original packet. This extra overhead could tip the size of the resulting packet over the egress MTU limit.

Fragmentation could be one solution in cases where the restriction in the mtu size on the packet's path from source to the destination cannot be avoided. IPv6 fragmentation is supported in NAT64 with some enriched capabilities, such as optional packet IPv6 fragmentation even in cases where DF-bit in corresponding IPv4 packet is set.

In general, the lengths of the fragments must be chosen such that resulting fragment packets fit within the MTU of the path to the packets destinations.

In downstream direction fragmentation can be implemented in two ways:

- IPv4 packet can be fragmented in the carrier IOM before it reaches ISA for any NAT function.
- IPv6 packet can be fragmented in the ISA, once the IPv4 packet is IPv6 encapsulated in IPv6 translated in NAT64.

In upstream direction, IPv4 packets can be fragmented once they are translated in NAT64. The fragmentation will occur in the IOM.

NAT64

For downstream fragmentation in NAT64, the configured `ipv6-mtu` represents the mtu size of the ipv6 packet. In addition, the IPv4 packet in NAT64 is not tunneled but instead IPv4/v6 headers are translated. Consequently, the fragmented IPv6 packet size will be 28 bytes larger than the translated IPv4 packet — 20 bytes difference in basic IP header sizes (40 bytes IPv6 header vs 20 byte IPv4 header) plus 8 bytes for extended fragmentation IPv6 header. Note that the only extended IPv6 header that NAT64 generates is the fragmentation header.

If the IPv4 packet is dropped due to the fragmentation not being allowed, the returned ICMP message will contain MTU size of `ipv6-mtu` minus 28 bytes.

```
configure
[router] | [service vprn]
  nat
    inside
      nat64
      ipv6-mtu bytes
  ip-fragmentation {disabled|fragment-ipv6|fragment-ipv6-unless-ipv4-df-set}
```

17.6 TCP MSS configuration for NAT

TCP MSS represents the maximum amount of data in a TCP message, not including the IP and TCP header lengths. The ability to change the TCP MSS for packets that transit the PGW/GGSN is supported when NAT is configured.

This can be used to avoid reassembly of GTP-U PDUs. By setting the TCP MSS to a low enough value, you control the size of the inner IP payload and avoid fragmentation that may occur when the GTP encapsulation overhead is added.

The TCP MSS field for NAT can be adjusted with a preconfigured value. This triggers the interception of transit TCP packets with the SYN bit set and consequently rewrites the MSS field of the intercepted packet. The MSS field will only be rewritten in cases where the configured TCP MSS adjust value is smaller than the one inside the intercepted packet. If the given TCP option field is not present in the TCP SYN packet, it will be inserted with the preconfigured value.

To configure TCP MSS, use the following command:

```
configure service nat nat-policy tcp-mss-adjust
```

For additional information about fragmentation and reassembly, refer to the *7750 SR Mobile Gateway Configuration Guide*.

17.7 NAT logging

LSN logging is extremely important to the Service Providers (SP) who are required by the government agencies to track source of suspicious Internet activities back to the users that are hidden behind the LSN device.

CMG supports several modes of logging for LSN applications. Choosing the right logging model will depend on the required scale, simplicity of deployment and granularity of the logged data.

For most purposes logging of allocation/de-allocation of outside port-blocks and outside IP address along with the corresponding LSN subscriber and inside service-id will suffice.

In certain cases port-block based logging is not satisfactory and per flow logging is required.

17.7.1 Syslog flow logging

The format of syslog messages for NAT flow logging in SR OS adheres to RFC 3164, *The BSD Syslog Protocol*:

`<PRI> <HEADER><MSG>`

where:

- `<PRI>` (the "<" and ">" are included in the syslog message) is the configured facility*8+severity (as described in the *7450 ESS*, *7750 SR*, *7950 XRS*, and *VSR System Management Guide* and RFC 3164).
- `<HEADER>` defines the MMM DD HH:MM:SS <hostname>. Two characters always appear for the day (DD) field. Single-digit days are preceded with a space character. Time is recorded as local time (and not UTC). The time zone designator is not shown in this example, but each event has its own timestamp where the time-zone designator is shown.
- `<MSG>` defines the `<log-prefix>: <seq> <application [<subject>]: <message>\n`

where:

- `<log-prefix>` is an optional 32-character string of text (default = 'TMNX') as configured in the `log-prefix` command.
- `<seq>` is the log event sequence number (always preceded by a colon and a space char).
- The `[<subject>]` field may be empty resulting in []:
- `<message>` displays a custom message relevant to the log event.
- `\n` is the standard ASCII new line character (hex 0A).

Table 69: Syslog message fields for NAT flow logging

Field name	Value	Comments
PRI	• Default: 6	• Configurable

Table 69: Syslog message fields for NAT flow logging (continued)

Field name	Value	Comments
- severity - facility	Default: 16	Configurable
Timestamps	MMM DD HH:MM:SS	—
<hostname>	—	The IP address of the SR OS system that is generating the message.
<log-prefix>	—	Configurable. This can be used as a field to differentiate between the vendors. For example, NOK(ia) in log-prefix indicates that this is a log format from a Nokia node so the operator can apply parsing logic accordingly.
<seq>	—	Sequence numbers can be used for tracking if loss in transit occurs.
<application>	NAT	The application that generated the log.
[<subject>]:	MDA ID	The BB-ISA on which the event occurred.
<message>	—	This is a custom part with specific information related to the event itself.

The message portion contains information relevant to the respective log event, even if this information is already repeated outside of the message (for example, timestamp). The fields in the message part are separated by a single whitespace for easier parsing and are placed in the order shown *Message fields*.

Table 70: Message fields

Field name	Value	Presence	Comments
NAT type	LSN44 NAT64	M(andatory)	—
Event name	SADD	M	SADD - session added event

Table 70: Message fields (continued)

Field name	Value	Presence	Comments
	SDEL		SDEL - session deleted event
Timestamp	<TimeStamp>: <Year> <Mon> <Day> <hh:mm:ss:cs> <TZ>, Year is 4-digit, Mon is 3-letter abbreviation, TZ is a 1-5 character time-zone designator.	M	Since events can be combined in the same syslog message, each event is uniquely timestamped with the local time (not UTC), including the time zone designator. During daylight saving's time (summer), the time zone designator is replaced by the DST designator, which is configurable.
Protocol ID	1, 6, 17	M	ICMP, UDP, TCMP
Inside router	0 to 2147483650	M	0 represents Base 1 to 2147483650 represents VPRNs
Source IP address	IPv4 address in LSN44 and IPv6 address in NAT64	M	—
Source port or ICMP identifier	0 to 65535	M	—
Outside router	0 to 2147483650	M	—
Outside (post NAT) IP address	IPv4 address	M	—
Outside (post NAT) port or ICMP identifier	0 to 65535	M	—
Foreign IP address	IPv4 address	O(ptional)	This is the original destination IPv4 address.

Table 70: Message fields (continued)

Field name	Value	Presence	Comments
Foreign port or ICMP identifier	0 to 65535	O	—
Destination IP address	IPv4 address	O	It represents the translated destination IP address.
Nat-policy	<name>	O	—
Sub-ID	<sub-name>	O	"-" if requested by the configuration (includes the sub-id statement) but the sub-aware NAT is not enabled. Otherwise, the sub-ID in the sub-aware NAT.

17.7.2 Syslog/SNMP/local-file logging

The simplest form of LSN NAT logging is via the logging facility in CMG, commonly called logger.

Each port-block allocation/de-allocation event is recorded and sent to the system logging facility (logger). Such an event can be:

- Recorded in the system memory as part of regular logs.
- Written to a local file.
- Sent to an external server via syslog facility.
- Sent to an SNMP trap destination.



NOTE:

NAT logging to a local compact flash must only be used in lab environments. For more information, refer to the section *Nat logging* in the *CMG Best Practices Guide*.

In this mode of logging, all applications in the system share the same logger.

Syslog/SNMP/Local-File logging on LSN is mutually exclusive with NAT RADIUS-based logging.

Syslog/SNMP/Local-File logging on LSN is mutually exclusive with the generation of NAT binding records in CSV format.

Syslog/SNMP/local-file logging must be separately enabled for LSN NAT in log even-control. The following displays relevant MIB events:

```
2012 tmnxNatPlBlockAllocationLsn
```

17.7.2.1 Filtering LSN events to system memory

In this example a single port-block [1884-1888] is allocated/de-allocated for the inside IP address 5.5.5.5 which is mapped to the outside IP address 80.0.0.1. Consequently the event is logged in the memory as.

```
2 2012/07/12 16:40:58.23 WEST MINOR: NAT #2012 Base NAT
"{2} Free 80.0.0.1 [1884-1888] -- vprn10 5.5.5.5 at 2012/07/12 16:40:58"

1 2012/07/12 16:39:55.15 WEST MINOR: NAT #2012 Base NAT
"{1} Map 80.0.0.1 [1884-1888] -- vprn10 5.5.5.5 at 2012/07/12 16:39:55"
```

Once the desired LSN events are enabled for logging via event-control configuration, they can be logged to memory via standard log-id 99 or be filtered via a custom log-id, such as in this example (log-id 5)

Example

```
*A:left-a20>config>log# info
-----
filter 1
  default-action drop
  entry 1
    action forward
    match
      application eq "nat"
      numbr eq 2012
    exit
  exit
exit
event-control "nat" 2001 suppress
event-control "nat" 2002 suppress
event-control "nat" 2003 suppress
event-control "nat" 2004 suppress
event-control "nat" 2005 suppress
event-control "nat" 2006 suppress
event-control "nat" 2007 suppress
event-control "nat" 2008 suppress
event-control "nat" 2009 suppress
event-control "nat" 2010 suppress
event-control "nat" 2011 suppress
```

```

event-control "nat" 2012 generate
event-control "nat" 2014 suppress
event-control "nat" 2015 suppress
event-control "nat" 2017 suppress
syslog 10
exit
log-id 5
    filter 1
    from main
    to memory
exit

```

```

-----
*A:left-a20# show log event-control "nat"

```

Log Events

Application

ID#	Event Name	P	g/s	Logged	Dropped
2002	tmnxNatIsaMemberSessionUsageHigh	WA	gen	0	0
2003	tmnxNatPlLsnMemberBlockUsageHigh	WA	gen	0	0
2004	tmnxNatLsnSubIcmpPortUsageHigh	WA	gen	0	0
2005	tmnxNatLsnSubUdpPortUsageHigh	WA	gen	0	0
2006	tmnxNatLsnSubTcpPortUsageHigh	WA	gen	0	0
2011	tmnxNatLsnSubSessionUsageHigh	WA	gen	0	0
2012	tmnxNatPlBlockAllocationLsn	MI	gen	2	0
2014	tmnxNatResourceProblemDetected	MI	gen	0	0
2015	tmnxNatResourceProblemCause	MI	gen	0	0
2016	tmnxNatPlAddrFree	MI	gen	0	0
2017	tmnxNatPlLsnRedActiveChanged	WA	gen	0	2
2019	tmnxNatFwdEntryAdded	MI	gen	0	0

Example: Event description

tmnxNatIsaMemberSessionUsageHigh

The tmnxNatIsaMemberSessionUsageHigh notification is sent when the session usage of a NAT ISA group member reaches its high watermark ('true') or when it reaches its low watermark again ('false').

tmnxNatPlLsnMemberBlockUsageHigh

The tmnxNatPlLsnMemberBlockUsageHigh notification is sent when the block usage of a Large Scale NAT address pool

reaches its high watermark ('true')
or when it reaches its low watermark again ('false')
on a particular member MDA of its ISA group.

tmnxNatLsnSubIcmpPortUsageHigh

The tmnxNatLsnSubIcmpPortUsageHigh notification is sent when the ICMP port usage of a Large Scale NAT subscriber reaches its high watermark ('true') or when it reaches its low watermark again ('false').

tmnxNatLsnSubUdpPortUsageHigh

The tmnxNatLsnSubUdpPortUsageHigh notification is sent when the UDP port usage of a Large Scale NAT subscriber reaches its high watermark ('true') or when it reaches its low watermark again ('false').

tmnxNatLsnSubTcpPortUsageHigh

The tmnxNatLsnSubTcpPortUsageHigh notification is sent when the TCP port usage of a Large Scale NAT subscriber reaches its high watermark ('true') or when it reaches its low watermark again ('false').

tmnxNatLsnSubSessionUsageHigh

The tmnxNatLsnSubSessionUsageHigh notification is sent when the session usage of a Large Scale NAT subscriber reaches its high watermark ('true') or when it reaches its low watermark again ('false').

tmnxNatPlBlockAllocationLsn

The tmnxNatPlBlockAllocationLsn notification is sent when an outside IP address and a range of ports is allocated to a NAT subscriber associated with a Large Scale NAT (LSN) pool, and when this allocation expires.

tmnxNatResourceProblemDetected

The tmnxNatResourceProblemDetected notification is sent when the value of the object tmnxNatResourceProblem changes.

tmnxNatResourceProblemCause

The tmnxNatResourceProblemCause notification is to describe the cause of a NAT resource problem.

tmnxNatPlAddrFree

The tmnxNatPlAddrFree notification is sent when a range of outside IP addresses becomes free at once.

tmnxNatPLSnRedActiveChanged

The tmnxNatPLSnRedActiveChanged notification is related to NAT Redundancy

sent when the value of the object tmnxNatPLSnRedActive changes. The cause is string.

17.7.2.2 NAT logging to a local file

In this case, the destination of log-id 5 in the following example is a local file instead of memory:

```
*A:left-a20>config>log# info
-----
    file-id 5
      description "nat logging"
      location cf3:
      rollover 15 retention 12
    exit

    log-id 5
      filter 1
      from main
      to file 5
    exit
```

The events are logged to a local file on the compact flash cf3 in a file under the /Log directory.



NOTE:

NAT logging to a local compact flash must only be used in lab environments. For more information, refer to the section *Nat logging* in the *CMG Best Practices Guide*.

17.7.3 SNMP trap logging

In case of SNMP logging to a remote node, the log destination should be set to SNMP destination. Allocation de-allocation of each port block will trigger sending a SNMP trap message to the trap destination.

```
*A:left-a20>config>log# info
-----
    filter 1
      default-action drop
    entry 1
```

```

        action forward
        match
            application eq "nat"
            number eq 2012
        exit
    exit
exit

snmp-trap-group 6
    trap-target "nat" address 114.0.1.10 port 9001 snmpv2c notify-
community "private"
    exit
    log-id 6
        filter 1
        from main
        to snmp
    exit
-----

```

```

+ Internet Protocol version 4, Src: 1.1.1.1 (1.1.1.1), Dst: 114.0.1.10
(114.0.0.10)
- User Datagram Protocol, Src Port: snmptrap (162), Dst Port: etlservicemgr
(9001)
    Source port: snmptrap (162)
    Destination port: etlservicemgr (9001)
    Length: 358
+ Checksum: 0x0e2c [correct]
Simple Network Management Protocol
    version: v2c (1)
    community: private
- data: snmpv2-trap (7)
- snmpV2-trap
    request-id: 1
    error-status: noError (0)
    error-index: 0
- variable-bindings: 14 items
+ 1.3.6.1.2.1.1.3.0: 19054240
+ 1.3.6.1.6.3.1.1.4.1.0: 1.3.6.1.4.1.6527.3.1.3.65.0.12
(iso.3.6.1.4.1.6527.3.1.3.65.0.12)
+ 1.3.6.1.4.1.6527.3.1.2.65.2.2.0:
+ 1.3.6.1.4.1.6527.3.1.2.65.2.4.0:
+ 1.3.6.1.4.1.6527.3.1.2.65.2.5.0: 50000001
+ 1.3.6.1.4.1.6527.3.1.2.65.2.8.0: 1894
+ 1.3.6.1.4.1.6527.3.1.2.65.2.9.0: 1898
+ 1.3.6.1.4.1.6527.3.1.2.65.2.10.0: 07dc070d00321b002b0000

```

```

+ 1.3.6.1.4.1.6527.3.1.2.65.2.13.0: 1
+ 1.3.6.1.4.1.6527.3.1.2.65.2.3.0:
+ 1.3.6.1.4.1.6527.3.1.2.65.2.6.0:
+ 1.3.6.1.4.1.6527.3.1.2.65.2.7.0: 1a000038
+ 1.3.6.1.4.1.6527.3.1.2.65.2.11.0:
+ 1.3.6.1.4.1.6527.3.1.2.65.2.17.0: 5

```

17.7.3.1 NAT syslog

NAT logs can be sent to a syslog remote facility. A separate syslog message is generated for every port-block allocation/deallocation.

```
*A:left-a20>config>log#info
```

```
-----
```

```
...
```

```

filter 1
  default-action drop
  entry 1
    action forward
    match
      application eq "nat"
      number eq 2012
    exit
  exit
exit
syslog 7
  address 114.0.1.10
exit

```

```

log-id 7
  filter 1
  from main
  to syslog 7
exit

```

```
-----
```

```

+ Internet Protocol version 4. Src: 1.1.1.1 (1.1.1.1), Dst: 114.0.1.10
  (114.0.1.10)
- User Datagram Protocol, Src Port: syslog (514), Dst Port: syslog (514)
  Source port: syslog (514)
  Destination port: syslog (514)
  Length: 184

```



```

- Checksum: 0x3539 [correct]
  [Good Checksum: True]
  [Bad Checksum: False]
- Syslog message: LOCAL7.INFO: Jul 13 15:04:53 1.1.1.1 TMNX: 35 Base NAT-
INDETERMINATE-tmNxNatplBlockAllocationLsn-2012 [NAT]: {45} Map 80.0.0.1
[1994-1998] -- vprn10 26.0.0.56 at 2012/
  1011 1... = Facility: LOCAL7 - reserved for local use (23)
  .... .110 = Level: INFO - informational (6)
  Message: Jul 13 15:04:53 1.1.1.1 TMNX: 35 Base NAT-INDETERMINATE-
tmNxNatplBlockAllocationLsn-2012 [NAT]: {45} Map 80.0.0.1 [1994-1998] --
vprn10 26.0.0.56 at 2012/07/13 08:04:53\n

```

Severity level for this event can be changed via CLI:

```

*A:left-a20# configure log event-control "nat" 2012 generate <severity-level>
cleared indeterminate critical major minor warning

```

17.7.4 LSN NAT flow logging

LSN NAT Flow logging is a facility that allows each BB-ISA card to export the creation and deletion of NAT flows to an external server.

A NAT flow or a Fully Qualified Flow consists of the following parameters: Inside IP, inside port, outside IP, outside port, foreign IP, foreign port, protocol (UDP, TCP, ICMP).

```

-----
Owner           : LSN-Host@10.0.0.15
Router          : 10
FlowType        : UDP                Timeout (sec)      : 11

Inside IP Addr   : 10.0.0.15         Inside Port       : 100
Outside IP Addr  : 80.0.0.1          Outside Port      : 1164
Foreign IP Addr  : 10.0.3.2          Foreign Port      : 5000
Dest IP Addr     : 10.0.3.2          Dest Port         : 5000
-----

```

In addition, the inside/outside service-id and subscriber string will be added to a flow record.

Flow logging can be deployed as an alternative to the port-range logging or can be complementary (providing a more granular log for offline reporting or compliance). Certain operators have legal and compliance requirements that require extremely detailed logs, created per flow, to be exportable from the NAT node.

Because the setup rate of new flows is excessive, logging to an internal facility (like compact flash) is not possible except in a debugging mode (which must specify match criteria down to the inside-IP and service level).

Flow logging can be enabled per nat-policy and consequently it is initiated from each BB-ISA card independently as a UDP stream, unlike a centralized Netflow/Cflowd application.

Flows are formatted according to IETF IPFIX RFC 5101, *Specification of the IP Flow Information Export (IPFIX) Protocol*, for the Exchange of IP Traffic Flow Information. Data structures are contained in RFC 5102, *Information Model for IP Flow Information Export*. NAT flow logging is sent to up to two different IP addresses both of which must be unicast IPv4 destinations. These UDP streams are stateless due to the significant volume of transactions. However they do contain sequence numbers such that packet loss can be identified. They egress the chassis at FC NC.

IPFIX defines two different type of messages that will be sent from the IPFIX exporter (CMG NAT node). The first contains Template Set - an IPFIX message that defines fields for subsequent IPFIX messages but contains no actual data of its own. The second IPFIX message type is that containing Data Sets - here the data is passed using the previous Template Set message to define the fields. This means an IPFIX message is NOT passed as sets of TLV, but instead data is encoded with a scheme defined through the Template Set message.

While an IPFIX message can contain both Template Set and Data Set, 7750 sends Template Set messages periodically without any data, whereas the Data Set messages are sent on demand and as required. When IPFIX is used over UDP, the default retransmission frequency of the Template Set messages defaults to 10 minutes. The interval for retransmission is configurable in CLI with a minimum interval of 1 minute and a maximum interval of 10 minutes. When the exporter first initializes, or when a configuration change occurs the Template Set is sent out three times, one second apart. Templates are sent before any data sets, assuming that the collector is enabled, so that an IPFIX collector can establish the data template set.

Although the UDP transport is unreliable, the IPFIX Sequence Number is a 32bit number that contains the total number of IPFIX Data Records sent for the UDP transport session prior to the receipt of the new IPFIX message. The sequence number starts with 0 and it will roll over once it reaches 4,294,967,268.

The default packet size is 1500B unless another value has been defined in config (range is 512B through 9212B inclusive). Traffic is originated from a random high-port to the collector on port 4739. Multiple create/delete flow records will be stuffed into a single IPFIX packet (although the mapping creates are not delayed) until stuffing an additional data record would exceed MTU or a timer expires. The timer is not configurable and is set to 250ms (that is, should any mapping occur a packet will be sent within 250ms of that mapping being created)

Each collector has a 50 packet buffering space. In case that due to excessive logging the buffering space becomes unavailable, new flows will be denied and the deletion of flows will be delayed until buffering space becomes available.

Two collector nodes can be defined in the same IPFIX export policy for redundancy purposes.

17.7.4.1 Large scale NAT44 flow logging configuration example

This section provides an example of how to configure large scale NAT44 flow logging.

1. Define a collector node along with other local transport parameters through an IPFIX export-policy.

```
*A:left-a20>config>service>ipfix# info detail
-----
      ipfix-export-policy "ipxif-policy" create
      description "external IPFIX collector"
      collector router "Base" ip 114.0.1.10 create
      mtu 1500
      source-address 114.0.1.20
      template-refresh-timeout min 10
      no shutdown
    exit
  exit
```

To export flow records via UDP stream, the BB-ISA card must be configured with an appropriate IPv4 address within a designated VPRN. This address (/32) will act as the source for sending all IPFIX records and is shared by all ISA.

2. After the IPFIX export policy is defined, apply it within the NAT policy.

```
*A:left-a20>config>service>nat>nat-policy# info
-----
      pool "base" router Base
      ipfix-export-policy "ipxif-policy"
```

The capture of IPFIX packet for an ICMP flow creation and deletion is shown in the following examples.

Flow creation and deletion template field descriptions describes the fields in the example flow creation and deletion templates.

Table 71: Flow creation and deletion template field descriptions

Field	Value	Description
Description	Size (B)	Description
Export Timestamp	n/a	Timestamp derived from chassis NTP, per RFC 5101
Sequence Id	n/a	Total number of IPFIX data records sent for the UDP transport session prior to the receipt of the new IPFIX message (modulo 232), per RFC 5101
Observation Domain Id	n/a	Unique ID set per ISA

Table 71: Flow creation and deletion template field descriptions (continued)

Field	Value	Description
FlowID	8	Unique ID (per observation domain ID) for this flow used for tracking purposes only (opaque value); flow ID in a create and a delete mapping record must be the same for a specific NAT mapping
IP_SRC_ADDR	4	Outside IP address used in the NAT mapping
IP_DST_ADDR	4	Destination or remote IP address used in the NAT mapping
L4_SRC_PORT	2	Outside source port used in the NAT mapping
L4_DST_PORT	2	Destination source port used in the NAT mapping
flowStartMilliseconds ⁵	8	Timestamp when the flow was created (chassis NTP derived) in milliseconds from epoch, per RFC 5102
flowEndMilliseconds ⁶	8	Timestamp when the flow was destroyed (chassis NTP derived) in milliseconds from epoch, per RFC 510
PROTOCOL	1	Protocol ID, TCP, UDP or ICMP. Per RFC 5102
PADDING	1	n/a
flowEndReason	1	Supported flow end reasons: 0x01: Idle Timeout A mapping expired (because of UDP or TCP timeout) 0x03: End of Flow Detected A mapping closed (only used for TCP after a FIN or RST) 0x04: Forced End Collects all other reasons included administrative or failure case
nokInsideServiceID	2	16-bit service ID representing the inside service ID

⁵ Flow Creation Template Set only⁶ Flow Deletion Template Set only

Table 71: Flow creation and deletion template field descriptions (continued)

Field	Value	Description
nokOutsideServiceId	2	16-bit service ID representing the outside service ID
nokNatSubString	var	A variable 8B aligned string that represents the NAT subscriber construct (as currently used in the tools dump service nat session commands)

Flow creation

```

+ Internet Protocol version 4, Src: 114.0.1.20 (114.0.1.20), Dst: 114.0.1.10 (114.0.0.10)
- User Datagram Protocol, Src Port 500000 (50000), Dst Port: ipfix (4739)
  Source port: 500000 (50000)
  Destination port: ipfix (4739)
  Length: 80
- Checksum: 0x0e6c [correct]
  [Good Checksum: True]
  [Bad Checksum: False]
- Cisco NetFlow/IPFIX
  Version: 10
  Length: 72
+ Timestamp: Jul 13, 2012 14:37:03.000000000 Pacific Daylight Time
  FlowSequence: 0
  Observation Domain Id: 1179650
- Set 1
  Flowset Id: (Data) (256)
  Flowset Length: 56
- Flow 1
  Flow Id: 285191984
  SrcAddr: 80.0.0.1 (80.0.0.1)
  DstAddr: 10.0.3.2 (10.0.3.2)
  SrcPort: 1031
  DstPort: 0
  Protocol: 1
  Padding (1 byte)
  Enterprise Private entry: (Nokia) Type 91: Value (hex bytes): 00 0a
  Enterprise Private entry: (Nokia) Type 92: Value (hex bytes): 00 00
  Padding (1 byte)
+ [Enterprise Private entry: (Nokia) Type 93: Value (hex bytes): 4c 53
4e 34 34 40 35 2e 35 2e 35 00 00 00 [Variable Length]

```

StartTime: Jul 13, 2012 14:37:03.277000000 Pacific Daylight Time

Flow deletion

```
+ Internet Protocol version 4, Src: 114.0.1.20 (114.0.1.20), Dst: 114.0.1.10
(114.0.0.10)
- User Datagram Protocol, Src Port 500000 (50000), Dst Port: ipfix (4739)
  Source port: 500000 (50000)
  Destination port: ipfix (4739)
  Length: 80
+ Checksum: 0x1357 [correct]
  [Good Checksum: True]
  [Bad Checksum: False]
- Cisco NetFlow/IPFIX
  Version: 10
  Length: 72
+ Timestamp: Jul 13, 2012 14:38:07.000000000 Pacific Daylight Time
  FlowSequence: 1
  Observation Domain Id: 1179650
- Set 1
  Flowset Id: (Data) (257)
  Flowset Length: 56
- Flow 1
  Flow Id: 285191984
  SrcAddr: 80.0.0.1 (80.0.0.1)
  DstAddr: 10.0.3.2 (10.0.3.2)
  SrcPort: 1031
  DstPort: 0
  Protocol: 1
  Flow End Reason: Idle Timeout (1)
  Enterprise Private entry: (Nokia) Type 91: Value (hex bytes): 00 0a
  Enterprise Private entry: (Nokia) Type 92: Value (hex bytes): 00 00
  Padding (1 byte)
+ [Enterprise Private entry: (Nokia) Type 93: Value (hex bytes): 4c 53 4e
34 34 40 35 2e 35 00 00 00 [Variable Length]
  EndTime: Not representable
```

17.7.5 NAT reporting EDRs

Mobile gateway supports for inclusion of mobility-specific AVPs to the RADIUS-based NAT binding.

Parameters such as IMSI, APN name, and PGW/GGSN IP address have been added to the RADIUS NAT binding record to provide the ability to determine subscriber identities, without offline processing of data.

The RADIUS NAT binding records can be converted into other formats, for example, CSV files, using mediation tools such as the Nokia CMD product.

In the accounting start/stop messages, the UE identifiers are populated in the following AVPs:

- 3GPP-IMSI
- Alc-Msisdn
- 3GPP-IMEISV

and the private and public IP addresses, and port ranges are populated in the following AVPs:

- Framed-IP-Address
- Alc-Nat-Outside-IP-Address
- Alc-Nat-Port-Range

17.8 Logging of NAT binding records in CSV format configuration

CMG supports logging of NAT binding records in the local file system. NAT binding records are saved in CSV format. To identify a subscriber, NAT binding records include the mobile subscriber identity parameters like IMSI, IMEI, MSISDN, and APN.

Logging is performed without offline processing of data to map each NAT record back to the permanent identity of the subscriber.



NOTE:

NAT logging to a local compact flash must only be used in lab environments. For more information, refer to the section *Nat logging* in the *CMG Best Practices Guide*.

This example shows the configuration for the logging of NAT binding records, with port-block level NAT binding specified for the accounting policy record type.

```
config log
  file-id 17
    description "NAT binding records"
    location cf1:
  exit
  accounting-policy 44
    description "NAT binding"
    record nat-binding-port-block
    to file 17
    no shutdown
  exit
```

To enable file-based logging, add the accounting policy to each NAT group configuration.

```
config isa
  nat-group 1
    active-mda-limit 1
    acct-policy-id 44
    mda 3/2
    mda 4/2
    no shutdown
```

Files with NAT binding records are generated based on the time configured in the accounting policy. Each event (alloc/dealloc of the NAT port block) creates a record with the following format:

```
#timestamp,alloc,imsi,apn,msisdn,imei,private-ip,public-ip,port-start,port-end
02/05/2019 12:33:00,1,123456789012301,ion.nokia.1.1.com,12345098761,987654321012399,
1.0.255.0,13.0.255.253,1024,3071
02/05/2019 12:33:05,0,123456789012301,ion.nokia.1.1.com,12345098761,987654321012399,
1.0.255.0,13.0.255.253,1024,3071
```

The following convention is applied for file naming:

<NB><Startdate>.<Starttime>- [<Enddate>.]<Endtime>[_<jobId>][_<UniqueId>].csv.gz

When new NAT binding record files are generated in the local file system, a trap is generated to alert the external collector system (for example, NFM-M) about the availability of new log files. The following example shows this availability.

```
2019/02/05 12:35:00.459 UTC MAJOR: LOGGER #2008 Base acct-log-id 44 file-id
17 "Log file cf3:\act\NB20190205.1230+0000-1235+0000_-441764_PGW-SF0.csv.gz on
compact flash cf3 has been rolled over"
```

Files that are saved and available for further analysis, are located in the \act directory on the configured storage device.

```
A:PGW-SF0# file dir act

Volume in drive cf3 on slot A is TIMOS_VM_CF.
Volume in drive cf3 on slot A is formatted as FAT32
Directory of cf3:\act
02/05/2019  12:32p          <DIR>          ./
02/05/2019  12:32p          <DIR>          ../
02/05/2019  12:35p             171 NB20190205.1230+0000-1235+0000_-441764_PGW-
SF0.csv.gz
02/05/2019  12:40p             83 NB20190205.1235+0000-1240+0000_-441764_PGW-
SF0.csv.gz
02/05/2019  12:45p             83 NB20190205.1240+0000-1245+0000_-441764_PGW-
SF0.csv.gz
                                3 File(s)                337 bytes.
```


2 Dir(s)

14482432 bytes free.

Files in progress are placed in the \act-collect directory on the configured storage device.

**NOTE:**

In the case of a no-IMSI attached UE (emergency call), the system reports 0000000000000000 in the IMSI CSV field.

```
A:PGW-SF0# file dir act-collect
```

```
Volume in drive cf3 on slot A is TIMOS_VM_CF.
```

```
Volume in drive cf3 on slot A is formatted as FAT32
```

```
Directory of cf3:\act-collect
```

```
02/05/2019  12:32p      <DIR>          . /
```

```
02/05/2019  12:32p      <DIR>          .. /
```

```
02/05/2019  12:45p               10 NB20190205.1245+0000-1250+0000_-441764_PGW-  
SF0.csv.gz
```

17.9 Deterministic NAT

In deterministic NAT the subscriber is deterministically mapped into an outside IP address and a port block.

The algorithm that performs this deterministic mapping is revertive, which means that a NAT subscriber can be uniformly derived from the outside IP address and the outside port (and the routing instance). Thus, logging in deterministic NAT is not needed.

The deterministic [subscriber <-> outside-ip, deterministic-port-block] mapping can be automatically extended by a dynamic port-block in case that deterministic port block becomes exhausted of ports. By extending the original deterministic port block of the NAT subscriber by a dynamic port block yields a satisfactory compromise between a deterministic NAT and a non-deterministic NAT. There will be no logging as long as the translations are in the domain of the deterministic NAT. Once the dynamic port block is allocated for port extension, logging will be automatically activated.

NAT subscribers in deterministic NAT are not assigned outside IP address and deterministic port-block on a first come first serve basis. Instead, deterministic mappings will be pre-created at the time of configuration regardless of whether the NAT subscriber is active or not. In other words we can say that overbooking of the outside address pool is not supported in deterministic NAT. Consequently, all configured deterministic subscribers (for example, inside IP addresses in LSN44) will be guaranteed access to NAT resources.

17.9.1 Supported deterministic NAT flavors

CMG supports Deterministic LSN44, in which a subscriber is an IPv4 address.

17.9.2 Number of subscribers per outside IP and per pool

The outside pools in deterministic NAT can contain an arbitrary number of address ranges, where each address range can contain an arbitrary number of IP addresses (up to the ISA maximum).

The maximum number of NAT subscribers that can be mapped to a single outside IP address is configurable using the `subscriber-limit` command under the pool hierarchy. For Deterministic NAT, this number is restricted to the power of 2 (2^n). The consequence of this is that the number of NAT subscribers must be configuration-wise organized in ranges with the boundary that must be power of 2.

For example, in LSN44 where the NAT subscriber is an IP address, the deterministic subscribers would be configured with prefixes (for example, 10.10.10.0/24 - 256 subscribers) rather than an IP address range that would contain an arbitrary number of addresses (for example, 10.10.10.10 - 10.10.10.50).

The number of subscribers per outside IP (the `subscriber-limit` command [2^n]) multiplied by the number of IP addresses over all address-range in an outside pool will determine the maximum number of subscribers that a deterministic pool can support.

17.9.3 Referencing a pool

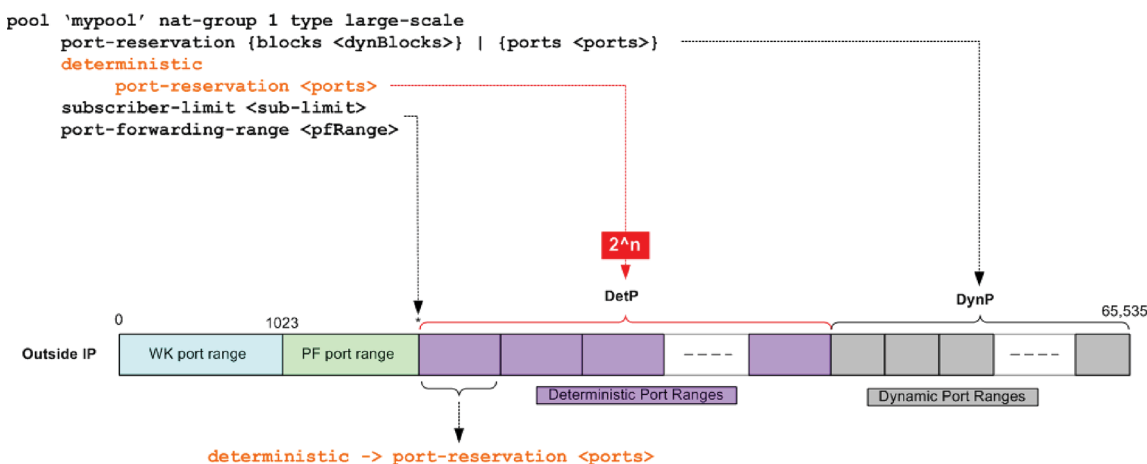
In deterministic NAT, the outside pool can be shared amongst subscribers from multiple routing instances. Also, NAT subscribers from a single routing instance can be selectively mapped to different outside pools.

17.9.4 Outside pool configuration

The number of deterministic mappings that a single outside IP address can sustain is determined through the configuration of the outside pool.

The port allocation per an outside IP is shown in *Outside pool configuration*.

Figure 207: Outside pool configuration



The well-known ports are predetermined and are in the range 0 - 1023.

The upper limit of the port range for static port forwards (wildcard range) is determined by the existing port-forwarding-range command.

The range of ports allocated for deterministic mappings (DetP) is determined by multiplying the number of subscribers per outside IP (subscriber-limit command) with the number of ports per deterministic block (deterministic>port-reservation command). Note that the number of subscribers per outside IP in deterministic NAT must be power of 2 (2^n).

The remaining ports, extending from the end of the deterministic port range to the end of the total port range (65,535) are used for dynamic port allocation. The size of each dynamic port block is determined with the existing port-reservation command.

The deterministic>port-reservation command enables deterministic mode of operation for the pool.

Examples:

The follow show three examples with deterministic Large Scale NAT44 where the requirements are:

- 300, 500 or 700 (three separate examples) ports in each deterministic port block.
- A subscriber (an inside IPv4 address in LSN44) can extend its deterministic ports by a minimum of one dynamic port-block and by a maximum of four dynamic port blocks.
- Each dynamic port-block contains 100 ports.
- Oversubscription of dynamic port blocks is 4:1. This means that 1/4th of inside IP addresses may be starved out of dynamic port blocks in worst case scenario.
- The wildcard (static) port range is 3000 ports.

In the first case, the ideal case will be examined where an arbitrary number of subscribers per outside IP address is allocated according to our requirements outlined above. Then the limitation of the number of subscribers being power of 2 will be factored in.

Contiguous number of subscribers shows how port ranges would be carved out in ideal scenario.

Table 72: Contiguous number of subscribers

Well-known ports*	Static port range*	Number of ports in deterministic block*	Number of deterministic blocks	Number of ports in dynamic block*	Number of dynamic blocks	Number of inside IP addresses per outside IP address*	Block limit per inside IP address*	Wasted ports
0-1023	1024-4023	300	153	100	153	153	5	312

Table 72: Contiguous number of subscribers (continued)

Well-known ports*	Static port range*	Number of ports in deterministic block*	Number of deterministic blocks	Number of ports in dynamic block*	Number of dynamic blocks	Number of inside IP addresses per outside IP address*	Block limit per inside IP address*	Wasted ports
0-1023	1024-4023	500	102	100	102	102	5	312
0-1023	1024-4023	700	76	100	76	76	5	712

**NOTE:**

The "*" signifies the fixed parameters (requirements). The other values are calculated according to the fixed requirements. The **port-block-limit** includes the deterministic port block plus all dynamic port-blocks.

Next, a more realistic example with the number of subscribers being equal to 2^n are considered. The ratio between the deterministic ports and the dynamic ports per port-block just like in the example above: 3/1, 5/1 and 7/1 are preserved. In this case, the number of ports per port-block is dictated by the number of subscribers per outside IP address.

Preserving Det/Dyn port ratio with 2^n subscribers lists the preserving Det/Dyn port ratio with 2^n subscribers.

Table 73: Preserving Det/Dyn port ratio with 2^n subscribers

Well-known ports*	Static port range*	Number of ports in deterministic block*	Number of deterministic blocks	Number of ports in dynamic block*	Number of dynamic blocks	Number of inside IP addresses per outside IP address*	Block limit per inside IP address*	Wasted ports
0-1023	1024-4023	180	256	60	256	256	5	72

Table 73: Preserving Det/Dyn port ratio with 2^n subscribers (continued)

Well-known ports*	Static port range*	Number of ports in deterministic block*	Number of deterministic blocks	Number of ports in dynamic block*	Number of dynamic blocks	Number of inside IP addresses per outside IP address*	Block limit per inside IP address*	Wasted ports
0-1023	1024-4023	400	128	80	128	128	5	72
0-1023	1024-4023	840	64	120	64	64	5	72

**NOTE:**

The "*" signifies the fixed parameters (requirements).

The example shown in *Fixed number of deterministic ports with $2n$ subscribers* is similar to *Preserving Det/Dyn port ratio with $2n$ subscribers*, with the difference that the number of deterministic port blocks fixed are kept, as in the original example (300, 500 and 700).

Table 74: Fixed number of deterministic ports with 2^n subscribers

Well-known ports	Static port range	Number of ports in deterministic block	Number of deterministic blocks	Number of ports in dynamic block	Number of dynamic blocks	Number of inside IP addresses per outside IP address	Block limit per inside IP address	Wasted ports
0-1023	1024-4023	300	128	180	128	128	5	72
0-1023	1024-4023	500	64	461	64	64	5	8
0-1023	1024-4023	700	64	261	64	64	5	8

The three examples from above should give us a perspective on the size of deterministic and dynamic port blocks in relation to the number of subscribers (2^n) per outside IP address. Operators should run a similar dimensioning exercise before they start configuring their deterministic NAT.

The CLI for the highlighted case in the *Fixed number of deterministic ports with $2n$ subscribers* is displayed:

```
configure
service
  vprn
    nat
      outside
        pool mypool
          port-reservation ports 180
          deterministic
        port-reservation 300
          subscriber-limit 128
          port-forwarding-range 4023
```

where:

$128 \text{ subs} * 300 \text{ ports} = 38,400$ deterministic port range

$128 \text{ subs} * 180 \text{ ports} = 23,040$ dynamic port range

Det+dyn available ports = $65,536 - 4024 = 61,512$

Det+dyn usable ports = $128 * 300 + 128 * 180 = 61,440$ ports

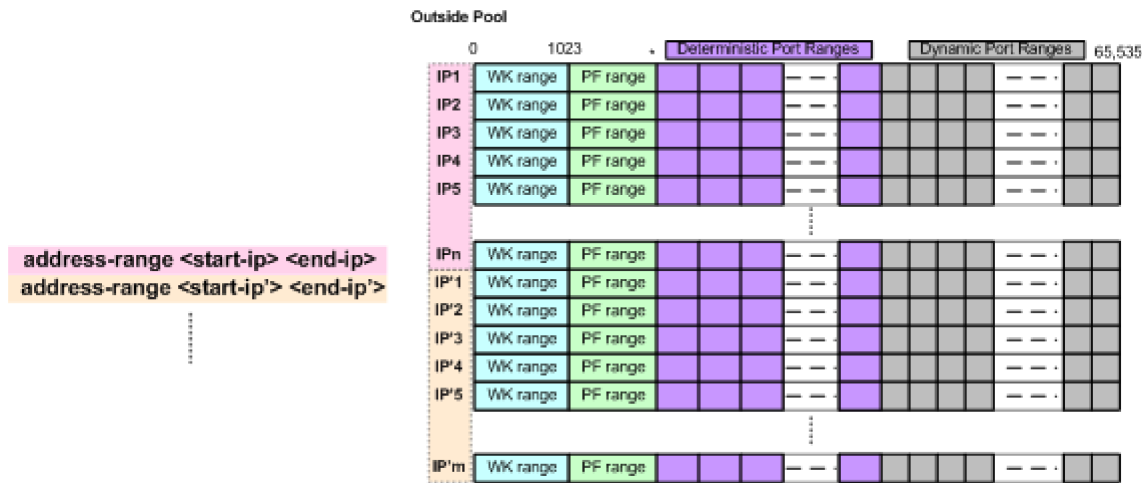
72 ports per outside-ip are wasted.

```
configure
service
  nat
    nat-policy mypolicy
      block-limit 5    ? 1 deterministic port block + 4 dynamic port blocks
```

This configuration will allow 128 subscribers (inside IP addresses in LSN44) for each outside address (compression ratio is 128:1) with each subscriber being assigned up to 1020 ports (300 deterministic and 720 dynamic ports over 4 dynamic port blocks).

The outside IP addresses in the pool and their corresponding port ranges are organized as shown in *Outside address ranges*.

Figure 208: Outside address ranges



Assuming that the above graph depicts an outside deterministic pool, the number of subscribers that can be accommodated by this deterministic pool is represented by purple squares (number of IP addresses in an outside pool * subscriber-limit). The number of subscribers across all configured prefixes on the inside that are mapped to the same deterministic pool must be less than the outside pool can accommodate. In other words, an outside address pool in deterministic NAT cannot be oversubscribed.

The following is a CLI representation of a deterministic pool definition including the outside IP ranges:

```
pool 'mypool' nat-group 1 type large-scale
port-reservation {blocks <dynBlocks>} | {ports <ports>}}
deterministic
port-reservation <ports>
subscriber-limit <sub-limit>
port-forwarding-range <pfRange>
address-range <start-ip-address> <end-ip-address>
address-range <start-ip-address> <end-ip-address>
```

17.9.5 Rule mapping and the map command in deterministic LSN44

The common building block on the inside in the deterministic LSN44 configuration is a IPv4 prefix. The NAT subscribers (inside IPv4 addresses) from the configured prefix will be deterministically mapped to the outside IP addresses and corresponding deterministic port-blocks. Any inside prefix in any routing instance can be mapped to any pool in any routing instance (including the one in which the inside prefix is defined).

The mapping between the inside prefix and the deterministic pool is achieved through a NAT policy that can be referenced per each individual inside IPv4 prefix. IPv4 addresses from the prefixes on the inside will be distributed over the IP addresses defined in the outside pool referenced by the NAT policy.

The mapping itself is represented by the map command under the prefix hierarchy.

```
router/service vprn
```

```

nat
  inside
    deterministic
      prefix ip-prefix/length subscriber-type nat-sub-type nat-policy nat-policy-name
      map start inside-ip-address end inside-ip-address to outside-ip-address

```

The purpose of the map statement is to split the number of subscribers within the configured prefix over available sequences of outside IP addresses. The key parameter that governs mappings between the inside IPv4 addresses and outside IPv4 addresses in deterministic LSN44 is defined by the `outside>pool>subscriber-limit` command. This parameter must be power of 2 and it limits the maximum number of NAT subscribers that can be mapped to the same outside IP address.

The follow are rules governing the configuration of the map statement:

If the number of subscribers per configured prefix is greater than the subscriber-limit per outside IP parameter (2^n), then the lowest n bits of the map `start inside-ip-address` must be set to 0.

If the number of subscribers per configured prefix is equal or less than the subscriber-limit per outside IP parameter (2^n), then only one map command for this prefix is allowed. In this case there is no restriction on the lower n bits of the map `start inside-ip-address`. The range of the inside IP addresses in such map statement represents the prefix itself.

The *outside-ip-address* in the map statements must be unique amongst all map statements referencing the same pool. In other words, two map statements cannot reference the same *outside-ip-address* in the pool.

In case that the number of subscribers (IP addresses in LSN44) in the map statement is larger than the subscriber-limit per outside IP, then the subscribers must be split over a block of consecutive outside IP addresses where the *outside-ip-address* in the map statement represent only the first outside IP address in that block.

The number of subscribers (range of inside IP addresses in LSN44) in the map statement does not have to be a power of 2. Rather it has to be a multiple of a power of two $m * 2^n$, where m is the number of consecutive outside IP addresses to which the subscribers are mapped and the 2^n is the subscriber-limit per outside IP.

Example: Map statement

```

router
  nat
    outside
      pool 'my-det-pool' nat-group 1 type large-scale
      subscriber-limit 128
      deterministic
        port-reservation 400
        address-range 128.251.0.0 128.251.0.10

  service vprn 10

```



```

nat
  inside
    deterministic
      prefix 10.0.0.0/24 subscriber-type classic-lsn-sub nat-policy det
      map start 10.0.0.0 end 10.0.0.255 to 128.251.0.1

```

In this case, the configured 10.0.0.0/24 prefix is represented by the range of IP addresses in the map statement (10.0.0.0-10.0.0.255). Since the range of 256 IP addresses in the map statement cannot be mapped into a single outside IP address (subscriber-limit=128), this range must be further implicitly split within the system and mapped into multiple outside IP addresses. The implicit split will create two IP address ranges, each with 128 IP addresses (10.0.0.0/25 and 10.0.0.128/25) so that addresses from each IP range are mapped to one outside IP address. The hosts from the range 10.0.0.0-10.0.0.127 will be mapped to the first IP address in the pool (128.251.0.1) as explicitly stated in the map statement (to statement). The hosts from the second range, 10.0.0.128-10.0.0.255 will be implicitly mapped to the next consecutive IP address (128.251.0.2).

Alternatively, the map statement can be configured as follows.

```

service vprn 10
nat
  inside
    deterministic
      prefix 10.0.0.0/24 subscriber-type classic-lsn-sub nat-policy det
      map start 10.0.0.0 end 10.0.0.127 to 128.251.0.1
      map start 10.0.0.128 end 10.0.0.255 to 128.251.0.5

```

In this case the IP address range in the map statement is split into two non-consecutive outside IP addresses. This gives the operator more freedom in configuring the mappings.

However, the following configuration is not supported.

```

service vprn 10
nat
  inside
    deterministic
      prefix 10.0.0.0/24 subscriber-type classic-lsn-sub nat-policy det
      map start 10.0.0.0 end 10.0.0.63 to 128.251.0.1
      map start 10.0.0.64 end 10.0.0.127 to 128.251.0.3
      map start 10.0.0.128 end 10.0.0.255 to 128.251.0.5

```

Considering that the subscriber-limit = 128 (2^n ; where $n=7$), the lower n bits of the start address in the second map statement (map start 10.0.0.64 end 10.0.0.127 to 128.251.0.3) are not 0. This is in violation of the rule #1 that governs the provisioning of the map statement.

Assuming that we use the same pool with 128 subscribers per outside IP address, the following scenario is also not supported (note that configured prefix in this example is different than in previous example).

```
service vprn 10
nat
  inside
    deterministic

    prefix 10.0.0.0/26 subscriber-type classic-lsn-sub nat-policy det
      map start 10.0.0.0 end 10.0.0.63 to 128.251.0.1

    prefix 10.0.0.0/26 subscriber-type classic-lsn-sub nat-policy det
      map start 10.0.1.0 end 10.0.1.63 to 128.251.0.1
```

Although the lower n bits in both map statements are 0, both statements are referencing the same outside IP (128.251.0.1). This is violating rule #2 that governs the provisioning of the map statement. Each of the prefixes in this case will have to be mapped to a different outside IP address, which will lead to underutilization of outside IP addresses (half of the deterministic port-blocks in each of the two outside IP addresses will be not be utilized).

In conclusion, considering that the number of subscribers per outside IP (subscriber-limit) must be 2^n , the inside IP addresses from the configured prefix will be split on the 2^n boundary so that every deterministic port-block of an outside IP is utilized. In case that the originally configured prefix contains less subscribers (IP addresses in LSN44) than an outside IP address can accommodate (2^n), all subscribers from such configured prefix will be mapped to a single outside IP. Since the outside IP cannot be shared with NAT subscribers from other prefixes, some of the deterministic port-blocks for this particular outside IP address will not be utilized.



NOTE:

Each configured prefix can evaluate into multiple map commands. The number of map commands will depend on the length of the configured prefix, the subscriber-limit command and fragmentation of outside address-range within the pool with which the prefix is associated.

17.9.6 Hashing considerations in deterministic LSN44

This feature supports multiple MS-ISAs in the NAT group calls for traffic hashing on the inside in the ingress direction.

This feature ensures fair load balancing of the traffic amongst multiple MS-ISAs. While hashing in non-deterministic LSN44 can be performed per source IP address, hashing in deterministic LSN44 is based on subnets instead of individual IP addresses.

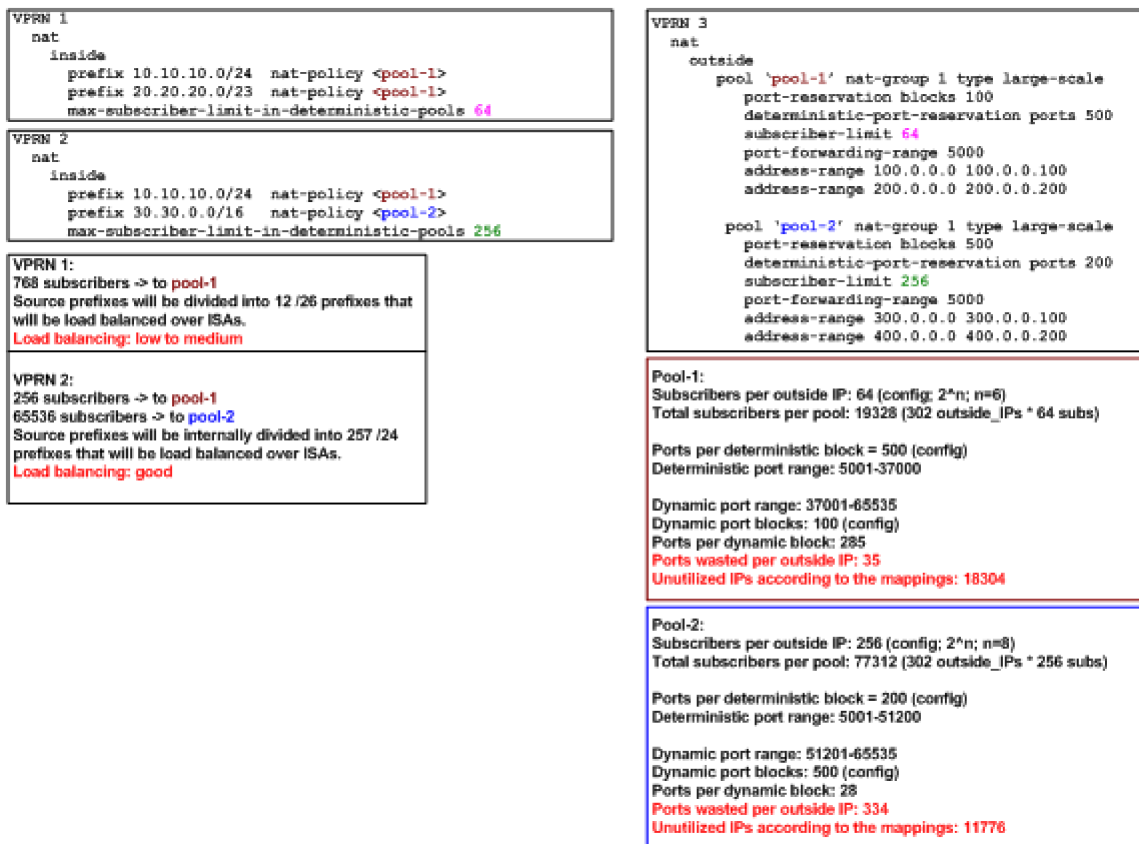
The length of the hashing subnet is common for all configured prefixes within an inside routing instance. In case that a prefixes from an inside routing instances is referencing multiple pools, the common hashing prefix length will be chosen according to the pool with the highest number of subscribers per outside IP

address. This will ensure that subscribers mapped to the same outside IP address will be always hashed to the same MS-ISA.

In general, load distribution based on hashing is dependent on the sample. Large and more diverse sample will ensure better load balancing. Therefore the efficiency of load distribution between the MS-ISAs is dependent on the number and diversity of subnets that hashing algorithm is taking into consideration within the inside routing context.

A simple rule for good load balancing is to configure a large number of subscribers relative to the largest `t` subscriber-limit parameter in any given pool that is referenced from this inside routing instance.

Figure 209: Deterministic LSN44 configuration example



The configuration example shown *Deterministic LSN44 configuration example* depicts a case in which prefixes from multiple routing instances are mapped to the same outside pool and at the same time the prefixes from a single inside routing instance are mapped to different pools (we do not support the latter with non-deterministic NAT).

Important to note in this example is the inside prefix 10.10.10.0/24 that is present in VPRN 1 and VPRN 2. In both VPRNs, this prefix is mapped to the same pool-pool-1 with the subscriber-limit of 64. Four outside IP addresses per prefix per VPRN (eight in total) are allocated to accommodate the mappings for all hosts in prefix 10.10.10.0/24. But the hashing prefix length in VPRN1 is based on the subscriber-limit 64 (VPRN1 references only pool-1) while the hashing prefix length in VPRN2 is based on the subscriber-limit 256 in pool-2 (VPRN2 references both pools, pool-1 and pool-2 and we must select the larger subscriber-limit).

The consequence of this is that the traffic from subnet 10.10.10.0/24 in VPRN 1 can be load balanced over 4 MS-ISA (hashing prefix length is 26) while traffic from the subnet 10.10.10.0/24 in VPRN 2 is always sent to the same MS-ISA (hashing prefix length is 24).

Distribution of outside IP addresses across MS-ISAs in an MS-ISA NA group

Distribution of outside IP addresses across the MS-ISAs is dependent on the ingress hashing algorithm. Since traffic from the same subscriber is always pre-hashed to the same MS-ISA, the corresponding outside IP address also must reside on the same ISA. CPM runs the hashing algorithm in advance to determine on which MS-ISA the traffic from particular inside subnet will land and then the corresponding outside IP address (according to deterministic NAT mapping algorithm) will be configured in that particular MS-ISA.

17.9.7 Simultaneous support of dynamic and deterministic NAT

Simultaneous support for deterministic and non-deterministic NAT inside of the same routing instance is supported. However, an outside pool can be only deterministic (although expandable by dynamic ports blocks) or non-deterministic at any given time.

Ingress hashing for all NATed traffic within the VRF will in this case be performed based on the subnets driven by the classic-lsn-max-subscriber-limit parameter.

17.9.8 Selecting traffic for NAT

Deterministic NAT does not change the way how traffic is selected for the NAT function but instead only defines a predictable way for translating subscribers into outside IP addresses and port-blocks.

Traffic is still diverted to NAT using the existing methods:

routing based

Traffic is forwarded to the NAT function if it matches a configured destination prefix that is part of the routing table. In this case inside and outside routing context must be separated.

filter based

Traffic is forwarded to the NAT function based on any criteria that can be defined inside an IP filter. In this case the inside and outside routing context can be the same.

17.9.9 Inverse mappings

The inverse mapping can be performed with a MIB locally on the 7x50 node or externally via a script sourced in 7x50.

In both cases, the input parameters are <outside routing instance, outside IP, outside port. The output from the mapping is the subscriber and the inside routing context in which the subscriber resides.

17.9.9.1 MIB approach

Reverse mapping information can be obtained using the following command:

```
tools dump nat deterministic-mapping outside-ip <ipv4-address> router <router-
instance> outside-port <[1..65535]>
<ipv4-address>      : a.b.c.d
<router-instance>   : <router-name>|<service-id>
                    router-name    - "Base"
                    service-id     - [1..2147483647]
```

Output example

```
tools dump nat deterministic-mapping outside-ip 85.0.0.2 router Base outside-
port 2333
```

```
Inside router 10 ip 20.0.5.171 -- outside router Base ip 85.0.0.2 port 2333 at
Mon Jan 7 10:02:02 PST 2013
```

17.9.9.2 Offline approach to obtain deterministic mappings

Instead of querying the system directly, you can generate a Python script on 7x50 and export it to an external node.

This Python script contains mapping logic for the configured deterministic NAT in 7x50. The script can be then queried offline to obtain mappings in either direction. The external node must have installed Python scripting language with the following modules: getopt, math, os, socket, and sys.

The purpose of such offline approach is to provide fast queries without accessing 7x50. Exporting the Python script for reverse querying is a manual operation that needs to be repeated every time there is configuration change in deterministic NAT.

The script is exported outside of the box to a remote location (assuming that writing permissions on the external node are correctly set).

The remote location is specified with the following command:

```
configure service nat deterministic-script location <remote-url>
<remote-url>      - [{ftp://|tftp://}<login>:<pswd>@<remote-locn>/][<file-
path>]
180 chars max
```

The status of the script is shown using the following command:

```
show service nat deterministic-script
=====
```

```
Deterministic NAT script data
=====
Location          : ftp://10.10.10.10/pub/det-nat-script/det-nat.py
Save needed       : yes
Last save result  : none
Last save time    : N/A
=====
```

Once the script location is specified, the script can be exported to that location with the following command:

```
admin nat save-deterministic-script
```

This needs to be repeated manually every time the configuration affecting deterministic NAT changes.

Once the script is exported (saved), the status of the script is changed as well:

```
show service nat deterministic-script
=====
Deterministic NAT script data
=====
Location          : ftp://10.10.10.10/pub/det-nat-script/det-nat.py
Save needed       : no
Last save result  : success
Last save time    : 2013/01/07 10:33:43
=====
```

The script itself can be run to obtain mapping in forward or backward direction:

```
user@external-server:/home/ftp/pub/det-nat-script$ ./det-nat.py
Usage: det-nat-.py {{DIRECTION PARAMS}} | -h[elp] }
where DIRECTION := { -f[success] | -b[ackward] }
where PARAMS := { -s[ervice] -a[ddress] -p[ort] }
```

Example: Mapping of source addresses

```
Router 10, Source-ip: 20.0.5.0-20.0.5.127 to router base, outside-ip
85.0.0.1
Router 10 Source-ip: 20.0.5.128-20.0.5.255 to router base outside-ip 85.0.0.2
```

The forward query for this example will be performed as follows:

```
user@external-server:/home/ftp/pub/det-nat-script$ ./det-nat.py -f -s 10 -a
20.0.5.10
```

Output:

```
subscriber has public ip address 85.0.0.1 from service 0 and is using ports  
[1324 - 1353]
```

The reverse query for this example will be performed as:

```
user@external-server:/home/ftp/pub/det-nat-script$ ./det-nat.py -b -s 0 -a  
85.0.0.1 -p 3020
```

Output:

```
subscriber has private ip address 20.0.5.66 from service 10
```

17.9.10 Logging

Every configuration change concerning the deterministic pool will be logged and the script (if configured for export) will be automatically updated (although not exported). This is needed to keep current track of deterministic mappings. In addition, every time a deterministic port-block is extended by a dynamic block, the dynamic block will be logged just as it is today in non-deterministic NAT. The same logic is followed when the dynamic block is de-allocated.

All static port forwards are also logged.

17.10 Destination-based NAT (DNAT)

Destination NAT (DNAT) is supported for LSN44 and L2-Aware NAT. DNAT can be used for traffic steering where the destination IP address of the packet is rewritten. In this fashion traffic can be redirected to an appliance or set of servers that are in control of the operator, without the need for a separate transport service (for example, PBR plus LSP).

Applications utilizing traffic steering via DNAT normally require some form of inline traffic processing, such as inline content filtering (parental control, antivirus/spam, firewalling), video caching, and so on.

Once the destination IP address of the packet is translated, traffic is naturally routed based on the destination IP address lookup. DNAT will translate the destination IP address in the packet while leaving the original destination port untranslated.

Similar to source based NAT (Source Network Address and Port Translation (SNAPT)), the 7750 SR MG will maintain state of DNAT translations so that the source IP address in the return (downstream) packet is translated back to the original address.

Traffic selection for DNAT processing in MS-ISA is performed via a NAT classifier.

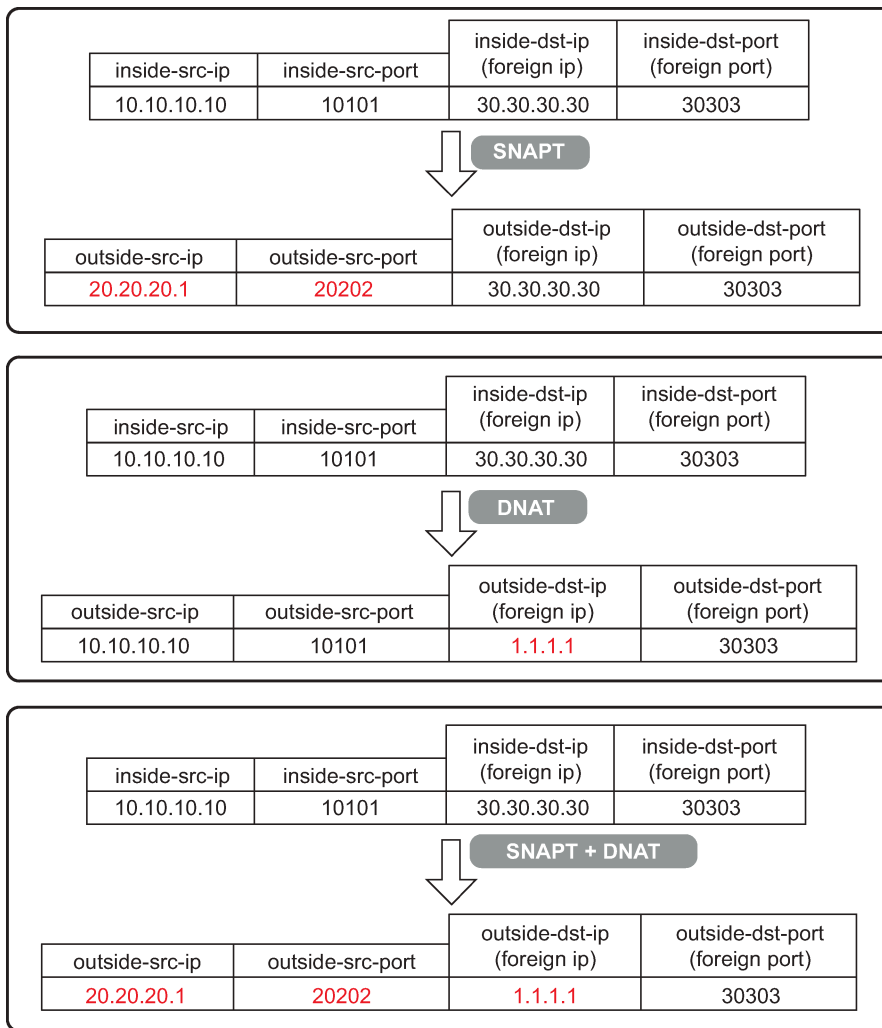
17.10.1 Combination of SNAPT and DNAT

In certain cases SNAPT is required along with DNAT. In other cases only DNAT is required without SNAPT.

Table 75: Supported combinations of SNAPT and DNAT

	SNAPT	DNAT-only	NAPT and DNAT
LSN44	✓	✓	✓
L2-Aware	✓		✓

Figure 210: IP address/port translation modes



0923

17.10.2 Forwarding model in DNAT

NAT forwarding in 7750 SR MG is implemented in two stages:

- Traffic is first directed toward the MS-ISA. This is performed via a routing lookup, via a filter or via a subscriber-management lookup (L2-Aware NAT). DNAT does not introduce any changes to the steering logic responsible for directing traffic from the I/O line card toward the MS-ISA.
- Once traffic reaches the MS-ISA, translation logic is performed. DNAT functionality will incur an additional lookup in the MS-ISA. This lookup is based on the protocol type and the destination port of the packets, as defined in the nat-classifier.

As part of the NAT state maintenance, the SR OS maintains the following fields for each DNATed flow: <inside host /port, outside IP/port, foreign IP address/port, destination IP address/ port, protocol (TCP,TCP,ICMP)> Note that the inside host in LSN is inside the IP address and in L2-Aware NAT it is the <inside IP address + subscriber-index>. The subscriber index is carried in session-id of the L2TP.

The foreign IP address represents the destination IP address in the original packet, while the destination IP address represents the DNAT address (translated destination IP address).

17.10.3 DNAT traffic selection via NAT classifier

Traffic intended for DNAT processing is selected via a nat classifier. The nat-classifier has configurable protocol and destination ports. The inclusion of the classifier in the NAT policy is the trigger for performing DNAT.

The configuration of the NAT classifier determines whether:

- a specific traffic defined in the match criteria is DNATed while the rest of the traffic is transparently passed through the nat classifier, OR
- a specific traffic defined in the match criteria is transparently passed through the nat classifier while the rest of the traffic is DNATed.

Classifier cannot drop traffic (no action drop). However, a non-reachable destination IP address in DNAT will cause traffic to be black-holed.

17.10.4 DNAT configuration

DNAT is enabled in the `config>service>nat>nat-policy` context.

```
configure service nat
nat-policy nat-policy-name create
dnat
dnat-only router router-instance nat-group nat-group-id
nat-classifier classifier-name
exit
```

The DNAT function is triggered by the presence of the NAT classifier (nat-classifier command), referenced in the NAT policy.

The DNAT-only option is configured when SNAPT is not required. This command is necessary to determine the outside routing context and the NAT group when SNAPT is not configured. The Pool (relevant to SNAPT) and DNAT-only configuration options within the NAT policy are mutually exclusive.

17.10.4.1 DNAT traffic selection and destination IP address configuration

DNAT traffic selection is performed via a nat-classifier. Nat-classifier is defined under config>service>nat hierarchy and is referenced within the nat-policy.

```
configure service nat
  nat-classifier name create
    default-DNAT-ip-address ip-addr
    default-action {DNAT|forward} [ip-addr ip-address]
    entry 1 create
      action {DNAT|forward}[ip-addr ipv4-address]
      match protocol {tcp | udp}
        dst-port range start port-number end port-number
      exit
    exit
  exit
exit
```

The default-dnat-ip-address is used in all match criteria that contain DNAT action without specific destination IP address. However, the default-dnat-ip-address is ignored in cases where IP address is explicitly configured as part of the action within the match criteria.

The default-action is applied to all packets that do not satisfy any match criteria. forward (forwarding action) has no effect on the packets and will transparently forward packets through the nat-classifier. By default, packets that do not match any matching criteria are transparently passed through the classifier.

17.10.4.2 Micro-netting original source (inside) IP space in DNAT-only case

To forward upstream and downstream traffic for the same NAT binding to the same MS-ISA, the original source IP address space must be known in advance and consequently hashed on the inside ingress toward the MS-ISAs and micro-netted on the outside. This will be performed with the following CLI:

```
router | service vprn id
nat
inside
classic-lsn-max-subscriber-limit max
dnat-only
source-prefix nat-prefix-list-name
```

```
service nat
nat-prefix-list name application dnat-only-subscribers create
prefix ip-prefix
```

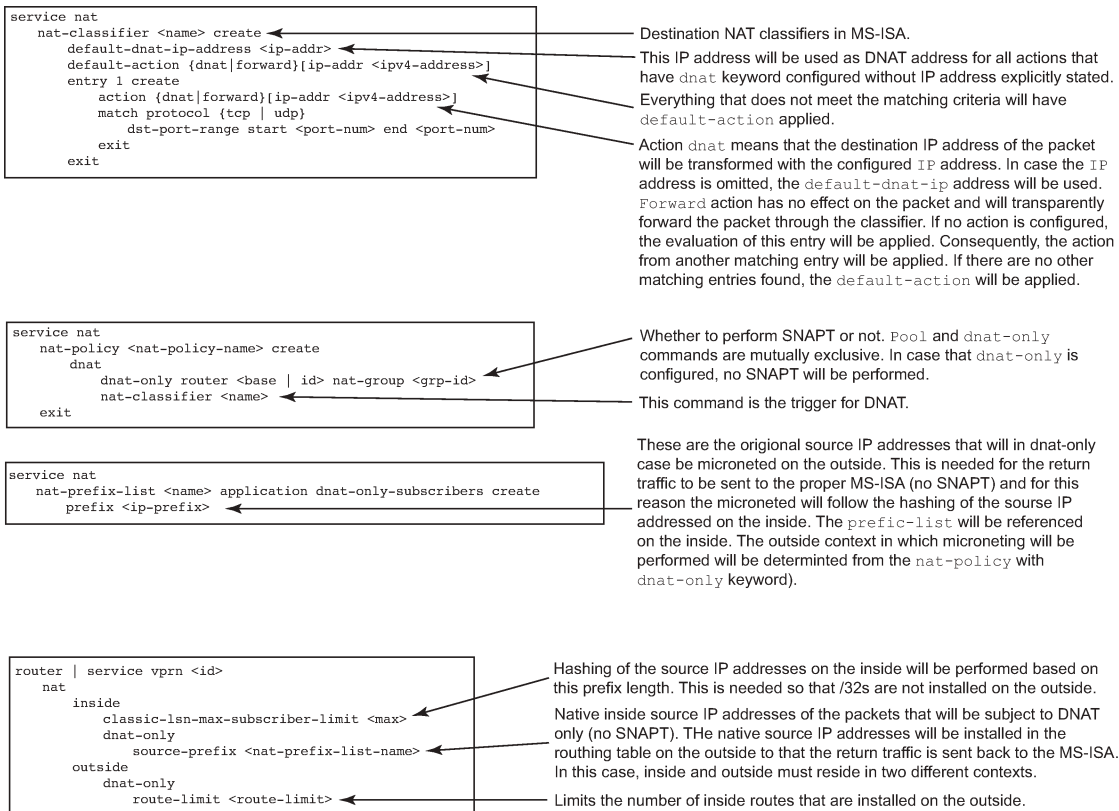
The `classic-lsn-max-subscriber-limit` parameter was introduced by deterministic NAT and it is reused here. This parameter affects the distribution of the traffic across multiple MS-ISA in the upstream direction traffic. Hashing mechanism based on source IPv4 addresses/prefixes is used to distribute incoming traffic on the inside (private side) across the MS-ISAs. Hashing based on the entire IPv4 address will produce the most granular traffic distribution, while hashing based on the IPv4 prefix (determined by prefix length) will produce less granular hashing. For more information about this command, see the CLI command description.

The source IP prefix is defined in the NAT prefix list and then applied under the DNAT only node in the inside routing context. This will instruct the SR OS to create micronets in the outside routing context. The number of routes installed in this fashion is limited by the following configuration:

```
router | service vprn id
nat
outside
dnat-only
route-limit route-limit
```

The configurable range is 1 to 128 Kb with the default value of 32 Kb.

Figure 211: DNAT provisioning model



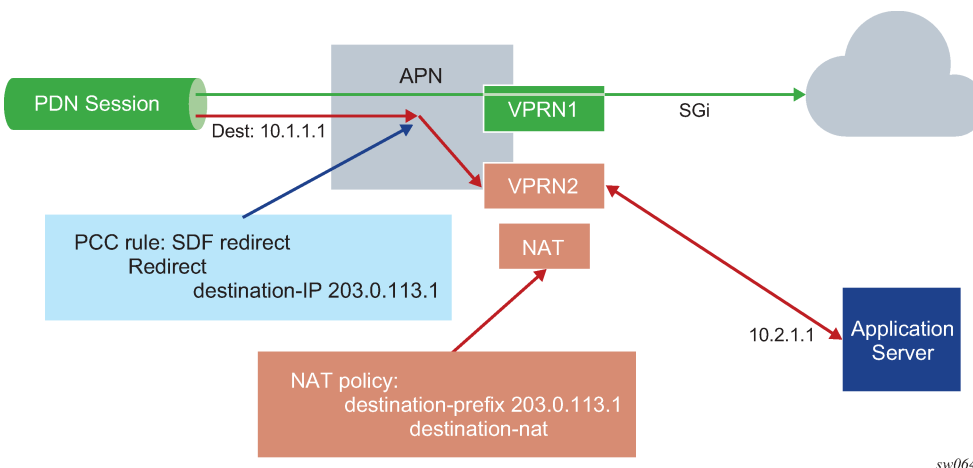
0924

17.10.5 Destination NAT with port override

This feature supports the DNAT function per flow basis.

The PCC rule is configured to redirect traffic to the NAT function. The NAT instance is configured to apply the destination NAT function. The flow with a matching PCC rule will be applied DNAT functionality while other flows will be routed based on the destination IP address.

Figure 212: Destination NAT based on PCC rule



sw0644

17.11 NAT optimizations for CMG

Improve UP performance with NAT-integrated MG-VMs as well as to avoid inter-VM traffic with LB-less architecture. In addition, UP performance can be improved on an LB-based architecture with the NAT function integrated with MG-VM, and inter-VM traffic between MG-VM, NAT, and LB-VM can be avoided.

17.11.1 Processing NAT inline with LB-less architecture

NAT on CMG with LB-less architecture is implemented using the NAT policy per MG group. Each MG group will have an active and standby MG-VM. Individual NAT groups will be configured to map each NAT group per MG group.

For example:

- MG-Group1 with VM1 (active) and VM3 (standby)
- MG-Group1 with VM2 (active) and VM4 (standby)
- NAT-Group1 with VM1 and VM3. Number of active cards = 1
- NAT-Group2 with VM2 and VM4. Number of active cards = 1

Figure 213: NAT and MG grouping in LB-less architecture

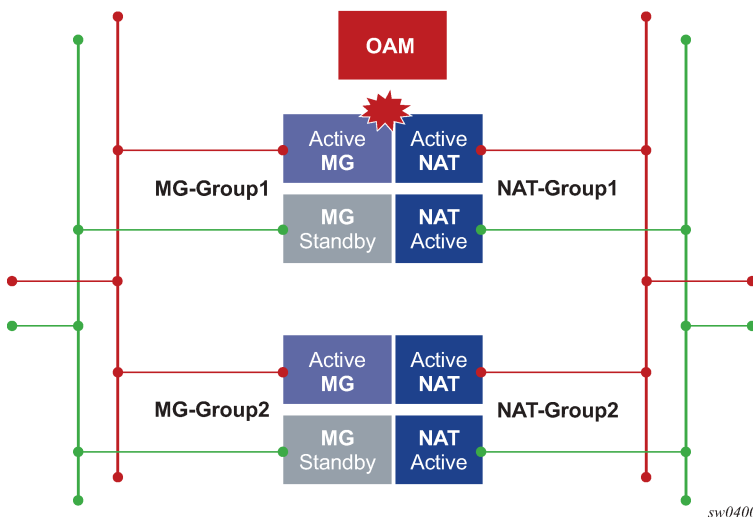
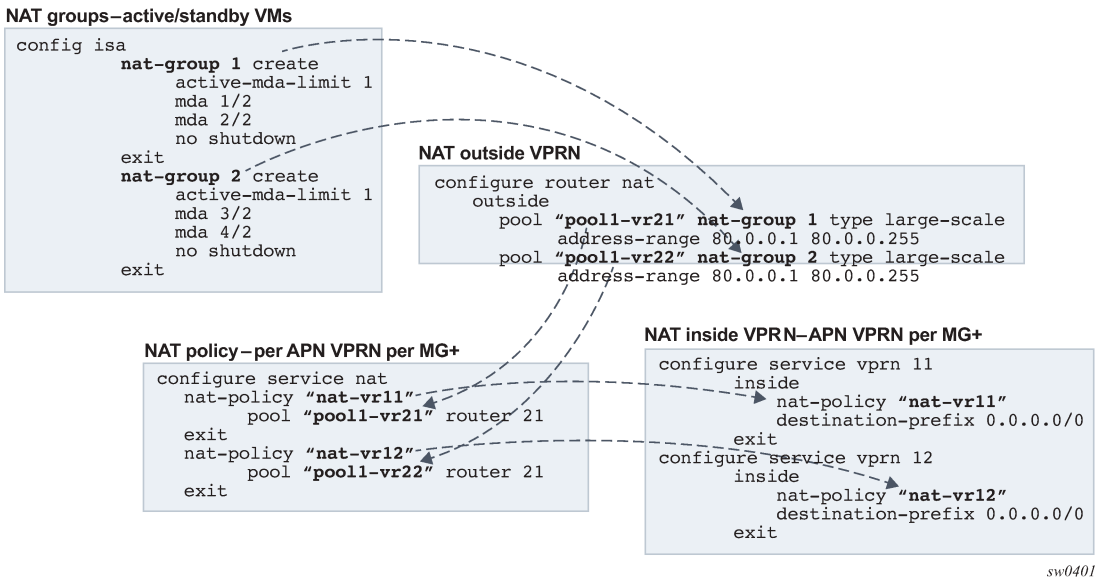
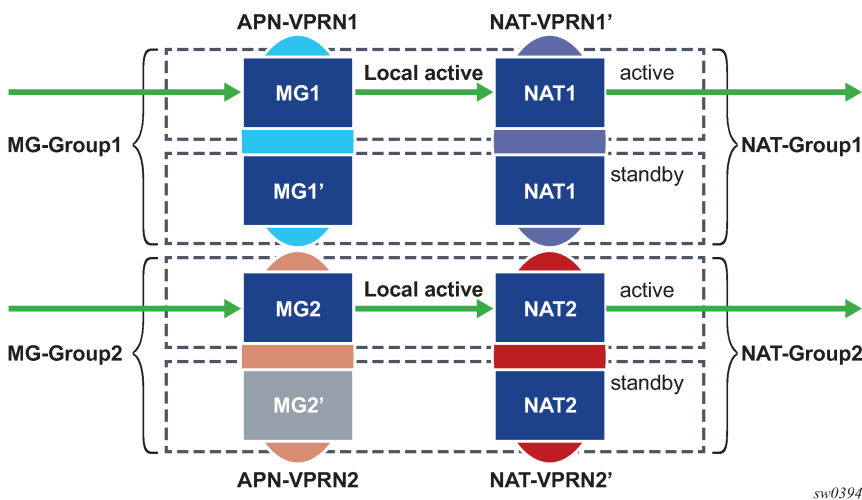


Figure 214: NAT configuration in LB-less architecture



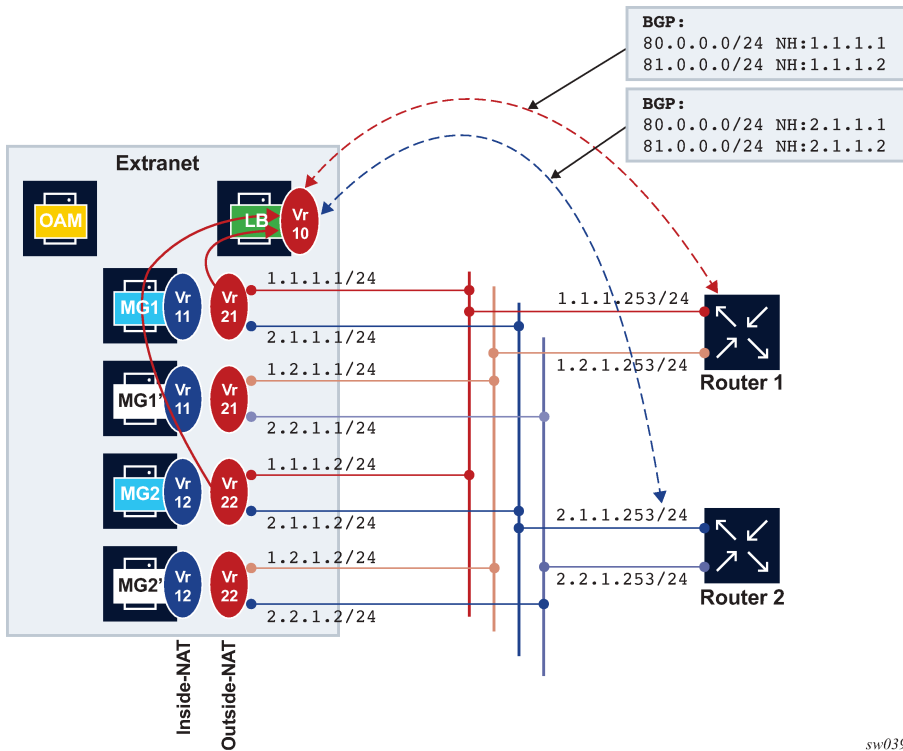
The active-mda-limit (configured with the `config>isa>nat-group>active-mda-limit` command) per NAT group must be set to 1 to allow only one VM to be used for NAT. When NAT is used locally, the state of the VM (whether it is active or standby) in a NAT group depends on the status of the MG group. For example, if MG1 is active in MG-Group1, then NAT on VM1 will be made active. On failover, MG1' will become active and NAT on VM3 will be made active.

Figure 215: Active VM in local NAT in LB-less architecture



The outside pool-prefix advertisement must be controlled so that the next-hop of the active MG-VM is announced to the peering router.

Figure 216: Local loopback to peering router in LB-less architecture



To enable NAT processing locally on the MG-VM, use the `config>isa>nat-group>local-nat` command.

17.11.2 Processing NAT inline on MG-VM with LB architecture

NAT on MG-VM with LB is implemented using a single NAT policy across all MG groups.

For example:

- MG-Group1 with VM1 (active) and VM3 (standby)
- MG-Group1 with VM2 (active) and VM4 (standby)
- NAT-Group1 with VM1, VM2, VM3, VM4. Number of active cards = 2 (based on number of MG groups)

Figure 217: NAT and MG grouping in LB architecture

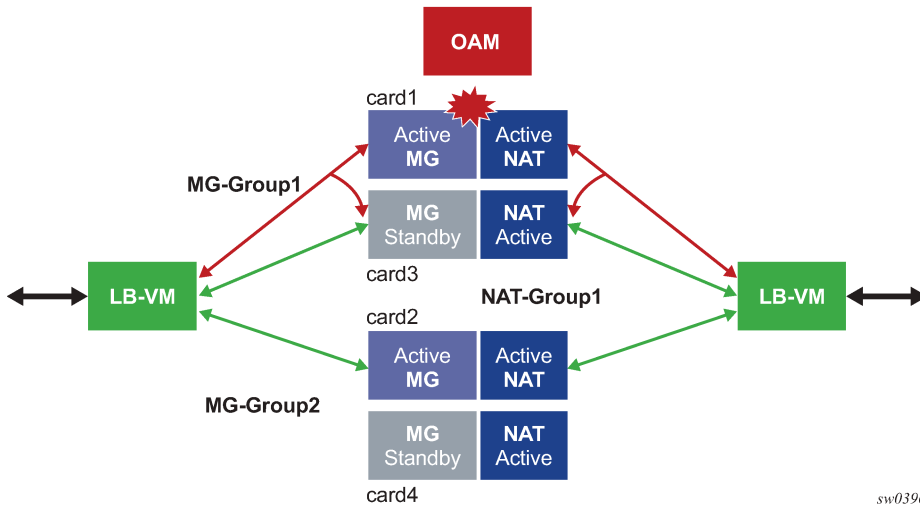
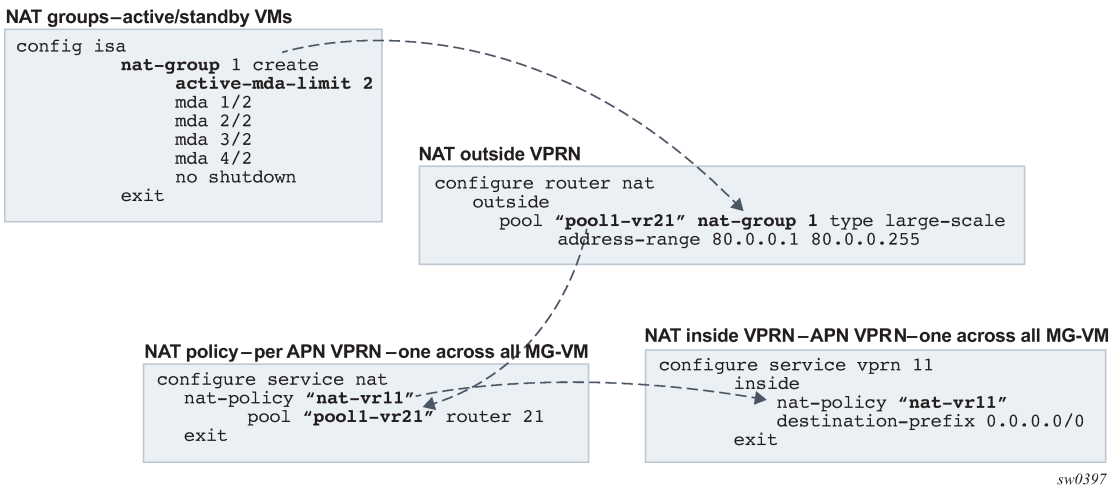
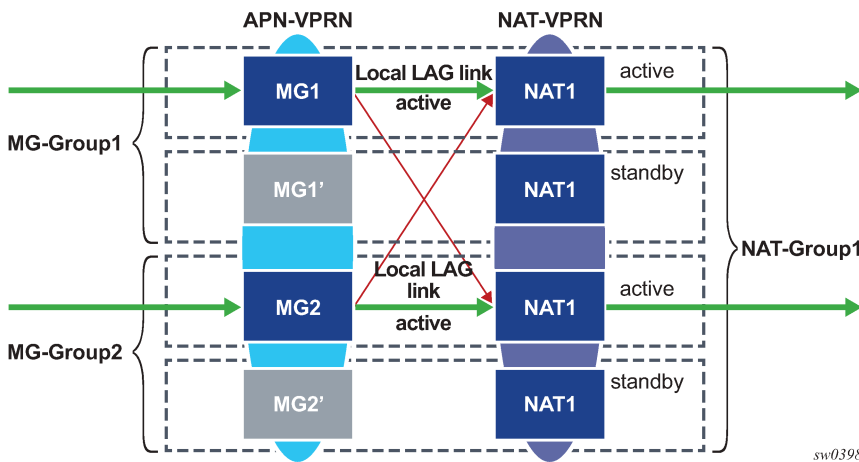


Figure 218: NAT configuration in LB architecture



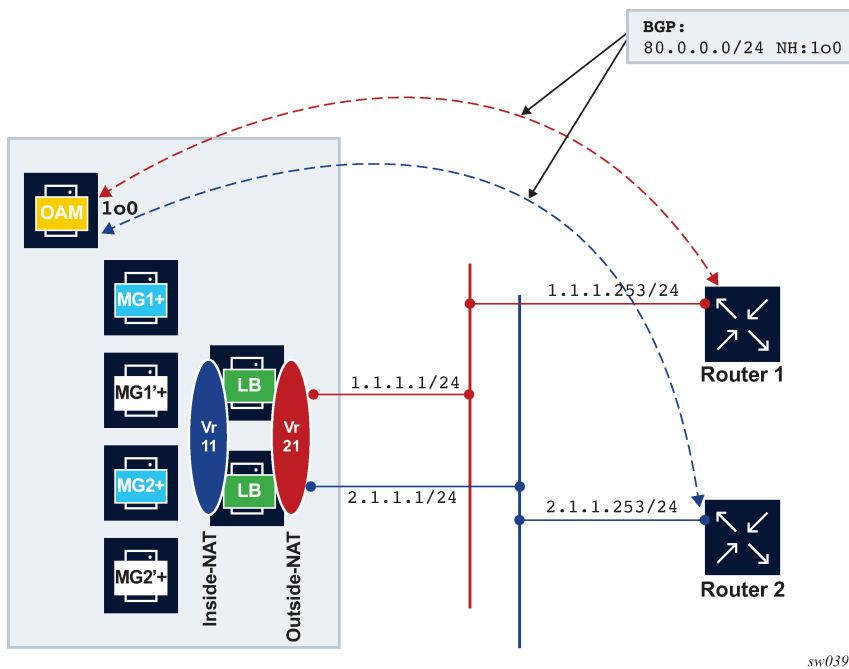
When the active-mds-limit (configured with the config>isa>nat-group>active-mds-limit command) per NAT group is set to the number of MG groups (in this example, 2) all VMs are allowed to be used for NAT. When NAT is used locally, hashing of traffic will pick the local instance of NAT instead of transmitting across multiple active VMs. For example, if two VMs are active, then the remote LAG (internal) link will be avoided. On failover, MG1' will become active and the local traffic hash will use the local NAT instance and avoid the remaining links.

Figure 219: Active VM in local NAT in LB architecture



The outside pool-prefix advertisement must be controlled so that the local loopback is advertised as a next-hop to the peering router. Traffic will be received on any LB-VM and based on internal micronets (NAT) will be forwarded to the respective VM that will process NAT traffic.

Figure 220: Local loopback to peering router in LB architecture



To enable NAT processing locally on the MG-VM use the `config>isa>nat-group>local-nat` command.

17.12 Enhanced statistics in NAT — Histogram

The NAT command histogram displays compartmentalized port distribution per protocol for an aggregated number of subscribers. This allows operators to trend port usage over time and consequently

adjust the configuration as the port demand per subscriber increase/decrease. For example, an operator may find that the port usage in a pools has increased over a period of time. Accordingly, the operator may plan to increase the number of ports per port block.

This feature is not applicable to pools that operate in one-to-one mode.

The output is organized in port buckets with the number of subscribers in each bucket.

```
tools dump nat histogram
  histogram router router-instance pool pool-name bucket-size <[1..65536]> num-buckets
  <[2..50]>
```

```
router-instance      : router-name|service-id
                        router-name      - "Base"
                        service-id      - [1..2147483647]
pool-name           : [32 chars max]
```

```
tools dump nat histogram router "Base" pool "det" bucket-size 20 num-buckets
20
```

```
=====
Usage histogram NAT pool "det" router "Base"
=====
```

```
Num-ports    Sub-TCP    Sub-UDP    Sub-ICMP
-----
```

0-19	0	0	0
20-39	0	0	0
40-59	0	0	0
60-79	0	0	0
80-99	0	0	0
100-119	0	0	0
120-139	0	0	0
140-159	0	0	0
160-179	0	0	0
180-199	0	0	0
200-219	0	0	0
220-239	0	0	0
240-259	0	0	0
260-279	0	0	0
280-299	0	0	0
300-319	0	0	0
320-339	0	0	0
340-359	0	0	0
360-379	0	0	0
380-	0	0	0

```
-----
```

The output of the `histogram` command can be periodically exported to an external destination via cron. The following is an example:

```
*A:CPM>config>cron# info
-----
script "nat_histogram"
    location "ftp://*: *@138.203.8.62/nat-histogram.txt"
    no shutdown
exit
action "dump_nat_histogram"
    results "ftp://*: *@138.203.8.62/nat_histogram_results.txt"
    script "nat_histogram"
    no shutdown
exit
schedule "nat_histogram_schedule"
    interval 600
    action "dump_nat_histogram"
    no shutdown
exit
-----
```

The `nat-histogram.txt` file contains the command execution line. For example:

```
tools dump nat histogram router 4 pool "deterministic" bucket-size 10 num-
buckets 10
```

This command will be executed every 10 minutes (600 seconds) and the output of the command will be written into a set of files on an external FTP server:

```
[root@ftp]# ls nat_histogram_results.txt*
nat_histogram_results.txt_20130117-153548.out
nat_histogram_results.txt_20130117-153648.out
nat_histogram_results.txt_20130117-153748.out
nat_histogram_results.txt_20130117-153848.out
nat_histogram_results.txt_20130117-153948.out
nat_histogram_results.txt_20130117-154048.out
[root@ftp]#
```

17.12.1 Configuration

The output of the following command displays the port usage in a given pool per protocol per subscriber. The output is organized in a configurable number of port-buckets.

```
tools dump nat histogram router router-instance pool pool-name bucket-size <[1..65536]>
num-buckets <[2..50]>
```

In the following example there is 1 subscriber that is using between 20 and 39 UDP ports in the pool named **det**. The pool is configured in the Base routing instance.

```
tools dump nat histogram router "Base" pool "det" bucket-size 20 num-buckets
40
```

```
=====
Usage histogram NAT pool "det" router "Base"
=====
```

```
Num-ports    Sub-TCP    Sub-UDP    Sub-ICMP
-----
```

0-19	0	0	0
20-39	0	1	0
40-59	0	0	0
60-79	0	0	0
80-99	0	0	0
100-119	0	0	0
120-139	0	0	0
140-159	0	0	0
160-179	0	0	0
180-199	0	0	0
200-219	0	0	0
220-239	0	0	0
240-259	0	0	0
260-279	0	0	0
280-299	0	0	0
300-319	0	0	0
320-339	0	0	0
340-359	0	0	0
360-379	0	0	0
380-399	0	0	0
400-419	0	0	0
420-439	0	0	0
440-459	0	0	0
460-479	0	0	0
480-499	0	0	0
500-519	0	0	0
520-539	0	0	0

540-559	0	0	0
560-579	0	0	0
580-599	0	0	0
600-619	0	0	0
620-639	0	0	0
640-659	0	0	0
660-679	0	0	0
680-699	0	0	0
700-719	0	0	0
720-739	0	0	0
740-759	0	0	0
760-779	0	0	0
780-	0	0	0

No. of entries: 40			
=====			

17.13 Multiple NAT policies per inside routing context

Restrictions

The following restrictions apply to multiple NAT policies per inside routing context

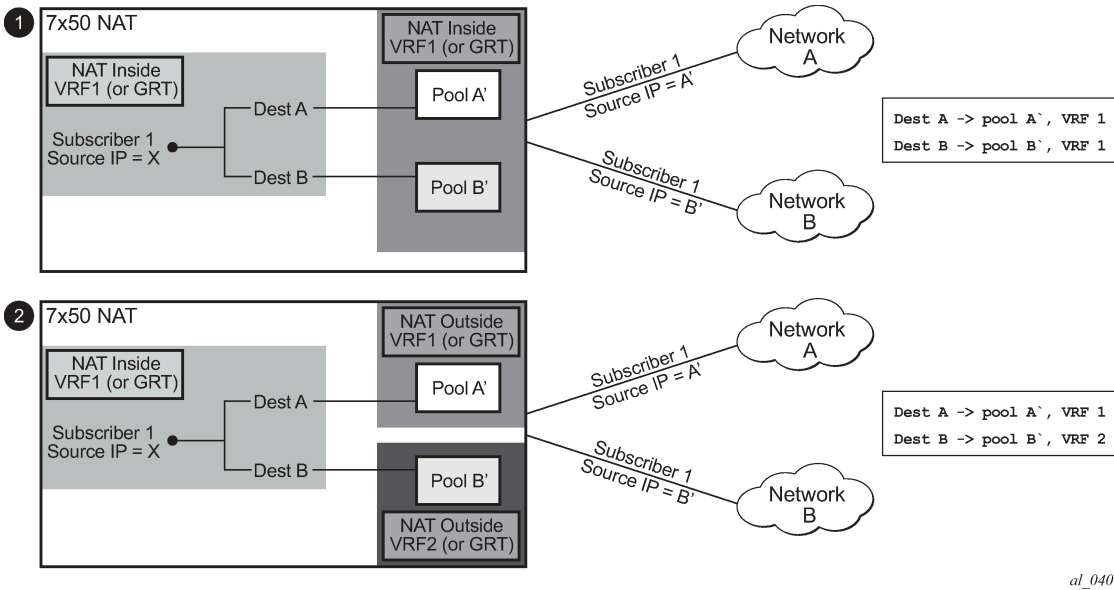
- There is no support for L2-aware NAT.
- A maximum of 8 different NAT policies per inside routing context are supported. For routing based NAT diversion, this limit is enforced during the configuration of the NAT policies within the inside routing context. In case of a filter-based NAT diversion, the filter instantiation will fail if the number of different nat-policies per inside routing context exceeds 8.
- The default NAT policy is counted toward this limit (8).

Overview

The selection of the NAT pool and the outside routing context is performed through the NAT policy. Multiple NAT policies can be used within an inside routing context. This feature effectively allows selective mapping of the incoming traffic within an inside routing context to different NAT pools with different mapping properties (Port-block size, subscriber-limit per pool, address-range, port-forwarding-range, deterministic vs non-deterministic behavior, port-block watermarks, and so on) and to different outside routing contexts. NAT policies can be configured:

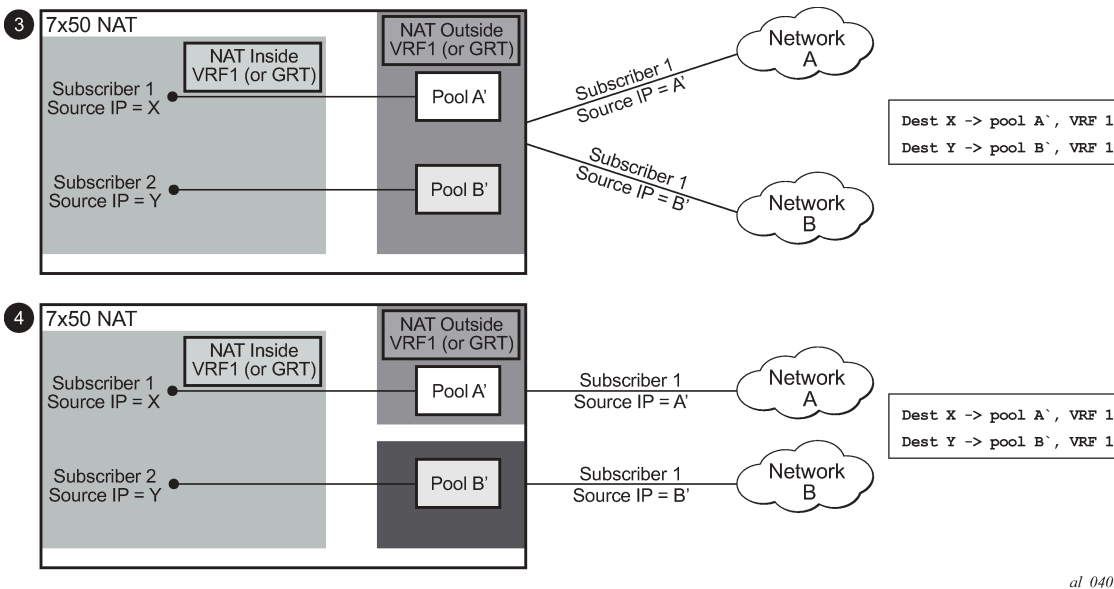
- via filters as part of the `action nat` command.
- via routing with the `destination-prefix` command within the inside routing context

Figure 221: Pool selection based on traffic destination



Only filter-based diversion solution is supported for this case. The filter-based solution can be extended to a 5 tuple matching criteria.

Figure 222: NAT pool selection based on the inside source IP address



The following considerations must be taken into account when deploying multiple NAT policies per inside routing context:

- The inside IP address can be mapped into multiple outside IP addresses based on the traffic destination. The relationship between the inside IP and the outside IP is 1:N.

- In case where the source IP address is selected as a matching criteria for a NAT policy (or pool) selection, the inside IP address will always stay mapped to the same outside IP address (relationship between the inside IP and outside IP address is, in this case, 1:1)
- Static Port Forwards (SPF) — Each SPF can be created only in one pool. This means that the pool (or NAT policy) must be an input parameter for SPF creation.

17.13.1 Routing approach for NAT diversion

The routing approach relies on upstream traffic being directed (or diverted) to the NAT function based on the `destination-prefix` command in the `configure>service>vprn/router>nat>inside` CLI context. In other words, the upstream traffic will be NATed only if it matches a preconfigured destination IP prefix. The `destination-prefix` command creates a static route in the routing table of the inside routing context. This static route will divert all traffic with the destination IP address that matches the created entry, toward the MS-ISA. The NAT function itself will be performed once the traffic is in the proper context in the MS-ISA.

The CLI for multiple NAT policies per inside routing context with routing based diversion to NAT is the following:

```
service vprn/router
nat
  inside
    destination-prefix ip-prefix/length nat-policy policy-name
```

Note that different destination prefixes can reference a single NAT policy (policy-1 in this case).

In case that the destination-policy does not directly reference the NAT policy, the default NAT policy will be used. The default nat-policy is configured directly in the `vprn/router>nat>inside` context.

Once that `destination-prefix` command referencing the `nat-policy` is configured, an entry in the routing table will be created that will direct the traffic to the MS-ISA.

```
service vprn/router
nat
  inside
    destination-prefix 20.20.10.0/24 nat-policy policy-1
    destination-prefix 30.30.30.0/24 nat-policy policy-1
    destination-prefix 40.40.40.0/24 nat-policy policy-2
```

17.13.2 Filter-based approach

A filter-based approach will divert traffic to NAT based on the IP matching criteria shown in the CLI below.

```
*A:right-a21>config>filter>ip-filter>entry# match
- match [protocol <protocol-id>]
```

```

- no match

<protocol-id>      : protocol numbers - [0..255] (Decimal,
                        Hexadecimal, or Binary representation).
                        Supported IANA IP protocol names -
                        none|crtp|crudp|egp|eigrp|encap|ether-ip|
                        gre|icmp|idrp|igmp|igp|ip|ipv6|ipv6-frag|
                        ipv6-icmp|ipv6-no-nxt|ipv6-opts|ipv6-route|
                        isis|iso-ip|l2tp|ospf-igp|pim|pnni|ptp|rdp|
                        rsvp|sctp|stp|tcp|udp|vrrp
                        * - udp/tcp wildcard

[no] dst-ip         - Configure dest. ip match condition
[no] dst-port       - Configure destination port match condition
[no] port           - Configure port match condition
[no] src-ip         - Configure source ip match condition
[no] src-port       - Configure source port match condition

```

17.13.3 Default NAT policy

The default nat-policy is always mandatory and must be configured under the router/vprn>nat>inside context. This default NAT policy can reference any configured pool in the desired ISA group. The pool referenced in the default nat-policy can be then overridden by the nat-policy associated with the destination-prefix in LSN44.

The NAT CLI nodes will fail to activate (be brought out of the no shutdown state), unless a valid nat-policy is referenced in the router/vprn>nat>inside context.

17.13.4 Scaling considerations

Each subscriber using multiple policies is counted as 1 subscriber for the inside resources scaling limits (such as the number of subscribers per MS-ISA), and counted as 1 subscriber per (subscriber + policy combination) for the outside limits (subscriber-limit subscribers per IP; port-reservation port/block reservations per subscriber).

17.13.5 Multiple NAT policies and SPF configuration considerations

Any given Static Port Forward (SPF) can be created only in one pool. This pool, which is referenced through the nat-policy, has to be specified at the SPF creation time, either explicitly through the configuration request or implicitly via defaults.

Explicit request will be submitted either via NFM-M or via CLI:

```
tools perform nat port-forwarding-action lsn
```



```
lsn create router router-instance [b4 ipv6-address] [aftr ipv6-address>]
ip <ipaddress> protocol {tcp|udp} [port port] lifetime lifetime [outside-
ip ipv4-address] [outside-port port] [nat-policy policy-name]
```

In the absence of the nat-policy referenced in the SPF creation request, the default nat-policy under the vprn/router>nat>inside context will be used.

The consequence of this is that the operator must know the nat-policy in which the SPF is to be created.

The static port forward creation and their use by the subscriber types must follow these rules:

Default nat-policy	Any subscriber type can use an SPF created in the pool referenced by the default nat-policy.
Deterministic LSN44 nat-policy	Only deterministic LSN44 subscribers matching the configured prefix can use the SPF created in the pool referenced by the deterministic LSN44 prefix nat-policy.
LSN44 filter based nat-policy	Only LSN44 subscribers matching the configured filter entry can use the SPF created in the pool referenced by the non-deterministic LSN44 nat-policy within the filter.
NAT64 filter based nat-policy	Only NAT64 subscribers matching the configured filter entry can use the SPF created in the pool referenced by the NAT64nat-policy within the filter.

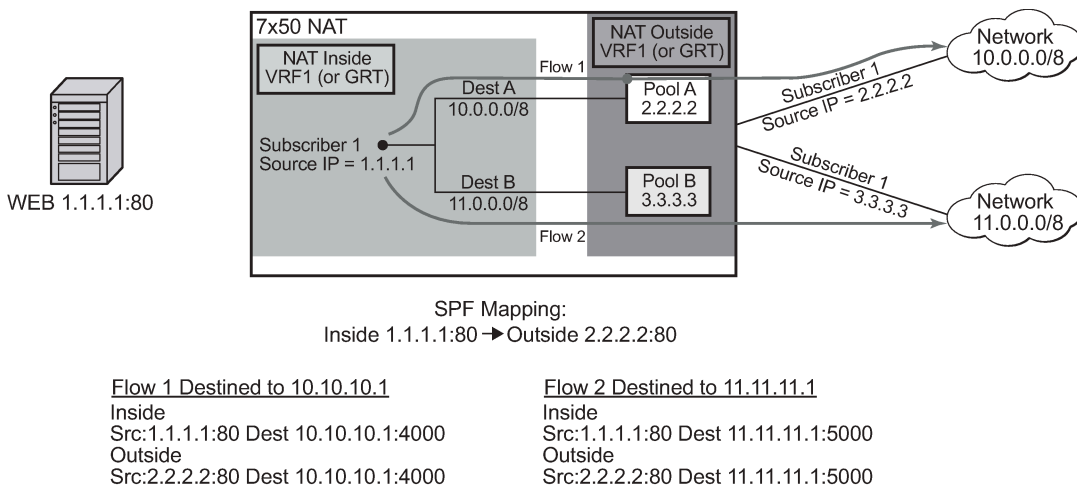
When the last relevant policy for a certain subscriber type is removed from the virtual router, the associated port forwards are automatically deleted.

17.13.5.1 Multiple NAT policies and forwarding considerations

SPF with multiple NAT policies and *Bypassing NAT policy rule* describe certain scenarios that are more theoretical and are less likely to occur in reality. However, they are described here for the purpose of completeness.

SPF with multiple NAT policies shows the case where traffic from the WEB server 1.1.1.1 is initiated toward the destined network 11.0.0.0/8. Such traffic will end up translated in the Pool B and forwarded to the 11.0.0.0/8 network even though the static port forward has been created in Pool A. In this case the nat-policy rule (dest 11.0.0.0/8 pool B) will determine the pool selection in the upstream direction (even though the SPF for the WEB server already exists in the Pool A).

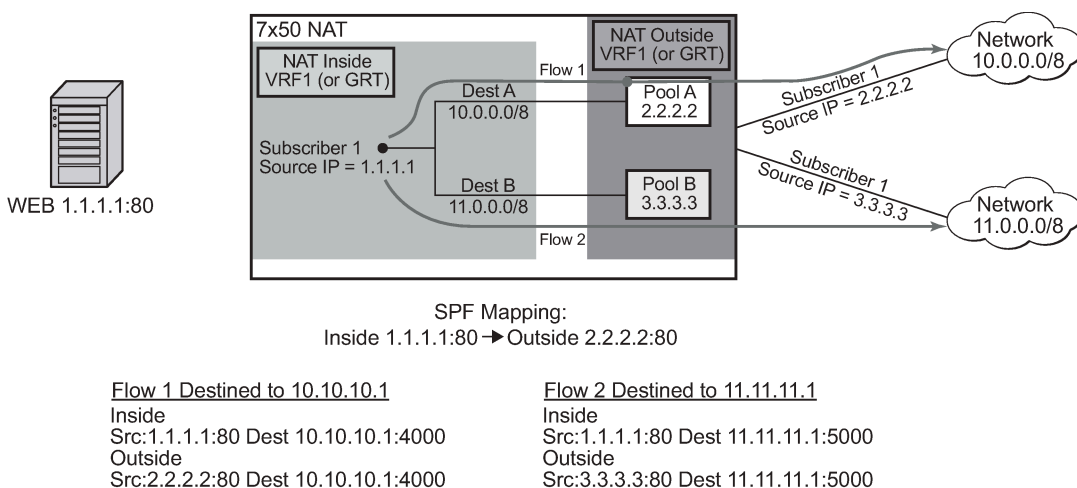
Figure 223: SPF with multiple NAT policies



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The next example in *Bypassing NAT policy rule* shows a case where the Flow 1 is initiated from the outside. Since the partial mapping matching this flow already exist (created by SPF) and there is no more specific match (FQF) present, the downstream traffic will be mapped according to the SPF (through Pool A to the Web server). A more specific match would be in this case fully qualified flows (FQF) that contains information about the foreign host: <host, inside IP/port, outside IP/port, foreign IP address/port, protocol>. At the same time, a more specific entry (FQF) will be created (initiated by the very same outside traffic). This FQF will now determine the forwarding path for all traffic originating from the inside that is matching this flow. This means that the Flow 2 (reverse of the Flow 1) will not be mapped to an IP address from the pool B (as the policy dictates) but instead to the Pool A which has a more specific match.

Figure 224: Bypassing NAT policy rule



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17.13.6 Logging

When multiple NAT policies per inside routing context are deployed, a new `policy-id` parameter is added to certain syslog messages. The format of the `policy-id` is:

```
plcy-id XX
```

where `XX` is an arbitrary unique number per inside routing context assigned by 7x50. This number, represents the corresponding `nat-policy`. Since the maximum number of NAT policies in the inside routing context is 8, the `policy-id` value is also a numerical value in the range 1 to 8.

Introduction of the `policy-id` in logs is necessary due to the bulk-operations associated with multiple NAT policies per inside routing context. A bulk operation, for example, represents the removal of the `nat-policy` from the configuration, shutting down the NAT pool, or removing an IP address range from the pool. Note that removing a NAT accounting policy in case of RADIUS NAT logging will not trigger a summarization log since an `acct-off` message is sent. Such operations have a tendency to be heavy on NAT logging since they affect a large number of NAT subscribers at once. Summarization logs are introduced to prevent excessive logging during bulk operations. For example, the `nat-policy` deletion can be logged with a single (summarized) entry containing the `policy-id` of the `nat-policy` that was removed and the inside `svrc-id`. Since all logs contain the `policy-id`, a single summarization `free` log (Map and Free logs are generated when the port-block for the subscribers are allocated and de-allocated) can be compared to all `map2` logs containing the same `policy-id` to determine for which subscribers the NAT mappings have ceased.

Summarization log is always generated on the CPM, regardless of whether the RADIUS logging is enabled or not. A summarization log simply cannot be generated via RADIUS logging since the RADIUS accounting message streams (start/interim-updates/stop) are always generated per subscriber. In other words, for RADIUS logging, the summarization log would need to be sent to each subscriber, which defeats the purpose of the summarization logs.

A summarization log on the CPM is generated:

- When the `nat-policy` is removed — With a single `nat-policy` per inside routing context, a summarization log is generated with only one field: inside `svrc-id` (**vprn** or **base**). This is sufficient since there is only one `nat-policy` per inside routing context. To determine subscribers for which NAT mappings are terminated, the operator should search all most recent map logs matching the service-id from the summarization log.

With multiple NAT policies per inside routing context, the inside `svrc-id` and the `policy-id` are included in the summarization log (no outside IPs, outside `svrc-id`, port-block or source IP).

A log search based on the `policy-id` and inside `svrc-id` should reveal all subscribers whose mappings were affected by the `nat-policy` removal.

- When the pool is shutdown — The 7x50 will send a summarization log that includes the outside `svrc-id` and all IP address ranges configured in the pool. No other parameters are included in the summarization log.

A log search based on the outside IP address and outside `svrc-id` should reveal all subscribers for which the NAT mappings have ceased.

- When an IP address-range is removed from the pool. The 7x50 will send a summarization log that includes the outside *svc-id* and the IP address range that has been removed. No other parameters are included in the summarization log.

A log search based on the outside IP addresses in the range and the outside *svc-id* should reveal all subscribers for which the NAT mappings have ceased.

- When the last AFTR address is removed.
- When NAT64 node is shutdown.
- When deterministic NAT prefixes are created or removed.

Summarization logs in RADIUS logging

The summarization log for bulk operation while RADIUS logging is in effect will be generated only in the CPM (syslog). This means that for bulk operations with RADIUS logging, the operator will have to rely on RADIUS logging as well as on the CPM logging.

An open log sequence in RADIUS, for example a map for the <inside IP 1, outside IP 1, port-block 1> followed at some later time with a map for <inside IP 2, outside IP 1, port-block 1>, is an indication that the free log for <inside IP 1, outside IP 1, port-block 1> is missing. This means that either the free log for <inside IP 1, outside IP 1, port-block 1> was lost or that a policy/pool/address-range was removed from the configuration. In the latter case, the operator should look in the CPM log for the summarization message.

The summarization logs are enabled via the event control 2021 *tmnxNatLsnSubBlksFree* which is by default suppressed. The even control 2021 is also used to report when all blocks for the subscriber are freed.

17.14 ISA feature interactions

The interaction between MS-ISA applications and other system features.

17.14.1 MS-ISA use with service mirrors

All MS-ISA uses include support for service mirroring running with no feature interactions or impacts. For example, any service diverted to AA, IPSec, NAT, LNS, or supported combinations of MS-ISA application also supports service mirroring simultaneously.

17.14.2 LNS, Application Assurance and NAT

Multiple uses of MS-ISAs can be combined at one time by daisy-chaining use of the MS-ISAs. Services and subscribers terminated on the LNS ISA are full supported by Application Assurance per AA subscriber and service capabilities, and by the full NAT capabilities.

When Application Assurance and NAT are used in combination (for both ESM and SAP service contexts):

- AA is always on subscriber of NAT to be able to see the original (inside) subscriber IP tuple (IP + port numbers).

- AA subscriber ID includes the VRF context from the service, so shared or private subscriber IP, as seen in Layer-2 Aware NAT (not supported), is compatible with AA subscriber contexts.

17.14.3 Subscriber-aware large scale NAT44

Subscriber aware Large Scale NAT44 attempts to combine the positive attributes of Large Scale NAT44 namely:

- The ability for some traffic to bypass the NAT function, such as IPTV traffic and VoIP traffic whenever a unique IP address per subscriber is used. This can be achieved using existing Large Scale NAT44 mechanisms (ingress IP-filters).
- The use of RADIUS Acct for logging of port-ranges, including multiple port-range blocks. Syslog, RADIUS, and CSV file based logging is supported.
- The use of subscriber-identification/RADIUS user-name to identify the customer to simplify management of Large Scale NAT44 subscribers.

Subscriber awareness in Large Scale NAT44 will facilitate release of NAT resources immediately after the BNG subscriber is terminated, without having to wait for the last flow of the subscriber to expire on its own (TCP timeout is 4hours by default).

The subscriber aware Large Scale NAT44 function leverages RADIUS accounting proxy built-in to the 7750SR. The RADIUS accounting proxy allows the 7750SR to inform Large Scale NAT44 application about individual BNG subscribers from the RADIUS accounting messages generated by a remote BNG and use this information in the management of Large Scale NAT44 subscribers. The combination of the two allows, for example, the 7750SR running as a Large Scale NAT44 to make the correlation between the BNG subscriber (represented in the Large Scale NAT44 by the Inside IP Address) and RADIUS attributes such as User-Name, Alc-Sub-Ident-String, Calling-Station-Id or Class. These attributes can subsequently be used for either management of the Large Scale NAT44 subscriber, or in the NAT RADIUS Accounting messages generated by the 7750SR Large Scale NAT44 application. Doing so will simplify both the administration of the Large Scale NAT44 and the logging function for port-range blocks.

As BNG subscribers authenticate and come online, the RADIUS accounting messages are 'snooped' via RADIUS accounting proxy which creates a cache of attributes from the BNG subscriber. BNG subscribers are correlated with the NAT subscriber via framed-ip address, and one of the following attributes that must be present in the accounting messages generated by BNG:

- User-name
- Subscriber id
- RADIUS Class attribute
- Calling-Station-id
- IMSI
- IMEI

Framed-ip address must also be present in the accounting messages generated by BNG.

Large Scale NAT44 Subscriber Aware application will receive a number of cached attributes which will then be used for appropriate management of Large Scale NAT44 subscribers, for example:

- Delete the Large Scale NAT44 subscriber when the BNG subscriber is terminated
- Report attributes in Large Scale NAT44 accounting messages according to configuration options

Creation and removal of RADIUS accounting proxy cache entries related to BNG subscriber is triggered by the receipt of accounting start/stop messages sourced by the BNG subscriber. Modification of entries can be triggered by interim-update messages carrying updated attributes. Cached entries can also be purged via CLI.

In addition to passing one of the above attributes in Large Scale NAT44 RADIUS accounting messages, a set of opaque BNG subscriber RADIUS attributes can optionally be passed in Large Scale NAT44 RADIUS accounting messages. Up to 128B of such opaque attributes will be accepted. The remaining attributes will be truncated.

Large Scale NAT44 subscriber instantiation can optionally be denied in case that corresponding BNG subscriber cannot be identified in Large Scale NAT44 via RADIUS accounting proxy.

Configuration guidelines

1. Configure the RADIUS accounting proxy functionality in a routing instance that will receive accounting messages from the remote or local BNG. Optionally forward received accounting message received by RADIUS accounting proxy to the final accounting destination (accounting server).
2. Point the BNG RADIUS accounting destination to the RADIUS accounting proxy - this way RADIUS accounting proxy will receive and 'snoop' BNG RADIUS accounting data.

BNG subscriber can be associated with two accounting policies, therefore pointing to two different accounting destinations. For example, one to the RADIUS accounting proxy, the other one to the real accounting server.

- Configure subscriber aware Large Scale NAT44. From Large Scale NAT44 Subscriber Aware application reference the RADIUS Proxy accounting server and define the string that will be used to correlate BNG subscriber with the Large Scale NAT44 subscriber.
- Optionally, enable NAT RADIUS accounting that includes BNG subscriber relevant data.

```
*A:left-a20>config>service>vprn#
radius-proxy
    server "proxy-acct" purpose accounting create
        default-accounting-server-policy "lsn-policy"
        description "two side server -interface:client ; default-
plcy:real server"
    interface "rad-proxy-loopback"
        secret "TEg1UEZzemRMyZXD1HvvQGkeGfoQ58MF" hash2
        no shutdown
    exit
exit
```

RADIUS accounting proxy will listen to accounting messages on interface rad-proxy-loopback.

The name proxy-acct as defined by the server command will be used to reference this proxy accounting server from Large Scale NAT44.

Received accounting messages can be relayed further from RADIUS accounting proxy to the accounting server which can be indirectly referenced in the default-accounting-policy lsn-policy.

The lsn-policy is defined as:

```
*A:left-a20>config>aaa#
radius-server-policy "lsn-policy" create
    servers
        router "Base"
        source-address 114.0.1.12
        server 1 name "114"
    exit
exit
```

This lsn-policy can then reference an external RADIUS accounting server with its own security credentials. This external accounting server can be configured in any routing instance.

```
*A:left-a20>config>router>radius-server# info
-----
server "114" address 114.0.1.10 secret "KRr7H.K3i0z90/
hj2BUSmdJUDl.zWrkE" hash2 port 1813 create
description "real radius or acct server"
exit
```

Two RADIUS accounting policies can be configured in BNG - one to the real RADIUS server, the other one to the RADIUS accounting proxy.

```
*A:left-a20>config>subscr-mgmt>sub-prof# info
-----
radius-accounting-policy "real-acct-srvr" duplicate "lsn"
egress
    agg-rate-limit 10000
exit
-----
*A:left-a20>config>subscr-mgmt>acct-plcy# info
-----
description "lsn radius-acct-policy"
update-interval 5
include-radius-attribute
    acct-authentic
    acct-delay-time
    called-station-id
    calling-station-id remote-id
```

```

        circuit-id
        framed-interface-id
        framed-ip-addr
        framed-ip-netmask
        mac-address
        nas-identifier
        nas-port-id
        nas-port-type
        nat-port-range
        remote-id
        sla-profile
        sub-profile
        subscriber-id
        user-name
        alc-acct-triggered-reason
    exit
    session-id-format number
    radius-accounting-server
        router 10 (service id where proxy radius is configured)
        server 1 address 5.5.5.5 secret "cVi1sidvgH28Pd9QoN1fLE"
hash2
(radius proxy IP address is 5.5.5.5 on interface "rad-proxy-loopback"; the
'secret' is the same as configured on RADIUS accounting proxy)
    exit

```

Sub-aware Large Scale NAT44 references the RADIUS accounting proxy server proxy-acct and defines the calling-station-id attribute from the BNG subscriber as the matching attribute:

```

*A:left-a20>config>service>vprn>nat>inside# info
-----
nat-policy "nat-base"
destination-prefix 10.0.0.0/16
subscriber-identification
    attribute vendor "standard" attribute-type "station-id"
        description "sub-aware CGN"
        radius-proxy-server router 10 name "proxy-acct"
        no shutdown
    exit
-----

```

Optionally RADIUS NAT accounting can be enabled:

```

*A:left-a20>config>isa>nat-group# info

```



```

-----
    active-mda-limit 1
    radius-accounting-policy "nat-acct-basic"
    mda 1/2
    no shutdown

*A:left-a20>config>aaa>nat-acct-plcy# info detail
-----
    description "nat-acct-basic policy"
    include-radius-attribute
        framed-ip-addr
        nas-identifier
        nat-subscriber-string
        user-name
        inside-service-id
        outside-service-id
        outside-ip
        port-range-block
        hardware-timestamp
        release-reason
        multi-session-id
        frame-counters
        octet-counters
        session-time
        called-station-id
        subscriber-data
    exit
    radius-accounting-server
        access-algorithm direct
        retry 3
        router "Base"
        source-address-range 114.0.1.20 114.0.1.20
        timeout sec 5
        server 1 address 114.0.1.10 secret "KlWIBi08CxTyM/
YXaU2gQitOu8GgfsD70j5hjese27A" hash2 port 1813
    exit
-----

```

Such setup would produce a stream of following Large Scale NAT44 RADIUS accounting messages:

```

Mon Jul 16 10:59:27 2012
  NAS-IP-Address = 1.1.1.1
  NAS-Identifier = "left-a20"
  NAS-Port = 37814272
  Acct-Status-Type = Start

```

```
Acct-Multi-Session-Id = "500456500365a4de7c29a9a07c29a9a0"  
Acct-Session-Id = "500456500365a4de6201d7b87c29a9a0"  
Called-Station-Id = "00-00-00-00-01-01"  
User-Name = "remote0"  
Calling-Station-Id = "remote0"  
Alc-Serv-Id = 10  
Framed-IP-Address = 26.0.0.7  
Alc-Nat-Outside-Ip-Addr = 80.0.0.1  
Alc-Nat-Port-Range = "80.0.0.1 1054-1058 router base"  
Acct-Input-Packets = 0  
Acct-Output-Packets = 0  
Acct-Input-Octets = 0  
Acct-Output-Octets = 0  
Acct-Input-Gigawords = 0  
Acct-Output-Gigawords = 0  
Acct-Session-Time = 0  
Event-Timestamp = "Jul 16 2012 10:58:40 PDT"  
NAS-IP-Address = 1.1.1.1  
User-Name = "cgn_1_ipoe"  
Framed-IP-Netmask = 255.255.255.0  
Class = 0x63676e2d636c6173732d7375622d6177617265  
NAS-Identifier = "left-a20"  
Acct-Session-Id = "D896FF0000000550045640"  
Event-Timestamp = "Jul 16 2012 10:58:24 PDT"  
NAS-Port-Type = Ethernet  
NAS-Port-Id = "1/1/5:5.10"  
Acct-Delay-Time = 0  
Acct-Authentic = RADIUS  
Acct-Unique-Session-Id = "10f8bce6e5e7eb41"  
Timestamp = 1342461567  
Request-Authenticator = Verified
```

Mon Jul 16 11:03:56 2012

```
NAS-IP-Address = 1.1.1.1  
NAS-Identifier = "left-a20"  
NAS-Port = 37814272  
Acct-Status-Type = Interim-Update  
Acct-Multi-Session-Id = "500456500365a4de7c29a9a07c29a9a0"  
Acct-Session-Id = "500456500365a4de6201d7b87c29a9a0"  
Called-Station-Id = "00-00-00-00-01-01"  
User-Name = "remote0"  
Calling-Station-Id = "remote0"  
Alc-Serv-Id = 10  
Framed-IP-Address = 26.0.0.7  
Alc-Nat-Outside-Ip-Addr = 80.0.0.1
```

```
Alc-Nat-Port-Range = "80.0.0.1 1054-1058 router base"
Acct-Input-Packets = 0
Acct-Output-Packets = 1168
Acct-Input-Octets = 0
Acct-Output-Octets = 86432
Acct-Input-Gigawords = 0
Acct-Output-Gigawords = 0
Acct-Session-Time = 264
Event-Timestamp = "Jul 16 2012 11:03:04 PDT"
Acct-Delay-Time = 5
NAS-IP-Address = 1.1.1.1
User-Name = "cgn_1_ipoe"
Framed-IP-Netmask = 255.255.255.0
Class = 0x63676e2d636c6173732d7375622d6177617265
NAS-Identifier = "left-a20"
Acct-Session-Id = "D896FF00000000550045640"
Acct-Session-Time = 279
Event-Timestamp = "Jul 16 2012 11:03:04 PDT"
NAS-Port-Type = Ethernet
NAS-Port-Id = "1/1/5:5.10"
Acct-Delay-Time = 0
Acct-Authentic = RADIUS
Acct-Unique-Session-Id = "10f8bce6e5e7eb41"
Timestamp = 1342461836
Request-Authenticator = Verified
```

Mon Jul 16 11:04:34 2012

```
NAS-IP-Address = 1.1.1.1
NAS-Identifier = "left-a20"
NAS-Port = 37814272
Acct-Status-Type = Stop
Acct-Multi-Session-Id = "500456500365a4de7c29a9a07c29a9a0"
Acct-Session-Id = "500456500365a4de6201d7b87c29a9a0"
Called-Station-Id = "00-00-00-00-01-01"
User-Name = "remote0"
Calling-Station-Id = "remote0"
Alc-Serv-Id = 10
Framed-IP-Address = 26.0.0.7
Alc-Nat-Outside-IP-Addr = 80.0.0.1
Alc-Nat-Port-Range = "80.0.0.1 1054-1058 router base"
Acct-Terminate-Cause = Host-Request
Acct-Input-Packets = 0
Acct-Output-Packets = 1321
Acct-Input-Octets = 0
Acct-Output-Octets = 97754
```

```
Acct-Input-Gigawords = 0
Acct-Output-Gigawords = 0
Acct-Session-Time = 307
Event-Timestamp = "Jul 16 2012 11:03:47 PDT"
NAS-IP-Address = 1.1.1.1
User-Name = "cgn_1_ipoe"
Framed-IP-Netmask = 255.255.255.0
Class = 0x63676e2d636c6173732d7375622d6177617265
NAS-Identifier = "left-a20"
Acct-Session-Id = "D896FF00000000550045640"
Acct-Session-Time = 279
Event-Timestamp = "Jul 16 2012 11:03:04 PDT"
NAS-Port-Type = Ethernet
NAS-Port-Id = "1/1/5:5.10"
Acct-Delay-Time = 0
Acct-Authentic = RADIUS
Acct-Unique-Session-Id = "10f8bce6e5e7eb41"
Timestamp = 1342461874
Request-Authenticator = Verified
```

The matching accounting stream generated on the BNG is given below:

```
Mon Jul 16 10:59:11 2012
Acct-Status-Type = Start
NAS-IP-Address = 1.1.1.1
User-Name = "cgn_1_ipoe"
Framed-IP-Address = 26.0.0.7
Framed-IP-Netmask = 255.255.255.0
Class = 0x63676e2d636c6173732d7375622d6177617265
Calling-Station-Id = "remote0"
NAS-Identifier = "left-a20"
Acct-Session-Id = "D896FF00000000550045640"
Event-Timestamp = "Jul 16 2012 10:58:24 PDT"
NAS-Port-Type = Ethernet
NAS-Port-Id = "1/1/5:5.10"
ADSL-Agent-Circuit-Id = "cgn_1_ipoe"
ADSL-Agent-Remote-Id = "remote0"
Alc-Subsc-ID-Str = "CGN1"
Alc-Subsc-Prof-Str = "nat"
Alc-SLA-Prof-Str = "tp_sla_prem"
Alc-Client-Hardware-Addr = "00:00:65:05:10:01"
Acct-Delay-Time = 0
Acct-Authentic = RADIUS
Acct-Unique-Session-Id = "9c1723d05e87c043"
Timestamp = 1342461551
```

Request-Authenticator = Verified

Mon Jul 16 11:03:51 2012

Acct-Status-Type = Interim-Update

NAS-IP-Address = 1.1.1.1

User-Name = "cgn_1_ipoe"

Framed-IP-Address = 26.0.0.7

Framed-IP-Netmask = 255.255.255.0

Class = 0x63676e2d636c6173732d7375622d6177617265

Calling-Station-Id = "remote0"

NAS-Identifier = "left-a20"

Acct-Session-Id = "D896FF00000000550045640"

Acct-Session-Time = 279

Event-Timestamp = "Jul 16 2012 11:03:04 PDT"

NAS-Port-Type = Ethernet

NAS-Port-Id = "1/1/5:5.10"

ADSL-Agent-Circuit-Id = "cgn_1_ipoe"

ADSL-Agent-Remote-Id = "remote0"

Alc-Subsc-ID-Str = "CGN1"

Alc-Subsc-Prof-Str = "nat"

Alc-SLA-Prof-Str = "tp_sla_prem"

Alc-Client-Hardware-Addr = "00:00:65:05:10:01"

Acct-Delay-Time = 0

Acct-Authentic = RADIUS

Nokia-IPD-Attr-163 = 0x00000001

Alc-Acct-I-Inprof-Octets-64 = 0x00010000000000000000

Alc-Acct-I-Outprof-Octets-64 = 0x000100000000000020468

Alc-Acct-I-Inprof-Pkts-64 = 0x00010000000000000000

Alc-Acct-I-Outprof-Pkts-64 = 0x00010000000000000052a

Alc-Acct-I-Inprof-Octets-64 = 0x00030000000000000000

Alc-Acct-I-Outprof-Octets-64 = 0x00030000000000000000

Alc-Acct-I-Inprof-Pkts-64 = 0x00030000000000000000

Alc-Acct-I-Outprof-Pkts-64 = 0x00030000000000000000

Alc-Acct-I-Inprof-Octets-64 = 0x00050000000000000000

Alc-Acct-I-Outprof-Octets-64 = 0x00050000000000000000

Alc-Acct-I-Inprof-Pkts-64 = 0x00050000000000000000

Alc-Acct-I-Outprof-Pkts-64 = 0x00050000000000000000

Alc-Acct-O-Inprof-Octets-64 = 0x00010000000000000000

Alc-Acct-O-Outprof-Octets-64 = 0x000100000000000003154

Alc-Acct-O-Inprof-Pkts-64 = 0x00010000000000000000

Alc-Acct-O-Outprof-Pkts-64 = 0x0001000000000000009a

Alc-Acct-O-Inprof-Octets-64 = 0x00030000000000000000

Alc-Acct-O-Outprof-Octets-64 = 0x00030000000000000000

Alc-Acct-O-Inprof-Pkts-64 = 0x00030000000000000000

Alc-Acct-O-Outprof-Pkts-64 = 0x00030000000000000000

```

Alc-Acct-0-Inprof-Octets-64 = 0x00050000000000000000
Alc-Acct-0-Outprof-Octets-64 = 0x00050000000000000000
Alc-Acct-0-Inprof-Pkts-64 = 0x00050000000000000000
Alc-Acct-0-Outprof-Pkts-64 = 0x00050000000000000000
Acct-Unique-Session-Id = "9c1723d05e87c043"
Timestamp = 1342461831
Request-Authenticator = Verified

```

Mon Jul 16 11:04:34 2012

```

Acct-Status-Type = Stop
NAS-IP-Address = 1.1.1.1
User-Name = "cgn_1_ipoe"
Framed-IP-Address = 26.0.0.7
Framed-IP-Netmask = 255.255.255.0
Class = 0x63676e2d636c6173732d7375622d6177617265
Calling-Station-Id = "remote0"
NAS-Identifier = "left-a20"
Acct-Session-Id = "D896FF00000000550045640"
Acct-Session-Time = 322
Acct-Terminate-Cause = User-Request
Event-Timestamp = "Jul 16 2012 11:03:47 PDT"
NAS-Port-Type = Ethernet
NAS-Port-Id = "1/1/5:5.10"
ADSL-Agent-Circuit-Id = "cgn_1_ipoe"
ADSL-Agent-Remote-Id = "remote0"
Alc-Subsc-ID-Str = "CGN1"
Alc-Subsc-Prof-Str = "nat"
Alc-SLA-Prof-Str = "tp_sla_prem"
Alc-Client-Hardware-Addr = "00:00:65:05:10:01"
Acct-Delay-Time = 0
Acct-Authentic = RADIUS
Alc-Acct-I-Inprof-Octets-64 = 0x00010000000000000000
Alc-Acct-I-Outprof-Octets-64 = 0x0001000000000000248c4
Alc-Acct-I-Inprof-Pkts-64 = 0x00010000000000000000
Alc-Acct-I-Outprof-Pkts-64 = 0x000100000000000005d9
Alc-Acct-I-Inprof-Octets-64 = 0x00030000000000000000
Alc-Acct-I-Outprof-Octets-64 = 0x00030000000000000000
Alc-Acct-I-Inprof-Pkts-64 = 0x00030000000000000000
Alc-Acct-I-Outprof-Pkts-64 = 0x00030000000000000000
Alc-Acct-I-Inprof-Octets-64 = 0x00050000000000000000
Alc-Acct-I-Outprof-Octets-64 = 0x00050000000000000000
Alc-Acct-I-Inprof-Pkts-64 = 0x00050000000000000000
Alc-Acct-I-Outprof-Pkts-64 = 0x00050000000000000000
Alc-Acct-0-Inprof-Octets-64 = 0x00010000000000000000
Alc-Acct-0-Outprof-Octets-64 = 0x00010000000000003860

```

```

Alc-Acct-0-Inprof-Pkts-64 = 0x00010000000000000000
Alc-Acct-0-Outprof-Pkts-64 = 0x00010000000000000000b0
Alc-Acct-0-Inprof-Octets-64 = 0x00030000000000000000
Alc-Acct-0-Outprof-Octets-64 = 0x00030000000000000000
Alc-Acct-0-Inprof-Pkts-64 = 0x00030000000000000000
Alc-Acct-0-Outprof-Pkts-64 = 0x00030000000000000000
Alc-Acct-0-Inprof-Octets-64 = 0x00050000000000000000
Alc-Acct-0-Outprof-Octets-64 = 0x00050000000000000000
Alc-Acct-0-Inprof-Pkts-64 = 0x00050000000000000000
Alc-Acct-0-Outprof-Pkts-64 = 0x00050000000000000000
Acct-Unique-Session-Id = "9c1723d05e87c043"
Timestamp = 1342461874
Request-Authenticator = Verified

```

17.15 Support for SSG with CGNAT and firewall/AA functions

The CGNAT function is supported on the CMG deployed as a Subscriber Services Gateway (SSG). The MG-VM can host both AA (firewall, video rate limiting, and so on) and CGNAT functions on a single MG-VM to reduce inter-VM traffic on the system and optimize the use of computer resources.

For general information about SSG functionality, refer to the *CMG Configuration Guide*.

17.16 Configuring NAT with CLI

Information about configuring NAT using the command line interface.

17.16.1 Provisioning a NAT ISA

To enable NAT on the router:

- Create a NAT ISA group.
- Provision and enable the group.



NOTE:

The card configuration described in this guide is applicable to the 7750 SR MG. For CMG VM configuration information, refer to the section *Provisioning a VM using CLI* in the *CMG Installation Guide*.

The following commands show the NAT ISA group configuration context.

```

configure card
card-type ism-mg-2pac-fp3

```

```
mda 2
    mda-type isa2-bb
exit
```

The following commands show the ISA group CLI configuration context.

```
configure isa nat-group nat-group-id create
description description
mda
[no] shutdown
```

17.16.2 ISA redundancy

The 7750 SR MG and CMG support ISA redundancy to provide reliable NAT even when an MDA fails.

The active-mda-limit allows you to specify how many MDAs are active in a specific NAT group. Any number of MDAs configured above the active-mda-limit are spare MDAs; they take over the NAT function if one of the current active MDAs fails.

```
configure
    isa
        nat-group 1 create
            active-mda-limit 1
            mda 1/2
            mda 2/2
            no shutdown
        exit
    exit
exit
```

Show commands are available to display the actual state of a nat-group and its corresponding MDAs:

```
show isa nat-group 1
=====
ISA NAT Group 1
=====
Admin state           : inService           Operational state : inService
Active MDA limit      : 1                   Reserved sessions : 0
High Watermark (%)    : (Not Specified)     Low Watermark (%) : (Not Specified)
Last Mgmt Change      : 01/11/2010 15:05:36
=====
ISA NAT Group 1 members
=====
```


Group	Member	State	Mda	Addresses	Blocks	Se-%	Hi	Se-Prio
1	1	active	1/2	0	0	0	N	0

No. of members: 1								
=====								

A maximum of four NAT groups can be configured. This gives the operator the ability to differentiate between different traffic types. Normal traffic could be routed to NAT group one, where a limited number of MDA without spare MDAs are available, while high priority traffic could make use of NAT group two, where several active MDAs and a spare MDA are configured. A maximum of six MDAs per NAT group can be configured.

A NAT group cannot become active (no shutdown) if the number of configured MDAs is lower that the active-mda-limit.

A given MDA can be configured in several NAT groups but it can only be active in a single NAT group at any moment in time. Spare MDAs can be shared in several NAT groups, but a spare can only become active in one NAT group at a time. Changing the active-mda-limit, adding or removing MDAs can only be done when the NAT group is shutdown.

NAT groups that share spare MDAs must be configured with the same list of MDAs. It is possible to remove/add spare MDAs to a NAT group while the NAT group is admin enabled.

```

configure
  isa
    nat-group 1 create
      active-mda-limit 1
      mda 1/2
      mda 2/2
      mda 3/1
      no shutdown
    exit
    nat-group 2 create
      active-mda-limit 1
      mda 1/2
      mda 2/2
      mda 3/1
      no shutdown
    exit
  exit
exit

```

Through show commands, it is possible to display an overview of all the NAT groups and MDAs.

```
show isa nat-group
```

```
=====
```

```
ISA NAT Group Summary
```

```
=====
```

Mda	Group 1	Group 2
1/1	active	busy
2/2	busy	active
3/1	standby	standby

```
=====
```

If an MDA fails, the spare (if available) will take over. All active sessions will be lost, but new incoming sessions will make use of the spare MDA.

In case of an MDA failure in a NAT group without any spare MDA, all traffic toward that MDA will be black-holed.

17.16.3 Large scale NAT configuration

The following sections provide Large Scale NAT configuration examples.

```
configure
```

```
#-----
```

```
echo "Card Configuration"
```

```
#-----
```

```
    card 3
```

```
        card-type iom3-xp
```

```
        mda 1
```

```
            mda-type isa2-bb
```

```
        exit
```

```
        mda 2
```

```
            mda-type isa2-bb
```

```
        exit
```

```
    exit
```

```
#-----
```

```
echo "ISA Configuration"
```

```
#-----
```

```
    isa
```

```
        nat-group 1 create
```

```
            active-mda-limit 2
```

```
            mda 3/1
```

```
            mda 3/2
```

```
            no shutdown
```

```

        exit
    exit
#-----
echo "NAT (Declarations) Configuration"
#-----
    service
        nat
            nat-policy "ls-outPolicy" create
        exit
    exit
exit
#-----
echo "Service Configuration"
#-----
    service
        customer 1 create
            description "Default customer"
        exit
        vprn 500 customer 1 create
            interface "ip-113.0.0.1" create
            exit
            nat
                outside
                    pool "nat1-pool" nat-group 1 type large-scale create
                    port-reservation ports 200
                    address-range 81.81.0.0 81.81.6.0 create
                    exit
                    no shutdown
                exit
            exit
        exit
    exit
    vprn 550 customer 1 create
        interface "ip-13.0.0.1" create
        exit
    exit
    nat
        nat-policy "ls-outPolicy" create
        pool "nat1-pool" router 500
        timeouts
            udp hrs 5
            udp-initial min 4
        exit
    exit
exit

```

```

vprn 500 customer 1 create
  router-id 10.21.1.2
  route-distinguisher 500:10
  vrf-target export target:500:1 import target:500:1
  interface "ip-113.0.0.1" create
    address 113.0.0.1/24
    static-arp 113.0.0.5 14:99:01:01:00:01
    sap 1/1/1:200 create
    exit
  exit
  no shutdown
exit
vprn 550 customer 1 create
  router-id 10.21.1.2
  route-distinguisher 550:10
  vrf-target export target:550:1 import target:550:1
  interface "ip-13.0.0.1" create
    address 13.0.0.1/8
    sap 1/2/1:900 create
      ingress
        filter ip 123
      exit
    exit
  exit
  nat
    inside
      nat-policy "ls-outPolicy"
    exit
  exit
  no shutdown
exit
exit
exit all

```

17.16.4 NAT configuration examples

The following output displays example configurations.

Example: VPRN Service

```

configure service vprn 100 nat
  inside
    nat-policy "priv-nat-policy"
    destination-prefix 0.0.0.0/0

```

```
        redundancy
            no peer
            no steering-route
        exit
        subscriber-identification
            shutdown
            no attribute
            no description
            no radius-proxy-server
        exit
    outside
        no mtu
    exit
```

Example: Router NAT

```
configure router nat
    outside
        no mtu
        pool "privpool" nat-group 3 type large-scale create
            no description
            port-reservation blocks 128
            port-forwarding-range 1023
            redundancy
                no export
                no monitor
            exit
            subscriber-limit 65535
            no watermarks
            mode auto
            address-range 13.0.0.5 13.0.0.6 create
                no description
                no drain
            exit
            no shutdown
        exit
        pool "pubpool" nat-group 1 type large-scale create
            no description
            port-reservation blocks 1
            port-forwarding-range 1023
            redundancy
                no export
                no monitor
            exit
            subscriber-limit 65535
```

```
        no watermarks
        mode auto
        address-range 138.203.8.241 138.203.8.247 create
        no description
        no drain
        exit
        no shutdown
    exit
exit
```

Example: Service NAT

```
configure service nat
    nat-policy "priv-nat-policy" create
        block-limit 4
        no destination-nat
        no description
        filtering endpoint-independent
        pool "privpool" router Base
        no ipfix-export-policy
        port-limits
            forwarding 64
            no reserved
            no watermarks
        exit
        priority-sessions
        exit
        session-limits
            max 65535
            no reserved
            no watermarks
        exit
        timeouts
            icmp-query min 1
            sip min 2
            no subscriber-retention
            tcp-established hrs 2 min 4
            tcp-syn sec 15
            no tcp-time-wait
            tcp-transitory min 4
            udp min 5
            udp-initial sec 15
            udp-dns sec 15
        exit
    no tcp-mss-adjust
```

```
        no udp-inbound-refresh
    exit
    nat-policy "pub-nat-policy" create
        block-limit 1
        no destination-nat
        no description
        filtering endpoint-independent
        pool "pubpool" router Base
        no ipfix-export-policy
        port-limits
            no forwarding
            no reserved
            no watermarks
        exit
        priority-sessions
        exit
        session-limits
            max 65535
            no reserved
            no watermarks
        exit
        timeouts
            icmp-query min 1
            sip min 2
            no subscriber-retention
            tcp-established hrs 2 min 4
            tcp-syn sec 15
            no tcp-time-wait
            tcp-transitory min 4
            udp min 5
            udp-initial sec 15
            udp-dns sec 15
        exit
        no tcp-mss-adjust
        no udp-inbound-refresh
    exit
```

18 Monitoring and troubleshooting tools

The following monitoring and troubleshooting tools are supported:

- *Call trace and packet capture*
- *Network trace*
- *Per Call Measurement Data*
- *Key Performance Indicator and Key Capacity Indicator*
- *Prometheus metrics*
- *Telemetry*
- *GTP ping*
- *UE ping*
- *SNMP traps*

18.1 Call trace and packet capture

The call trace feature supports call trace and packet capture functionality.

Call trace is a real-time troubleshooting and debugging tool that provides the ability to capture CP messages from the mobile gateway. Call trace also supports capture of data plane traffic on the SGW, PGW/GGSN, ePDG, SMF and UPF. Capturing UP traffic packets from the gateway provides an option to debug forwarding plane issues in a live production network. Packet capture is done using industry-standard PCAP format. Capturing control and user plane messages from a live network without requiring a tap on a physical link supports quick response to failure, by narrowing down the root cause without having to recreate the scenarios in a laboratory environment. Complex issues can be debugged without impacting other subscribers.

Control and user plane messages received on a mobile gateway or sent to a peering node are copied to a data collector node over UDP stream in a PCAP format. The PCAP file for the Service Based Interfaces in 5G includes the HTTP2 and JSON packets for the traced user. The TCP and HTTP2 information are reconstructed from the available data, including IP addresses and ports. The TLS layer is not included in the PCAP file. TCP sequence numbers are zero.

Streaming of call data enables performance analysis and root cause analysis of network issues in real time.

Call trace functionality for CP capture is supported on the following gateways:

- SGW, SGW-C
- PGW/GGSN, PGW-C/GGSN-C
- combined SGW/PGW/GGSN
- ePDG

- combined TWAG/ePDG
- SMF
- UPF

User plane call trace in 7750 hardware captures all user data packets from all reference points once, as they are processed in the gateway. The user data packets are captured without the tunneling and transport headers, for example, only the inner IP packet is captured.

User plane call trace in cloud gateways captures user data packets in ingress and egress LB-VMs (using the **ingress** and **egress** keywords) and in MG-VMs (using the **internal** keyword). User plane packets captured in LB-VMs contain tunneling and transport headers while the user plane packets captured in MG-VM contain only the inner IP packet without tunneling and transport headers.

User plane call trace supports apart from the actual raw packets, metadata which contain information how the gateway has identified and analyzed the packet. Packet capture is done using industry-standard pcapng format when 'metadata' attribute is selected in user-traffic Call Trace profile command for the UE. Currently, the supported metadata includes policy-rule-base, policy-rule, rating-group and service-id from charging-rule-unit, precedence and corresponding actions. This information provides more insights on the packet handling besides the packet itself. The PCC rules information is added as a comment in each user plane packet. The packets that are dropped due to a specific policy cannot be captured by call trace. But they can be detected with call trace statistics which capture packets in ingress and egress LB-VMs. Metadata in user traffic must not be configured with live output and is recommended for up to 5 UEs.

Packet content is available in User Plane PCAP files to provide information about where the packet was captured. For example, in LB-VM or MG-VM with vm:lb or vm:mg and in ingress or egress side of the VM, for instance, dir:in or dir:out respectively.

Figure 225: Metadata inside data capture files

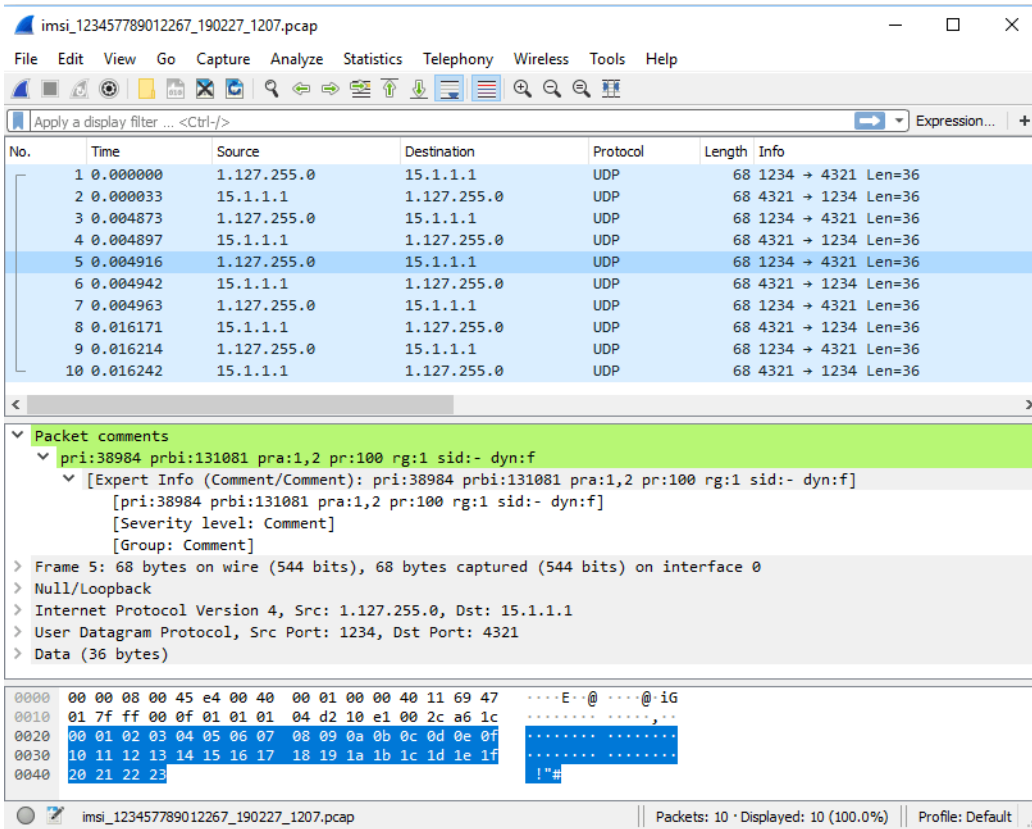


Table 76: Metadata acronyms

Metadata fields	Acronyms
pcc-rulebase-id	prbi
pcc-rule-id	pri
rating group	rg
service-id	sid
dynamic rules	dyn The possible values are: f false t true

Table 76: Metadata acronyms (continued)

Metadata fields	Acronyms												
	It separates static rules without rule bases (dyn:f) and dynamic rules (dyn:t)												
precedence	pr												
pcc-rule-actions	pra Comma separated list of values. The possible values are: <table> <tr> <td>1</td><td>pass</td></tr> <tr> <td>2</td><td>volume charge</td></tr> <tr> <td>3</td><td>time charge</td></tr> <tr> <td>4</td><td>packet is going to be redirected</td></tr> <tr> <td>6</td><td>radius-start-accounting</td></tr> <tr> <td>9</td><td>update-eps-session (the dedicated bearer will be created after matching this rule)</td></tr> </table>	1	pass	2	volume charge	3	time charge	4	packet is going to be redirected	6	radius-start-accounting	9	update-eps-session (the dedicated bearer will be created after matching this rule)
1	pass												
2	volume charge												
3	time charge												
4	packet is going to be redirected												
6	radius-start-accounting												
9	update-eps-session (the dedicated bearer will be created after matching this rule)												
packet content	vm and dir The possible values are: • mg or lb for vm (for example, vm:mg) • in or out for dir vm defines the vm type, for example, lb or mg dir defines the direction of the packets, for example, ingress or egress												

The correlation of the internal IDs which are displayed in the User Plane PCAP file (for example, pri:38984 prbi:otfPrb1) (see [Metadata inside data capture files](#)) with the configured PCC Rule bases and PCC rules can be done with the output of the CLI command `show>mobile>pdn>bearer -context`, refer to section *Example Output Bearer Context IMSI Detail Statistics (PGW)* in the *CMG CLI Reference Guide*.

User plane call trace in Cloud MG, collects debugging statistics per UE at the:

- MG-VM
- ingress points on an LB-VM
- egress points on an LB-VM

User plane debugging statistics helps in troubleshooting and debugging inter-VM connectivity issues or packet drops. It allows the operator to identify whether there are any packet losses in the transport path from one VM to the other.

The reset of UE's user traffic statistics is supported without disabling the call trace job. The UP debugging statistics and PCAP capture files in LB-VMs are supported only in CMG and ePDG. PCC rule metadata information in PCAP files are supported only in no CUPS CMG.

Call trace functionality for UP capture is supported on the following gateways:

- SGW, SGW-U
- PGW/GGSN, PGW-U/GGSN-U
- ePDG
- UPF

Call trace can be enabled using CLI. Options exist to enable packet capture for a subscriber based on IMSI, MSISDN, IMEI, and MAC address. CP packet capture can also be enabled on specific reference point interfaces. Flexibility to either see CP messages for all subscribers on a reference point or for all reference points for a specific subscriber makes call trace a powerful debugging tool. Similarly, CP packet capture can be enabled on specific Service Based Interfaces in 5G SMF.

CP path management messages are also supported per reference points for all subscribers providing a better flexibility and usability in troubleshooting suspicious interfaces per reference point and/or Service Based Interfaces with NRF in 5G.

The following example shows the call trace CLI configuration commands.

```
config>mobile# profile
call-insight
  all-ues
  events
  format
  live-output
  path-management
  ref-point
  sba-service
  size-limit
  time-limit
```

```
config>mobile# profile
call-insight
```

```

ue
  events
  format
  live-output
  ref-point
  sba-service
  size-limit
  time-limit
  user-traffic

```

A call trace profile can be configured for either a specific UE or for all UEs using the `call-insight` command. The latter is configured under the `all-ues` context and is later assigned to a call trace job that will capture the configured information (events and signaling) for all UEs without matching to specific IMSI, IMEI, and so on. Having both `all-ues` and `ue` specific profiles provides the flexibility to define different jobs for various monitoring and troubleshooting scenarios.

In an `all-ues` profile and debug job, when a `live-output` is used, the configuration of a limit (either time or size limit) on the amount of traffic to be captured is not mandatory. When local storage on CF is used to store the PCAP file a `time-limit` and/or a `size-limit` configuration is mandatory. A profile configuration by default specifies some limits. For more information on how to configure or disable the limits, see the `config>mobile>profile>call-insight>all-ues` command.

An `all-ues` debug job is not allowed on 7750 SR MG, when no `live-output` is configured in the call trace profile, even if the time and/or size limits are configured.



NOTICE:

Nokia does not recommend to use the `debug>mobile>call-insight>all-ues` command in live deployments with more than 50 UEs because it may cause performance degradation.

Local storage on CF is allowed only for path management messages which are supported only in `all-ues` call trace profile in 7750 hardware.

A maximum of 10 different `all-ues` profiles can be configured. The `user-traffic` capturing is not supported with `all-ues` profiles/jobs.

In selective UE call trace, up to 50 different UEs can be traced by IMSI, IMEI, MSISDN, and/or MAC address for CP traffic and up to 15 different UEs for UP traffic. When `ue-mask` is used, the UP tracing starts only for the 15 UEs even if the mask configured in call trace matches more than 15 UEs, regardless if the call trace mask matches more than the 15 UEs. A maximum 10 different selective `ue` profiles can be configured (1 default included). User traffic capturing is independently configured in each profile and by default it is disabled.

User plane capturing supports the troubleshooting of VoLTE calls.

A total throughput of 4Mbit/s user data or approximately 270kb/s per UE for a max of 15 UEs can be captured by mobile gateway, which is adequate to troubleshoot VoLTE calls. In Cloud gateways call trace also supports unlimited data plane capturing for a specific subscriber when the **live-output**, **simulated-pcap** and the **internal** attribute in `user-traffic` parameter are enabled in call trace for the UE profile.

Events capturing via the events configuration provides advanced debugging capability.

**NOTE:**

The events configuration should only be used if needed and as instructed by Nokia Technical Support.

Capturing CP packets can be done for all signaling reference points or for a specific set of them. The ref-point/sba-service configuration is used to specify which reference point in 3G/4G and/or Service Based Interfaces in 5G will be captured.

Capturing path management packets can be done for all supported signaling reference points or for a specific set of them. The ref-point/sba-service configuration specifies which reference point in 3G/4G and/or Service Based Interfaces in 5G will be captured. Path management messages are supported only for all-ues call trace profile.

User plane capturing is enabled by setting the user traffic configuration. User plane capturing is not supported in an all-ues profile.

The end marker messages are captured in user traffic call trace, only if the user-traffic ingress, the user-traffic egress, or the user-traffic ingress egress command is configured in the call trace profile.

To capture outgoing end markers from the gateway (SGW, PGW or combined SGW/PGW) the user-traffic egress command must be configured.

To capture incoming end markers into any gateway (SGW, or combined SGW/PGW) the user-traffic ingress command must be configured.

To include both incoming and outgoing end markers, the user-traffic ingress egress command must be configured.

**NOTE:**

- Any other configuration does not have an impact to the end marker messages captures in user plane call trace
- When you enable user traffic capture in a call trace profile to capture end markers, the actual user data traffic is also captured. The operator must be cautious with the memory requirements for the call trace PCAP file storage. For end markers debugging, the appropriate end marker filters must be used to filter out end markers from the user data traffic

The following example shows the call trace CLI system commands which enable the context to configure the global call trace settings.

```
config>mobile# system
call-insight
location
disable
limit
```

```
primary
max-files-number
```

The following example shows the basic call trace CLI debug commands.

```
debug>mobile#
call-insight
  all-ues
  ue
  ue mac-address
  ue-mask
```

The debug commands create a call trace job for either a specific UE, a set of UEs, or for all UEs by linking an already created call insight profile to the job. At the same time, call trace is capturing statistics.

```
show>mobile#
call-insight
  all-ues
  files
  ue
  ue-mask
```

**NOTE:**

The call-insight profiles and debug jobs are not synchronized between the master and slave gateway in an ICR deployment. To avoid losing packet capturing in case of an SWO, the operator must configure call-insight profiles and debug jobs in both the master and slave gateway.

For detailed descriptions of all the call trace CLI commands and parameters, see the following sections in the *CMG CLI Reference Guide*.

- Call Trace Profile Commands
- Mobile-Gateway System Configuration Commands
- Call Trace Debug Commands
- Profile Show Commands

Refer to the following sections in the *CMG and CMG-a Troubleshooting Guide* about configuring call trace:

- Configuring a Call Trace Profile
- Configuring CF Storage Parameters
- Executing a Call Trace Debug Session
- Disabling Call Trace Debug Sessions
- Accessing Call Trace PCAP Files

- Viewing Call Trace Debug Logs via Live Output

18.1.1 Capturing of SWu ESP drop reason

The call trace functionality supports the capability to include the reason for an ESP packet drop in the Swu reference Point. If **swu-esp-drop** is included in the list of interfaces of a **call-insight** profile (only all UEs) and the selected format is PCAP, then the IMSI of the subscriber and the reason for the ESP packet drop is included in the Nokia call-trace header of the captured packet.

Table 77: List of supported packet drop reasons

Encoded drop reason	Description
IP_FRAG	Received ESP packet is fragmented
INV_IPV4_HDR_LEN	Invalid IPv4 header length in the received ESP packet
INV_IPV6_HDR_PLEN	Received ESP packet has invalid IPv6 header length
INV_UDP_HDR_LEN	Received ESP packet has invalid UDP header length
INV_UDP_PORT	Received ESP packet has invalid UDP port
INV_IP_PROTO	Received ESP packet has invalid IPv4 protocol
INV_ESP_LEN	Received ESP packet has invalid ESP length
INV_ESP_SPI	Received ESP packet has invalid ESP SPI
INV_ESP_SEQNUM	Received ESP packet has invalid ESP sequence number
INV_SADB_ENTRY	Received ESP packet has invalid SADB entry
SADB_MISS	Received ESP packet has ESP SPI pointing to inactive SA

Table 77: List of supported packet drop reasons (continued)

Encoded drop reason	Description
INV_TEP_TYPE	Received ESP packet has invalid tunnel endpoint address type
INV_REMOTE_TEP_AD	Received ESP packet has invalid remote tunnel endpoint addr
INV_LOCAL_TEP_ADD	Received ESP packet has invalid local tunnel endpoint addr
STALE_ESP_SEQNUM	Received ESP packet has stale ESP sequence number
DUP_ESP_SEQNUM	Received ESP packet has duplicate ESP sequence number
CRYPTO_ERROR	Received IP packet has a crypto error
INV_PAD_LEN	Received IP packet has invalid pad length
INV_PAD_BYTE	Received IP packet has invalid pad byte
INV_IPV4_HDR_CHKS	Received IP packet has invalid header checksum
INV_SRC_IP_ADDR	Received IP packet has invalid source IP address
INV_DST_IP_ADDR	Received IP packet has invalid destination IP address
CRYPT_MISS_ARGS	Received IP packet has missing crypto function args
CRYPT_INPR_LEN	Received IP packet has improper crypto packet len
CRYPT_INPR_CONF_K	Received IP packet has improper crypto conf key len
CRYPT_INPR_MAC_SE	Received IP packet has improper crypto MAC secret

Table 77: List of supported packet drop reasons (continued)

Encoded drop reason	Description
CRYPT_INB_MAC_MSM	Received IP packet has an authentication error
CRYPT_INB_OPENSSL	Received IP packet has other decrypt/auth failure
INV_REMOTE_TEP_PO	Received ESP packet has invalid remote tunnel endpoint port

18.2 Network trace

CMG can collect and stream 3G, 4G and 5G signaling messages to a third-party broker.

Network trace (vTAP) in the CMG collects and streams received and sent 3G, 4G and 5G external (SBI) signaling messages to a third-party broker (probe solution or network trace entity). The unencrypted data payloads are encapsulated using the HEP3 format. The HEP3 messages are then sent to the broker using HTTP/2.

The network trace feature monitors all SBI messages in the SMF, UPF, 4G and 5G signaling messages in SMF/PGW-C and UPF/PGW-U, and 4G CUPS and no-CUPS deployments. Network trace is supported in CMG CNF, CMG VNF, and CMG-a deployments.

The CMG encapsulates the received L7 payload, along with any additional information (for example, vendor-specific chunks), to the HEP3 format and does not mirror the original transport-level information (only the source IP address and port and destination IP address and port). In case of connection disruption with the 3rd party trace collector, only a limited amount (memory) of buffering is supported. The network trace module can handle connection disruption of few milliseconds with 3rd party trace collector. When the buffer is full, the CMG drops outgoing trace events. To communicate with the third-party probe, the CMG uses an IP connection.

For information about the vTAP functionality, the supported signaling interfaces, and the HEP3 interface specification, refer to the *CMG and NRD vTAP Interface Specification*.



NOTE:

- For 5G, all external (SBI) signaling messages are supported. The 3G/4G messages currently supported are described in Interface types table in the *CMG and NRD vTAP Interface Specification*
- The network trace feature significantly increases the overall HTTP/2 message throughput of the system (each incoming/outgoing message on each SBI interface is packaged with a HEP3 header and sent to the network trace broker). HTTP/2 LI traffic is not captured in network trace.

- The CPU impact of the network trace feature for SBA interfaces is between 5% to 15%. The CPU impact for the supported non-SBA interfaces is between 20% and 25%. The overall CPU impact from both SBA and non-SBA interfaces is approximately 20% to 25%. The exact value depends on the current usage and load patterns within the system (for example, attach rates). For systems running close to CPU overload thresholds, additional hardware capacity may be necessary to activate this feature. The network trace CPU overload threshold is 80% and when exceeded, network trace stops tracing SBI messages. Tracing resumes when the CPU level drops below 60%.

Related information

Configuring a network trace profile

Before enabling the network trace feature, you must create a network trace profile that specifies the configuration used by the network trace client service.

Enabling the network trace feature

Enable network trace to mirror all 3G, 4G and 5G external signaling messages and encapsulate them in Homer Encapsulation Protocol version 3 (HEP3) format.

18.2.1 Configuring a network trace profile

Before enabling the network trace feature, you must create a network trace profile that specifies the configuration used by the network trace client service.

Procedure

1. Create a network trace profile.

```
configure mobile-gateway profile network-trace-profile profile-name
```

Step example

```
configure mobile-gateway profile network-trace-profile network_trace_1
```

2. Configure the Packet Broker server.



NOTE:

In the current release, communication with only one Packet Broker server is supported.

```
configure mobile-gateway profile network-trace-profile profile-name packet-  
broker-server {ip-address} [port port]
```

Step example

```
configure mobile-gateway profile network-trace-profile network_trace_1  
packet-broker-server 10.10.1.255 80
```

Sub-steps

- a) Configure the Packet Broker server path.

```
configure mobile-gateway profile network-trace-profile profile-name
packet-broker-server server-path server-path-string
```

Step example

```
configure mobile-gateway profile network-trace-profile network_trace_1
packet-broker-server server-path server_path_1
```

Result

The network trace profile is created. To view the profile configuration, use the following command:

```
# show mobile-gateway profile network-trace-profile
```

```
=====
Network Trace Profile
=====
```

```
Profile Name           : network_trace_1
```

```
Packet Broker Server Options
```

```
Packet Broker Server Address : 10.10.1.255
```

```
Packet Broker Server Port   : 80
```

```
Packet Broker Server Path   : server_path_1
```

```
HTTP/2 Options
```

```
HTTP/2 Profile Name       : http2prof1
```

```
-----
Number of profiles        : 1
=====
```

Related information**Network trace**

CMG can collect and stream 3G, 4G and 5G signaling messages to a third-party broker.

Enabling the network trace feature

Enable network trace to mirror all 3G, 4G and 5G external signaling messages and encapsulate them in Homer Encapsulation Protocol version 3 (HEP3) format.

18.2.2 Enabling the network trace feature

Enable network trace to mirror all 3G, 4G and 5G external signaling messages and encapsulate them in Homer Encapsulation Protocol version 3 (HEP3) format.

Before you start

A network trace profile is configured.

Procedure

1. Configure the network trace interface in a network trace instance.

Configure the interface of the network trace client service that connects to the Packet Broker server.

```
configure mobile-gateway pdn gw-id network-trace instance instance-name  
interface interface-name [router router-name]
```

Step example

```
configure mobile-gateway pdn 1 network-trace instance test interface  
netw_trace_intf_1 router Base|2112
```

2. Associate a network trace profile.

Associate an already configured network trace profile with the network trace client service.

```
configure mobile-gateway pdn gw-id network-trace instance instance-name  
network-trace-profile profile-name
```

Step example

```
configure mobile-gateway pdn 1 network-trace instance test network-trace-  
profile network_trace_1
```

3. Enable the network trace tapping on SBA, non-SBA, or both interfaces.

```
configure mobile-gateway gw-id network-trace instance instance-name tapping-  
interface {all|sba|non-sba}
```

Step example

```
configure mobile-gateway pdn 1 network-trace instance test tapping-interface  
non-sba
```

4. Enable the network trace client service.

```
configure mobile-gateway pdn gw-id network-trace instance instance-name no  
shutdown
```

Step example

```
configure mobile-gateway pdn 1 network-trace instance test no shutdown
```

Related information

Configuring a network trace profile

Before enabling the network trace feature, you must create a network trace profile that specifies the configuration used by the network trace client service.

Network trace

CMG can collect and stream 3G, 4G and 5G signaling messages to a third-party broker.

18.3 Per Call Measurement Data

Per Call Measurement Data (PCMD) provides the ability to do performance analysis of mobile gateway functions on a per call session basis. Unlike KPI/KCI accounting that generates cumulative statistics over a specific time period, the PCMD feature supports analysis of each procedure at per call session granularity.

PCMD is a real-time diagnostics and troubleshooting tool that provides call and connection records for information such as call duration, quality, and disposition, and other important call events that occur during a call, such as call setup, failure, handover, and termination. PCMD is used to evaluate UE and network node performance. For example, the call flow procedure failure for the majority of UEs from a specific vendor can be used to provide a quick conclusion on the root cause of a problem at the UE level.

When used for a specific scenario, PCMD provides the ability to find the root cause of the problem due to non-compliance on the network node. It can help determine failure scenarios using end-to-end network-wide per call measurements from multiple nodes. End-to-end call session data from a live network allows quick response to failure by narrowing down the root cause without recreating the scenarios in a lab environment. Analysis of UP traffic volume is also supported. Uplink and downlink traffic patterns can be studied on a per subscriber basis or a pool of subscribers to evaluate subscriber experience.

PCMD data is generated on the gateway and sent to a data collector node over UDP stream. Call data streaming supports performance and root cause analysis of real-time network issues.

PCMD is supported on the following gateways:

- SGW
- PGW/GGSN
- combined SGW/PGW/GGSN
- SMF
- ePDG

PCMD can be enabled using CLI commands. A PCMD record is created when any of the supported 3GPP procedures are triggering (Attach, Detach, PDU Session Create, Service Request, and so on). The PCMD record is streamed to the provisioned real-time port when the call flow procedure ends (normally or abnormally). Each PCMD record contains fields for Subscriber information (IMSI, SUPI, MSISDN, GPSI, IMEI,

PEI), Procedure ID, Procedure Result, Procedure Failure Cause, and other data relevant to the procedure, such as Bearer ID, QoS Flow ID, APN, slice, default or dedicated bearer.

For information about PCMD configuration, CLI commands, record generation, and record format, and content, refer to the *CMG PCMD Guide*.

18.4 Key Performance Indicator and Key Capacity Indicator

Information about the Key Performance Indicator (KPI) and Key Capacity Indicator (KCI) counters that are used for post processing of the system behavior with respect to CP signaling, bearer traffic pattern, and system resource usage during peak and non-peak hours.



NOTE:

For more information about KPI-KCI counters including XML file naming conventions, storage, and formats, record types and counter module mappings, measurement object groups and indexes, and configuration guidelines, refer to the *CMG KPI-KCI Counters Guide*.

Statistical data about the system behavior helps to understand the network activity and helps the operators to plan their network to handle the signaling, data bandwidth demands, and system resource needs.

The mobile gateway periodically generates XML files containing performance and capacity measurements for the past collection period. The XML file generation is controlled by the configuration options under the `config>log>acct-policy` context. Statistics are organized in records and an accounting policy defines which records should be reported and at which period.

Measurements are reported in various dimensions, for example per peer or per message type. The dimension is a per counter specific record. Most of the statistics are reported per MG-VM in CMG. For more information about the dimensions, refer to the *CMG KPI-KCI Counters Guide*. For the exact dimension per record, refer to the *CMG KPI-KCI Counters Spreadsheet*.

The generated XML files are stored in the compact flash card and can be retrieved by any external management system using FTP. It is recommended to use the cf2 card for storing KPI/KCI, as the primary usage of cf1 is to keep configuration files and images. The files are compressed using the gzip format.

The XML files are temporarily being written in the `mob-gw-stats` directory and once they are closed, they are moved to the `mob-gw` directory. An SNMP trap is then sent to the configured SNMP trap sink.

The filename convention used for the XML files follows the 3GPP standard specification (32.432):

```
<Type><Startdate>.<Starttime>- [<Enddate>.]<Endtime>[_-<jobId>][_<UniqueId>][_<RC>].xml.gz
```

The NFM-M/MantaRay NM retrieves the XML files via FTP once it receives the relevant trap. After a successful FTP transfer, the NFM-M/MantaRay NM deletes the XML from the local mobile gateway storage.

The XML files contain a large set of raw statistics, covering all functional areas and features of the mobile gateway. These raw statistics can be used arbitrarily in post-processing analytics tools to calculate and report various KPIs and KCIs, for example:

- Success % = (Success) / (Success + Failures)
- Attach Rate = (Success) / Collection interval

The performance measurements conveyed in the XML files and the XML file format complies to the following 3GPP standards:

- The PM XML file follows the naming conventions specified in 3GPP TS 32.432
- The PM XML file follows the format specified in 3GPP TS 32.401 v5.5.0 Annex B, and Release 11.8.0 December 2013
- TS 32.401 Concept and requirements
- TS 32.404 Performance measurements - Definitions and template
- TS 32.406 Performance measurements Core Network (CN) Packet Switched (PS)

For more information about the KPI-KCI statistics, refer to the *CMG KPI-KCI Counters Spreadsheet*.

18.4.1 KPI-KCI collection configuration

The KPI-KCI XML file generation is controlled by the CLI configuration options under the context `config>log>acct -policy`. The accounting policy defines which KPI-KCI records should be reported and at which period.

For more information about configuring accounting policies, refer to the event and accounting logs information in the *7450 ESS, 7750 SR, 7950 XRS, and VSR System Management Guide*.

For information about the available records for CMG, and guidelines and examples for configuring accounting policies for KPI-KCI record generation, refer to the *CMG KPI-KCI Counters Guide*.

18.4.2 Statistics reporting

CMG supports the following KPI-KCI statistics reporting. For more information on configuring the following items, refer to the *CMG KPI-KCI Counters Guide*.

- QCI Profile Lists
- Statistics Reporting per ARP
- Statistics Reporting per PLMN
- Statistics Reporting per APN
- User Plane Statistics per Reference Point

18.5 Prometheus metrics

CMG when deployed as a CNF supports the use of Prometheus to scrape a set of KPI and KCI counters.

The NASC periodically polls the main OAM and MG containers. Based on the scraping interval, configured by the Kubernetes cluster administrator, Prometheus scrapes the Kubernetes workloads at an endpoint and port specified by the `prometheus.io/scrape`, `prometheus.io/path`, and `prometheus.io/port` annotations.

For more information about the use of Prometheus, refer to the section *Prometheus metrics* in the *CMG KPI-KCI Counters Guide*.

18.6 Telemetry

The Telemetry interface is used for reporting a limited number of KPI (counters) and KCI (gauges) for real-time monitoring of the mobile gateway (for example, when an immediate action is needed).

The Telemetry interface uses gRPC, a high-performance open source framework for streaming data, to report the KPIs and KCIs. For more information about configuring and using telemetry, refer to the *gRPC* section of the *SR OS System Management Guide*.

The KPI-KCI accounting XMLs described in *KPI-KCI collection configuration*, are generated with a minimum reporting interval of five minutes. Nokia recommends setting the reporting period to 15 minutes for system-level trend monitoring throughout the day.

The mobile gateway application counters supported by the Telemetry interface are modeled in YANG state models. The mobile gateway YANG models are used by the Telemetry infrastructure and by the gRPC clients and include the descriptions of each object (counter). These object descriptions serve as a reference for the user. The YANG state models are bundled with the software build.

For more information about the real-time counters for mobile gateway, refer to *CMG KPI-KCI Counters Guide*.

Related information

[Nokia Support Portal](#)

18.7 GTP ping

The reachability of GTP end-points, such as newly commissioned ENBs, MMEs and PGWs, can be checked using GTP ping commands.

Firewall policies block the ICMP ping packet, and therefore it cannot be used to check the end-point reachability in that case.

GTP end-points respond to the GTP echo requests, even when there are no bearer resources associated with the GTP end-point.

The GTP ping utility generates GTP (v1 or v2) echo requests based on the reference points listed in the command below.

```
oam gtp-ping {s11|s5c|s1u|s2b|s5u|s8c|s8u|gnc|gpc|s2bc|s2bu|s4c|s4u|s12|s2ac|s2au}
[router router-instance] destination ip-address port port-number gateway gw-id [retry-
count count] [time-out seconds]
```



NOTE:

Additional supported OAM commands are described in the *7750 SR OAM and Diagnostics Guide*.

18.8 UE ping

The **ue-ping** command checks the IP connectivity toward a UE that has an IP PDN session on the PGW/GGSN.

The tool uses Internet Control Message Protocol (ICMP) messages to contact the UE and displays the status of the connectivity test. ICMP Echo Request messages are encapsulated in GTP-U packets and are transmitted in the S1-U interface toward the specified UE.

The syntax of the **ue-ping** command is the following:

```
ue-ping {imsi|msisdn|ue-ip} value apn apn source-ip ip-address [size bytes]
[count requests] [timeout timeout] gateway gw-id
```

The source IP address of the ICMP Echo Request must be set by the user and it can be any IP address of the used IP type. The tunnel used to convey the ICMP messages is that of the default bearer of the PDN connection that matches the IP type of the specified source IP address for the specified APN. When the UE has more than one session with the same IP type in the same APN (multi-PDN connectivity), the UE IP address must be specified in the command.

The ICMP Echo Requests and the ICMP Echo Replies initiated from **ue-ping** command are not charged as user data.



NOTE:

- PMIP is not supported for UE ping.
- ICMP Echo Request and Response packets when initiated from the **ue-ping** tool, are not captured in the UP call trace.
- In CUPS deployments, the UE ping tool in the UPF only supports the UE IP address as the type of ping destination.

18.9 SNMP traps

CMG supports SNMP notifications (traps), which are sent to any registered SNMP Network Manager or NFM-M/MantaRay NM.

The SNMP notifications are grouped into two separate types:

alarms

These SNMP notifications provide the Fault Management application of the gateway. External management systems may treat these as notifications about state changes of stateful alarm objects; for example, an alarm becomes active. The list of active alarms, which may be maintained by an external management system, represents the list of present problems in the running system.

event notifications

These are SNMP notifications about various system events that may be of interest to external managers; for example, a new CDR file is closed and ready for pulling.

For information about CMG-specific notifications, refer to the *CMG Log Events Guide*. For information about the 7750 SR notifications that may also be supported by CMG, refer to the 7750 SR documentation.

Appendix A: Messages and information element reference

This appendix provides a list of the supported Information Element messages for the various protocols with a brief description of the attributes of each message.

RADIUS interface messages

Note that the abbreviations listed in the Presence column in the subsequent tables in this section represent the following characteristics.

- M=Mandatory
- C=Conditional
- O=Optional

Access-Request

Table 78: Generic Access-Request attributes

Attr no.	Attribute name	Presence	Description	Content type
1	User-Name	M	May be set to the IMSI, MSISDN, APN ⁷ , fixed value or user name received in the PCO ⁸ according to configuration. The maximum is 128 octets. For more information about the User-Name attribute, see APN override based on PCO .	String
2	User-Password	C	If PAP is used, this arrives in the PCO ⁸ . Alternatively, a locally configured password can be used.	String
3	CHAP-Password	C	If CHAP is used, this arrives in the PCO ⁸ .	String
4	NAS-IP-Address	C	The source IP address that the mobile gateway is using to communicate with the RADIUS server.	IPv4

⁷ APN is supported only in the authentication and not in the pre-authentication phase.

⁸ For information about the supported PCO IE, see [Protocol Configuration Options \(PCO\) IE](#).

Table 78: Generic Access-Request attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type
6	Service-Type	O	=2 "Framed"	Unsigned 32 bits
7	Framed-Protocol	O	=7 "GPRS PDP Context". This value is set for EPS bearers and PDP contexts.	Unsigned 32 bits
8	Framed-IP-Address	C	This should be set to the IP address received from the UE. If no IP address was received, this should be set to 0.	IPv4
9	Framed-IP-Netmask	C	Should be set to 255.255.255.255.	IPv4
30	Called-Station-Id	M	If this Access-Request message is part of pre-authentication, this attribute shall be set to the APN received via GTP. If part of the IP address allocation process (after the optional pre-authentication step), this attribute shall be set to either the APN received via GTP (the real APN) or the PGW APN to which the bearer/context has been mapped (the virtual APN). This is controlled via an APN-level configuration command.	UTF-8 encoded string
31	Calling-Station-Id	O	MSISDN in international format according to 3GPP TS 23.003, UTF-8 encoded decimal character.	UTF-8 encoded string
32	NAS-Identifier	O	The RADIUS client's IP address in UTF-8 encoded hexadecimal character.	String
44	Acct-Session-Id	M	The PGW/GGSN IP address (IPv4 or IPv6) and 3GPP-Charging-ID concatenated in a UTF-8 encoded hexadecimal character. If both addresses are configured, IPv4 is used.	UTF-8 encoded string

Table 78: Generic Access-Request attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type
55	Event-Timestamp	O	Sent by the TWAG or the 3GPP2 Home Agent or the PGW/GGSN with the time that this event occurred, in seconds since January 1, 1970 00:00 UTC. The Value field is four octets encoding an unsigned integer.	UTF-8 encoded string
60	CHAP-Challenge	C	Present if CHAP is used. It will arrive in the PCO ⁸ .	String
61	NAS-Port-Type	C	Port type for the PGW/GGSN. Per RFC 2865, value use is set to "Ethernet".	Unsigned 32 bits
95	NAS-IPv6-Address	C	This shall be the source IPv6 address when IPv6 transport is used.	IPv6
96	Framed-Interface-Id	C	This shall be set to the value received in the UE.	IPv6
97	Framed-IPv6-Prefix	C	This shall be set if an IPv6 address is provided.	IPv6

Table 79: VSAs 26/10415 - 3GPP vendor-specific Access-Request attributes

Attr no.	Sub-attribute name	Presence	Description	Content type
1	3GPP-IMSI	O	The IMSI.	UTF-8 encoded string
2	3GPP-Charging-Id	O	The Charging-ID for the IP-CAN bearer.	Unsigned 32

Table 79: VSAs 26/10415 - 3GPP vendor-specific Access-Request attributes (continued)

Attr no.	Sub-attribute name	Presence	Description	Content type
3	3GPP-PDP-Type	C	For an EPS bearer this indicates the PDN type (IPv4, IPv6, IPv4/IPv6, or non-IP). For a PDP context this indicates PDP type IP.	Unsigned 32
5	3GPP-GPRS-Negotiated-QoS-Profile	O	For a PDP context, this is the requested QoS-profile as specified in 24.008 Release 7. For an EPS bearer, the format differs for GBR and non-GBR bearers. For a GBR bearer, the QoS-profile shall contain the ARP, QCI, UL/DL MBR and UL/DL GBR. For a non-GBR bearer the QoS-profile shall contain the ARP, QCI and UL/DL APN-AMBR.	UTF-8 encoded string
6	3GPP-SGSN-Address	O	For a GGSN session, this is the IPv4 address of the SGSN. For a PGW session, this is the IPv4 address of the SGW. For a combined SGW/PGW session, this is by default the IPv4 address of the MME, but it can also be configured to be the IPv4 address of the SGW.	IPv4
7	3GPP-GGSN-Address	O	This is the IP address of the interface that the session is coming in from. For GGSN, this is the IPv4 address of the GTP control plane (Gn/Gp). For PGW, this is the IPv4 address of the GTP control plane (S5/S8/S2a/S2b) or PMIP control plane (S2a).	IPv4
8	3GPP-IMSI-MCC-MNC	O	This shall be set to the MCC and MNC extracted from the IMSI.	UTF-8 encoded string

Table 79: VSAs 26/10415 - 3GPP vendor-specific Access-Request attributes (continued)

Attr no.	Sub-attribute name	Presence	Description	Content type
9	3GPP-GGSN-MCC-MNC	O	This shall be set to the PGW/GGSN MCC and MNC (otherwise known as the PLMN ID.) Since we support the configuration of multiple PLMN IDs, we should use the first one in the list.	UTF-8 encoded string
10	3GPP-NSAPI	O	For GGSN, this is the NSAPI received via GTP. For PGW, this is the EPS Bearer Id received via GTP.	UTF-8 encoded string
12	3GPP-Selection-Mode	O	For PGW/GGSN this is the Selection Mode received via GTP.	UTF-8 encoded string
13	3GPP-Charging-Characteristics	O	For PGW/GGSN this is the Charging-Characteristics value received via GTP or produced internally from selected charging profile.	UTF-8 encoded string
15	GPP-SGSN-Ipv6-Address	O	For a GGSN session, this is the IPv6 address of the SGSN. For a PGW session, this is the IPv6 address of the SGW. For a combined SGW/PGW session, this is by default the IPv6 address of the MME, but it can also be configured to be the IPv6 address of the SGW.	Octet String
16	3GPP-GGSN-IPv6-Address	O	This is the IP address of the interface from which the session is coming. For GGSN, this is the IPv6 address of the GTP control plane (Gn/Gp). For PGW, this is the IPv6 address of the GTP control plane (S5/S8/S2a/S2b) or PMIP control plane (S2a).	Octet String

Table 79: VSAs 26/10415 - 3GPP vendor-specific Access-Request attributes (continued)

Attr no.	Sub-attribute name	Presence	Description	Content type
18	3GPP-SGSN-MCC-MNC	O	MCC and MNC extracted from the RAI within the Create PDP Context Request or Update PDP Context Request message.	UTF-8 encoded string
20	3GPP-IMEISV	O	International Mobile Equipment Id and its Software Version.	UTF-8 encoded string
21	3GPP-RAT-Type	O	Indicate which Radio Access Technology is currently serving the UE.	Octet String
22	3GPP-User-Location-Info	O	For PGW/GGSN this shall be set to the ULI received via GTP. Sent only when the UE is connected via 3GPP access.	Octet String
23	3GPP-MS-Time Zone	O	For GGSN this shall be set to the MS Time Zone received via GTP. For PGW this shall be set to the UE Time Zone received via GTP.	Octet String
27	3GPP-Allocate-IP-Type	O	Indicates whether the Access-Request is sent for user authentication only and/or for allocation of IPv4 and/or IPv6 address. The following values will be supported: • 0 — Do not allocate IPv4 address or IPv6 prefix, • 1 — Allocate IPv4 address, • 2 — Allocate IPv6 prefix, and • 3 — Allocate IPv4 address and IPv6 prefix. The RADIUS server is expected to interpret this as whether or not to allocate an IP address and decide based on other information what types of IP addresses to allocate.	Octet String

Access-Accept

Table 80: Generic Access-Accept attributes

Attr no.	Attribute name	Presence	Description	Content type	7750 SR MG release
1	User-Name	O	The RADIUS server may return the User-Name received in the Access-Request or it may return a substitute User-Name. If the RADIUS server changes the User-Name, this substitute User-Name shall be used in subsequent RADIUS messages. Max is 128 octets.	String	3.0R1
6	Service-Type	O	=2 "Framed"	Unsigned 32 bits	3.0R1
7	Framed-Protocol	O	=7 "GPRS PDP Context". This value is set for EPS bearers and PDP contexts.	Unsigned 32 bits	3.0R1
8	Framed-IP-Address	C	The IPv4 address allocated by the RADIUS server.	IPv4	3.0R1
9	Framed-IP-Netmask	C	The IPv4 netmask to be applied.	IPv4	3.0R1
22	Framed-Route	C	This attribute contains an IP subnet that should be reachable behind the UE. Multiple subnets require multiple framed-route attributes. The framed-route attribute also contains the gateway address, which may be an actual address or may be 0.0.0.0 meaning using the UE address as the gateway. Only a real gateway address is accepted if the same address is received in the Framed-IP-Address attribute. This attribute also supports an optional numeric metric after the addresses: "7.0.2.	UTF-8 encoded string (up to 4)	3.0R1

Table 80: Generic Access-Accept attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type	7750 SR MG release
			0/24 0.0.0.0 5". Additional numeric metrics are ignored, and any other non-numeric strings will cause authentication failure. The numeric metric should be [0, 65535]. If more than four framed-route attributes are received in an access-accept, then the additional attributes will be ignored.		
25	Class	O	This class identifier shall be used in subsequent RADIUS accounting messages if present in the Access-Accept.	String	3.0R1
27	Session-Timeout	O	The absolute timeout value to be enforced for the PDN connection. The value overrides the local configuration. The following values are supported: • value 0 disables the session timeout timer • a value greater than 0 but less than 1800 will be increased to 1800 and used • a value greater than or equal to 1800 up to 4294967295 will be used	Unsigned 32 bits	3.0R1
28	Idle-Timeout	O	The idle timeout value to be enforced for the PDN connection. The value overrides the local configuration. The following values are supported: • value 0 disables the idle timeout timer • a value greater than 0 but less than 300 will be increased to 300 and used • a value greater than or equal to 300 up to 2592000 will be used	Unsigned 32 bits	3.0R1

Table 80: Generic Access-Accept attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type	7750 SR MG release
			a value greater than 2592000 is rounded down to 2592000 and used		
64	Tunnel-Type	O	This attribute may be used for L2TP/PPP regen deployments and indicates the tunneling protocol. Only L2TP (3) type supported.	Integer	3.0R1
65	Tunnel-Medium-Type	O	This attribute may be used for L2TP/PPP regen deployments and indicates which transport medium to use when creating a tunnel. Only IPv4 (1) is supported.	Integer	3.0R1
67	Tunnel-Server-Endpoint	O	This attribute may be used for L2TP/PPP regen deployments and indicates the address of the server end of the tunnel. Only IPv4 Addresses supported.	String	3.0R1
83	Tunnel-Preference	O	This attribute may be used for L2TP/PPP regen deployments. If more than one set of tunneling attributes is returned by the RADIUS server to the tunnel initiator, this attribute should be included in each set to indicate the relative preference assigned to each tunnel.	Integer	3.0R1
88	Framed-Pool	C	The PGW/GGSN shall accept only one Framed-Pool attribute.	String	3.0R1
97	Framed-IPv6-Prefix	C	The IPv6 prefix allocated by the RADIUS server.	IPv6 Prefix	3.0R1

Table 80: Generic Access-Accept attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type	7750 SR MG release
99	Framed-IPv6-Route	C	This attribute contains an IPv6 subnet that must be reachable behind the UE. Multiple subnets require multiple Framed-IPv6-Route attributes. The Framed-IPv6-Route attribute also contains the gateway/next-hop address and optional metric. The gateway/next-hop address must be set to "::", "0:0:0:0:0:0:0:0" or to the IP address of the PDN session's IP address (the one returned in Framed-IPv6-Prefix attribute). This optional numeric metric is supported with first "metric-1". Additional numeric metrics are ignored and any other non-numeric strings will cause authentication failure. The numeric Metric-1 must be in the range of 0 to 65535. If more than four Framed-IPv6-Route attributes are received in an Access-Accept message, then the additional attributes are ignored.	IPv6 Prefix	10.0R7
100	Framed-IPv6-Pool	C	The PGW/GGSN shall accept only one Framed-IPv6-Pool attribute.	String	3.0R1

Table 81: VSAs 26/6527 - Nokia vendor-specific Access-Accept attributes

Sub-attr no.	Sub-attribute name	Presence	Description	Content type	7750 SR MG release
46	Alc-Tunnel-Group	O	This attribute may be used for L2TP/PPP regen deployments and includes the tunnel-group-name configured locally on the system to be used for APN and L2TP configuration.	String	4.0R1

Table 81: VSAs 26/6527 - Nokia vendor-specific Access-Accept attributes (continued)

Sub-attr no.	Sub-attribute name	Presence	Description	Content type	7750 SR MG release
104	Alc-Tunnel-Serv-Id	O	This attribute may be used for L2TP/PPP regen deployments and includes the VPRN service ID configured locally on the system to be used for the APN and L2TP configuration.	Integer	4.0R1
142	Alc-APN-Password	O	If returned as part of pre-authentication, this password shall be set in the User-Password attribute in the subsequent RADIUS Access-Request (that is, if RADIUS authentication is configured for the target APN). If returned as part of the IP address allocation, this attribute will be ignored.	RADIUS salt-encrypted string	4.0R1
143	Alc-APN-Name	O	If returned as part of pre-authentication, this attribute maps the session to this PGW APN (Virtual APN). If returned as part of the IP address allocation, this attribute will be ignored.	String	4.0R1
150	Alc-Charging-Prof-ID	O	This attribute is used for selecting a charging profile. If present, this value overrides any locally configured value. If a charging profile with the received value is not configured, or value is out of range, the charging profile selection is done via other available methods. If returned as part of pre-authentication, the attribute is ignored. Range is 0 to 255.	Unsigned 32 bit	4.0R1

Table 82: VSAs 26/94 - Nokia vendor-specific Access-Accept attributes

Sub-attr no.	Sub-attribute name	Presence	Description	Content type	7750 SR MG release
3	Nokia-Service-Name	O	This attribute indicates the name of one rulebase. The value field (UTF-8 encoded string) contains the rulebase name, which is installed in addition to any local default rulebases configured in the APN. The received rulebase is always session-specific, not bearer-specific. Modifying the QoS of the default bearer does not remove the AAA rulebases. The rulebases (and the rules contained within) must be configured with wildcard QCI/ARP rules, or a dedicated bearer should be created instead and the rulebases can remain. Up to 10 Nokia-Service-Name attributes are supported in the same Access-Accept message.	String, 1 to 247 octets The value must contain a 1-octet tag field preceding the actual value The tag field can contain any character excluding the NULL character	11.0.R1
11	Nokia-Session-Charging-Type	O	This attribute is used for selecting a charging profile. If present, this value overrides any locally configured value. If a charging profile with the received value is not configured, or value is out of range, the charging profile selection is done via other available methods. If returned as part of pre-authentication, the attribute is ignored. Range is 0 to 255.	Integer (1-octet)	11.0.R1

Table 83: VSAs 26/311 - Microsoft vendor-specific Access-Accept attributes

Sub-attr no.	Sub-attribute name	Presence	Description	Content type	7750 SR MG release
28	MS-Primary-DNS-Server	O	Contains the primary DNS server IPv4 address.	IPv4	3.0R1
29	MS-Secondary-DNS-Server	O	Contains the secondary DNS server IPv4 address.	IPv4	3.0R1
30	MS-Primary-NBNS-Server	O	Contains the primary NetBIOS name server IPv4 address.	IPv4	3.0R1
31	MS-Secondary-NBNS-Server	O	Contains the secondary NetBIOS server IPv4 address.	IPv4	3.0R1

Table 84: VSAs 26/10415 - 3GPP vendor-specific Access-Accept attributes

Sub-attr no.	Sub-attribute name	Presence	Description	Content type	7750 SR MG release
5	3GPP-GPRS-Negotiated-QoS-Profile	O	29.061 defines this VSA for Access-Request and Accounting-Request messages. However, MG also supports this in Access-Response messages as a way to set the QoS for a bearer/PDP context. The RADIUS server must send this in Release 8 format as described in 29.061, even if the request was for a PDP context rather than an EPS bearer. For an EPS bearer, the format differs for GBR and non-GBR bearers. For a GBR bearer, the QoS-profile shall contain the ARP, QCI, UL/DL MBR and UL/DL GBR. For a non-GBR bearer the QoS-profile shall contain the ARP, QCI and UL/DL APN-AMBR.	UTF-8 encoded string	3.0R1

Table 84: VSAs 26/10415 - 3GPP vendor-specific Access-Accept attributes (continued)

Sub-attr no.	Sub-attribute name	Presence	Description	Content type	7750 SR MG release
17	3GPP-IPv6-DNS-Servers	O	This shall contain the list of IPv6 DNS server addresses to be provided to the UE.	IPv6	3.0R1

Table 85: VSAs 26/2011 - Huawei vendor-specific Access-Accept attributes

Sub-attr no.	Sub-attribute name	Presence	Description	Content type	7750 SR MG release
231	HW-Transparent-Data	O	This attribute defines the transparent data, which may be needed to transfer to Ga/Gx/Gy Interfaces.	String	10.0R9
228	HW-Firewall-User-Profile-Names	O	This attribute defines the firewall user profile name.	String	10.0R9
229	HW-QoS-Profile-Name	O	This attribute defines the Qos profile name.	String	10.0R9

Accounting-Request (Start)

Table 86: Generic Accounting-Request (Start) attributes

Attr no.	Attribute name	Presence	Description	Content type
1	User-Name	O	May be set to the IMSI, MSISDN, APN ⁹ , fixed value, or username received in the	String

⁹ APN is supported only in the authentication and not in the pre-authentication phase.

Table 86: Generic Accounting-Request (Start) attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type
			PCO ¹⁰ according to configuration. The maximum is 128 octets. For TWAG open SSID (captive portal access) this attribute contains the user name received in the CoA. For more information about the User-Name attribute, see APN override based on PCO .	
4	NAS-IP-Address	O	The source IP address that the PGW/GGSN is using to communicate with the RADIUS server. For TWAG open SSID (captive portal access) this attribute contains the source IP address of WLAN AP.	IPv4
6	Service-Type	O	Indicates the type of service for this user. "Framed" = 2	Unsigned 32 bits
7	Framed-Protocol	O	This value is set for EPS bearers and PDP contexts.	Unsigned 32 bits
8	Framed-IP-Address	C	Set to the IPv4 address assigned to the UE. If no IPv4 address was assigned this attribute will sent with value = 0.	IPv4
25	Class	C	This attribute will be set to the value received in the Access-Accept.	String
30	Called-Station-Id	M	This attribute shall be set to either the APN received via GTP (the real APN) or the PGW/GGSN APN to which the bearer/context has been mapped (the virtual APN, or the defined public APN string). This is	UTF-8 encoded string

¹⁰ For information about the supported PCO IE, see [Protocol Configuration Options \(PCO\) IE](#).

Table 86: Generic Accounting-Request (Start) attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type
			controlled via an APN-level configuration command. For TWAG open SSID (captive portal access) this contains the MAC address of the WLAN AP. It has the format: AA-BB-CC-DD-EE-FF:SSID	
31	Calling-Station-Id	O	MSISDN in international format according to 3GPP TS 23.003. For TWAG open SSID (captive portal access) this contains the "ueid@apn-name" id string.	UTF-8 encoded string
32	NAS-Identifier	O	The RADIUS client's IP address in dot-decimal notation.	String
40	Acct-Status-Type	M	=1 "Start"	Unsigned 32 bits
41	Acct-Delay-Time	O	Indicates how many seconds the PGW/GGSN has been trying to send this record for, and can be subtracted from the time of arrival on the AAA server to find the approximate time (in seconds) of the event generating this Accounting-Request.	Unsigned 32 bits
44	Acct-Session-Id	M	PGW/GGSN IP address (IPv4 or IPv6) and Charging-ID concatenated in a UTF-8 encoded hexadecimal character. If both addresses are configured, IPv4 is used. With TWAG SaMOG RADIUS Accounting, this is TWAGs s2a interface address concatenated with the 3GPP-Charging-ID.	UTF-8 encoded string

Table 86: Generic Accounting-Request (Start) attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type
45	Acct-Authentic	O	"RADIUS" or "Local". The PGW/GGSN should set this to "RADIUS" if RADIUS authentication is used or to "Local" if RADIUS authentication is disabled.	Unsigned 32 bits
55	Event-Timestamp	O	Sent by the PGW/GGSN with the time that this event occurred, in seconds since January 1, 1970 00:00 UTC. The Value field is four octets encoding an unsigned integer.	UTF-8 encoded string
61	NAS-Port-Type	O	Port type for the PGW/GGSN. Per RFC 2865, value use is set to "Ethernet". For TWAG open SSID (captive portal access) this is set to "Wireless - 802.11".	Unsigned 32 bits
87	NAS-Port-ID	O	For TWAG open SSID (captive portal access) this contains the value received in the CoA.	String
95	NAS-IPv6-Address	O	Indicates the IPv6 address of the PGW/GGSN for communication with the AAA server.	IPv6
96	Framed-Interface-Id	O	Indicates the User IPv6 Interface Identifier. It is set to the value received in the PCO ¹⁰ .	IPv6
97	Framed-IPv6-Prefix	O	Indicates the IPv6 address prefix allocated for this user.	IPv6

Table 87: VSAs 26/10415 - 3GPP vendor-specific Accounting-Request (Start) attributes

Sub-attr no.	Sub-attribute name	Presence	Description	Content type
1	3GPP-IMSI	O	The IMSI for this user.	UTF-8 encoded string
2	3GPP-Charging-Id	O	The Charging-ID for the IP-CAN bearer.	Unsigned 32
3	3GPP-PDP-Type	C	For an EPS bearer this indicates the PDN type (IPv4, IPv6, IPv4/IPv6, or non-IP). For a PDP context this indicates PDP type IP.	Unsigned 32
5	3GPP-GPRS-Negotiated-QoS-Profile	O	For a PDP context, this is the requested QoS-profile as specified in 24.008 Release 7. For an EPS bearer, the format differs for GBR and non-GBR bearers. For a GBR bearer, the QoS-profile shall contain the ARP, QCI, UL/DL MBR and UL/DL GBR. For a non-GBR bearer the QoS-profile shall contain the ARP, QCI and UL/DL APN-AMBR.	UTF-8 encoded string
6	3GPP-SGSN-Address	O	For a GGSN session, this is the IPv4 address of the SGSN. For a PGW session, this is the IPv4 address of the SGW. For a combined SGW/PGW session, this is by default the IPv4 address of the MME, but it can also be configured to be the IPv4 address of the SGW.	IPv4
7	3GPP-GGSN-Address	O	This is the IP address of the interface that the session is coming in from. For GGSN, this is the IPv4 address of the GTP control plane (Gn/Gp).	IPv4

Table 87: VSAs 26/10415 - 3GPP vendor-specific Accounting-Request (Start) attributes (continued)

Sub-attr no.	Sub-attribute name	Presence	Description	Content type
			For PGW, this is the IPv4 address of the GTP control plane (S5/S8/S2a/S2b) or PMIP control plane (S2a). For TWAG, this is the IPv4 address of the PGW incoming s2a address.	
8	3GPP-IMSI-MCC-MNC	O	This is set to the MCC and MNC extracted from the IMSI.	UTF-8 encoded string
9	3GPP-GGSN-MCC-MNC	O	This is set to the PGW/GGSN MCC and MNC (otherwise known as the PLMN ID.) If multiple PLMN IDs are configured, the first one in the list will be used.	UTF-8 encoded string
10	3GPP-NSAPI	O	For GGSN, this is the NSAPI of the primary PDP context. For PGW, this is EPS Bearer Id of default bearer.	UTF-8 encoded string
12	3GPP-Selection-Mode	O	For PGW/GGSN this is the Selection Mode received via GTP.	UTF-8 encoded string
13	3GPP-Charging-Characteristics	O	For PGW/GGSN this is the Charging-Characteristics value received via GTP or produced internally from selected charging profile.	UTF-8 encoded string
15	3GPP-SGSN-IPv6-Address	O	For a GGSN session, this is the IPv6 address of the SGSN. For a PGW session, this is the IPv6 address of the SGW. For a combined SGW/PGW session, this is by default the IPv6 address of the MME, but it can also be configured to be the IPv6 address of the SGW.	Octet String

Table 87: VSAs 26/10415 - 3GPP vendor-specific Accounting-Request (Start) attributes (continued)

Sub-attr no.	Sub-attribute name	Presence	Description	Content type
16	3GPP-GGSN-IPv6-Address	O	This is the IP address of the interface from which the session is coming. For the GGSN, this is the IPv6 address of the GTP control plane (Gn/Gp). For the PGW, this is the IPv6 address of the GTP control plane (S5/S8/S2a/S2b) or the PMIP control plane (S2a). For the TWAG, this is the IPv6 address of the PGW incoming s2a address.	Octet String
18	3GPP-SGSN-MCC-MNC	O	MCC and MNC extracted from the RAI within the Create PDP Context Request or Update PDP Context Request message.	UTF-8 encoded string
20	3GPP-IMEISV	O	International Mobile Equipment Id and its Software Version.	UTF-8 encoded string
21	3GPP-RAT-Type	O	Indicate which Radio Access Technology is currently serving the UE.	Octet String
22	3GPP-User-Location-Info	O	For PGW/GGSN this is set to the ULI received via GTP. Sent only when the UE is connected via 3GPP access.	Octet String
23	3GPP-MS-Time Zone	O	For GGSN this is set to the MS Time Zone received via GTP. For PGW this is set to the UE Time Zone received via GTP.	Octet String

Table 88: VSAs 26/6527 - Nokia vendor-specific Accounting-Request (Start) attributes

Sub-attr no.	Sub-attribute name	Presence	Description	Content type
11	Alc-Subsc-ID-Str	O	For TWAG open SSID (captive portal access) this contains the value from the CoA.	String
13	Alc-SLA-Prof-Str	O	For TWAG open SSID (captive portal access) this contains the value from the CoA.	String

Accounting-Request (Stop)

Table 89: Generic Accounting-Request (Stop) attributes

Attr no.	Attribute name	Presence	Description	Content type
1	User-Name	O	May be set to the IMSI, MSISDN, APN ¹¹ , fixed value or username received in the PCO ¹² according to configuration. Max is 128 octets. For TWAG open SSID (captive portal access) this contains the user name received in the CoA. For more information about the User-Name attribute, see APN override based on PCO .	String
4	NAS-IP-Address	C	The source IP address that the PGW/GGSN is using to communicate with the RADIUS server.	IPv4

¹¹ APN is supported only in the authentication and not in the pre-authentication phase.

¹² For information about the supported PCO IE, see [Protocol Configuration Options \(PCO\) IE](#).

Table 89: Generic Accounting-Request (Stop) attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type
			For TWAG open SSID (captive portal access) this contains the source IP address of WLAN AP.	
6	Service-Type	O	Indicates the type of service for this user. =2 "Framed"	Unsigned 32 bits
7	Framed-Protocol	O	Indicates the type of protocol for this user. This value is set for EPS bearers and PDP contexts.	Unsigned 32 bits
8	Framed-IP-Address	C	Indicates the IPv4 address assigned to the UE. If no IPv4 address was assigned this attribute is sent with value = 0.	IPv4
25	Class	O	This attribute is set to the value received in the Access-Accept.	String
30	Called-Station-Id	M	This attribute is set to either the APN received via GTP (the real APN, or the defined public APN string) or the PGW APN to which the bearer/context has been mapped (the virtual APN). This is controlled via an APN-level configuration command. For TWAG open SSID (captive portal access) this contains the MAC address of the WLAN AP.	UTF-8 encoded string
31	Calling-Station-Id	O	MSISDN in international format according to 3GPP TS 23.003. For TWAG open SSID (captive portal access) this contains the "ueid@apn-name" id string.	UTF-8 encoded string

Table 89: Generic Accounting-Request (Stop) attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type
32	NAS-Identifier	O	The RADIUS client's IP address in dot-decimal notation.	String
40	Acct-Status-Type	O	=2 "Stop"	Unsigned 32 bits
41	Acct-Delay-Time	O	Indicates how many seconds the PGW/GGSN has been trying to send this record for, and can be subtracted from the time of arrival on the AAA server to find the approximate time (in seconds) of the event generating this Accounting-Request.	Unsigned 32 bits
42	Acct-Input-Octets	C	The octets received from the UE on this IP CAN bearer.	Unsigned 32 bits
43	Acct-Output-Octets	C	The octets sent to the UE on this IP CAN bearer.	Unsigned 32 bits
44	Acct-Session-Id	M	PGW/GGSN IP address (IPv4 or IPv6) and Charging-ID concatenated in a UTF-8 encoded hexadecimal character. If both addresses are configured, IPv4 is used. With TWAG SaMOG RADIUS Accounting, this is TWAGs s2a interface address concatenated with the 3GPP-Charging-ID.	UTF-8 encoded string
45	Acct-Authentic	O	"RADIUS" or "Local". The PGW/GGSN should set this to "RADIUS" if RADIUS authentication is used or to "Local" if RADIUS authentication is disabled.	Unsigned 32 bits
46	Acct-Session-Time	C	The duration of the session.	Unsigned 32 bits

Table 89: Generic Accounting-Request (Stop) attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type
47	Acct-Input-Packets	C	The number of packets received from the UE on this IP CAN bearer.	Unsigned 32 bits
48	Acct-Output-Packets	C	The number of packets sent to the UE on this IP CAN bearer.	Unsigned 32 bits
49	Acct-Terminate-Cause	C	Per RFC 2866. This indicates how the session was terminated.	Unsigned 32 bits
52	Acct-Input-Gigawords	C	Indicates how many times the Acct-Input-Octets counter has wrapped around 2^{32} over the course of this service being provided.	The Value field is four octets
53	Acct-Output-Gigawords	C	Indicates how many times the Acct-Output-Octets counter has wrapped around 2^{32} in the course of delivering this service.	The Value field is four octets
55	Event-Timestamp	O	Sent by the TWAG, the 3GPP2 Home Agent or the PGW/GGSN with the time that this event occurred, in seconds since January 1, 1970 00:00 UTC. The Value field is four octets encoding an unsigned integer.	UTF-8 encoded string
61	NAS-Port-Type	O	Port type for the PGW/GGSN. Per RFC 2865, value use is set to "Ethernet". For TWAG open SSID (captive portal access) this is set to "Wireless - 802.11"	Unsigned 32 bits
87	NAS-PORT-ID	O	For TWAG open SSID (captive portal access) this will have the value received in the CoA.	String

Table 89: Generic Accounting-Request (Stop) attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type
95	NAS-IPv6-Address	O	Indicates the IPv6 address of the PGW/ GGSN for communication with the AAA server.	IPv6
96	Framed-Interface-Id	O	Indicates the User IPv6 Interface Identifier. Set to the value received in the PCO ¹² .	IPv6
97	Framed-IPv6-Prefix	O	Indicates the IPv6 address prefix allocated for this user.	IPv6

Table 90: VSAs 26/10415 - 3GPP-Vendor-Specific Accounting-Request (Stop) Attributes

Sub-attr no.	Sub-attribute name	Presence	Description	Content type
1	3GPP-IMSI	O	Indicates the IMSI for this user.	UTF-8 encoded string
2	3GPP-Charging-Id	O	The Charging-ID for the IP-CAN bearer.	Unsigned 32
3	3GPP-PDP-Type	O	For an EPS bearer this indicates the PDN type (IPv4, IPv6, IPv4/IPv6, or non-IP). For a PDP context this indicates PDP type IP.	Unsigned 32
5	3GPP-GPRS-Negotiated-QoS-Profile	O	For a PDP context, this is the requested QoS-profile as specified in 24.008 Release 7. For an EPS bearer, the format differs for GBR and non-GBR bearers. For a GBR bearer, the QoS-profile shall contain the ARP, QCI, UL/DL MBR and UL/DL GBR. For a non-GBR bearer the QoS-profile shall	UTF-8 encoded string

Table 90: VSAs 26/10415 - 3GPP-Vendor-Specific Accounting-Request (Stop) Attributes (continued)

Sub-attr no.	Sub-attribute name	Presence	Description	Content type
			contain the ARP, QCI and UL/DL APN-AMBR.	
6	3GPP-SGSN-Address	O	For a GGSN session, this is the IPv4 address of the SGSN. For a PGW session, this is the IPv4 address of the SGW. For a combined SGW/PGW session, this is by default the IPv4 address of the MME, but it can also be configured to be the IPv4 address of the SGW.	IPv4
7	3GPP-GGSN-Address	O	This is the IP address of the interface that the session is coming in from. For GGSN, this is the IPv4 address of the GTP control plane (Gn/Gp). For PGW, this is the IPv4 address of the GTP control plane (S5/S8/S2a/S2b) or PMIP control plane (S2a). For TWAG, this is the IPv4 address of the PGW incoming s2a address.	IPv4
8	3GPP-IMSI-MCC-MNC	O	Set to the MCC and MNC extracted from the IMSI.	UTF-8 encoded string
9	3GPP-GGSN-MCC-MNC	O	Includes the PGW/GGSN MCC and MNC (otherwise known as the PLMN ID.) If multiple PLMN IDs are configured the first one in the list is used.	UTF-8 encoded string
10	3GPP-NSAPI	O	For GGSN, this is the NSAPI of the primary PDP context. For PGW, this is EPS Bearer Id of default bearer.	UTF-8 encoded string

Table 90: VSAs 26/10415 - 3GPP-Vendor-Specific Accounting-Request (Stop) Attributes (continued)

Sub-attr no.	Sub-attribute name	Presence	Description	Content type
11	3GPP-Session-Stop-Indicator	O	Indicates that the last PDP context or EPS bearer of an IP-CAN session has been released and therefore this IP-CAN session has been terminated. Value set to all 1.	One byte of bit string
12	3GPP-Selection-Mode	O	For PGW/GGSN this is the Selection Mode received via GTP.	UTF-8 encoded string
13	3GPP-Charging-Characteristics	O	For PGW/GGSN this is the Charging-Characteristics value received via GTP or produced internally from selected charging profile.	UTF-8 encoded string
15	3GPP-SGSN-IPv6-Address	O	For a GGSN session, this is the IPv6 address of the SGSN. For a PGW session, this is the IPv6 address of the SGW. For a combined SGW/PGW session, this is by default the IPv6 address of the MME, but it can also be configured to be the IPv6 address of the SGW.	Octet String
16	3GPP-GGSN-IPv6-Address	O	This is the IP address of the interface from which the session is coming. For GGSN, this is the IPv6 address of the GTP control plane (Gn/Gp). For PGW, this is the IPv6 address of the GTP control plane (S5/S8/S2a/S2b) or PMIP control plane (S2a). For TWAG, this is the IPv6 address of the PGW incoming s2a address.	Octet String

Table 90: VSAs 26/10415 - 3GPP-Vendor-Specific Accounting-Request (Stop) Attributes (continued)

Sub-attr no.	Sub-attribute name	Presence	Description	Content type
18	3GPP-SGSN-MCC-MNC	O	MCC and MNC extracted from the RAI within the Create PDP Context Request or Update PDP Context Request message.	UTF-8 encoded string
20	3GPP-IMEISV	O	International Mobile Equipment Id and its Software Version.	UTF-8 encoded string
21	3GPP-RAT-Type	O	Indicate which Radio Access Technology is currently serving the UE.	Octet String
22	3GPP-User-Location-Info	O	For PGW/GGSN this shall be set to the ULI received via GTP. Sent only when the UE is connected via 3GPP access.	Octet String
23	3GPP-MS-Time Zone	O	For GGSN this shall be set to the MS Time Zone received via GTP. For PGW this shall be set to the UE Time Zone received via GTP.	Octet String

Table 91: VSAs 26/6527 - Nokia-Vendor-Specific Accounting-Request (Stop) Attributes

Sub-attr no.	Sub-attribute name	Presence	Description	Content type
11	Alc-Subsc-ID-Str	O	For TWAG open ssid (captive portal access) this will have the value from the CoA.	String
13	Alc-SLA-Prof-Str	O	For TWAG open ssid (captive portal access) this will have the value from the CoA.	String

Accounting-Request (Interim-Update)

Table 92: Generic Accounting-Request (Interim-Update) attributes

Attr no.	Attribute name	Presence	Description	Content type
1	User-Name	O	May be set to the IMSI, MSISDN, APN¹³, fixed value or username received in the PCO¹⁴ according to configuration. Max is 128 octets. For TWAG open SSID (captive portal access) this contains the user name received in the CoA. For more information about the User-Name attribute, see APN override based on PCO.	String
4	NAS-IP-Address	C	The source IP address that the PGW/GGSN is using to communicate with the RADIUS server. For TWAG open SSID (captive portal access) this contains the source IP address of WLAN AP.	IPv4
6	Service-Type	O	Indicates the type of service for this user. =2 "Framed"	Unsigned 32 bits
7	Framed-Protocol	O	Indicates the type of protocol for this user. This value is set for EPS bearers and PDP contexts.	Unsigned 32 bits
8	Framed-IP-Address	C	This is set to the IPv4 address assigned to the UE. If no IPv4 address was assigned this attribute should be sent with value = 0.	IPv4

¹³ APN is supported only in the authentication and not in the pre-authentication phase.

¹⁴ For information about the supported PCO IE, see [Protocol Configuration Options \(PCO\) IE](#).

Table 92: Generic Accounting-Request (Interim-Update) attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type
25	Class	O	This attribute shall be set to the value received in the Access-Accept.	String
30	Called-Station-Id	M	This attribute shall be set to either the APN received via GTP (the real APN) or the PGW APN to which the bearer/context has been mapped (the virtual APN, or the defined public APN string). This is controlled via an APN-level configuration command. For TWAG open SSID (captive portal access) this contains the MAC address of the WLAN AP.	UTF-8 encoded string
31	Calling-Station-Id	O	MSISDN in international format according to 3GPP TS 23.003. For TWAG open SSID (captive portal access) this contains the "ueid@apn-name" id string.	UTF-8 encoded string
32	NAS-Identifier	O	The RADIUS client's IP address in dot-decimal notation.	String
40	Acct-Status-Type	M	=3 "Interim-Update"	Unsigned 32 bits
41	Acct-Delay-Time	O	Indicates how many seconds the PGW/GGSN has been trying to send this record for, and can be subtracted from the time of arrival on the AAA server to find the approximate time (in seconds) of the event generating this Accounting-Request.	Unsigned 32 bits

Table 92: Generic Accounting-Request (Interim-Update) attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type
42	Acct-Input-Octets	C	The octets received from the UE on this IP CAN bearer.	Unsigned 32 bits
43	Acct-Output-Octets	C	The octets sent to the UE on this IP CAN bearer.	Unsigned 32 bits
44	Acct-Session-Id	M	PGW/GGSN IP address (IPv4 or IPv6) and Charging-ID concatenated in a UTF-8 encoded hexadecimal character string. If both addresses are configured, IPv4 is used. With TWAG SaMOG RADIUS Accounting, this is TWAGs s2a interface address concatenated with the 3GPP-Charging-ID.	UTF-8 encoded string
45	Acct-Authentic	O	"RADIUS" or "Local". The PGW/GGSN should set this to "RADIUS" if RADIUS authentication is used or to "Local" if RADIUS authentication is disabled.	Unsigned 32 bits
46	Acct-Session-Time	O	The duration of the session.	Unsigned 32 bits
47	Acct-Input-Packets	O	The number of packets received from the UE on this IP CAN bearer.	Unsigned 32 bits
48	Acct-Output-Packets	O	The number of packets sent to the UE on this IP CAN bearer.	Unsigned 32 bits
52	Acct-Input-Gigawords	C	Indicates how many times the Acct-Input-Octets counter has wrapped around 2 ³² over the course of this service being provided	The Value field is four octets

Table 92: Generic Accounting-Request (Interim-Update) attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type
53	Acct-Output-Gigawords	C	Indicates how many times the Acct-Output-Octets counter has wrapped around 2^{32} in the course of delivering this service	The Value field is four octets
55	Event-Timestamp	O	Sent by the PGW/GGSN with the time that this event occurred, in seconds since January 1, 1970 00:00 UTC. The Value field is four octets encoding an unsigned integer.	UTF-8 encoded string
61	NAS-Port-Type	O	Port type for the PGW/GGSN. Per RFC 2865, value use is set to "Ethernet".	Unsigned 32 bits
87	NAS-PORT-ID	O	For TWAG open SSID (captive portal access) this contains the value received in the CoA.	String
95	NAS-IPv6-Address	O	IPv6 address of the PGW/GGSN for communication with the AAA server.	IPv6
96	Framed-Interface-Id	O	User IPv6 Interface Identifier. Set to the value received in the PCO ¹⁴ .	IPv6
97	Framed-IPv6-Prefix	O	IPv6 address prefix allocated for this user.	IPv6

Table 93: VSAs 26/10415 - 3GPP vendor-specific Accounting-Request (Interim-Update) attributes

Sub-attr no.	Sub-attribute name	Presence	Description	Content type
1	3GPP-IMSI	O	Indicates the IMSI for this user.	UTF-8 encoded string
2	3GPP-Charging-Id	O	The Charging-ID for the IP-CAN bearer.	Unsigned 32
3	3GPP-PDP-Type	C	For an EPS bearer this indicates the PDN type (IPv4, IPv6, IPv4/IPv6, or non-IP). For a PDP context this indicates PDP type IP.	Unsigned 32
5	3GPP-GPRS-Negotiated-QoS-Profile	O	For a PDP context, this is the requested QoS-profile as specified in 24.008 Release 7. For an EPS bearer, the format differs for GBR and non-GBR bearers. For a GBR bearer, the QoS-profile shall contain the ARP, QCI, UL/DL MBR and UL/DL GBR. For a non-GBR bearer the QoS-profile shall contain the ARP, QCI and UL/DL APN-AMBR.	UTF-8 encoded string
6	3GPP-SGSN-Address	O	For a GGSN session, this is the IPv4 address of the SGSN. For a PGW session, this is the IPv4 address of the SGW. For a combined SGW/PGW session, this is by default the IPv4 address of the MME, but it can also be configured to be the IPv4 address of the SGW.	IPv4
7	3GPP-GGSN-Address	O	This is the IP address of the interface that the session is coming in from. For GGSN, this is the IPv4 address of the GTP control plane (Gn/Gp).	IPv4

Table 93: VSAs 26/10415 - 3GPP vendor-specific Accounting-Request (Interim-Update) attributes (continued)

Sub-attr no.	Sub-attribute name	Presence	Description	Content type
			For PGW, this is the IPv4 address of the GTP control plane (S5/S8/S2a/S2b) or PMIP control plane (S2a). For TWAG, this is the IPv4 address of the PGW incoming s2a address.	
8	3GPP-IMSI-MCC-MNC	O	Includes the MCC and MNC extracted from the IMSI.	UTF-8 encoded string
9	3GPP-GGSN-MCC-MNC	O	Includes the PGW/GGSN MCC and MNC (otherwise known as the PLMN ID.) If multiple PLMN IDs are configured the first one in the list will be used.	UTF-8 encoded string
10	3GPP-NSAPI	O	For GGSN, this is the NSAPI of the primary PDP context. For PGW, this is EPS Bearer Id of default bearer.	UTF-8 encoded string
12	3GPP-Selection-Mode	O	For PGW/GGSN this is the Selection Mode received via GTP.	UTF-8 encoded string
13	3GPP-Charging-Characteristics	O	For PGW/GGSN this is the Charging-Characteristics value received via GTP or produced internally from selected charging profile.	UTF-8 encoded string
15	3GPP-SGSN-IPv6-Address	O	For a GGSN session, this is the IPv6 address of the SGSN. For a PGW session, this is the IPv6 address of the SGW. For a combined SGW/PGW session, this is by default the IPv6 address of the MME,	Octet String

Table 93: VSAs 26/10415 - 3GPP vendor-specific Accounting-Request (Interim-Update) attributes (continued)

Sub-attr no.	Sub-attribute name	Presence	Description	Content type
			but it can also be configured to be the IPv6 address of the SGW.	
16	3GPP-GGSN-IPv6-Address	O	This is the IP address of the interface that the session is coming in from. For GGSN, this is the IPv6 address of the GTP control plane (Gn/Gp). For PGW, this is the IPv6 address of the GTP control plane (S5/S8/S2a/S2b) or PMIP control plane (S2a). For TWAG, this is the IPv6 address of the PGW incoming s2a address.	Octet String
18	3GPP-SGSN-MCC-MNC	O	MCC and MNC extracted from the RAI within the Create PDP Context Request or Update PDP Context Request message.	UTF-8 encoded string
20	3GPP-IMEISV	O	International Mobile Equipment Id and its Software Version.	UTF-8 encoded string
21	3GPP-RAT-Type	O	Indicate which Radio Access Technology is currently serving the UE.	Octet String
22	3GPP-User-Location-Info	O	For PGW/GGSN this shall be set to the ULI received via GTP. Sent only when the UE is connected via 3GPP access.	Octet String
23	3GPP-MS-Time Zone	O	For GGSN this shall be set to the MS Time Zone received via GTP. For PGW this shall be set to the UE Time Zone received via GTP.	Octet String

Table 94: VSAs 26/6527 - Nokia vendor-specific Accounting-Request (Interim-Update) attributes

Sub-attr no.	Sub-attribute name	Presence	Description	Content type
11	Alc-Subsc-ID-Str	O	For TWAG open SSID (captive portal access) this will have the value from the CoA.	String
13	Alc-SLA-Prof-Str	O	For TWAG open SSID (captive portal access) this will have the value from the CoA.	String

Accounting-Request (Accounting-On)

Table 95: Generic Accounting-Request (Accounting-On) attributes

Attr no.	Attribute name	Presence	Description	Content type
4	NAS-IP-Address	C	IPv4 address of the PGW/GGSN for communication with the AAA server.	IPv4
30	Called-Station-Id	O	This attribute shall be set to either the APN received via GTP (the real APN) or the PGW/GGSN APN to which the bearer/context has been mapped (the virtual APN, or the defined public APN string). This is controlled via an APN-level configuration command	UTF-8 encoded string
32	NAS-Identifier	O	The RADIUS client's IP address in dot-decimal notation.	String
40	Acct-Status-Type	M	=7 "Accounting-On"	Unsigned 32 bits

Table 95: Generic Accounting-Request (Accounting-On) attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type
44	Acct-Session-Id	O	PGW/GGSN IP address (IPv4 or IPv6) and Charging-ID (all zeros) concatenated in a UTF-8 encoded hexadecimal character string.If both addresses are configured, IPv4 is used. With TWAG SaMOG RADIUS Accounting, this is TWAGs s2a interface address concatenated with the 3GPP-Charging-ID.	UTF-8 encoded string
55	Event-Timestamp	O	Sent by the PGW/GGSN with the time that this event occurred, in seconds since January 1, 1970 00:00 UTC. The Value field is four octets encoding an unsigned integer.	UTF-8 encoded string
95	NAS-IPv6-Address	C	IPv6 address of the PGW/GGSN for communication with the AAA server.	IPv6

Accounting-Request (Accounting-Off)

Table 96: Generic Accounting-Request (Accounting-Off) attributes

Attr no.	Attribute name	Presence	Description	Content type
4	NAS-IP-Address	C	IPv4 address of the PGW/GGSN for communication with the AAA server.	IPv4
30	Called-Station-Id	O	This attribute shall be set to either the APN received via GTP (the real APN) or the PGW/GGSN APN to which the bearer/context has been mapped (the virtual APN, or the defined public APN string). This is	UTF-8 encoded string

Table 96: Generic Accounting-Request (Accounting-Off) attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type
			controlled via an APN-level configuration command.	
32	NAS-Identifier	O	The RADIUS client's IP address in dot-decimal notation.	String
40	Acct-Status-Type	M	=8 "Accounting-Off"	Unsigned 32 bits
44	Acct-Session-Id	O	PGW/GGSN IP address (IPv4 or IPv6) and Charging-ID (all zeros) concatenated in a UTF-8 encoded hexadecimal character string. If both addresses are configured, IPv4 is used. With TWAG SaMOG RADIUS Accounting, this is TWAGs s2a interface address concatenated with the 3GPP-Charging-ID.	UTF-8 encoded string
55	Event-Timestamp	O	Sent by the PGW/GGSN with the time that this event occurred, in seconds since January 1, 1970 00:00 UTC. The Value field is four octets encoding an unsigned integer.	UTF-8 encoded string
95	NAS-IPv6-Address	C	IPv6 address of the PGW/GGSN for communication with the AAA server.	IPv6

RADIUS CoA-Request

Table 97: RADIUS CoA-Request attributes

Attr no.	Attribute name	Presence	Description	Content type
1	User-Name	O	This will be the user-name that the PGW/ GGSN used in the Access-Request. For TWAG open SSID (captive portal access) this will be user-name that the subscriber used in the portal auth. The length of the user-name cannot exceed 64 bytes.	String
8	Framed-IP-Address	O	IPv4 address assigned to the UE.	IPv4
44	Acct-Session-Id	O	PGW/GGSN IP address (IPv4 or IPv6) and Charging-ID concatenated in a UTF-8 encoded hexadecimal character string.	UTF-8 encoded string
87	NAS-Port-ID	O	For TWAG open SSID (captive portal access) this will have the "ueid@apn-name" name string.	String

Table 98: VSAs 2626/6527/11 - Nokia vendor-specific RADIUS CoA-Request attributes

Sub-attr no.	Sub-attribute name	Presence	Description	Content type
11	Alc-Subsc-ID-Str	O	For TWAG open SSID (captive portal access) this will have a subscriber-id string (typically the MSISDN when UE is a mobile)	String
13	Alc-SLA-Prof-Str	O	For TWAG open SSID (captive portal access) this will have the policy name that maps to access and redirect policies	String

Table 98: VSAs 2626/6527/11 - Nokia vendor-specific RADIUS CoA-Request attributes (continued)

Sub-attr no.	Sub-attribute name	Presence	Description	Content type
			locally configured for the UE. This attribute will also trigger RADIUS START or STOP depending on the policy value change.	

Disconnect-Request

Table 99: Generic Disconnect-Request attributes

Attr no.	Attribute name	Presence	Description	Content type	7750 SR MG release
1	User-Name	O	This will be the user-name that the PGW/GGSN used in the Access-Request.	String	3.0R1
8	Framed-IP-Address	C	Set to the IPv4 address assigned to the UE. If no IPv4 address was assigned this attribute will be sent with value = 0.	IPv4	3.0R1
33	Proxy-State	C	The Proxy-State attribute is defined in RFC 2865. The value is not interpreted by the GGSN/PGW, but only echoed in the response. Up to 10 Proxy-State attributes can be copied from the request to the response packet.	String	10.0R1
44	Acct-Session-Id	M	PGW/GGSN IP address (IPv4 or IPv6) and Charging-ID concatenated in a UTF-8 encoded hexadecimal character string.	UTF-8 encoded string	3.0R1

Table 99: Generic Disconnect-Request attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type	7750 SR MG release
			Must match the Acct-Session-Id sent with Accounting-Request.		
96	Framed-Interface-Id	C	Will have the IPv6 interface identifier if it was provided to the RADIUS server.	Octet String	3.0R1
97	Framed-IPv6-Prefix	C	Will have the IPv6 prefix known for the PDN connection if it was provided to the RADIUS server.	Octet String	3.0R1

Table 100: VSAs 26/10415 - 3GPP vendor-specific Disconnect-Request attributes

Sub-attr no.	Sub-attribute name	Presence	Description	Content type	7750 SR MG release
19	3GPP-Teardown-Indication	O	Indicates to the PGW/GGSN that all IP CAN bearers or PDP contexts shall be released for this IP-CAN session.	One byte with 1 bit defined and 7 spare bits	3.0R1

Disconnect-Response (ACK)

Table 101: Disconnect-Response (ACK) attributes

Attr no.	Attribute name	Presence	Description	Content type	7750 SR MG release
33	Proxy-State	C	The Proxy-State attribute is defined in RFC 2865. The value is the same as received in earlier Disconnect-Request in the response. Up to 10 Proxy-State attributes can be copied from the request to the response packet.	String	10.0R1
49	Acct-Terminate-Cause	O	This indicates how the session was terminated. Per RFC 5176, the value set to 6 for Admin-Reset.	Unsigned 32 bits	10.0R1
55	Event-Time-Stamp	O	Time that this event occurred, in seconds since January 1, 1970 00:00 UTC. The Value field is four octets encoding an unsigned integer.	UTF-8 encoded string	10.0R1

Disconnect-Response (NAK)

Table 102: Disconnect-Response (NAK) attributes

Attr no.	Attribute name	Presence	Description	Content type	7750 SR MG release
33	Proxy-State	C	The Proxy-State attribute is defined in RFC 2865. The value is the same as received in earlier Disconnect-Request in the response. Up to 10 Proxy-State attributes can be copied from the request to the response packet.	String	10.0R1

Table 102: Disconnect-Response (NAK) attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type	7750 SR MG release
101	Error-Cause	O	Contains a detailed error cause. The value field is four octets, containing an integer specifying the cause of the error.	UTF-8 encoded string	10.0R1

GTPv2 interface messages

Note that the abbreviations listed in the Presence column in the subsequent tables in this section represent the following characteristics.

- M=Mandatory
- C=Conditional
- O=Optional
- CO=Conditional-Optional, where Conditional applies to sender and Optional applies to receiver

For more information on presence requirements of information elements, refer to 3GPP TS 29.274.

Path management messages (S11, S5, S8, S4, S2a, S2b)

Table 103: Echo Request attributes

Attribute name	Presence	Notes and comments	Supported
Recovery	M	—	Yes
Private Extension	O	For MME pool extension. For more information, see Network-triggered service restoration .	Yes
Sending Node Features	CO	—	Yes

Table 104: Echo Response attributes

Attribute name	Presence	Notes and comments	Supported
Recovery	M	—	Yes
Private Extension	O	For MME pool extension. For more information, see Network-triggered service restoration .	Yes
Sending Node Features	CO	—	Yes

Tunnel management messages (S11, S5, S8, S4, S2a, S2b)

Table 105: Create Session Request

Attribute name	Presence	Notes and comments	Supported
IMSI	C	—	Yes
MSISDN	C	—	Yes
ME Identity (MEI)	C	—	Yes
User Location Information (ULI)	C Co	—	Yes
Serving Network	C Co	—	Yes
RAT Type	M	—	Yes
Indication Flags	C	—	Yes

Table 105: Create Session Request (continued)

Attribute name	Presence	Notes and comments	Supported
Sender F-TEID for Control Plane	M	—	Yes
PGW S5/S8 Address for Control Plane or PMIP	C Co	—	Yes
Access Point Name (APN)	M	—	Yes
Selection Mode	C Co	—	Yes
PDN-Type	C	—	Yes
PDN Address Allocation	C Co	—	Yes
Maximum APN Restriction	C	—	Yes
Aggregate Maximum Bit Rate (APN-AMBR)	C	—	Yes
Linked EPS Bearer ID	C	—	Yes
Protocol Configuration Options (PCO)	C	See <i>Protocol Configuration Options (PCO) IE</i>	Yes
Bearer Contexts to be created	M	—	Yes
Bearer Contexts to be removed	C	—	Yes
ePDG Overload Control Information	O	—	Yes

Table 105: Create Session Request (continued)

Attribute name	Presence	Notes and comments	Supported
MME Overload Control Information	O	—	Yes
SGW Overload Control Information	O	—	Yes
Trace Information	C	Tracing functionality is not supported	No
Recovery	C	—	Yes
MME-FQ-CSID	C	Partial MG-ISM Failure is not supported	No
SGW-FQ-CSID	C	Partial MG-ISM Failure is not supported	No
ePDG-FQ-CSID	C	Partial MG-ISM Failure is not supported	No
TWAN-FQ-CSID	C	Partial MG-ISM Failure is not supported	No
UE Time Zone	O C	—	Yes
User CSG Information (UCI)	CO	—	Yes
Charging Characteristics	C	—	Yes
MME/S4-SGSN LDN	O	Tracing functionality is not supported	No
SGW LDN	O	Tracing functionality is not supported	No
ePDG LDN	O	—	No
TWAN LDN	O	—	No

Table 105: Create Session Request (continued)

Attribute name	Presence	Notes and comments	Supported
Signaling Priority Indicator	CO	—	No
UE Local IP Address	CO	—	Yes
UE UDP Port	CO	—	Yes
Additional Protocol Configuration Options (APCO)	CO	For information about the PCO/APCO IE, see Protocol Configuration Options (PCO) IE .	Yes
H(e)NP Local IP Address	CO	—	No
H(e)NP UDP Port	CO	—	No
MME/S4-SGSN Identifier	CO	—	Yes
TWAN Identifier	O	—	No
ePDG IP Address	O	—	No
CN Operator Selection Entity	CO	—	Yes
Presence Reporting Area Information	CO	—	Yes
Extended Protocol Configuration Options (ePCO)	CO	For information about the PCO IE, see Protocol Configuration Options (PCO) IE .	Yes
Serving PLMN Rate Control	CO	—	Yes
MO Exception Data Counter	CO	—	—
Private Extension	O	—	No

Table 105: Create Session Request (continued)

Attribute name	Presence	Notes and comments	Supported
Origination Time Stamp	CO	—	Yes
Maximum Wait Time	CO	—	Yes
Secondary RAT Usage Data Report	CO	—	Yes
UP Function Selection Indication Flags	CO	Depending on the operator policy, the MME/ S4-SGSN includes this IE on the S4/S11 interface, if any applicable flags are set to 1 Applicable Flags: Bit 1 DCNR (Dual connectivity with NR) — when set to 1, indicates to the SGW-C and PGW-C to select a specific SGW-U and PGW-U for UEs that support dual connectivity with NR and are not restricted from using NR by user subscription This information is used for SGW-U, PGW-U or combined SGW-U/PGW-U selection (as specified in Annex B.2 of 3GPP TS 29.244)	Yes

Table 106: Bearer Context to be Created IE (for Create Session Request)

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes
TFT	O	—	Yes
S1-U eNodeB F-TEID	C	—	Yes
S4-U SGSN F-TEID	C	—	Yes

Table 106: Bearer Context to be Created IE (for Create Session Request) (continued)

Attribute name	Presence	Notes and comments	Supported
S5/S8-U SGW F-TEID	C	—	Yes
S5/S8-U PGW F-TEID	C	—	Yes
S12 RNC F-TEID	Co	—	Yes
S2b-U ePDG F-TEID	C	—	Yes
S2a-U TWAN F-TEID	C	—	No
Bearer Level QoS	M	—	Yes
S11-U MME F-TEID	CO	—	Yes

Table 107: Bearer Context to be Removed IE (for Create Session Request)

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes
S4-U SGSN F-TEID	C	—	Yes

Table 108: Create Session Response

Attribute name	Presence	Notes and comments	Supported
Cause	M	—	Yes
Change Reporting Action	C	—	Yes

Table 108: Create Session Response (continued)

Attribute name	Presence	Notes and comments	Supported
CSG Information Reporting Action	C	—	Yes
Sender F-TEID for PMIP-based interface or for GTP-based Control Plane interface	M	—	Yes
PGW S5/S8 F-TEID for PMIP-based interface or for GTP-based Control Plane interface	C	—	Yes
PDN Address Allocation (PAA)	C	—	Yes
APN Restriction	C	—	Yes
Aggregate Maximum Bit Rate (APN-AMBR)	C	—	Yes
Linked EPS Bearer ID	C	—	Yes
Protocol Configuration Options (PCO)	C	For information about the PCO IE, see <i>Protocol Configuration Options (PCO) IE</i> .	Yes
Bearer Contexts created	M	—	Yes
Bearer Contexts marked for removal	C	—	Yes
Recovery	C	—	Yes
Charging Gateway Name	C	—	Yes
Charging Gateway Address	C	—	Yes

Table 108: Create Session Response (continued)

Attribute name	Presence	Notes and comments	Supported
Indication Flags	CO	—	Yes
PGW node level Load Control Information	O	—	Yes
PGW Overload Control Information	O	—	Yes
SGW node level Load Control Information	O	—	Yes
SGW Overload Control Information	O	—	Yes
PGW-FQ-CSID	C	Partial MG-ISM failure is not supported.	No
SGW-FQ-CSID	C	Partial MG-ISM failure is not supported.	No
SGW LDN	O	—	No
PGW LDN	O	—	No
PGW Back-off Time	O	—	Yes
PGW APN Load Control Information (LCI)	O	—	Yes
Additional Protocol Configuration Options (APCO)	CO	For information about the PCO/APCO IE, see Protocol Configuration Options (PCO) IE .	Yes
	O		Yes
Trusted WLAN IPv4 Parameters	CO	—	No

Table 108: Create Session Response (continued)

Attribute name	Presence	Notes and comments	Supported
Presence Reporting Area Action	CO	—	Yes
Extended Protocol Configuration Options (ePCO)	CO	For information about the PCO IE, see Protocol Configuration Options (PCO) IE .	Yes
Private Extension	CO	—	No

Table 109: Bearer Context to be Created IE (for Create Session Response)

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes
Cause	M	—	Yes
S1-U SGW F-TEID	C	—	Yes
S4-U SGW F-TEID	C	—	Yes
S5/S8-U PGW F-TEID	C	—	Yes
S2b-U PGW F-TEID	C	—	Yes
S2a-U PGW F-TEID	C	—	No
Bearer Level QoS	C	—	Yes
Charging Id	C	—	Yes
	O	—	—

Table 109: Bearer Context to be Created IE (for Create Session Response) (continued)

Attribute name	Presence	Notes and comments	Supported
	CO	—	—
Bearer Flags	O	—	No
S11-U SGW F-TEID	C	—	Yes

Table 110: Bearer Context to be Removed IE (for Create Session Response)

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes
Cause	M	—	Yes

Table 111: Create Bearer Request

Attribute name	Presence	Notes and comments	Supported
Procedure Transaction ID (PTI)	C	—	Yes
ePDG Overload Control Information	O	—	Yes
Linked Bearer Identity (LBI)	M	—	Yes
Protocol Configuration Options (PCO)	O	For information about the PCO IE, see Protocol Configuration Options (PCO) IE .	Yes
Bearer Contexts	M	—	Yes
Indication Flags	CO	—	Yes

Table 111: Create Bearer Request (continued)

Attribute name	Presence	Notes and comments	Supported
PGW node level Load Control Information	O	—	Yes
PGW Overload Control Information	O	—	Yes
SGW node level Load Control Information	O	—	Yes
SGW Overload Control Information	O	—	Yes
PGW APN Load Control Information (LCI)	O	—	Yes
PGW-FQ-CSID	C	Partial failure is not supported.	No
SGW-FQ-CSID	C	Partial failure is not supported.	No
Change Reporting Action	C	—	Yes
CSG Information Reporting Action	CO	—	Yes
H(e)NB Information Reporting	CO	—	Yes
Presence Reporting Area Action	CO	—	Yes
Private Extension	O	—	No

Table 112: Bearer Context to be Created IE (for Create Bearer Request)

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes
TFT	M	—	Yes
S1-U SGW F-TEID	C	—	Yes
S5/8-U PGW F-TEID	C	—	Yes
S12 SGW F-TEID	C	—	Yes
S4-U SGW F-TEID	C	—	Yes
S2b-U PGW F-TEID	C	—	Yes
S2a-U PGW F-TEID	C	—	No
	M	—	Yes
Charging Id	C	—	Yes
	O	—	Yes
Bearer Flags	O	—	No
Protocol Configuration Options (PCO)	O	For information about the PCO IE, see Protocol Configuration Options (PCO) IE .	Yes
Extended Protocol Configuration Options (ePCO)	O	For information about the PCO IE, see Protocol Configuration Options (PCO) IE .	Yes

Table 113: Create Bearer Response

Attribute name	Presence	Notes and comments	Supported
Cause	M	—	Yes
Bearer Contexts	M	—	Yes
Recovery	C	—	Yes
MME Identifier	CO	—	Yes
MME Overload Control Information	O	—	Yes
SGW Overload Control Information	O	—	Yes
MME-FQ-CSID	C	Partial failure is not supported.	No
SGW-FQ-CSID	C	Partial failure is not supported.	No
ePDG-FQ-CSID	C	—	No
TWAN-FQ-CSID	C	—	No
Protocol Configuration Options (PCO)	C	For information about the PCO IE, see Protocol Configuration Options (PCO) IE .	Yes
UE Time Zone	O	—	Yes
	CO	—	Yes
User Location Information (ULI)	O	—	Yes
	CO	—	Yes

Table 113: Create Bearer Response (continued)

Attribute name	Presence	Notes and comments	Supported
Presence Reporting Area Information	CO	—	Yes
Private Extension	O	—	No
RAN/NAS Release Cause	CO	—	Yes

Table 114: Bearer Context to be Created IE (for Create Bearer Response)

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes
Cause	M	—	Yes
S1-U eNB F-TEID	C	—	Yes
S1-U SGW F-TEID	C	—	Yes
S5/S8-U SGW F-TEID	C	—	Yes
S5/8-U PGW F-TEID	C	—	Yes
S12 RNC F-TEID	C	—	Yes
S12 SGW F-TEID	C	—	Yes
S4-U SGSN F-TEID	C	—	Yes
S4-U SGW F-TEID	C	—	Yes
S2b-U ePDG F-TEID	C	—	Yes

Table 114: Bearer Context to be Created IE (for Create Bearer Response) (continued)

Attribute name	Presence	Notes and comments	Supported
S2b-U PGW F-TEID	C	—	Yes
S2a-U TWAN F-TEID	C	—	Yes
S2a-U PGW F-TEID	C	—	No
Protocol Configuration Options (PCO)	O	For information about the PCO IE, see <i>Protocol Configuration Options (PCO) IE</i> .	No
Extended Protocol Configuration Options (ePCO)	CO	For information about the PCO IE, see <i>Protocol Configuration Options (PCO) IE</i> .	Yes

Table 115: Bearer Resource command

Attribute name	Presence	Notes and comments	Supported
Linked EPS Bearer ID (LBI)	M	—	Yes
Procedure Transaction ID (PTI)	M	—	Yes
Flow Quality of Service (Flow QoS)	C	—	Yes
Traffic Aggregate Description (TAD)	M	—	Yes
RAT Type	C	—	Yes
Serving Network	O	—	Yes
User Location Information (ULI)	O	—	Yes

Table 115: Bearer Resource command (continued)

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	C	—	Yes
Indication Flags	O	For change reporting support.	Yes
S4-U SGSN F-TEID	C	—	Yes
S12 RNC F-TEID	C	—	Yes
Protocol Configuration Options (PCO)	O	For information about the PCO IE, see Protocol Configuration Options (PCO) IE .	Yes
Signaling Priority Indicator	CO	—	Yes
MME Identifier	CO	—	Yes
MME Overload Control Information	O	—	Yes
SGW Overload Control Information	O	—	Yes
Sender F-TEID for Control Plane	CO	—	Yes
Extended Protocol Configuration Options (ePCO)	O	For information about the PCO IE, see Protocol Configuration Options (PCO) IE .	Yes
Private Extension	O	—	Yes

Table 116: Bearer Resource Command Failure Indication

Attribute name	Presence	Notes and comments	Supported
Cause	M	—	Yes
Linked EPS Bearer ID	M	—	Yes
Procedure Transaction ID (PTI)	M	—	Yes
Recovery	O	—	Yes
Private Extension	O	—	No

Table 117: Modify Bearer Request

Attribute name	Presence	Notes and comments	Supported
ME Identity (MEI)	C	—	Yes
User Location Information (ULI)	C	—	Yes
Serving Network	C	—	Yes
RAT Type	C	—	Yes
Indication Flags	C	—	Yes
Sender F-TEID for Control Plane	C	—	Yes
Aggregate Maximum Bit Rate (APN-AMBR)	C	—	Yes

Table 117: Modify Bearer Request (continued)

Attribute name	Presence	Notes and comments	Supported
Delay Downlink Packet Notification Request	C	—	Yes
Bearer Contexts to be modified	C	—	Yes
Bearer Contexts to be removed	C	—	Yes
Recovery	C	—	Yes
UE Time Zone	O	—	Yes
	C	—	Yes
Indication Flags	CO	—	Yes
PGW node level Load Control Information	O	—	Yes
PGW Overload Control Information	O	—	Yes
SGW node level Load Control Information	O	—	Yes
SGW Overload Control Information	O	—	Yes
MME-FQ-CSID	C	—	No
SGW-FQ-CSID	C	—	No
User CSG Information (UCI)	CO	—	Yes

Table 117: Modify Bearer Request (continued)

Attribute name	Presence	Notes and comments	Supported
UE Local IP Address	CO	—	Yes
UE UDP Port	CO	—	Yes
MME/SGSN LDN	O	—	No
SGW LDN	O	—	No
H(e)NB Local IP Address	CO	—	No
H(e)NB UDP Port	CO	—	No
MME/S4-SGSN Identifier	CO	—	No
CN Operator Selection Entity	CO	—	Yes
Presence Reporting Area Information	CO	—	Yes
Serving PLMN Rate Control	CO	—	Yes
MO Exception Data Counter	CO	—	—
IMSI	O	—	Yes
Private Extension	O	—	No
Secondary RAT Usage Data Report	CO	—	Yes

Table 118: Bearer Context to be Modified IE (for Modify Bearer Request)

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes
S1 eNodeB F-TEID	C	—	Yes
S5/8-U SGW F-TEID	C	—	Yes
S12 RNC F-TEID	C	—	Yes
S4-U SGSN F-TEID	C	—	Yes
S11-U MME F-TEID	CO	—	Yes

Table 119: Bearer Context Deleted IE (for Modify Bearer Request)

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes

Table 120: Modify Bearer Response

Attribute name	Presence	Notes and comments	Supported
Cause	M	—	Yes
MSISDN	C	—	Yes
Linked EPS Bearer ID	C	—	Yes
Aggregate Maximum Bit Rate (APN-AMBR)	C	—	Yes

Table 120: Modify Bearer Response (continued)

Attribute name	Presence	Notes and comments	Supported
APN Restriction	C	—	Yes
Protocol Configuration Options (PCO)	C	For information about the PCO IE, see <i>Protocol Configuration Options (PCO) IE</i> .	Yes
Bearer Contexts modified	C	—	Yes
Bearer Contexts marked for removal	C	—	Yes
Change Reporting Action	C	—	Yes
CSG Information Reporting Action	CO	—	Yes
Charging Gateway Name	C	—	Yes
Charging Gateway Address	C	—	Yes
MME Identifier	CO	—	Yes
MME Overload Control Information	O	—	Yes
SGW Overload Control Information	O	—	Yes
PGW APN Load Control Information (LCI)	O	—	Yes
PGW-FQ-CSID	C	—	No
SGW-FQ-CSID	C	—	No

Table 120: Modify Bearer Response (continued)

Attribute name	Presence	Notes and comments	Supported
Recovery	C	—	Yes
SGW LDN	O	—	No
PGW LDN	O	—	No
Indication Flags	CO	—	No
Presence Reporting Area Action	CO	—	Yes
Private Extension	O	—	No

Table 121: Bearer Context to be Modified IE

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes
Cause	M	—	Yes
S1 SGW F-TEID	C	—	Yes
S12 SGW F-TEID	C	—	Yes
S4-U SGW F-TEID	C	—	Yes
Charging ID	C	—	Yes
	O	—	Yes
Bearer Flags	CO	—	No

Table 121: Bearer Context to be Modified IE (continued)

Attribute name	Presence	Notes and comments	Supported
S11-U SGW F-TEID	C	—	Yes

Table 122: Bearer Context Deleted (for Modify Bearer Response)

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes

Table 123: Delete Session Request

Attribute name	Presence	Notes and comments	Supported
Cause	C	—	Yes
Linked EPS Bearer ID (LBI)	C	—	Yes
User Location Information (ULI)	C	—	Yes
ePDG Overload Control Information	O	—	Yes
Indication Flags	C	—	Yes
Protocol Configuration Options (PCO)	C	For information about the PCO IE, see <i>Protocol Configuration Options (PCO) IE</i> .	Yes
MME Identifier	CO	—	Yes
MME Overload Control Information	O	—	Yes

Table 123: Delete Session Request (continued)

Attribute name	Presence	Notes and comments	Supported
SGW Overload Control Information	O	—	Yes
Originating Node	C	—	No
Sender F-TEID for Control Plane	CO	—	Yes
UE Time Zone	CO	—	Yes
ULI Timestamp	CO	—	Yes
RAN/NAS Release Cause	CO	—	Yes
Extended Protocol Configuration Options (ePCO)	CO	For information about the PCO IE, see Protocol Configuration Options (PCO) IE .	Yes
Private Extension	O	—	No
Secondary RAT Usage Data Report	CO	—	Yes

Table 124: Delete Session Response

Attribute name	Presence	Notes and comments	Supported
Cause	M	—	Yes
Recovery	C	Included if contacting the peer for the first time	Yes
Protocol Configuration Options (PCO)	C	The PGW includes the PCO IE, if available	Yes

Table 124: Delete Session Response (continued)

Attribute name	Presence	Notes and comments	Supported
		For information about the PCO IE, see <i>Protocol Configuration Options (PCO) IE</i>	
Indication Flags	CO	—	Yes
PGW node level Load Control Information	O	—	Yes
PGW Overload Control Information	O	—	Yes
SGW node level Load Control Information	O	—	Yes
SGW Overload Control Information	O	—	Yes
PGW APN Load Control Information (LCI)	O	—	Yes
Extended Protocol Configuration Options (ePCO)	CO	For information about the PCO IE, see <i>Protocol Configuration Options (PCO) IE</i> .	Yes
Private Extension	O	—	No

Table 125: Delete Bearer Request

Attribute name	Presence	Notes and comments	Supported
Linked EPS Bearer ID (LBI)	C	—	Yes
EPS Bearer IDs	C	—	Yes
Failed Bearer Contexts	C	—	Yes

Table 125: Delete Bearer Request (continued)

Attribute name	Presence	Notes and comments	Supported
Procedure Transaction ID (PTI)	C	—	Yes
Protocol Configuration Options (PCO)	C	For information about the PCO IE, see <i>Protocol Configuration Options (PCO) IE</i> .	Yes
Indication Flags	CO	—	Yes
PGW node level Load Control Information	O	—	Yes
PGW Overload Control Information	O	—	Yes
SGW node level Load Control Information	O	—	Yes
SGW Overload Control Information	O	—	Yes
PGW APN Load Control Information (LCI)	O	—	Yes
PGW-FQ-CSID	C	—	No
SGW-FQ-CSID	C	—	No
Cause	C	—	Yes
Extended Protocol Configuration Options (ePCO)	CO	For information about the PCO IE, see <i>Protocol Configuration Options (PCO) IE</i> .	Yes
Private Extension	O	—	No

Table 126: Bearer Context within Delete Bearer Request IE

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes
Cause	M	—	Yes
Protocol Configuration Options (PCO)	Co	For information about the PCO IE, see <i>Protocol Configuration Options (PCO) IE</i> .	Yes

Table 127: Delete Bearer Response

Attribute name	Presence	Notes and comments	Supported
Cause	M	—	Yes
ePDG Overload Control Information	O	—	Yes
Linked EPS Bearer ID (LBI)	C	—	Yes
Bearer Contexts	C	—	Yes
Recovery	C	—	Yes
MME Identifier	CO	—	Yes
MME Overload Control Information	O	—	Yes
SGW Overload Control Information	O	—	Yes
MME-FQ-CSID	C	—	No

Table 127: Delete Bearer Response (continued)

Attribute name	Presence	Notes and comments	Supported
SGW-FQ-CSID	C	—	No
ePDG-FQ-CSID	C	—	No
TWAN-FQ-CSID	C	—	No
Protocol Configuration Options (PCO)	C	For information about the PCO IE, see <i>Protocol Configuration Options (PCO) IE</i> .	Yes
UE Time Zone	O	—	Yes
	CO	—	Yes
User Location Information (ULI)	O	—	Yes
	CO	—	Yes
ULI Timestamp	CO	—	Yes
RAN/NAS Release Cause	CO	—	Yes
Private Extension	O	—	No
Secondary RAT Usage Data Report	CO	—	Yes

Table 128: Bearer Context within Delete Bearer Response IE

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes

Table 128: Bearer Context within Delete Bearer Response IE (continued)

Attribute name	Presence	Notes and comments	Supported
Cause	M	—	Yes
Protocol Configuration Options (PCO)	CO	For information about the PCO IE, see Protocol Configuration Options (PCO) IE .	Yes
Extended Protocol Configuration Options (ePCO)	CO	For information about the PCO IE, see Protocol Configuration Options (PCO) IE .	Yes

Table 129: Downlink Data Notification Request

Attribute name	Presence	Notes and comments	Supported
Cause	O	—	Yes
EPS Bearer ID	CO	—	Yes
Allocation/Retention Priority	CO	—	Yes
IMSI	CO	—	Yes
Sender F-TEID for Control Plane	O	—	Yes
Paging and Service Information	CO	—	Yes
Private Extension	O	—	No

Table 130: Downlink Data Notification ACK

Attribute name	Presence	Notes and comments	Supported
Cause	O	—	Yes
Data Notification Delay	C	—	Yes
Recovery	C	—	Yes
DL Low Priority Traffic Throttling	O	—	Yes
IMSI	CO	—	Yes
DL Buffering Duration	CO	If the received value exceeds the configured maximum duration, the configured maximum duration will be applied. If the received value is less than 30 seconds, it will be ignored and the buffering duration will be set to 30 seconds.	Yes
DL Buffering Suggested Packet Count	O	If the received value is more than 8 packets or the DL buffering duration in the message is less than 30 seconds, the suggested packet count is ignored. The DL Buffering Suggested Packet Count will be ignored also if the DL Buffering Duration IE is not included in the DDN Ack message.	Yes
Private Extension	O	—	No

Table 131: Downlink Data Notification Failure Indication

Attribute name	Presence	Notes and comments	Supported
Cause	M	—	Yes

Table 131: Downlink Data Notification Failure Indication (continued)

Attribute name	Presence	Notes and comments	Supported
Originating Node	CO	—	No
IMSI	CO	—	No
Private Extension	O	—	No

Table 132: Delete Indirect Data Forwarding Tunnel Request

Attribute name	Presence	Notes and comments	Supported
Private Extension	O	—	No

Table 133: Delete Indirect Data Forwarding Tunnel Response

Attribute name	Presence	Notes and comments	Supported
Cause	M	—	Yes
Recovery	C	—	Yes
Private Extension	O	—	No

Table 134: Modify Bearer command

Attribute name	Presence	Notes and comments	Supported
APN-Aggregate Maximum Bit Rate (APN-AMBR)	M	—	Yes
Bearer Context	M	—	Yes

Table 134: Modify Bearer command (continued)

Attribute name	Presence	Notes and comments	Supported
ePDG Overload Control Information	O	—	Yes
MME Overload Control Information	O	—	Yes
Sender F-TEID for Control Plane	CO	—	Yes
Private Extension	O	—	Yes
SGW Overload Control Information	O	—	Yes

Table 135: Bearer Context IE within the commands

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID Cause	M	—	Yes
Bearer Level QoS	C	—	Yes
	CO	—	Yes

Table 136: Modify Bearer Command Failure Indication

Attribute name	Presence	Notes and comments	Supported
Cause	M	—	Yes
Recovery	C	—	Yes

Table 136: Modify Bearer Command Failure Indication (continued)

Attribute name	Presence	Notes and comments	Supported
Private Extension	O	—	No

Table 137: Update Bearer Request

Attribute name	Presence	Notes and comments	Supported
Bearer Contexts	M	—	Yes
Procedure Transaction ID (PTI)	C	—	Yes
Protocol Configuration Options (PCO)	C	For information about the PCO IE, see Protocol Configuration Options (PCO) IE .	Yes
Aggregate Maximum Bit Rate (APN-AMBR)	M	—	Yes
Change Reporting Action	C	—	Yes
CSG Information Reporting Action	CO	—	Yes
Indication Flags	O	—	Yes
PGW node level Load Control Information	O	—	Yes
PGW Overload Control Information	O	—	Yes
SGW node level Load Control Information	O	—	Yes

Table 137: Update Bearer Request (continued)

Attribute name	Presence	Notes and comments	Supported
SGW Overload Control Information	O	—	Yes
PGW APN Load Control Information (LCI)	O	—	Yes
H(e)NB Information Reporting	CO	—	No
Indication flags	CO	—	No
PGW-FQ-CSID	C	—	No
SGW-FQ-CSID	C	—	No
Presence Reporting Area Action	CO	—	Yes
Private Extension	O	—	No

Table 138: Bearer Context IE to be Modified

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes
TFT	C	—	Yes
Bearer Level QoS	C	—	Yes
Bearer Flags	O	—	No

Table 138: Bearer Context IE to be Modified (continued)

Attribute name	Presence	Notes and comments	Supported
Protocol Configuration Options (PCO)	CO	For information about the PCO IE, see Protocol Configuration Options (PCO) IE .	Yes
Extended Protocol Configuration Options (ePCO)	CO	For information about the PCO IE, see Protocol Configuration Options (PCO) IE .	Yes

Table 139: Update Bearer Response

Attribute name	Presence	Notes and comments	Supported
Cause	M	—	Yes
Bearer Contexts	M	—	Yes
ePDG Overload Control Information	O	—	Yes
Protocol Configuration Options (PCO)	C	For information about the PCO IE, see Protocol Configuration Options (PCO) IE .	Yes
Recovery	C	—	Yes
MME Identifier	CO	—	Yes
MME Overload Control Information	O	—	Yes
SGW Overload Control Information	O	—	Yes
MME-FQ-CSID	C	—	No
SGW-FQ-CSID	C	—	No

Table 139: Update Bearer Response (continued)

Attribute name	Presence	Notes and comments	Supported
ePDG-FQ-CSID	C	—	No
TWAN-FQ-CSID	O	—	No
Indication Flags	O	—	No
UE Time Zone	O	—	Yes
	CO	—	Yes
User Location Information (ULI)	O	—	Yes
	CO	—	Yes
RAN/NAS Release Cause	CO	—	Yes
Presence Reporting Area Information	CO	—	Yes
Private Extension	O	—	No

Table 140: Bearer Context IE to be Modified

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes
Cause	M	—	Yes
S4-U SBSN F-TEID	C	—	Yes
S12 RNC F-TEID	C	—	Yes

Table 140: Bearer Context IE to be Modified (continued)

Attribute name	Presence	Notes and comments	Supported
Protocol Configuration Options	CO	—	Yes
Extended Protocol Configuration Options (ePCO)	CO	For information about the PCO IE, see Protocol Configuration Options (PCO) IE .	Yes

Table 141: Delete Bearer Command

Attribute name	Presence	Notes and comments	Supported
Bearer Contexts	M	—	Yes
User Location Information (ULI)	M	—	Yes
ULI Timestamp	CO	—	Yes
UE Time Zone	CO	—	Yes
MME Identifier	CO	—	Yes
MME Overload Control Information	O	—	Yes
SGW Overload Control Information	O	—	Yes
RAN/NAS Release Cause	CO	—	Yes
Sender F-TEID for Control Plane	CO	—	Yes
Private Extension	O	—	No

Table 142: Bearer Context IE to be Deleted

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes
Bearer Flags	CO	—	No

Table 143: Delete Bearer Failure Indication

Attribute name	Presence	Notes and comments	Supported
Cause	M	—	Yes
Bearer Context	M	—	Yes
Recovery	C	—	Yes
Private Extension	O	—	No

Table 144: Bearer Context IE to be Deleted

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes
Cause	M	—	Yes

Table 145: Create Indirect Data Forwarding Tunnel Request

Attribute name	Presence	Notes and comments (S11, S4)	Supported
IMSI	C	—	No

Table 145: Create Indirect Data Forwarding Tunnel Request (continued)

Attribute name	Presence	Notes and comments (S11, S4)	Supported
ME Identity (MEI)	C	—	—
Indication Flags	CO	—	—
Sender F-TEID for Control Plane	C	—	No
Bearer Contexts	M	—	Yes
Recovery	CO	—	Yes
Private Extension	O	—	No

Table 146: Bearer Context IE within CIDFTR

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes
eNodeB F-TEID for DL data forwarding	C	—	Yes
SGW F-TEID for DL data forwarding	C	—	Yes
SGSN F-TEID for DL data forwarding	C	—	Yes
	—	—	—
RNC F-TEID for DL forwarding	C	—	Yes
	—	—	—

Table 146: Bearer Context IE within CIDFTR (continued)

Attribute name	Presence	Notes and comments	Supported
eNodeB F-TEID for UL data forwarding	O	UL temporary tunnels for data forwarding are not supported.	No
SGW F-TEID for UL data forwarding	O	UL temporary tunnels for data forwarding are not supported.	No

Table 147: Create Indirect Data Forwarding Tunnel Response

Attribute name	Presence	Notes and comments (S11, S4)	Supported
Cause	M	—	Yes
Sender F-TEID for Control Plane	C	—	Yes
Bearer Contexts	M	—	Yes
Recovery	CO	—	Yes
Private Extension	O	—	No

Table 148: Bearer Context IE with CIDFTR

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes
Cause	M	—	Yes
S1-U SGW F-TEID for DL data forwarding	C	—	Yes

Table 148: Bearer Context IE with CIDFTR (continued)

Attribute name	Presence	Notes and comments	Supported
S12 SGW F-TEID for DL data forwarding	C	—	Yes
S4-U SGW F-TEID for DL data forwarding	C	—	Yes
SGW F-TEID for DL data forwarding	C	—	Yes
S1-U SGW F-TEID for UL data forwarding	O	—	No
SGW F-TEID for UL data forwarding	O	—	No

Table 149: Release Access Bearer Request

Attribute name	Presence	Notes and comments (S11, S4)	Supported
List of RABs	C	All bearers of the UE must be listed	Yes
Originating Node	CO	—	No
Private Extension	O	—	No
Secondary RAT Usage Data Report	CO	—	Yes

Table 150: Release Access Bearer Response

Attribute name	Presence	Notes and comments (S11, S4)	Supported
Cause	M	—	Yes
Recovery	O	—	Yes
Private Extension	O	—	No

Table 151: Modify Access Bearer Request

Attribute name	Presence	Notes and comments (S11, S5, S8, S4)	Supported
Indication Flags	C	—	No
Sender F-TEID for Control Plane	C	—	Yes
Delay Downlink Packet Notification Request	C	—	Yes
Bearer Contexts to be modified	C	—	Yes
Bearer Contexts to be removed	C	—	Yes
Recovery	C	—	Yes
Private Extension	O	—	No
Secondary RAT Usage Data Report	CO	—	Yes

Table 152: Bearer Context to be Modified IE

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes
S1eNodeB F-TEID	C	—	Yes

Table 153: Bearer Context Deleted IE

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes

Table 154: Modify Access Bearer Response

Attribute Name	Presence	Notes and comments (S11, S5, S8, S4)	Supported
Cause	M	—	Yes
Bearer Contexts modified	C	—	Yes
Bearer Contexts marked for removal	C	—	Yes
Recovery	C	—	Yes
Private Extension	O	—	No

Table 155: Bearer Context to be Modified IE (for Modify Access Bearer Response)

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes

Table 155: Bearer Context to be Modified IE (for Modify Access Bearer Response) (continued)

Attribute name	Presence	Notes and comments	Supported
Cause	M	—	Yes
S1 eNodeB F-TEID	C	—	Yes

Table 156: Bearer Context Deleted IE (for Modify Access Bearer Response)

Attribute name	Presence	Notes and comments	Supported
EPS Bearer ID	M	—	Yes

Mobility management messages (S11, S5, S8, S4)

Table 157: Change Notification Request IE

Attribute name	Presence	Notes and comments	Supported
IMSI	C	—	Yes
ME Identity (MEI)	C	—	Yes
Indication Flags	CO	—	Yes
RAT Type	M	—	Yes
User Location Information (ULI)	C	—	Yes
	CO	—	Yes
User CSG Information (UCI)	CO	—	Yes

Table 157: Change Notification Request IE (continued)

Attribute name	Presence	Notes and comments	Supported
LBI	CO	—	Yes
Presence Reporting Area Information	CO	—	Yes
Secondary RAT Usage Data Report	CO	—	Yes

Table 158: Change Notification Response IE

Attribute name	Presence	Notes and comments	Supported
IMSI	C	—	Yes
ME Identity (MEI)	C	—	Yes
Cause	M	—	Yes
Change Reporting Action	C	—	Yes
CSG Information Reporting Action	CO	—	Yes
Presence Reporting Area Action	CO	—	Yes

CSFB messages (S11, S5, S8)

Table 159: Suspend Notification

Attribute name	Presence	Notes and comments	Supported
IMSI	C	—	Yes
Routing Area Identity (RAI)	C	—	No
	CO	—	—
Link Bearer Identity (LBI)	CO	—	Yes
Packet TMSI (P-TMSI)	C	—	No
	CO	—	—
Originating Node	CO	—	No
Address for Control Plane	CO	—	No
UDP Source Port Number	CO	—	No
Hop Counter	O	—	No
Private Extension	O	—	No

Table 160: Suspend Acknowledge

Attribute name	Presence	Notes and comments	Supported
Cause	M	—	Yes
Private Extension	O	—	No

Table 161: Resume Notification

Attribute name	Presence	Notes and comments	Supported
IMSI	M	—	Yes
Linked Bearer Identity (LBI)	CO	—	Yes
Originating Node	CO	—	No
Private Extension	O	—	No

Table 162: Resume Acknowledge

Attribute name	Presence	Notes and comments	Supported
Cause	M	—	Yes
Private Extension	O	—	No

PGW Restart (S11, S4)

Table 163: PGW Restart Notification

Attribute name	Presence	Notes and comments	Supported
PGW S5/S8 IP Address for Control Plane or PMIP	M	—	Yes
SGW S11/S4 IP Address for Control Plane	M	—	Yes
Cause	CO	—	Yes

Table 163: PGW Restart Notification (continued)

Attribute name	Presence	Notes and comments	Supported
Private Extension	O	—	No

Table 164: PGW Restart Notification Acknowledge

Attribute name	Presence	Notes and comments	Supported
Cause	M	—	Yes
Private Extension	O	—	No

Gx interface messages

Note that the abbreviations listed in the Presence column in the subsequent tables in this section represent the following characteristics.

- M=Mandatory
- C=Conditional
- O=Optional

CC-Request

Table 165: CC-Request attributes

Attribute name	Presence	Notes and comments	Reference
3GPP-MS-TimeZone	O	Indicates the UE timezone; may not be available for PMIP-S2a	29.061
3GPP-Charging-Characteristics	O	For PGW/GGSN this is the Charging-Characteristics value received via GTP or produced internally from selected charging profile.	29.061

Table 165: CC-Request attributes (continued)

Attribute name	Presence	Notes and comments	Reference
3GPP-SGSN-Address	O	Indicates the SGSN address	29.061
3GPP-SGSN-Ipv6-Address	O	Indicates the IPv6 address of the SGSN	29.212
3GPP-SGSN-MCC-MNC	O	For LTE, MCC+MNC of SGW May not be available for PMIP-S2a	29.061
3GPP-User-Location-Info	O	Indicates the UE user location information	29.061 / § 16.4.7, 29.274 / § 8.24
Access-Network-Charging-Address	O	Indicates the IP address of the network entity within the access network that is performing charging (for example the GGSN IP address).	29.212
Access-Network-Charging-Identifier-Gx	O	Applicable Charging Identifier	29.214 / § 5.3.22
AN-GW-Address	O	Indicates the SGW address	29.212 / § 5.3.49
AN-Trusted	M	Indicates the 3GPP AAA Server's decision on handling the non-3GPP access network; that is, trusted or untrusted. For the SWa, the value "UNTRUSTED" will be used.	29.212
Auth-Application-Id	M	Gx application-id as specified in § 5.1 of 29.212	RFC 6733 / § 6.8
Bearer-Usage	O	Indicates how the bearer is being used (General / IMS-Signaling)	29.212 / § 5.3.1

Table 165: CC-Request attributes (continued)

Attribute name	Presence	Notes and comments	Reference
Called-Station-ID	O	Indicates the APN (Access Point Network)	RFC 4005 / § 4.5
CC-Request-Number	M	Monotonically increasing positive integer starting at 0	RFC 4006 / § 8.2
CC-Request-Type	M	Can have one of the following values: • INITIAL • UPDATE • TERMINATION	RFC 4006 / § 8.3
Charging-Rule-Report	O	Indicates the status of the PCC rules.	29.212 / § 5.3.18
Default-EPS-Bearer-QoS	O	Indicates the subscribed QoS for the default EPS bearer	29.212 / § 5.3.48
Destination-Host	O	FQDN of PCRF. Origin-Host received in CCA is used to populate Dest-Host	RFC 6733 / § 6.5
Destination-Realm	M	FQDN of PCRF realm	RFC 6733 / § 6.6
Event-Report-Indication	O	Indicates the event triggers that the PCEF is interested in. (applicable for eHRPD deployments)	29.212 / § 5.3.30
Event-Trigger	O	Indicates the specific event that occurred at the Gateway	29.212 / § 5.3.7
Framed-IP-Address	O	UE IP address	RFC 4005 / § 6.11.1
Framed-IPv6-Prefix	O	UE IP address	RFC 4005 / § 6.11.6

Table 165: CC-Request attributes (continued)

Attribute name	Presence	Notes and comments	Reference
IP-CAN-Type	O	3GPP-EPS for EPS Access and for GGSN Gx (Gn/Gp support) Note that if custom-1 is enabled then the IP CAN for eHRPD access is "3GPP2".	29.212 / § 5.3.27
Maximum-Wait-Time	O	Indicates the number of milliseconds since the Origination-Time-Stamp during which the originator of a request waits for a response	29.273 / § 9.2.3.2.7
Network-Request-Support	O	Indicates UE and the network support for network initiated procedures	29.212 / § 5.3.24
Offline	O	Indicates the default charging method for offline preconfigured at the PCEF	29.212 / § 5.3.9
Online	O	Indicates the default charging method for online preconfigured at the PCEF	29.212 / § 5.3.10
Origin-Host	M	FQDN of PGW	RFC 6733 / § 6.3
Origin-Realm	M	FQDN of PGW realm	RFC 6733 / § 6.4
Origin-State-Id	O	Indicates the state of the origin-host (MG-ISM) generating this message	RFC 6733 / § 8.16
Origination-Time-Stamp	O	Indicates the UTC time when the originating entity initiated the request	29.273 / § 9.2.3.2.6
Packet-Filter-Information	O	Contains information for packet filters sent from the PCEF to the PCRF. Each instance corresponds to a single packet filter	29.212 / § 5.3.56

Table 165: CC-Request attributes (continued)

Attribute name	Presence	Notes and comments	Reference
Packet-Filter-Operation	O	Indicates a UE initiated resource operation (Deletion/Addition/Modification)	29.212 / § 5.3.57
PDN-Connection-ID	O	Indicates the PDN connection Identifier	29.212 / § 5.3.58
QoS-Information	O	Indicates the requested QoS information associated with resources requested by UE, an IP CAN bearer or subscribed QoS at APN level	29.212 / § 5.3.16
RAI	O	Indicates the Routing Area Identifier	29.061
RAT-Type	O	For E-UTRAN for LTE and EHRPD for eHRPD Note that if custom-1 is enabled, then the RAT type for eHRPD access will be "HRPD".	29.212 / § 5.3.31
Subscription-Id	O	IMSI, MSISDN, UE NAI	RFC 4006 / § 8.46
Supported-Features	O	Indicates required support for REI-10 Gx	29.212 § 5.4.1 / 29.229 § 7.2.1
Session-Id	M	PCEF generate this.	RFC 6733 / § 8.8
TCP-Source-Port	O	Indicates the TCP source port number within as received from the access network.	29.212
Termination-Cause	O	Indicates the reason why a session was terminated	RFC 6733 / § 8.15
TWAN-Identifier	O	Indicates the UE location in a trusted or untrusted WLAN Access Network	3GPP TS 29.061

Table 165: CC-Request attributes (continued)

Attribute name	Presence	Notes and comments	Reference
UDP-Source-Port	O	Indicates the UDP source port number within as received from the access network	29.212
UE-Local-IP-Address	O	Indicates the source IPv4 or IPv6 address of the IKE_SA_AUTH message from the UE. The ePDG will include this IE based on the local policy for fixed broadband access network interworking as specified in 3GPP TS 23.139. The ePDG may also include this IE, regardless of fixed broadband access network interworking.	29.212
Usage-Monitoring-Information	O	Indicates information pertaining to a Usage Monitoring Control Instance (for example, measured usage for a monitoring key instance)	RFC 29.212/ § 5.3.60
User-Equipment-Info	O	IMEI for LTE, MEID for eHRPD, MAC for S2b if AP MAC address comes in Private IE	RFC 4006 / § 8.49
User-Location-Info-Time	C	Indicates the NTP time at which the UE was last known to be in the location reported in the WLAN Location Information. This IE must be present if the WLAN Location Information IE is present.	29.212
Presence-Reporting-Area-Information	O	This AVP (AVP code 2822) is of type Grouped and contains the information which describes a Presence Reporting Area.	3GPP TS 29.212 § 5.3.109

CC-Answer

Table 166: CC-Answer attributes

Attribute name	Presence	Notes and comments	Reference
3GPP-User-Location-Info	O	Indicates the UE user location information	29.061 / § 16.4.7, 29.274 / § 8.24
Auth-Application-Id	M	Gx application-id as specified in § 5.1 of 29.212	RFC 6733 / § 6.8
Bearer-Control-Mode	O	Indicates the PCRF selected bearer control mode (UE-only / UE-NW)	29.212 / § 5.3.23
Bearer-Usage	O	Indicates how the bearer is being used	29.212 / § 5.3.1
CC-Request-Number	M	Same as provided by PGW in the CCR	RFC 4006 / § 8.2
CC-Request-Type	M	Same as provided by PGW in the CCR	RFC 4006 / § 8.3
Charging-Rule-Install	O	Used to activate, install or modify PCC rules as instructed from PCRF to PCEF	29.212 / § 5.3.2
Charging-Rule-Remove	O	Used to deactivate or remove PCC rules from an IP-CAN session	29.212 / § 5.3.3
Conditional-Policy-Information	O	Up to four Conditional-Policy-Information AVPs may be provisioned to change the authorized QoS per APN based on time conditions. Each instance includes an Execution-Time AVP.	29.212 / Table 5.3.0.1 / Clause 5.3.128
Default-EPS-Bearer-QoS	O	Indicates the authorized QoS for the default EPS bearer	29.212 / § 5.3.48

Table 166: CC-Answer attributes (continued)

Attribute name	Presence	Notes and comments	Reference
Error-Message	O	May accompany a result-code as a human readable error message	RFC 6733 / § 7.3
Error-Reporting-Host	O	Identity of the host that sent the result-code (value other than success) only if the host is different from origin-host	RFC 6733 / § 7.4
Event-Trigger	O	Indicates event triggers, PCRF is interested in and subscribes to.	29.212 / § 5.3.7
Experimental-Result	O	Indicates if a request was completed successfully or an error occurred. Either Result-Code or this AVP is present. Can be used to indicate result codes specific to Gx application	RFC 6733 / § 7.6
Failed-AVP	O	Indicates a failed AVP for debugging	RFC 6733 / § 7.5
Offline	O	Indicates the default charging method for offline. PCRF provided value takes precedence over any preconfigured method at the PCEF	29.212 / § 5.3.9
Online	O	Indicates the default charging method for online. PCRF provided value takes precedence over any preconfigured method at the PCEF	29.212 / § 5.3.10
Origin-Host	M	FQDN of PCRF	RFC 6733 / § 6.3
Origin-Realm	M	FQDN of PCRF realm	RFC 6733 / § 6.4
Origin-State-Id	O	Indicates the state of the origin-host (PCRF) generating this message	RFC 6733 / § 8.16

Table 166: CC-Answer attributes (continued)

Attribute name	Presence	Notes and comments	Reference
Proxy-Info	O	Indicates the identity and state of the host that added the proxy-info AVP	RFC 6733 / § 6.7.2
QoS-Information	O	Indicates the authorized QoS from the PCRF	29.212 / § 5.3.16
Result-Code	O	Indicates if the request was completed successfully or an error occurred	RFC 6733 / § 7.1
Route-Record	O	Indicates the identity of the peer the request was received from. Indicates the route traversed with multiple hops.	RFC 6733 / § 6.7.1
Session-Id	M	Session-Id provided by PGW in the CCR.	RFC 6733 / § 8.8
Supported-Features	O	Indicates Release supported for Gx	29.212 § 5.4.1 / 29.229 § 7.2.1
Usage-Monitoring-Information	O	Contains the usage monitoring control information (Usage Monitoring-Key, Granted-Service-Unit, Usage-Monitoring-level and so on)	29.212/ § 5.3.60
Presence-Reporting-Area-Information	O	The Presence-Reporting-Area-Information AVP (AVP code 2822) is of type Grouped, and contains the information which describes a PRA.	3GPP TS 29.212
PRA-Install	O	This AVP (AVP code 2845) is of type Grouped, and is used to provision a list of new or updated PRAs for an IP-CAN session.	3GPP TS 29.274
PRA-Remove	O	This AVP (AVP code 2846) is of type Grouped, and is used to stop the reporting of a list of PRAs for an IP-CAN session.	3GPP TS 29.274

Table 167: VSAs 26/28458 - Nokia vendor-specific CC-Answer attributes

Sub-attr no.	Sub-attribute name	Presence	Description	Content type	7750 SR MG release
22625	Nokia-Transparent-Data	O	This attribute defines the transparent data, which may be needed to transfer to Ga/Gx/Gy Interfaces.	String	10.0R9
22626	Nokia-Bandwidth-Policy	O	This attribute defines the Qos profile name.	String	10.0R9
22627	Nokia-Firewall-Policy	O	This attribute defines the firewall user profile name.	String	10.0R9

Re-Auth-Request (RAR)

Table 168: Re-Auth-Request attributes

Attribute name	Presence	Notes and comments	29.212 section
Auth-Application-Id	M	Gx application-id as specified in §5.1 of 29.212	RFC 6733 / §6.8
Charging-Rule-Install	O	Used to activate, install or modify PCC rules as instructed from PCRF to PCEF	29.212 / §5.3.2
Charging-Rule-Remove	O	Used to deactivate or remove PCC rules from an IP-CAN session	29.212 / §5.3.3
Conditional-Policy-Information	O	Up to four Conditional-Policy-Information AVPs may be provisioned to change the authorized QoS per APN based on time conditions. Each instance shall include an Execution-Time AVP.	29.212 / Table 5.3.0.1 / Clause 5.3.128

Table 168: Re-Auth-Request attributes (continued)

Attribute name	Presence	Notes and comments	29.212 section
Default-EPS-Bearer-QoS	O	Indicates the authorized QoS for the default EPS bearer	29.212 / § 5.3.48
Destination-Host	M	FQDN of PGW	RFC 6733 / §6.5
Destination-Realm	M	FQDN of PGW realm	RFC 6733 / §6.6
Event-Report-Indication	O	Used to report an event coming from the Access Network Gateway (for example, HSGW in eHRPD)	29.212 / §5.3.30
Event-Trigger	O	Indicates event triggers, PCRF is interested in and subscribes to.	29.212 / §5.3.7
Origin-Host	M	FQDN of PCRF	RFC 6733 / §6.3
Origin-Realm	M	FQDN of PCRF realm	RFC 6733 / §6.4
Origin-State-Id	O	Indicates the state of the origin-host (PCRF) generating this message	RFC 6733 / §8.16
Proxy-Info	O	Indicates the identity and state of the host that added the proxy-info AVP	RFC 6733 / §6.7.2
QoS-Information	O	Indicates the authorized QoS from the PCRF	29.212 / §5.3.16
PCSCF-Restoration-Indication	O	Indicates that a P-CSCF Restoration is requested	29.212 / §5.3.113

Table 168: Re-Auth-Request attributes (continued)

Attribute name	Presence	Notes and comments	29.212 section
Re-Auth-Request-Type	M	AUTHORIZE_ONLY	RFC 4006 / §8.3
Route-Record	O	Indicates the identity of the peer the request was received from. Indicates the route traversed with multiple hops.	RFC 6733 / §6.7.1
Session-Id	M	PCEF generates this, when the DCC session is set up	RFC 6733 / §8.8
Session-Release-Cause	O	Indicates the reason for PCRF initiating a session termination	29.212 / § 5.3.44
Usage-Monitoring-Information	O	Contains the usage monitoring control information (Usage Monitoring-Key, Granted-Service-Unit, Usage-Monitoring-level and so on)	29.212/ § 5.3.60
PRA-Install	O	This AVP (AVP code 2845) is of type Grouped, and is used to provision a list of new or updated PRAs for an IP-CAN session.	3GPP TS 29.274
PRA-Remove	O	This AVP (AVP code 2846) is of type Grouped, and is used to stop the reporting of a list of PRAs for an IP-CAN session.	3GPP TS 29.274

Re-Auth-Answer (RAA)

Table 169: Re-Auth-Answer attributes

Attribute name	Presence	Notes and comments	29.212 section
3GPP-MS-TimeZone	O	Time zone where the UE is currently located	29.061
3GPP-SGSN-Address	O	IPv4 address of the SGSN	29.061
3GPP-SGSN-Ipv6-Address	O	The IPv6 address of the SGSN	29.212
3GPP-SGSN-MCC-MNC	O	MCC & MNC of the SGSN (Gn/Gp) or that of the Access node Gateway (for example, SGW)	29.061
3GPP-User-Location-Info	O	Indicates details of where the UE is currently located	29.061 / § 16.4.7, 29.274 / § 8.24
AN-GW-Address	O	Indicates the address(es) of access node gateway (for example, SGW or HSGW)	29.212 / § 5.3.49
Charging-Rule-Report	O	Indicates the status of the PCC rules.	29.212 / § 5.3.18
Error-Message	O	May accompany a result-code as a human readable error message	RFC 6733 / § 7.3
Error-Reporting-Host	O	Identity of the host that sent the result-code (value other than success) only if the host is different from origin-host	RFC 6733 / § 7.4
Experimental-Result	O	Indicates if a request was completed successfully or an error occurred. Either Result-Code or this AVP is present. Can be used to indicate result codes specific to Gx application	RFC 6733 / § 7.6

Table 169: Re-Auth-Answer attributes (continued)

Attribute name	Presence	Notes and comments	29.212 section
Failed-AVP	O	Indicates a failed AVP for debugging	RFC 6733 / § 7.5
IP-CAN-Type	O	Indicates the type of connectivity access network in which user is connected (for example, 3gpp-EPS)	29.212 / §5.3.27
Origin-Host	M	FQDN of PGW	RFC 6733 / § 6.3
Origin-Realm	M	FQDN of PGW realm	RFC 6733 / § 6.4
Origin-State-Id	O	Indicates the state of the origin-host (MG-ISM) generating this message	RFC 6733 / § 8.16
Proxy-Info	O	Included by PGW if received in RAR	RFC 6733 / § 6.7.2
RAI	O	Routing Area Identity of the SGSN where the UE is registered	29.061
RAT-Type	O	Indicates the Radio Access Technology serving the UE	29.212 / §5.3.31
Result-Code	O	Indicates if the request was completed successfully or an error occurred	RFC 6733 / § 7.1
Session-Id	M	PCEF generate this, when the DCC session is set up.	RFC 6733 / § 8.8

Gy interface messages

Note that the abbreviations listed in the Presence column in the subsequent tables in this section represent the following characteristics.

- M=Mandatory
- C=Conditional
- O=Optional

Credit Control Request (CCR)

Table 170: Credit Control Request (CCR) attributes

Attribute name	Presence	Notes and comments	7750 SR MG release
Session-Id	M	This field identifies the operation session.	5.0R5
Auth-Application-Id	M	The field corresponds to the application ID of the Diameter Credit Control Application and is defined with the value 4.	5.0R5
Origin-Host	M	This field contains the identification of the source point of the operation and the realm of the operation originator.	5.0R5
Origin-Realm	M	This field contains the realm of the operation originator. The "origin-realm" received in CCA-I from the OCS peer is used as the "destination-realm" for subsequent messages (CCR-U and CCR-T) from the gateway for this Gy session.	5.0R5
Destination-Realm	M	This field contains the realm of the operator domain. The realm will be addressed with the domain address of the corresponding public URI. The "origin-realm" received in CCA-I from the OCS peer is used as the "destination-realm" for subsequent messages (CCR-U and CCR-T) from the gateway for this Gy session. Note that CCR-I will	5.0R5

Table 170: Credit Control Request (CCR) attributes (continued)

Attribute name	Presence	Notes and comments	7750 SR MG release
		include the value configured for 'destination-realm' within the Diameter Peer profile configuration.	
Service-Context-Id	M	This field indicates the supported protocol version. (32.299 § 7.1.12)	5.0R5
CC-Request-Type	M	This field defines the transfer type: event for event-based charging and initial, update, terminate for session-based charging.	5.0R5
CC-Request-Number	M	This field contains the sequence number of the transferred messages.	5.0R5
GGSN-Address	O	This field defines the IP address of the P-GW that generated the GPRS/EPC Charging ID.	11.0R1
SGSN-Address	O	This field defines the IP address of the SGSN/ Serving Node (that is S-GW, AGW, ePDG, TWAG from the P-GW/TDF and S4-SGSN, MME from the S-GW) that was used during a report.	11.0R1
Destination-Host	Oc	This field contains the destination peer address of the OCS identity. Not sent in CCR(Initial). Sent in subsequent CCR(U) and CCR(Final) based on Origin-Host received in CCA (Initial)	5.0R5
User-Name	Oc	Contains the username determined by the domain: bearer, sub-system or service as described in middle tier TS. (32.299 § 7.1.15) Username in the format of NAI	5.0R5
Origin-State-Id	Oc	This field contains the state associated to the CTF.	5.0R5

Table 170: Credit Control Request (CCR) attributes (continued)

Attribute name	Presence	Notes and comments	7750 SR MG release
Event-Timestamp	Oc	This field corresponds to the exact time the quota is requested.	5.0R5
Subscription-Id	Om	This field contains the identification of the user that is going to access the service to be identified by the OCS. MSISDN, IMSI & NAI of the UE are populated by default.	5.0R5
Subscription-Id-Type	M	This field determines the type of the identifier, for example value 0 is used for the international E.164 format according to ITU-T E.164 numbering plan.	5.0R5
Subscription-Id-Data	M	This field contains the user data content for example the MSISDN.	5.0R5
Termination-Cause	Oc	This field contains the reason the credit control session was terminated (RFC 6733 § 8.15). For example, If the OCS sends an invalid CCA, the CC session is terminated with a Termination-Cause = Diameter-Bad-Answer.	5.0R5
Multiple-Services-Indicator	Om	This field indicates whether the CTF is capable of handling multiple services independently. This is set to a value of 1 = Multiple-Services-Supported.	5.0R5
Multiple-Services-Credit Control	Oc	This field contains all parameters for the CTF quota management and defines the quotas to allow traffic to flow. There is one instance of MSCC per Rating Group.	5.0R5
Requested-Service-Unit	Oc	This field contains the amount of requested service units for a particular category or an indication	5.0R5

Table 170: Credit Control Request (CCR) attributes (continued)

Attribute name	Presence	Notes and comments	7750 SR MG release
		that units are needed for a particular category, as defined in DCCA. Will be empty.	
Used-Service-Unit	Oc	This field contains the amount of used non-monetary service units measured for a particular category to a particular quota type. Only sent in CCR(U) and CCR(T).	5.0R5
Reporting-Reason	Oc	Used as defined in clause (32.299 § 7.2.175).	5.0R5
Tariff-Change-Usage	Oc	This field identifies the reporting period for the used service unit, that is, before, after or during tariff change.	5.0R5
CC-Time	Oc	This field contains the amount of used time.	5.0R5
CC-Total-Octets	Oc	This field contains the amount of sent and received octets.	5.0R5
Service-Identifier	Oc	This field contains identity of the used service. This ID with the Service-Context-ID together forms an unique identification of the service.	5.0R5
Rating-Group	Oc	This field contains the identifier of a rating group.	5.0R5
Reporting-Reason	Oc	Used as defined in clause (32.299 § 7.2.175).	5.0R5
Trigger	Oc	Used as defined in clause (32.299 § 7.2.235).	5.0R5
Trigger-Type	Oc	Used as defined in clause (32.299 § 7.2.236).	5.0R5
AF-Correlation-Information	Oc	Used as defined in clause (32.299 § 7.2.11).	5.0R5

Table 170: Credit Control Request (CCR) attributes (continued)

Attribute name	Presence	Notes and comments	7750 SR MG release
QoS-Information	Oc	This field contains authorized QoS applicable for service data flow, which either initially triggers the activation of this MSCC instance or was re-authorized. If service data flow specific QoS control is in use, see TS 29.212 for more information. See (29.212 § 5.3.16) for the definition of this AVP.	5.0R5
QoS-Class-Identifier	M	—	5.0R5
Max-Requested-Bandwidth-UL	Oc	—	5.0R5
Max-Requested-Bandwidth-DL	Oc	—	5.0R5
Guaranteed-Bitrate-UL	Oc	—	5.0R5
Guaranteed-Bitrate-DL	Oc	—	5.0R5
Bearer-Identifier	Oc	—	5.0R5
Allocation-Retention-Priority	Oc	—	5.0R5
Priority-Level	Oc	—	5.0R5
Pre-emption-Capability	Oc	—	5.0R5
Pre-emption-Vulnerability	Oc	—	5.0R5
APN-Aggregate-Max-Bitrate-UL	Oc	—	5.0R5

Table 170: Credit Control Request (CCR) attributes (continued)

Attribute name	Presence	Notes and comments	7750 SR MG release
APN-Aggregate-Max-Bitrate-DL	Oc	—	5.0R5
Extended-Max-Requested-BW-UL	Oc	Defines the maximum authorized bandwidth in kbps for uplink	10.0.R7
Extended-Max-Requested-BW-DL	Oc	Defines the maximum authorized bandwidth in kbps for downlink	10.0.R7
Extended-GBR-UL	Oc	Indicates the guaranteed bit rate in kbits per second for a uplink service data flow. The bandwidth contains all the overhead coming from the IP-layer and the layers above, for example, IP, UDP, RTP and RTP payload.	10.0.R7
Extended-GBR-DL	Oc	Indicates the guaranteed bit rate in kbits per second for a downlink service data flow. The bandwidth contains all the overhead coming from the IP-layer and the layers above, for example, IP, UDP, RTP and RTP payload.	10.0.R7
Extended-APN-AMBR-UL	Oc	Defines the total bandwidth usage for the uplink direction of non-GBR QCI at the APN	10.0.R7
Extended-APN-AMBR-DL	Oc	Defines the total bandwidth usage for the downlink direction of non-GBR QCI at the APN	10.0.R7
CC-Correlation-Id	Oc	This field contains information to correlate credit-control requests generated for different components of the service, for example, transport and service level.	5.0R5

Table 170: Credit Control Request (CCR) attributes (continued)

Attribute name	Presence	Notes and comments	7750 SR MG release
User-Equipment-Info	Oc	This field contains the identification of the identity and terminal capability the subscriber is using for the connection to mobile network if available.	5.0R5
User-Equipment-Info-Type	M	This field determines the type of the identifier. The used value is 0 for the international mobile equipment identifier and software version according to 3GPP TS 23.003.	5.0R5
User-Equipment-Info-Value	M	This field contains the user IMEISV.	5.0R5
Proxy-Info	Oc	This field contains information of the host. Not included by the gateway.	5.0R5
Proxy-Host	M	This field contains the identity of the host that added the Proxy-Info field.	5.0R5
Proxy-State	M	This field contains state local information.	5.0R5
Route-Record	Oc	This field contains an identifier inserted by a relaying or proxying node to identify the node it received the message from. Not included by the gateway.	5.0R5
Service-Information	Om	This parameter holds the individual service specific parameters as defined in the corresponding 'middle tier' TS. See 32.251 section 6.3.1.1, 6.3.1.2, and 6.3.3 (Table 6.3.3.1 and Table 6.3.3.2)	5.0R5
PS-Information	Oc	Used as defined in clause 32.299 section 7	5.0R5
3GPP-Charging-Id	Oc	Used as defined in clause 32.299 section 7	5.0R5

Table 170: Credit Control Request (CCR) attributes (continued)

Attribute name	Presence	Notes and comments	7750 SR MG release
Node-Id	Oc	Used as defined in clause 32.299 section 7	5.0R5
3GPP-PDP-Type	Oc	Used as defined in clause 32.299 section 7	5.0R5
PDP-Address	Oc	Used as defined in clause 32.299 section 7	5.0R5
PDN-Address-Prefix-Length	Oc	Used as defined in clause 32.299 section 7	5.0R5
Dynamic-Address-Flag	Oc	Used as defined in clause 32.299 section 7	5.0R5
Dynamic-Address-Flag-Extension	Oc	Used as defined in clause 32.299 section 7	5.0R5
QoS-Information	Oc	Used as defined in clause 32.299 section 7	5.0R5
3gpp-gprs-negotiated-QoS-profile	Oc	Used as defined in clause 32.299 section 7	5.0R5
SGW-Address	Oc	Used as defined in clause 32.299 section 7	5.0R5
Serving-Node-Type	Oc	Used as defined in clause 32.299 section 7	5.0R5
3GPP-IMSI-MCC-MNC	Oc	Used as defined in clause 32.299 section 7	5.0R5
3GPP-GGSN-MCC-MNC	Oc	Used as defined in clause 32.299 section 7	5.0R5
3GPP-NSAPI	Oc	Used as defined in clause 32.299 section 7	5.0R5
Called-Station-Id	Oc	Used as defined in clause 32.299 section 7	5.0R5
3GPP-Session-Stop-Indicator	Oc	Used as defined in clause 32.299 section 7	5.0R5

Table 170: Credit Control Request (CCR) attributes (continued)

Attribute name	Presence	Notes and comments	7750 SR MG release
3GPP-Charging-Characteristics AVP	Oc	Used as defined in clause 32.299 section 7	5.0R5
Charging-Characteristics-Selection-Mode AVP	Oc	Used as defined in clause 32.299 section 7	5.0R5
3GPP-SGSN-MCC-MNC AVP	Oc	Used as defined in clause 32.299 section 7	5.0R5
3GPP-MS-TimeZone AVP	Oc	Used as defined in clause 32.299 section 7	5.0R5
3GPP-User-Location-Info AVP	Oc	Used as defined in clause 32.299 section 7	5.0R5
PDP-Context-Type AVP	Oc	Used as defined in clause 32.299 section 7	5.0R5
CG-Address AVP	Oc	Used as defined in clause 32.299 section 7 The CG-Address AVP (AVP code 846) is of type Address and holds the IP-address of the charging gateway	8.0.R3
Charging-Rule-Base-Name AVP	OcM	Used as defined in clause 32.299 section 7 The Charging-Rule-Base-Name AVP (AVP code 1004) is of type UTF8String, and it indicates the name of a predefined group of PCC rules residing at the PCEF	8.0.R3
UWAN-User-Location-Info	Oc	Used as defined in clause 32.299 section 7	–
User-Location-Info-Time	Oc	Used as defined in clause 32.299 section 7	–
Presence-Reporting-Area-Information	Oc	This AVP (AVP code 2822) is of type Grouped, and contains the information which describes a Presence Reporting Area.	10.0.R7

Credit Control Answer (CCA)

Table 171: Credit Control Answer (CCA) attributes

Attribute name	Presence	Notes and comments	7750 SR MG release
Session-Id	M	This field identifies the operation session.	5.0R5
Result-Code	M	This field contains the result of the specific query. (32.299 § 7.1.11).	5.0R5
Auth-Application-Id	M	The field corresponds to the application ID of the Diameter Credit Control Application and is defined with the value 4.	5.0R5
Origin-Host	M	This field contains the identification of the source point of the operation and the realm of the operation originator.	5.0R5
Origin-Realm	M	This field contains the realm of the operation originator. The 'origin-realm' received in CCA-I from the OCS peer is used as the 'destination-realm' for subsequent messages (CCR-U and CCR-T) from the mobile gateway for this Gy session.	5.0R5
CC-Request-Type	M	This field defines the transfer type: event for event-based charging and initial, update, terminate for session-based charging.	5.0R5
CC-Request-Number	M	This field contains the sequence number of the transferred messages.	5.0R5
CC-Session Failover	Oc	This field contains an indication to the CTF whether or not a failover handling is to be used when necessary.	5.0R5

Table 171: Credit Control Answer (CCA) attributes (continued)

Attribute name	Presence	Notes and comments	7750 SR MG release
Multiple-Services-Credit-Control	Oc	This field contains all parameters for the CTF quota management and defines the quotas to allow traffic to flow. (32.299 § 7.1.9).	5.0R5
Granted-Service-Unit	Oc	This field contains the amount of granted service units for a particular category.	5.0R5
Tariff-Time-Change	Oc	This field identifies the reporting period for the granted service units, that is, before, after or during tariff change.	5.0R5
CC-Time	Oc	This field contains the amount of granted time.	5.0R5
CC-Total-Octets	Oc	This field contains the amount for sent and received octets.	5.0R5
Service-Identifier	Oc	This field contains identity of the used service. This ID with the Service-Context-ID together forms an unique identification of the service.	5.0R5
Rating-Group	Oc	This field contains the identifier of a rating group. (32.299 § 7.1.10).	5.0R5
Validity-Time	Oc	This field defines the time to limit the validity of the granted quota for a given category instance.	5.0R5
Result-Code	Oc	This field contains the result of the query.	5.0R5
Final-Unit-Indication	Oc	This field indicates that the Granted-Service-Unit containing the final units for the service.	5.0R5
Final-Unit-Action	M	—	5.0R5

Table 171: Credit Control Answer (CCA) attributes (continued)

Attribute name	Presence	Notes and comments	7750 SR MG release
Redirect-Server	Oc	—	5.0R5
Redirect-Address-Type	M	Only Redirect-Address-Type = URL supported.	5.0R5
Redirect-Server-Address	M	—	5.0R5
Time-Quota-Threshold	Oc	Used as defined in clause (32.299 § 7.2.229).	5.0R5
Volume-Quota-Threshold	Oc	Used as defined in clause (32.299 § 7.2.243).	5.0R5
Quota-Holding-Time	Oc	Used as defined in clause (32.299 § 7.2.160).	5.0R5
Quota-Consumption-Time	Oc	Used as defined in clause (32.299 § 7.2.159).	5.0R5
Trigger	Oc	Used as defined in clause (32.299 § 7.2.235).	5.0R5
Trigger-Type	Oc	Used as defined in clause (32.299 § 7.2.236). Trigger-Type values 1,2,3,4 and 61 supported.	5.0R5
PS-Furnish-Charging-Information	Oc	Used as defined in clause (32.299 § 7.2.157).	5.0R5
Credit-Control-Failure-Handling	Oc	Used as defined in DCCA	5.0R5
Proxy-Info	Oc	This field contains information of the host.	5.0R5
Proxy-Host	M	This field contains the identity of the host that added the Proxy-Info field.	5.0R5
Proxy-State	M	This field contains state local information.	5.0R5

Table 171: Credit Control Answer (CCA) attributes (continued)

Attribute name	Presence	Notes and comments	7750 SR MG release
Route-Record	Oc	This field contains an identifier inserted by a relaying or proxying node to identify the node it received the message from.	5.0R5
Failed-AVP	Oc	—	5.0R5
Service-Information	Oc	This parameter holds the individual service specific parameters as defined in the corresponding 'middle tier' TS. See 32.251 section 6.3.1.1, 6.3.1.2, and 6.3.3 (Table 6.3.3.1 and Table 6.3.3.2)	5.0R5
PS-Furnish-Charging-Information	Oc	Used as defined in clause 32.299 section 7.2)	5.0R5
Presence-Reporting-Area-Information	Oc	This AVP (AVP code 2822) is of type Grouped, and contains the information which describes a Presence Reporting Area.	10.0.R7
NSN-Header Enrichment (NSN/2013MV)	Oc	This grouped AVP contains the NSN-Header-Enrichment-ID and NSN-Header-Enrichment-Data AVPs. Supported values are 0 to 4,294,967, and 296 (2 ³²).	10.0.R7
NSN-Header-Enrichment-ID (NSN/2014V)	Oc	This AVP is ignored and is of type Unsigned32.	10.0.R7
NSN-Header-Enrichment-Data (NSN/2015MV)	Oc	This AVP contains the customer ID AVP sent from the PCRF toward the mobile gateway in CCA-I. The AVP is a 10-byte string which is encoded from PCRF in the UTF-8 format (UTF8String). For example, if the customer ID value in PCRF is 1234123412, the value 31323334313233343132 is sent to the mobile gateway.	10.0.R7

Table 172: VSAs 26/6527 - Nokia vendor-specific CC-Answer attributes

Sub-attr no.	Sub-attribute name	Presence	Description	Content type	7750 SR MG release
–	URL-Append	O	This value is used to indicate to the PCEF whether to append the URL received from the user to the redirect URL received in the Redirect-Server AVP.	Enumerated	10.0R9

Re-Auth-Request (RAR)

Table 173: Re-Auth-Request Gy attributes

Attribute name	Presence	Notes and comments	7750 SR MG release
Session-Id	M	PCEF generates this, when the DCC session is set up	5.0R5
Origin-Host	M	FQDN of OCS	5.0R5
Origin-Realm	M	FQDN of OCS realm	5.0R5
Destination-Realm	M	FQDN of PGW realm	5.0R5
Destination-Host	M	FQDN of PGW	5.0R5
Auth-Application-Id	M	—	5.0R5
Re-Auth-Request-Type	M	AUTHORIZE_ONLY	5.0R5
User-Name	Oc	—	5.0R5
Origin-State-Id	Oc	—	5.0R5

Table 173: Re-Auth-Request Gy attributes (continued)

Attribute name	Presence	Notes and comments	7750 SR MG release
Proxy-Info	Oc	—	5.0R5
Route-Record	Oc	—	5.0R5
Service-Identifier	Oc	—	5.0R5
Rating-Group	Oc	—	5.0R5

Re-Auth-Answer (RAA)

Table 174: Re-Auth-Answer Gy attributes

Attribute name	Presence	Notes and comments	7750 SR MG release
Session-Id	M	PCEF generate this, when the DCC session is set up.	5.0R5
Origin-Host	M	FQDN of PGW	5.0R5
Origin-Realm	M	FQDN of PGW realm	5.0R5
Result-Code	M	—	5.0R5
User-Name	Oc	—	5.0R5
Origin-State-Id	Oc	—	5.0R5
Error-Message	Oc	—	5.0R5

Table 174: Re-Auth-Answer Gy attributes (continued)

Attribute name	Presence	Notes and comments	7750 SR MG release
Error-Reporting-Host	Oc	—	5.0R5
Failed-AVP	Oc	—	5.0R5
Proxy-Info	Oc	Included by PGW if received in RAR	5.0R5

Abort-Session-Request (ASR)

Table 175: Abort-Session-Request Gy Attributes

Attribute name	Presence	Notes and comments	7750 SR MG release
Session-Id	M	PCEF generates this, when the DCC session is set up	5.0R5
Auth-Application-Id	M	—	5.0R5
Origin-Host	M	FQDN of OCS	5.0R5
Origin-Realm	M	FQDN of OCS realm	5.0R5
Destination-Realm	M	FQDN of PGW realm	5.0R5
Destination-Host	M	FQDN of PGW	5.0R5
Origin-State-Id	Oc	—	5.0R5
User-Name	Oc	—	5.0R5

Table 175: Abort-Session-Request Gy Attributes (continued)

Attribute name	Presence	Notes and comments	7750 SR MG release
Proxy-Info	Oc	—	5.0R5
Route-Record	Oc	—	5.0R5

Abort-Session-Answer (ASA)

Table 176: Abort-Session-Answer Gy Attributes

Attribute name	Presence	Notes and comments	7750 SR MG release
Session-Id	M	PCEF generate this, when the DCC session is set up.	5.0R5
Origin-Host	M	FQDN of PGW	5.0R5
Origin-Realm	M	FQDN of PGW realm	5.0R5
Result-Code	M	—	5.0R5
User-Name	Oc	—	5.0R5
Origin-State-Id	Oc	—	5.0R5
Error-Message	Oc	—	5.0R5
Error-Reporting-Host	Oc	—	5.0R5
Failed-AVP	Oc	—	5.0R5

Table 176: Abort-Session-Answer Gy Attributes (continued)

Attribute name	Presence	Notes and comments	7750 SR MG release
Proxy-Info	Oc	Included by PGW if received in RAR	5.0R5

TDF Credit Control Request message AVPs

Table 177: TDF Debit / Reserve Units Request message attributes

Attribute name	Presence	Notes and comments	TDF I/ U/T/E
Session-Id	M	This field identifies the operation session.	IUT-
Originator-Host	M	This field contains the identification of the source point of the operation and the realm of the operation originator.	IUT-
Originator-Domain	M	This field contains the identification of the source point of the operation and the realm of the operation originator.	IUT-
Destination-Domain	M	—	IUT-
Operation-Identifier	M	This field contains the realm of the operator domain. The realm will be addressed with the domain address of the corresponding public URI.	IUT-
Operation-Token	M	—	IUT-
Operation-Type	M	—	IUT-
Operation-Number	M	—	IUT-

Table 177: TDF Debit / Reserve Units Request message attributes (continued)

Attribute name	Presence	Notes and comments	TDF I/ U/T/E
Destination-Host	Oc	This field contains the destination peer address of the OCS identity.	IUT-
User-Name	Oc	Contains the username determined by the domain: bearer, sub-system or service as described in middle tier TS. (32.299 §7.1.15) Username in the format of NAI	IUT-
Origination-State	Oc	This field contains the state associated to the CTF.	IUT-
Origination-timestamp	Oc	—	IUT-
Subscriber-Identifier	Om	This field contains the identification of the user that is going to access the service to be identified by the OCS. MSISDN, IMSI & NAI of the UE populated by default.	IUT-
Termination-Cause	Oc	This field contains the reason the credit control session was terminated (RFC 6733 § 8.15). For example, If the OCS sends an invalid CCA, the CC session is terminated with a Termination-Cause = Diameter-Bad-Answer.	--T-
Requested-Action	—	—	IUT-
Multiple-Operation	—	—	IU--
Multiple-Unit-Operation	—	—	IUT-
Subscriber-Equipment-Number	Oc	This field contains the identification of the identity and terminal capability the subscriber is using for the connection to mobile network if available.	IUT-

Table 177: TDF Debit / Reserve Units Request message attributes (continued)

Attribute name	Presence	Notes and comments	TDF I/ U/T/E
Route-Information	Oc	This field contains an identifier inserted by a relaying or proxying node to identify the node it received the message from. Not included by the gateway.	IUT-
Service-Information	Om	This parameter holds the individual service specific parameters as defined in the corresponding 'middle tier' TS. See 32.251 section 6.3.1.1, 6.3.1.2, and 6.3.3 (Table 6.3.3.1 and Table 6.3.3.2)	IUT-
Service Information with PS Information			
Charging-Id	Oc	Used as defined in clause of 32.299 section 7)	—
Service-Context-Id	M	This field indicates the supported protocol version. (32.299 §7.1.12)	IUT-
Node-Id	Oc	Used as defined in clause of 32.299 section 7)	IUT-
PDN-Connection-charging-id	Oc	Specifies the PDN connection charging identifier, used to identify different records belonging to the same PDN connection. This field includes Charging ID of first IP-CAN bearer activated within the PDN connection.	IUT-
PDN-Type	Oc	Indicates the PDN Type, that is, IPv4, IPv6, IPv4v6.	—
PDP-PDN-Address	Oc	Used as defined in clause of 32.299 section 7)	IUT-
PDN-Address-Prefix-Length	Oc	Used as defined in clause of 32.299 section 7)	IUT-
Dynamic-Address-Flag	Oc	Used as defined in clause of 32.299 section 7)	IUT-

Table 177: TDF Debit / Reserve Units Request message attributes (continued)

Attribute name	Presence	Notes and comments	TDF I/ U/T/E
Dynamic-Address-Flag-Extension	Oc	Used as defined in clause of 32.299 section 7)	IUT-
QoS-Information	Oc	This field contains authorized QoS applicable for service data flow, which either initially triggers the activation of this MSCC instance or was re-authorized. If service data flow specific QoS control is in use, see TS 29.212 for more information. See (29.212 §5.3.16) for the definition of this AVP.	IUT-
Serving-Node-Address	Oc	Used as defined in clause of 32.299 section 7)	IUT-
Serving-Node-Type	Oc	Used as defined in clause of 32.299 section 7)	IUT-
SGW-change	Oc	Indicates this is the first Accounting Request (ACR) [Start] due to an inter Node change (from SGW, e PDG, HSGW). If this AVP is not present, this means this ACR [Start] is not due to an inter Node change. The following values are defined: • 0 — ACR_Start_NOT_due_to_SGW_Change • 1 — ACR_Start_due_to_SGW_Change	—
PGW-Address	Oc	Indicates this is the first Accounting Request (ACR) [Start] due to an inter Node change (from PGW, e PDG, HSGW). If this AVP is not present, this means this ACR [Start] is not due to an inter Node change.	IUT-
TDF-Address	Oc	—	IUT-
CG-Address	Oc	—	IUT-
IMSI-MCC-MNC	Oc	MCC and MNC extracted from the user's IMSI (first 5 or 6 digits, as applicable from the presented IMSI).	IUT-

Table 177: TDF Debit / Reserve Units Request message attributes (continued)

Attribute name	Presence	Notes and comments	TDF I/ U/T/E
IMSI-Unauthenticated-Flag	Oc	The IMSI-Unauthenticated-Flag AVP is of type Enumerated, and indicates the served IMSI is not authenticated. This may occur when emergency bearer service is provided (refer to TS 23.060 and TS 29.274). If this flag is not present, this means the served IMSI is authenticated. The following values are defined: • 0 — Authenticated • 1 — Unauthenticated	IUT-
PGW-MCC-MNC	Oc	MCC-MNC of the network the PGW belongs to.	IUT-
TDF-MCC-MNC	Oc	MCC-MNC of the network the TDF belongs to.	IUT-
NSAPI	Oc	Used as defined in clause 32.299 section 7)	IUT-
Called-Station-Id	Oc	Used as defined in clause 32.299 section 7)	IUT-
Session-Stop-Indicator	Oc	Used as defined in clause 32.299 section 7)	--T-
Selection-Mode	Oc	Used as defined in clause 32.299 section 7)	IUT-
Charging-Characteristics	Oc	Used as defined in clause 32.299 section 7)	IUT-
Charging-Characteristics Selection Mode	Oc	Used as defined in clause 32.299 section 7)	IUT-
Serving-Node-MCC-MNC	Oc	Selected Serving Network ID	IUT-
MS-TimeZone	Oc	Used as defined in clause 32.299 section 7)	IUT-

Table 177: TDF Debit / Reserve Units Request message attributes (continued)

Attribute name	Presence	Notes and comments	TDF I/ U/T/E
Charging-Rule-Base-Name	Oc	Indicates the name of a pre-defined group of PCC rules residing at the PCEF.	—
ADC-Rule-Base-Name	Oc	Indicates the name of a pre-defined group of ADC rules residing at the PCEF.	IUT-
User-Location-Info	Oc	Used as defined in clause 32.299 section 7)	IUT-
User-Location-Info-Time	Oc	Used as defined in clause 32.299 section 7)	IUT-
User-CSG-Information	Oc	Used as defined in clause 32.299 section 7)	IUT-
3GPP-User-Location-Info	Oc	Used as defined in clause 32.299 section 7)	IUT-
TWAN-User-Location-Info	Oc	Used as defined in clause 32.299 section 7)	IUT-
Presence-Reporting-Area-Info	—	Presence Area Identity where the UE is registered	IUT-
RAT-Type	Oc	Indicate which Radio Access Technology is currently serving the UE.	IUT-
Furnish-Charging-Information	—	—	—
Offline-Charging-Information	O	Indicates the information for offline for offline preconfigured at the PCEF	IUT-
PDP-Context-Type	Oc	Used as defined in clause 32.299 section 7)	—
Traffic-Data-Volumes	Oc	Its purpose is to allow the transmission of the IP-CAN bearer container, on encountering change on charging condition for this IP-CAN bearer.This	—

Table 177: TDF Debit / Reserve Units Request message attributes (continued)

Attribute name	Presence	Notes and comments	TDF I/ U/T/E
		container reports the volume count (separated for uplink and downlink).	
Service-Data-Container	Oc	Its purpose is to allow the transmission of the container to be reported for Flow-based Charging. On encountering change on charging condition, this container identifies the volume count (separated for uplink and downlink), elapsed time or number of events, per service data flow identified per rating group or combination of the rating group and service-id within an IP-CAN bearer.	—
User-Equipment-Info	Oc	This field contains the identification of the identity and terminal capability the subscriber is using for the connection to mobile network if available.	—
Terminal-Information	Oc	—	—
Start-Time	Oc	This AVP contains a time-stamp (in UTC format) which represents the start of a PDN session or a PDN session at a specific QCI at the SGW.	—
Stop-Time	Oc	This AVP contains a time-stamp (in UTC format) which represents the termination of a PDN or a PDN at a specific QCI at the SGW.	—
Change-Condition	Oc	Indicates the change in charging condition; for example, Volume Limit	—
Diagnostics	Oc	Provides a more detailed cause value for sending Accounting-request from PCN node.	—
CN Operator Selection Entity	Oc	—	—

TDF Credit Control Response message AVPs

Table 178: TDF Debit / Reserve Units Response message attributes

Attribute name	Presence	Notes and comments	TDF I/ U/T/E
Session-Id	M	This field identifies the operation session.	IUT-
Result-Code	Oc	This field contains the result of the query.	IUT-
Originator-Host	M	This field contains the identification of the source point of the operation and the realm of the operation originator.	IUT-
Originator-Domain	M	This field contains the identification of the source point of the operation and the realm of the operation originator.	IUT-
Operation-Identifier	M	This field contains the realm of the operator domain. The realm will be addressed with the domain address of the corresponding public URI.	IUT-
Operation-Type	M	—	IUT-
Operation-Number	M	—	IUT-
Operation-Failover	Oc	This field contains an indication of whether or not a failover handling is to be used when necessary.	IUT-
Multiple-Unit-Operation	Oc	This field contains an indication of whether or not a multiple unit is to be used when necessary.	IUT-
Operation-Failover-Action	Oc	This field contains an indication of the failover handling action to be used when necessary.	IUT-
Redirect-Host	Oc	Redirect host	IUT-

Table 178: TDF Debit / Reserve Units Response message attributes (continued)

Attribute name	Presence	Notes and comments	TDF I/ U/T/E
Redirect-Host-Usage	Oc	Redirect host usage	IUT-
Redirect-Cache-Time	Oc	Redirect cache time	IUT-
Route-Information	Oc	This field contains an identifier inserted by a relaying or proxying node to identify the node it received the message from. Not included by the gateway.	IUT-
Failed-Parameter	Oc	—	IUT-
Service-Information	Oc	This parameter holds the individual service specific parameters as defined in the corresponding 'middle tier' TS. See 32.251 section 6.3.1.1, 6.3.1.2, and 6.3.3 (Table 6.3.3.1 and Table 6.3.3.2)	IUT-
Service Information with PS Information			
Furnish-Charging-Information	Oc	Used as defined in clause 32.299 section 7.2)	IUT-
Offline-Charging	O	Indicates offline preconfigured at the PCEF	IUT-
Presence-Reporting-Area-Information	—	Presence Area Identity where the UE is registered	IUT-

Rf interface messages

Note that the abbreviations listed in the Presence column in the subsequent tables in this section represent the following characteristics.

- M=Mandatory
- C=Conditional

- O=Optional

Accounting-Request (ACR)

The generic attributes for the Accounting-Request message apply to all gateway types.

Population of the AVPs for ACR message types S (Start)/I (Interim)/S (Stop)/- is as per 3GPP TS 32.251 section 6.3.2.

Table 179: Accounting-Request generic attributes

Attribute name	Presence	Notes and comments	Reference
Accounting-Record-Number	M	The Accounting-Record-Number AVP identifies this record within one session	RFC 6733 / §9.8.3
Accounting-Record-Type	M	This field defines the transfer type: event for event-based charging and start, interim, stop for session-based charging.	RFC 6733 / §9.8.1
Acct-Application-Id	Om	The field corresponds to the application ID of the Diameter Accounting Application and is defined with the value 3.	RFC 6733 / §6.9
Acct-Interim-Interval	Oc	Uses information in this AVP to decide how and when to produce accounting records.	RFC 6733 / §9.8.2
Destination-Host	O	This field contains the destination peer address of the CCF identity. This AVP is not defined in the ACR in 3GPP. The AVP is defined in RFC 4005.	RFC 4005
Destination-Realm	M	This field contains the realm of the operation originator.	RFC 6733 / §6.6
Event-Timestamp	Oc	This field corresponds to the exact time the accounting is requested.	RFC 6733 / §8.21
Origin-Host	M	This field contains the identification of the source point of the operation and the realm of the operation originator.	RFC 6733 / §6.3

Table 179: Accounting-Request generic attributes (continued)

Attribute name	Presence	Notes and comments	Reference
Origin-Realm	M	This field contains the realm of the operation originator.	RFC 6733 / §6.4
Origin-State-Id	Oc	This field contains the state associated to the CTF.	RFC 6733 / §8.16
GGSN-Address	O	This field defines the IP address of the P-GW that generated the GPRS/EPC Charging ID.	3GPP 32.299 V14.6.0
SGSN-Address	O	This field defines the IP address of the SGSN/ Serving Node (that is S-GW, AGW, ePDG, TWAG from the P-GW/TDF and S4-SGSN, MME from the S-GW) that was used during a report.	3GPP 32.299 V14.6.0
Proxy-Info	Oc	This field contains information of the host.	RFC 6733 / §6.7.2
Route-Record	Oc	This field contains an identifier inserted by a relaying or proxying node to identify the node it received the message from.	RFC 6733 / §6.7.1
Service-Context-Id	Om	This field indicates the service and the corresponding 'middle tier' TS. 32.251@3gpp.org; For example, for 32.251 v9.4.0 => 4.mnc.mcc.9.32251@3gpp.org For 32.251 v10.5.0 => 5.mnc.mcc.10.32251@3gpp.org	TS 32.299 / §7.1.7
Service-Information	Om	Its purpose is to allow the transmission of additional service specific information elements.	TS 32.299 / §7.2.149, TS 32.251 / §6.3.1
Session-Id	M	This field identifies the operation session.	RFC 6733 / §8.8
User-Name	Oc	Contains the user name determined by the domain: bearer, sub-system or service as	RFC 6733 / §8.14

Table 179: Accounting-Request generic attributes (continued)

Attribute name	Presence	Notes and comments	Reference
		described in middle tier TS. For LTE this will contain IMSI.	

Accounting-Answer (ACA)

The generic attributes for the Accounting-Answer message apply to all gateway types.

Population of the AVPs for ACA message types S (Start)/I (Interim)/S (Stop)/- is as per 3GPP TS 32.251 section 6.3.2.

Table 180: Accounting-Answer generic attributes

Attribute name	Presence	Notes and comments	Reference
Accounting-Record-Number	M	The Accounting-Record-Number AVP identifies this record within one session.	RFC 6733 / §9.8.3
Accounting-Record-Type	M	This field defines the transfer type: event for event-based charging and start, interim, stop for session-based charging.	RFC 6733 / §9.8.1
Acct-Application-Id	Om	The field corresponds to the application ID of the Diameter Accounting Application and is defined with the value 3.	RFC 6733 / §6.9
Acct-Interim-Interval	Oc	Uses information in this AVP to decide how and when to produce accounting records.	RFC 6733 / §9.8.2
Destination-Realm	M	This field contains the realm of the operatioAccn originator.	RFC 6733 / §6.6
Error-Reporting-Host	Oc	This field contains the identity of the Diameter host that sent the Result-Code AVP to a value other than 2001 (Success) if the host setting the	RFC 6733 / §7.4

Table 180: Accounting-Answer generic attributes (continued)

Attribute name	Presence	Notes and comments	Reference
		Result-Code is different from the one encoded in the Origin-Host AVP.	
Event-Timestamp	Oc	This field corresponds to the exact time the accounting is requested.	RFC 6733 / §8.21
Origin-Host	M	This field contains the identification of the source point of the operation and the realm of the operation originator.	RFC 6733 / §6.3
Origin-Realm	M	This field contains the realm of the operation originator.	RFC 6733 / §6.4
Origin-State-Id	Oc	This field contains the state associated to the CTF.	RFC 6733 / §8.16
Proxy-Info	Oc	This field contains information of the host.	RFC 6733 / §6.7.2
Result-Code	M	This field contains the result of the specific query.	RFC 6733 / §7.1, TS 32.299 / §7.1.6
Route-Record	Oc	This field contains an identifier inserted by a relaying or proxying node to identify the node it received the message from.	RFC 6733 / §6.7.1
Session-Id	M	This field identifies the operation session.	RFC 6733 / §8.8
User-Name	Oc	Contains the user name determined by the domain: bearer, sub-system or service as described in middle tier TS. For LTE this will contain IMSI.	RFC 6733 / §8.14

ePDG Rf interface attributes

Table 181: ePDG Rf attributes

Attribute Name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
3GPP-Charging-Characteristics	SIS-	Oc	Contains the charging characteristics for the IP-CAN bearer.	MG 7.0. R1
3GPP-Charging-Id	SIS-	Oc	Charging ID for this PDN Session (this together with the PGW IP address constitutes a unique identifier for the PDN session).	MG 7.0. R1
3GPP-GGSN-MCC-MNC	SIS-	Oc	MCC-MNC of the network the PGW belongs to.	MG 7.0. R1
3GPP-IMSI-MCC-MNC	SIS-	Oc	MCC and MNC extracted from the user's IMSI (first 5 or 6 digits, as applicable from the presented IMSI).	MG 7.0. R1
3GPP-PDP-Type	SIS-	Oc	For SGW, it indicates the PDN Type, that is, IPv4, IPv6, IPv4v6.	MG 7.0. R1
3GPP-RAT-Type	SIS-	Oc	Indicate which Radio Access Technology is currently serving the UE.	MG 7.0. R1
3GPP-Selection-Mode	SIS-	Oc	For SGW it contains the Selection mode for this EPS Bearer received in the Create Session Request message.	MG 7.0. R1
Accounting-Input-Octets	-IS-	Oc	This AVP contains the number of octets in IP packets received from the user (upstream).	MG 7.0. R1

Table 181: ePDG Rf attributes (continued)

Attribute Name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
Accounting-Input-Packets	-IS-	Oc	This AVP contains the number of packets received from the user (upstream).	MG 7.0. R1
Accounting-Output-Octets	-IS-	Oc	This AVP contains the number of octets in IP packets sent to the user (downstream).	MG 7.0. R1
Accounting-Output-Packets	-IS-	Oc	This AVP contains the number of packets sent to the user (downstream).	MG 7.0. R1
Accounting-Record-Number	SIS-	M	The Accounting-Record-Number AVP identifies this record within one session	MG 7.0. R1
Accounting-Record-Type	SIS-	M	This field defines the transfer type: event for event-based charging and start, interim, stop for session-based charging.	MG 7.0. R1
Acct-Application-Id	SIS-	Om	The field corresponds to the application ID of the Diameter Accounting Application and is defined with the value 3.	MG 7.0. R1
Acct-Interim-Interval	SIS-	Oc	Uses information in this AVP to decide how and when to produce accounting records.	MG 7.0. R1
Allocation-Retention-Priority	-IS-	Oc	It is used to indicate the priority of allocation and retention, the pre-emption capability and pre-emption vulnerability for the TDV if provided within the QoS-Information-AVP.	MG 7.0. R1

Table 181: ePDG Rf attributes (continued)

Attribute Name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
			Allocation-Retention-Priority::= < AVP Header: 1034 > {Priority-Level} [Pre-emption-Capability] [Pre-emption-Vulnerability]	
APN-Aggregate-Max-Bitrate-DL	-IS-	Oc	It indicates the maximum aggregate bit rate in bits per seconds for the downlink direction across all non-GBR bearers related with the same APN.	MG 7.0. R1
APN-Aggregate-Max-Bitrate-UL	-IS-	Oc	It indicates the maximum aggregate bit rate in bits per seconds for the uplink direction across all non-GBR bearers related with the same APN.	MG 7.0. R1
Called-Station-Id	SIS-	Oc	The address the user is connected to. This is the APN name.	MG 7.0. R1
Change-Condition	-IS-	Oc	Indicates the change in charging condition: (for example, QoS change).	MG 7.0. R1
Change-Condition	-IS-	Oc	Indicates the change in charging condition; for example, Volume Limit	MG 7.0. R1
Change-Time	-IS-	Oc	In EPC Charging, it holds the time in UTC format when the volume counts associated to the IP-CAN bearer, or the service data container, is closed and reported due to Charging condition change.	MG 7.0. R1
Charging-Characteristics-Selection-Mode	SIS-	Oc	This AVP is of type Enumerated and indicates how the applied Charging-Characteristics was selected.	MG 7.0. R1

Table 181: ePDG Rf attributes (continued)

Attribute Name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
			It may take following values: • 0 — Serving-Node-Supplied • 1 — Subscription-specific • 2 — APN-specific • 3 — Home-Default • 4 — Roaming-Default • 5 — Visiting-Default	
Charging-Gateway-Function-Host	SIS-	Oc	This AVP is of type OctetString and represents a fully qualified domain name (FQDN) for the CGF. CCF assumes its FQDN as the default FQDN if this AVP is absent. The value for this AVP is provisioned at the HSS. FQDN is in normal dotted format encoded in ASCII.	MG 7.0. R1
Charging-Group-ID	SIS-	Oc	Identify subscriber charging group. The value for this AVP is provisioned at the HSS.	MG 7.0. R1
Connected-Duration	-IS-	Oc	This AVP provides the duration of time in seconds the UE is in ECM-CONNECTED State.	MG 7.0. R1
Destination-Host	SIS-	Oc	This field contains the destination peer address of the CCF identity. This AVP is not defined in the ACR in 3GPP. The AVP is defined in RFC 4005.	MG 7.0. R1

Table 181: ePDG Rf attributes (continued)

Attribute Name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
Destination-Realm	SIS-	M	This field contains the realm of the operator domain. The realm will be addressed with the domain address of the corresponding public URI.	MG 7.0. R1
Diagnostics	--S-	Oc	Provides a more detailed cause value for sending Accounting-request from PCN node.	MG 7.0. R1
Dynamic-Address-Flag	SIS-	Oc	Indicates whether the PDP context/ PDN address is statically or dynamically allocated. If not present, this means that the address is statically allocated. The following values are defined: • 0 — Static • 1 — Dynamic The Dynamic-Address-Flag will be reported as Dynamic for sessions whose addresses are allocated through S6b. In cases where the SGW does not know how the address was allocated, the SGW will not populate this AVP.	MG 7.0. R1
Dynamic-Address-Flag-Extension	SIS-	Oc	Indicates whether served IPv4 PDP/ PDN address is dynamic. This field is not populated if IPv4 address is static.	MG 7.0. R1
ePDG-Address	SIS-	Oc	ePDG-Address holds the IP-Address of the ePDG-Node for the control plane. If both an IPv4 and an IPv6	MG 7.0. R1

Table 181: ePDG Rf attributes (continued)

Attribute Name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
			address are available, ePDG shall include the IPv4 address.	
Event-Timestamp	SIS-	Oc	This field corresponds to the exact time the accounting is requested.	MG 7.0. R1
Guaranteed-Bitrate-DL	-IS-	Oc	Indicates the guaranteed bitrate in bits per second for a downlink service data flow. The bandwidth contains all the overhead coming from the IP-layer and the layers above, for example, IP, UDP, RTP, and RTP payload. The AVP will reflect the value SGW receives from the PGW. For Non-GBR traffic, this AVP is not sent from SGW.	MG 7.0. R1
Guaranteed-Bitrate-UL	-IS-	Oc	Indicates the guaranteed bitrate in bits per second for an uplink service data flow. The bandwidth contains all the overhead coming from the IP-layer and the layers above, for example, IP, UDP, RTP, and RTP payload. The AVP will reflect the value SGW receives from the PGW. For Non-GBR traffic, this AVP is not sent from SGW.	MG 7.0. R1
Idle-To-Connected-Transition-Count	-IS-	Oc	This AVP provides the number of transitions between ECM-IDLE and ECM-CONNECTED state of the LTE UE.	MG 7.0. R1
IMS-Information	SIS-	Oc	IMS-Information Grouped AVP for Node-functionality and for future IMS interaction with LTE/eHRPD components.	MG 7.0. R1

Table 181: ePDG Rf attributes (continued)

Attribute Name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
Low-Priority-Indicator	S---	Oc	The AVP indicates if the PDN connection has a low priority. It can be one of the following values: NO 0 YES 1	MG 7.0. R1
Max-Requested-Bandwidth-DL	-IS-	Oc	It indicates the maximum bandwidth in bits per second for a downlink IP flow. The bandwidth contains all the overhead coming from the IP-layer and the layers above, for example, IP, UDP, RTP, and RTP payload. The AVP will reflect the value SGW receives from the PGW. For Non-GBR traffic, this AVP is not sent from SGW.	MG 7.0. R1
Max-Requested-Bandwidth-UL	-IS-	Oc	It indicates the maximum requested bandwidth in bits per second for an uplink IP flow. The bandwidth contains all the overhead coming from the IP-layer and the layers above, for example, IP, UDP, RTP, and RTP payload. The AVP will reflect the value SGW receives from the PGW. For Non-GBR traffic, this AVP is not sent from SGW.	MG 7.0. R1
Node-functionality	SIS-	M	Type Enumerated and includes the functionality identifier of the node. The functionality identifier can be one of the following: • S-CSCF — 0 • P-CSCF — 1 • I-CSCF — 2 • MRFC — 3	MG 7.0. R1

Table 181: ePDG Rf attributes (continued)

Attribute Name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
			• MGCF — 4 • BGCF — 5 • AS — 6 • IBCF — 7 • SGW — 8 • PGW — 9 • HSGW — 10	
Node-Id	SIS-	Oc	Includes an optional, operator configurable identifier string for the node that had generated the Accounting Request (ACR).). The value of this AVP is limited to 20 characters as specified in 32.298.	MG 7.0. R1
Origin-Host	SIS-	M	This field contains the identification of the source point of the operation and the realm of the operation originator.	MG 7.0. R1
Origin-Realm	SIS-	M	This field contains the realm of the operation originator.	MG 7.0. R1
Origin-State-Id	SIS-	Oc	This field contains the state associated to the CTF.	MG 7.0. R1
PDN-Address-prefix-length	SIS-	Oc	PDP/PDN Address prefix length of an IPv6 typed Served PDP Address. This field is not populated if the prefix is 64 bits (default).	MG 7.0. R1
PDN-Connection-charging-id	SIS-	Oc	Specifies the PDN connection charging identifier, used to identify	MG 7.0. R1

Table 181: ePDG Rf attributes (continued)

Attribute Name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
			different records belonging to the same PDN connection. This field includes Charging ID of first IP-CAN bearer activated within the PDN connection.	
PDP-Address	SIS-	Oc	This AVP holds the IP-address associated with the IP CAN bearer session (PDN connection). Note: Multiple occurrences are needed to send both IPv4 and IPv6 addresses of the UE if available.	MG 7.0. R1
Priority-Level	-IS-	Oc	Must be present if Allocation-Retention-Priority AVP is present. The AVP is used for deciding whether a bearer establishment or modification request can be accepted or needs to be rejected in case of resource limitations. The AVP can also be used to decide which existing bearers to pre-empt during resource limitations. The priority level defines the relative importance of a resource request. Values 1 to 15 are defined, with value 1 as the highest level of priority.	MG 7.0. R1
Proxy-Host	SIS-	M	This field contains the identity of the host that added the Proxy-Info field.	MG 7.0. R1
Proxy-Info	SIS-	Oc	This field contains information of the host.	MG 7.0. R1

Table 181: ePDG Rf attributes (continued)

Attribute Name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
Proxy-State	SIS-	M	This field contains state local information.	MG 7.0. R1
PS-Information	SIS-	Oc	PS-Information Grouped AVP.	MG 7.0. R1
QoS-Class-Identifier	-IS-	Oc	It identifies a set of IP-CAN specific QoS parameters that define the authorized QoS, excluding the applicable bitrates and ARP for the IP-CAN bearer or service flow.	MG 7.0. R1
QoS-information	-IS-	Oc	The QoS-Information AVP (AVP code 1016) is of type Grouped, and it defines the QoS information for an IP-CAN bearer, PCC rule or QCI.	MG 7.0. R1
Route-Record	SIS-	Oc	This field contains an identifier inserted by a relaying or proxying node to identify the node it received the message from.	MG 7.0. R1
Service-Context-Id	SIS-	Om	This field indicates the service and the corresponding 'middle tier' TS. 32.251@3gpp.org; For example, for 32.251 v9.4.0 => 4.mnc.mcc.9.32251@3gpp.org For 32.251 v10.5.0 => 5.mnc.mcc.10.32251@3gpp.org	Pre-Release MG 5.0
Service-Information	SIS-	Oc	Its purpose is to allow the transmission of additional service specific information elements.	MG 7.0. R1

Table 181: ePDG Rf attributes (continued)

Attribute Name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
Session-Id	SIS-	M	This field identifies the operation session.	MG 7.0. R1
SGW-Change	S---	Oc	Indicates this is the first Accounting Request (ACR)[Start] due to an inter Node change (from SGW, ePDG, HSGW). If this AVP is not present, this means this ACR [Start] is not due to an inter Node change. The following values are defined: • 0 — ACR_Start_NOT_due_to_SGW_Change • 1 — ACR_Start_due_to_SGW_Change	MG 7.0. R1
Start-Time	S---	Oc	This AVP contains a time-stamp (in UTC format) which represents the start of a PDN session or a PDN session at a specific QCI at the SGW.	MG 7.0. R1
Stop-Time	--S-	Oc	This AVP contains a time-stamp (in UTC format) which represents the termination of a PDN or a PDN at a specific QCI at the SGW.	MG 7.0. R1
Subscription-Id	SIS-	Om	This field contains the identification of the user that is going to access the service. MSISDN is sent.	MG 7.0. R1
Subscription-Id-Data	SIS-	M	MSISDN.From 32.299, the Subscription-Id-Data contains the user data content for example, the MSISDN.	MG 7.0. R1

Table 181: ePDG Rf attributes (continued)

Attribute Name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
			So the contents of the Subscription-Id-Data should just be the E.164 number (that is, CC + National significant number; for example, for USA 1 + NPA + NXX +XXXX for example, 19085821234). The MSISDN complies with the CR0416 (Rel-8) of 3GPP TS 29.274	
Subscription-Id-Type	SIS-	M	This field determines the type of the identifier. For LTE this will be MSISDN. From 32.299, the Subscription-Id-Type determines the type of the identifier, for example, the value 0 is used for the international E.164 format according to ITU-T E.164 numbering plan.	MG 7.0. R1
Traffic-Data-Volumes	-IS-	Oc	Its purpose is to allow the transmission of the IP-CAN bearer container, on encountering change on charging condition for this IP-CAN bearer. This container reports the volume count (separated for uplink and downlink).	MG 7.0. R1
UDP-Source-Port	SIS-	Oc	The UDP-Source-Port AVP (AVP Code 2806) is of type Unsigned32. When the Untrusted WLAN over S2b access is supported, it contains the UDP source port number used for the IKEv2 tunnel, if a NAT is detected. The IKEv2 messages exchanged between the UE and the ePDG are encapsulated in the UDP messages according to IETF RFC 3948.	—

Table 181: ePDG Rf attributes (continued)

Attribute Name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
UE-Local-IP-Address	SIS-	Oc	The UE-Local-IP-Address AVP (AVP code 2805) is of type Address and contains the UE local IP address. The UE local IP address type may be Ipv4 or Ipv6.	—
User-Location-Info-Time	SIS-	Oc	This field contains the NTP time at which the UE was last known to be in the location which is reported during a bearer deactivation or IP-CAN session termination procedure.	—
User-Name	SIS-	Oc	Contains the username determined by the domain: bearer, sub-system or service as described in middle tier TS. For LTE this will contain IMSI.	MG 7.0. R1
UWAN-User-Location-Info	SIS-	Oc	The UWAN-User-Location-Info AVP (AVP code 3918) is of type Grouped and contains the UE location in an Untrusted Wireless Access Network (UWAN) which includes the UE local IP address and optionally the UDP source port number (if NAT is detected) as defined in TS 29.274. It may also include WLAN location information (SSID and, when available, BSSID of the access point) when provided by the AAA server.	—

SGW Rf interface attributes

Table 182: SGW Rf attributes

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
3GPP-Charging-Characteristics	SIS-	Oc	Contains the charging characteristics for the IP-CAN bearer.	Pre-Release MG 5.0
3GPP-Charging-Id	SIS-	Oc	Charging ID for this PDN Session (this together with the PGW IP address constitutes a unique identifier for the PDN session).	Pre-Release MG 5.0
3GPP-GGSN-MCC-MNC	SIS-	Oc	MCC-MNC of the network the PGW belongs to.	Pre-Release MG 5.0
3GPP-IMSI-MCC-MNC	SIS-	Oc	MCC and MNC extracted from the user's IMSI (first 5 or 6 digits, as applicable from the presented IMSI).	Pre-Release MG 5.0
3GPP-MS-TimeZone	SIS-	Oc	Indicate the offset between universal time and local time in steps of 15 minutes of where the MS/UE currently resides.	Pre-Release MG 5.0
3GPP-PDP-Type	SIS-	Oc	For SGW, it indicates the PDN Type, that is, IPv4, IPv6, IPv4v6.	Pre-Release MG 5.0
3GPP-RAT-Type	SIS-	Oc	Indicate which Radio Access Technology is currently serving the UE.	Pre-Release MG 5.0

Table 182: SGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
3GPP-Selection-Mode	SIS-	Oc	For SGW it contains the Selection mode for this EPS Bearer received in the Create Session Request message.	Pre-Release MG 5.0
3GPP-SGSN-MCC-MNC	SIS-	Oc	Selected Serving Network ID received in S11 will be populated for this AVP for LTE.	Pre-Release MG 5.0
3GPP-User-Location-Info	SIS-	Oc	Indicate details of where the UE was at the time of sending the ACR message. The value of this AVP is encoded as specified in TS 29.061.	Pre-Release MG 5.0
3GPP-User-Location-Info	-IS-	Oc	Indicate details of where the UE is currently located (for example, TAI or ECGI). This AVP in the container level is only populated when the previous ACR interim trigger had a ULI change trigger occurrence.	Pre-Release MG 5.0
3GPP2-BSID	SIS-	Oc	For eHRPD indicates the BSID of where the UE was initially attached (for example, Cell-Id, SID, NID). 3GPP2-BSID at initial attachment provided in each SIS ACR message.	Pre-Release MG 5.0
3GPP2-BSID	-IS-	Oc	For eHRPD indicates the BSID of where the UE is currently located (for example, Cell-Id, SID, NID). This AVP in the container level is only populated when the previous ACR	Pre-Release MG 5.0

Table 182: SGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
			interim trigger had a ULI change trigger occurrence.	
Accounting-Input-Octets	-IS-	Oc	This AVP contains the number of octets in IP packets received from the user (upstream).	Pre-Release MG 5.0
Accounting-Input-Packets	-IS-	Oc	This AVP contains the number of packets received from the user (upstream).	Pre-Release MG 5.0
Accounting-Output-Octets	-IS-	Oc	This AVP contains the number of octets in IP packets sent to the user (downstream).	Pre-Release MG 5.0
Accounting-Output-Packets	-IS-	Oc	This AVP contains the number of packets sent to the user (downstream).	Pre-Release MG 5.0
Accounting-Record-Number	SIS-	M	The Accounting-Record-Number AVP identifies this record within one session	Pre-Release MG 5.0
Accounting-Record-Type	SIS-	M	This field defines the transfer type: event for event-based charging and start, interim, stop for session-based charging.	Pre-Release MG 5.0
Acct-Application-Id	SIS-	Om	The field corresponds to the application ID of the Diameter Accounting Application and is defined with the value 3.	Pre-Release MG 5.0

Table 182: SGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
Acct-Interim-Interval	SIS-	Oc	Uses information in this AVP to decide how and when to produce accounting records.	Pre-Release MG 5.0
Allocation-Retention-Priority	SIS-	Oc	It is used to indicate the priority of allocation and retention, the pre-emption capability and pre-emption vulnerability for the TDV if provided within the QoS-Information-AVP. Allocation-Retention-Priority:= < AVP Header: 1034 > {Priority-Level} [Pre-emption-Capability] [Pre-emption-Vulnerability]	Pre-Release MG 5.0
Allocation-Retention-Priority	-IS-	Oc	It is used to indicate the priority of allocation and retention, the pre-emption capability and pre-emption vulnerability for the TDV if provided within the QoS-Information-AVP. Allocation-Retention-Priority:= < AVP Header: 1034 > {Priority-Level} [Pre-emption-Capability] [Pre-emption-Vulnerability]	Pre-Release MG 5.0
APN-Aggregate-Max-Bitrate-DL	SIS-	Oc	It indicates the maximum aggregate bit rate in bits per seconds for the downlink direction across all non-GBR bearers related with the same APN.	Pre-Release MG 5.0

Table 182: SGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
APN-Aggregate-Max-Bitrate-DL	-IS-	Oc	It indicates the maximum aggregate bit rate in bits per seconds for the downlink direction across all non-GBR bearers related with the same APN.	Pre-Release MG 5.0
APN-Aggregate-Max-Bitrate-UL	SIS-	Oc	It indicates the maximum aggregate bit rate in bits per seconds for the uplink direction across all non-GBR bearers related with the same APN.	Pre-Release MG 5.0
APN-Aggregate-Max-Bitrate-UL	-IS-	Oc	It indicates the maximum aggregate bit rate in bits per seconds for the uplink direction across all non-GBR bearers related with the same APN.	Pre-Release MG 5.0
Called-Station-Id	SIS-	Oc	The address the user is connected to. This is the APN name.	Pre-Release MG 5.0
Change-Condition	-IS-	Oc	Indicates the change in charging condition: (for example, QoS change)	Pre-Release MG 5.0
Change-Condition	-IS-	Oc	Indicates the change in charging condition: (for example, Volume Limit)	Pre-Release MG 5.0
Change-Time	-IS-	Oc	In EPC Charging, it holds the time in UTC format when the volume counts associated to the IP-CAN bearer, or the service data container, is closed and reported due to Charging condition change.	Pre-Release MG 5.0

Table 182: SGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
Charging-Characteristics-Selection-Mode	SIS-	Oc	This AVP is of type Enumerated and indicates how the applied Charging-Characteristics was selected. It may take following values: • 0 — Serving-Node-Supplied • 1 — Subscription-specific • 2 — APN-specific • 3 — Home-Default • 4 — Roaming-Default • 5 — Visiting-Default	Pre-Release MG 5.0
Charging-Gateway-Function-Host	SIS-	Oc	This AVP is of type OctetString and represents a fully qualified domain name (FQDN) for the CGF. CCF assumes its FQDN as the default FQDN if this AVP is absent. The value for this AVP is provisioned at the HSS. FQDN is in normal dotted format encoded in ASCII.	Pre-Release MG 5.0
Charging-Group-ID	SIS-	Oc	Identify subscriber charging group. The value for this AVP is provisioned at the HSS.	Pre-Release MG 5.0
Connected-Duration	-IS-	Oc	This AVP provides the duration of time in seconds the UE is in ECM-CONNECTED State.	Pre-Release MG 5.0
CP-CIoT-EPS-Optimisation-Indicator	SIS-	Oc	Indicates if S11-U is currently used.	MG 9. OR1

Table 182: SGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
Destination-Host	SIS-	Oc	This field contains the destination peer address of the CCF identity. This AVP is not defined in the ACR in 3GPP. The AVP is defined in RFC 4005.	Pre-Release MG 5.0
Destination-Realm	SIS-	M	This field contains the realm of the operator domain. The realm will be addressed with the domain address of the corresponding public URI.	Pre-Release MG 5.0
Diagnostics	--S-	Oc	Provides a more detailed cause value for sending Accounting-request from PCN node.	Pre-Release MG 5.0
Dynamic-Address-Flag	SIS-	Oc	Indicates whether the PDP context/ PDN address is statically or dynamically allocated. If not present, this means that the address is statically allocated. The following values are defined: • 0 — Static • 1 — Dynamic The Dynamic-Address-Flag will be reported as Dynamic for sessions whose addresses are allocated through S6b. In cases where the SGW does not know how the address was allocated, the SGW will not populate this AVP.	Pre-Release MG 5.0
Dynamic-Address-Flag-Extension	SIS-	Oc	Indicates whether served IPv4 PDP/ PDN address is dynamic. This field is not populated if IPv4 address is static.	MG 5.0R1

Table 182: SGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
Event-Timestamp	SIS-	Oc	This field corresponds to the exact time the accounting is requested.	Pre-Release MG 5.0
GGSN-Address	SIS-	Oc	PGW Address	Pre-Release MG 5.0
Guaranteed-Bitrate-DL	-IS-	Oc	Indicates the guaranteed bitrate in bits per second for a downlink service data flow. The bandwidth contains all the overhead coming from the IP-layer and the layers above, for example, IP, UDP, RTP, and RTP payload. The AVP will reflect the value SGW receives from the PGW. For Non-GBR traffic, this AVP is not sent from SGW.	Pre-Release MG 5.0
Guaranteed-Bitrate-UL	-IS-	Oc	Indicates the guaranteed bitrate in bits per second for an uplink service data flow. The bandwidth contains all the overhead coming from the IP-layer and the layers above, for example, IP, UDP, RTP, and RTP payload. The AVP will reflect the value SGW receives from the PGW. For Non-GBR traffic, this AVP is not sent from SGW.	Pre-Release MG 5.0
Idle-To-Connected-Transition-Count	-IS-	Oc	This AVP provides the number of transitions between ECM-IDLE and ECM-CONNECTED state of the LTE UE.	Pre-Release MG 5.0

Table 182: SGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
IMS-Information	SIS-	Oc	IMS-Information Grouped AVP for Node-functionality and for future IMS interaction with LTE/eHRPD components.	Pre-Release MG 5.0
IMSI-Unauthenticated-Flag	SIS-	Oc	The IMSI-Unauthenticated-Flag AVP is of type Enumerated, and indicates the served IMSI is not authenticated. This may occur when emergency bearer service is provided (refer to TS 23.060 and TS 29.274). If this flag is not present, this means the served IMSI is authenticated. The following values are defined: • 0 — Authenticated • 1 — Unauthenticated	Pre-Release MG 5.0
Low-Priority-Indicator	S---	Oc	The AVP indicates if the PDN connection has a low priority. It can be one of the following values: NO 0 YES 1	Pre-Release MG 5.0
Max-Requested-Bandwidth-DL	-IS-	Oc	It indicates the maximum bandwidth in bits per second for a downlink IP flow. The bandwidth contains all the overhead coming from the IP-layer and the layers above, for example, IP, UDP, RTP, and RTP payload. The AVP will reflect the value SGW receives from the PGW. For Non-GBR traffic, this AVP is not sent from SGW.	Pre-Release MG 5.0

Table 182: SGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
Max-Requested-Bandwidth-UL	-IS-	Oc	It indicates the maximum requested bandwidth in bits per second for an uplink IP flow. The bandwidth contains all the overhead coming from the IP-layer and the layers above, for example, IP, UDP, RTP, and RTP payload. The AVP will reflect the value SGW receives from the PGW. For Non-GBR traffic, this AVP is not sent from SGW.	Pre-Release MG 5.0
Node-functionality	SIS-	M	Type Enumerated and includes the functionality identifier of the node. The functionality identifier can be one of the following: • S-CSCF — 0 • P-CSCF — 1 • I-CSCF — 2 • MRFC — 3 • MGCF — 4 • BGCF — 5 • AS — 6 • BCF — 7 • SGW — 8 • PGW — 9 • HSGW — 10	Pre-Release MG 5.0
Node-Id	SIS-	Oc	Includes an optional, operator configurable identifier string for the node that had generated the Accounting Request (ACR).). The	Pre-Release MG 5.0

Table 182: SGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
			value of this AVP is limited to 20 characters as specified in 32.298.	
Origin-Host	SIS-	M	This field contains the identification of the source point of the operation and the realm of the operation originator.	Pre-Release MG 5.0
Origin-Realm	SIS-	M	This field contains the realm of the operation originator.	Pre-Release MG 5.0
Origin-State-Id	SIS-	Oc	This field contains the state associated to the CTF.	Pre-Release MG 5.0
PDN-Address-prefix-length	SIS-	Oc	PDP/PDN Address prefix length of an IPv6 typed Served PDP Address. This field is not populated if the prefix is 64 bits (default).	MG 5. OR1
PDN-Connection-charging-id	SIS-	Oc	Specifies the PDN connection charging identifier, used to identify different records belonging to the same PDN connection. This field includes Charging Id of first IP-CAN bearer activated within the PDN connection.	MG 5. OR1
PDP-Address	SIS-	Oc	This AVP holds the IP-address associated with the IP CAN bearer session (PDN connection). Note: Multiple occurrences are needed to send both IPv4 and IPv6 addresses of the UE if available.	Pre-Release MG 5.0

Table 182: SGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
Priority-Level	SIS-	Oc	Must be present if Allocation-Retention-Priority AVP is present. The AVP is used for deciding whether a bearer establishment or modification request can be accepted or needs to be rejected in case of resource limitations. The AVP can also be used to decide which existing bearers to pre-empt during resource limitations. The priority level defines the relative importance of a resource request. Values 1 to 15 are defined, with value 1 as the highest level of priority.	Pre-Release MG 5.0
Priority-Level	-IS-	Oc	Must be present if Allocation-Retention-Priority AVP is present. The AVP is used for deciding whether a bearer establishment or modification request can be accepted or needs to be rejected in case of resource limitations. The AVP can also be used to decide which existing bearers to pre-empt during resource limitations. The priority level defines the relative importance of a resource request. Values 1 to 15 are defined, with value 1 as the highest level of priority.	Pre-Release MG 5.0
Proxy-Host	SIS-	M	This field contains the identity of the host that added the Proxy-Info field.	Pre-Release MG 5.0

Table 182: SGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
Proxy-Info	SIS-	Oc	This field contains information of the host.	Pre-Release MG 5.0
Proxy-State	SIS-	M	This field contains state local information.	Pre-Release MG 5.0
PS-Information	SIS-	Oc	PS-Information Grouped AVP	Pre-Release MG 5.0
QoS-Class-Identifier	SIS-	Oc	It identifies a set of IP-CAN specific QoS parameters that define the authorized QoS, excluding the applicable bitrates and ARP for the IP-CAN bearer or service flow.	Pre-Release MG 5.0
QoS-Class-Identifier	-IS-	Oc	It identifies a set of IP-CAN specific QoS parameters that define the authorized QoS, excluding the applicable bitrates and ARP for the IP-CAN bearer or service flow.	Pre-Release MG 5.0
QoS-information	SIS-	Oc	The QoS-Information AVP (AVP code 1016) is of type Grouped, and it defines the QoS information for an IP-CAN bearer, PCC rule or QCI.	Pre-Release MG 5.0
QoS-information	-IS-	Oc	The QoS-Information AVP (AVP code 1016) is of type Grouped, and it defines the QoS information for an IP-CAN bearer, PCC rule or QCI.	Pre-Release MG 5.0

Table 182: SGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
RAN-Secondary-RAT-Usage-Report	-IS-	Oc	AVP code 1302 is of type Grouped and contains the volume count as reported by the RAN for the secondary RAT (separated for uplink and downlink) including the time of the report.	Release 10.0.R9
[Secondary-RAT-Type]	-IS-	Oc	AVP code 1304 is of type OctetString and holds the following value of secondary RAT Type, as provided by the RAN for transfer to the SGW/PGW in the RAN Traffic Volume element: • 0 — Reserved • 1 — 5G NR	Release 10.0.R9
[RAN-Start-Timestamp]	-IS-	Oc	AVP code 1303 is of type Time and holds in UTC format the start time for the reported usage for the volume container collected.	Release 10.0.R9
[RAN-End-Timestamp]	-IS-	Oc	AVP code 1301 is of type Time and holds in UTC format the end time of the reported usage for the volume container collected.	Release 10.0.R9
[Accounting-Output-Octets]	-IS-	Oc	AVP code 363 contains the number of octets transmitted during the data container recording interval, reflecting the volume counts for uplink traffic for a data flow.	Release 10.0.R9
[Accounting-Input-Octets]	-IS-	Oc	AVP code 364 contains the number of octets transmitted during the data container recording interval,	Release 10.0.R9

Table 182: SGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
			reflecting the volume count for downlink traffic for a data flow.	
Route-Record	SIS-	Oc	This field contains an identifier inserted by a relay or proxy node to identify the node from which it received the message.	Pre-Release MG 5.0
Service-Context-Id	SIS-	Om	This field indicates the service and the corresponding 'middle tier' TS. 32.251@3gpp.org For example, for 32.251 v9.4.0 => 4. mnc.mcc.9.32251@3gpp.org For 32.251 v10.5.0 => 5.mnc.mcc.10.32251@3gpp.org	Pre-Release MG 5.0
Service-Information	SIS-	Oc	Its purpose is to allow the transmission of additional service specific information elements.	Pre-Release MG 5.0
Serving-Node-Type	SIS-	Oc	Type of Serving Node. It may take the following values: • 0 — SGSN • 1 — PMIPSGW • 2 — GTPSGW • 3 — ePDG • 4 — hSGW • 5 — MME For LTE, this will be 5(MME).	Pre-Release MG 5.0

Table 182: SGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
Session-Id	SIS-	M	This field identifies the operation session.	Pre-Release MG 5.0
SGSN-Address	SIS-	Oc	MME Address in LTE.	Pre-Release MG 5.0
SGW-Address	SIS-	Oc	SGW-Address holds the IP-Address of the SGW-Node for the control plane. If both an IPv4 and an IPv6 address of the SGW are available, SGW shall include the IPv4 address.	MG 5.0R1
SGW-Change	S---	Oc	Indicates this is the first Accounting Request (ACR)[Start] due to an inter Node change (from SGW, ePDG, HSGW). If this AVP is not present, this means this ACR [Start] is not due to an inter Node change. The following values are defined: • 0 — ACR_Start_NOT_due_to_SGW_Change • 1 — ACR_Start_due_to_SGW_Change	Pre-Release MG 5.0
Start-Time	S---	Oc	This AVP contains a time-stamp (in UTC format) which represents the start of a PDN session or a PDN session at a specific QCI at the SGW.	Pre-Release MG 5.0
Stop-Time	--S-	Oc	This AVP contains a time-stamp (in UTC format) which represents the	Pre-Release MG 5.0

Table 182: SGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
			termination of a PDN or a PDN at a specific QCI at the SGW.	
Subscription-Id	SIS-	Om	This field contains the identification of the user that is going to access the service. MSISDN is sent.	Pre-Release MG 5.0
Subscription-Id-Data	IS-	M	MSISDN. From 32.299, the Subscription-Id-Data contains the user data content for example, the MSISDN. So the contents of the Subscription-Id-Data should just be the E.164 number (that is, CC + National significant number; for example, for USA 1 + NPA + NXX +XXXX for example, 19085821234). The MSISDN complies with the CR0416 (Rel-8) of 3GPP TS 29.274	Pre-Release MG 5.0
Subscription-Id-Type	IS-	M	This field determines the type of the identifier. For LTE this will be MSISDN. From 32.299, the Subscription-Id-Type determines the type of the identifier, for example, the value 0 is used for the international E.164 format according to ITU-T E.164 numbering plan.	Pre-Release MG 5.0
Traffic-Data-Volumes	IS-	Oc	Its purpose is to allow the transmission of the IP-CAN bearer container, on encountering change on charging condition for this IP-CAN bearer. This container reports the	Pre-Release MG 5.0

Table 182: SGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
			volume count (separated for uplink and downlink).	
User-Equipment-Info	S---	Oc	his field contains the identification of the identity and terminal capability the subscriber is using for the connection to mobile network if available hardware ID. This contains a non-standard enumeration for MEID from RFC 4006.	Pre-Release MG 5.0
User-Equipment-Info-Type	S---	Oc	Used for mobile hardware ID type. This field determines the type of the identifier. IMEI will be the hardware Identification for LTE UE. For non-LTE UE, additional ESN and MEID equipment types are defined for this AVP.	Pre-Release MG 5.0
User-Equipment-Info-Value	S---	Oc	Used for mobile hardware ID value. The value of this AVP is encoded as UTF-8 (ASCII).	Pre-Release MG 5.0
User-Name	SIS-	Oc	Contains the user name determined by the domain: bearer, sub-system or service as described in middle tier TS. For LTE this will contain IMSI.	Pre-Release MG 5.0

PGW Rf interface attributes

Table 183: PGW Rf attributes

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
3GPP-Charging-Characteristics	SIS-	Oc	For PGW/GGSN this is the Charging-Characteristics value received via GTP or produced internally from selected charging profile.	Pre-Release MG 5.0
3GPP-Charging-Id	SIS-	Oc	Charging ID for this PDN Session (this together with the PGW IP address constitutes a unique identifier for the PDN session).	Pre-Release MG 5.0
3GPP-GGSN-MCC-MNC	SIS-	Oc	MCC-MNC of the network the PGW belongs to.	Pre-Release MG 5.0
3GPP-IMSI-MCC-MNC	SIS-	Oc	MCC and MNC extracted from the user's IMSI (first 5 or 6 digits, as applicable from the presented IMSI).	Pre-Release MG 5.0
3GPP-MS-TimeZone	SIS-	Oc	Indicate the offset between universal time and local time in steps of 15 minutes of where the MS/UE currently resides. Not supported for eHRPD.	Pre-Release MG 5.0
3GPP-PDP-Type	SIS-	Oc	For SGW, it indicates the PDN Type, that is, IPv4, IPv6, IPv4v6.	Pre-Release MG 5.0
3GPP-RAT-Type	SIS-	Oc	Indicate which Radio Access Technology is currently serving the UE.	Pre-Release MG 5.0

Table 183: PGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
3GPP-Selection-Mode	SIS-	Oc	For PGW it contains the Selection mode for this EPS Bearer received in the Create Session Request message. Not supported for eHRPD.	Pre-Release MG 5.0
3GPP-SGSN-MCC-MNC	SIS-	Oc	MCC-MNC of the SGW in LTE. Not supported for eHRPD.	Pre-Release MG 5.0
3GPP-User-Location-Info	SIS-	Oc	Indicate details of where the UE was at the time of sending the ACR message. The value of this AVP is encoded as specified in TS 29.061. Not supported for eHRPD.	Pre-Release MG 5.0
3GPP-User-Location-Info	-IS-	Oc	Indicate details of where the UE is currently located (for example, TAI or ECGI). This AVP in the container level is only populated when the previous ACR interim trigger had a ULI change trigger occurrence.	Pre-Release MG 5.0
3GPP2-BSID	SIS-	Oc	For eHRPD indicates the BSID of where the UE was initially attached (for, Cell-Id, SID, NID). 3GPP2-BSID at initial attachment provided in each SIS ACR message.	Pre-Release MG 5.0
3GPP2-BSID	-IS-	Oc	For eHRPD indicates the BSID of where the UE is currently located (for example, Cell-Id, SID, NID). This AVP in the container level is only populated when the previous ACR	Pre-Release MG 5.0

Table 183: PGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
			interim trigger had a ULI change trigger occurrence.	
Accounting-Input-Octets	-IS-	Oc	This AVP contains the number of octets in IP Packets received from the user (upstream).	Pre-Release MG 5.0
Accounting-Input-Packets	-IS-	Oc	This AVP contains the number of packets received from the user (upstream).	Pre-Release MG 5.0
Accounting-Output-Octets	-IS-	Oc	This AVP contains the number of octets in IP Packets sent to the user (downstream).	Pre-Release MG 5.0
Accounting-Output-Packets	-IS-	Oc	This AVP contains the number of packets sent to the user (downstream).	Pre-Release MG 5.0
Accounting-Record-Number	SIS-	M	The Accounting-Record-Number AVP identifies this record within one session	Pre-Release MG 5.0
Accounting-Record-Type	SIS-	M	This field defines the transfer type: event for event-based charging and start, interim, stop for session-based charging.	Pre-Release MG 5.0
Acct-Application-Id	SIS-	Om	The field corresponds to the application ID of the Diameter Accounting Application and is defined with the value 3.	Pre-Release MG 5.0

Table 183: PGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
Acct-Interim-Interval	SIS-	Oc	Uses information in this AVP to decide how and when to produce accounting records.	Pre-Release MG 5.0
AF-Correlation-Information	-IS-	Oc	This AVP includes the "AF Charging Identifier" (ICID for IMS) and associated flow Identifiers. This AVP is only applicable when the charging is set at Rating Group & Service-Identifier. This AVP is not sent if the charging is based on Rating Group.	Pre-Release MG 5.0
Allocation-Retention-Priority	SIS-	Oc	It is used to indicate the priority of allocation and retention, the pre-emption capability and pre-emption vulnerability for the TDV if provided within the QoS-Information-AVP. Allocation-Retention-Priority:= < AVP Header: 1034 > {Priority-Level} [Pre-emption-Capability] [Pre-emption-Vulnerability]	Pre-Release MG 5.0
Allocation-Retention-Priority	-IS-	Oc	It is used to indicate the priority of allocation and retention, the pre-emption capability and pre-emption vulnerability for the TDV if provided within the QoS-Information-AVP. Allocation-Retention-Priority:= < AVP Header: 1034 > {Priority-Level} [Pre-emption-Capability] [Pre-emption-Vulnerability]	Pre-Release MG 5.0

Table 183: PGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
APN-Aggregate-Max-Bitrate-DL	SIS-	Oc	It indicates the maximum aggregate bit rate in bits per seconds for the downlink direction across all non-GBR bearers related with the same APN.	Pre-Release MG 5.0
APN-Aggregate-Max-Bitrate-DL	-IS-	Oc	It indicates the maximum aggregate bit rate in bits per seconds for the downlink direction across all non-GBR bearers related with the same APN.	Pre-Release MG 5.0
APN-Aggregate-Max-Bitrate-UL	SIS-	Oc	It indicates the maximum aggregate bit rate in bits per seconds for the uplink direction across all non-GBR bearers related with the same APN.	Pre-Release MG 5.0
APN-Aggregate-Max-Bitrate-UL	-IS-	Oc	It indicates the maximum aggregate bit rate in bits per seconds for the uplink direction across all non-GBR bearers related with the same APN.	Pre-Release MG 5.0
Called-Station-Id	SIS-	Oc	The address the user is connected to. This is the APN name.	Pre-Release MG 5.0
Change-Condition	-IS-	Oc	Indicates the change in charging condition: (for example, QoS change,)	Pre-Release MG 5.0
Change-Condition	-IS-	Oc	Indicates the change in charging condition; for example, Volume Limit	Pre-Release MG 5.0

Table 183: PGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
Change-Time	-IS-	Oc	In EPC Charging, it holds the time in UTC format when the volume counts associated to the IP-CAN bearer, or the service data container, is closed and reported due to Charging condition change.	Pre-Release MG 5.0
Charging-Characteristics-Selection-Mode	SIS-	Oc	This AVP is of type Enumerated and indicates how the applied Charging-Characteristics was selected. It may take following values: • 0 — Serving-Node-Supplied • 1 — Subscription-specific • 2 — APN-specific • 3 — Home-Default • 4 — Roaming-Default • 5 — Visiting-Default	Pre-Release MG 5.0
Charging-Rule-Base-Name	-IS-	Oc	Indicates the name of a pre-defined group of PCC rules residing at the PCEF.	Pre-Release MG 5.0
Destination-Host	SIS-	Oc	This field contains the destination peer address of the CCF identity. This AVP is not defined in the ACR in 3GPP. The AVP is defined in RFC 4005.	Pre-Release MG 5.0
Destination-Realm	SIS-	M	This field contains the realm of the operator domain. The realm will be addressed with the domain address of the corresponding public URI.	Pre-Release MG 5.0

Table 183: PGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
Diagnostics	--S-	Oc	Provides a more detailed cause value for sending Accounting-request from PCN node.	Pre-Release MG 5.0
Dynamic-Address-Flag	SIS-	Oc	Indicates whether the PDP context/ PDN address is statically or dynamically allocated. If not present, this means that the address is statically allocated. The following values are defined: • 0 — Static • 1 — Dynamic The Dynamic-Address-Flag will be reported as Dynamic for sessions whose addresses are allocated through S6b.	Pre-Release MG 5.0
Dynamic-Address-Flag-Extension	SIS-	Oc	Indicates whether served IPv4 PDP/ PDN address is dynamic. This field is not populated if IPv4 address is static.	MG 5. OR1
Event-Timestamp	SIS-	Oc	This field corresponds to the exact time the accounting is requested.	Pre-Release MG 5.0
GGSN-Address	SIS-	Oc	PGW Address	Pre-Release MG 5.0
Guaranteed-Bitrate-DL	-IS-	Oc	The Guaranteed-Bitrate-DL AVP (AVP code 1025) is of type Unsigned32, and it indicates the guaranteed	Pre-Release MG 5.0

Table 183: PGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
			bitrate in bits per second for a downlink service data flow.	
Guaranteed-Bitrate-UL	-IS-	Oc	(AVP code 1026) is of type Unsigned32, and it indicates the guaranteed bitrate in bits per second for an uplink service data flow.	Pre-Release MG 5.0
IMS-Information	SIS-	Oc	IMS-Information Grouped AVP for Node-functionality and for future IMS interaction with LTE/eHRPD components.	Pre-Release MG 5.0
IMSI-Unauthenticated-Flag	SIS-	Oc	The IMSI-Unauthenticated-Flag AVP is of type Enumerated, and indicates the served IMSI is not authenticated. This may occur when emergency bearer service is provided (refer to TS 23.060 and TS 29.274). If this flag is not present, this means the served IMSI is authenticated. The following values are defined: • 0 — Authenticated • 1 — Unauthenticated	Pre-Release MG 5.0
Local-Sequence-Number	-IS-	Oc	Holds the service data container sequence number: increased by 1 for each service data container closed in 1 PDN session.	Pre-Release MG 5.0
Low-Priority-Indicator	S---	Oc	The AVP indicates if the PDN connection has a low priority. It can be one of the following values: NO 0 YES 1	Pre-Release MG 5.0

Table 183: PGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
Max-Requested-Bandwidth-DL	-IS-	Oc	(AVP code 1026) is of type Unsigned32, and it indicates the guaranteed bitrate in bits per second for an uplink service data flow.	Pre-Release MG 5.0
Max-Requested-Bandwidth-UL	-IS-	Oc	The Max -Bandwidth-UL AVP (AVP code 516) is of type Unsigned32, and it indicates the maximum requested bandwidth in bits per second for an uplink IP flow [29.214 section 5.3.15]	Pre-Release MG 5.0
Node-functionality	SIS-	M	Type Enumerated and includes the functionality identifier of the node. The functionality identifier can be one of the following: • S-CSCF — 0 • P-CSCF — 1 • I-CSCF — 2 • MRFC — 3 • MGCF — 4 • BGCF — 5 • AS — 6 • IBCF — 7 • SGW — 8 • PGW — 9 • HSGW — 10	Pre-Release MG 5.0
Node-Id	SIS-	Oc	Includes an optional, operator configurable identifier string for the node that had generated the Accounting Request (ACR).). The value	Pre-Release MG 5.0

Table 183: PGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
			of this AVP is limited to 20 characters as specified in 32.298.	
Origin-Host	SIS-	M	This field contains the identification of the source point of the operation and the realm of the operation originator.	Pre-Release MG 5.0
Origin-Realm	SIS-	M	This field contains the realm of the operation originator.	Pre-Release MG 5.0
Origin-State-Id	SIS-	Oc	This field contains the state associated to the CTF.	Pre-Release MG 5.0
PDN-Address-prefix-length	SIS-	Oc	PDP/PDN Address prefix length of an IPv6 typed Served PDP Address. This field is not populated if the prefix is 64 bits (default).	MG 5. OR1
PDN-Connection-charging-id	SIS-	Oc	Specifies the PDN connection charging identifier, used to identify different records belonging to the same PDN connection. This field includes Charging Id of first IP-CAN bearer activated within the PDN connection.	MG 5. OR1
PDP-Address	SIS-	Oc	This AVP holds the IP-address associated with the IP CAN bearer session (PDN connection). Note: Multiple occurrences are needed to send both IPv4 and IPv6 addresses of the UE if available.	Pre-Release MG 5.0

Table 183: PGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
Priority-Level	SIS-	Oc	Must be present if Allocation-Retention-Priority AVP is present. The AVP is used for deciding whether a bearer establishment or modification request can be accepted or needs to be rejected in case of resource limitations. The AVP can also be used to decide which existing bearers to pre-empt during resource limitations. The priority level defines the relative importance of a resource request. Values 1 to 15 are defined, with value 1 as the highest level of priority.	Pre-Release MG 5.0
Priority-Level	-IS-	Oc	Must be present if Allocation-Retention-Priority AVP is present. The AVP is used for deciding whether a bearer establishment or modification request can be accepted or needs to be rejected in case of resource limitations. The AVP can also be used to decide which existing bearers to pre-empt during resource limitations. The priority level defines the relative importance of a resource request. Values 1 to 15 are defined, with value 1 as the highest level of priority.	Pre-Release MG 5.0
Proxy-Host	SIS-	M	This field contains the identity of the host that added the Proxy-Info field.	Pre-Release MG 5.0

Table 183: PGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
Proxy-Info	SIS-	Oc	This field contains information of the host.	Pre-Release MG 5.0
Proxy-State	SIS-	M	This field contains state local information.	Pre-Release MG 5.0
PS-Information	SIS-	Oc	PS-Information Grouped AVP.	Pre-Release MG 5.0
QoS-Class-Identifier	SIS-	Oc	It identifies a set of IP-CAN specific QoS parameters that define the authorized QoS, excluding the applicable bitrates and ARP for the IP-CAN bearer or service flow.	Pre-Release MG 5.0
QoS-Class-Identifier	-IS-	Oc	It identifies a set of IP-CAN specific QoS parameters that define the authorized QoS, excluding the applicable bitrates and ARP for the IP-CAN bearer or service flow.	Pre-Release MG 5.0
QoS-information	SIS-	Oc	The QoS-Information AVP (AVP code 1016) is of type Grouped, and it defines the QoS information for an IP-CAN bearer, PCC rule or QCI.	Pre-Release MG 5.0
QoS-information	-IS-	Oc	The QoS-Information AVP (AVP code 1016) is of type Grouped, and it defines the QoS information for an IP-CAN bearer, PCC rule or QCI.	Pre-Release MG 5.0

Table 183: PGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
RAN-Secondary-RAT-Usage-Report	-IS-	Oc	AVP code 1302 is of type Grouped and contains the volume count as reported by the RAN for the secondary RAT (separated for uplink and downlink) including the time of the report.	Release 10.0.R9
[Secondary-RAT-Type]	-IS-	Oc	AVP code 1304 is of type OctetString and holds the following value of Secondary RAT Type, as provided by the RAN for transfer to the SGW/PGW in the RAN Traffic Volume element: • 0 — Reserved • 1 — 5G NR	Release 10.0.R9
[RAN-Start-Timestamp]	-IS-	Oc	AVP code 1303 is of type Time and holds in UTC format the start time for the reported usage for the volume container collected.	Release 10.0.R9
[RAN-End-Timestamp]	-IS-	Oc	AVP code 1301 is of type Time and holds in UTC format the end time of the reported usage for the volume container collected.	Release 10.0.R9
[Accounting-Output-Octets]	-IS-	Oc	AVP code 363 contains the number of octets transmitted during the data container recording interval, reflecting the volume counts for uplink traffic for a data flow.	Release 10.0.R9
[Accounting-Input-Octets]	-IS-	Oc	AVP code 364 contains the number of octets transmitted during the data container recording interval,	Release 10.0.R9

Table 183: PGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
			reflecting the volume count for downlink traffic for a data flow.	
Rating-Group	-IS-	Oc	Contains the charging key (defined in 3GPP TS 23.203).	Pre-Release MG 5.0
Route-Record	SIS-	Oc	This field contains an identifier inserted by a relaying or proxying node to identify the node it received the message from.	Pre-Release MG 5.0
Service-Context-Id	SIS-	Om	This field indicates the service and the corresponding 'middle tier' TS. 32.251@3gpp.org For example, for 32.251 v9.4.0 => 4. mnc.mcc.9.32251@3gpp.org For 32.251 v10.5.0 => 5.mnc.mcc.10.32251@3gpp.org	Pre-Release MG 5.0
Service-Data-Container	-IS-	Oc	Its purpose is to allow the transmission of the container to be reported for Flow-based Charging. On encountering change on charging condition, this container identifies the volume count (separated for uplink and downlink), elapsed time or number of events, per service data flow identified per rating group or combination of the rating group and service-id within an IP-CAN bearer.	Pre-Release MG 5.0
Service-Identifier	-IS-	Oc	Contains identity of the used service. This ID with Service-	Pre-Release MG 5.0

Table 183: PGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
			Context-ID together forms an unique identification of the service.	
Service-Information	SIS-	Oc	Its purpose is to allow the transmission of additional service specific information elements.	Pre-Release MG 5.0
Serving-Node-Type	SIS-	Oc	Type of Serving Node. It may take the following values: • 0 — SGSN • 1 — PMIPSGW • 2 — GTPSGW • 3 — ePDG • 4 — hSGW • 5 — MME For LTE, this will be 2(GTPSGW). For e HRPD this will be 4 (hSGW)	Pre-Release MG 5.0
Session-Id	SIS-	M	This field identifies the operation session.	Pre-Release MG 5.0
SGSN-Address	SIS-	Oc	SGW Address in LTE. HSGW address in eHRPD.	Pre-Release MG 5.0
SGSN-Address	-IS-	Oc	SGW Address in LTE. HSGW Address in eHRPD.	Pre-Release MG 5.0
SGW-Address	SIS-	Oc	SGW-Address holds the IP-Address of the SGW-Node for the control plane. If both an IPv4 and an IPv6 address	MG 5. OR1

Table 183: PGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
			of the SGW are available, SGW shall include the IPv4 address.	
Start-Time	S---	Oc	This AVP contains a time-stamp (in UTC format) which represents the start of a PDN session or a PDN session at a specific QCI at the PGW.	Pre-Release MG 5.0
Stop-Time	--S-	Oc	This AVP contains a time-stamp (in UTC format) which represents the termination of a PDN or a PDN at a specific QCI at the PGW.	Pre-Release MG 5.0
Subscription-Id	SIS-	Om	This field contains the identification of the user that is going to access the service. MSISDN is sent.	Pre-Release MG 5.0
Subscription-Id-Data	SIS-	M	MSISDN.From 32.299, the Subscription-Id-Data contains the user data content for example, the MSISDN. So the contents of the Subscription-Id-Data should just be the E.164 number (that is, CC + National significant number; for example, for USA 1 + NPA + NXX +XXXX for example, 19085821234). The MSISDN complies with the CR0416 (Rel-8) of 3GPP TS 29.274	Pre-Release MG 5.0
Subscription-Id-Type	SIS-	M	This field determines the type of the identifier. For LTE this will be MSISDN. From 32.299, the Subscription-Id-Type determines the type of the identifier, for, the value 0 is used	Pre-Release MG 5.0

Table 183: PGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
			for the international E.164 format according to ITU-T E.164 numbering plan.	
Time-First-Usage	-IS-	Oc	Holds the time in UTC format for the first IP packet to be transmitted and mapped to the current service data container.	Pre-Release MG 5.0
Time-Last-Usage	-IS-	Oc	Holds the time in UTC format for the last IP packet to be transmitted and mapped to the current service data container	Pre-Release MG 5.0
Time-Usage	-IS-	Oc	Indicates the length of the current Flow in seconds	Pre-Release MG 5.0
UDP-Source-Port	SIS-	Oc	The UDP-Source-Port AVP (AVP Code 2806) is of type Unsigned32. When the Untrusted WLAN over S2b access is supported, it contains the UDP source port number used for the IKEv2 tunnel, if a NAT is detected. The IKEv2 messages exchanged between the UE and the ePDG are encapsulated in the UDP messages according to IETF RFC 3948.	—
UE-Local-IP-Address	SIS-	Oc	The UE-Local-IP-Address AVP (AVP code 2805) is of type Address and contains the UE local IP address. The UE local IP address type may be Ipv4 or Ipv6.	—

Table 183: PGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
User-Equipment-Info	S---	Oc	This field contains the identification of the identity and terminal capability the subscriber is using for the connection to mobile network if available hardware ID. This contains a non-standard enumeration for MEID from RFC 4006.	Pre-Release MG 5.0
User-Equipment-Info-Type	S---	Oc	Used for mobile hardware ID type. This field determines the type of the identifier. IMEI will be the hardware Identification for LTE UE. For non-LTE UE, additional ESN and MEID equipment types are defined for this AVP.	Pre-Release MG 5.0
User-Equipment-Info-Value	S---	Oc	Used for mobile hardware ID value. The value of this AVP is encoded as UTF-8 (ASCII).	Pre-Release MG 5.0
User-Name	SIS-	Oc	Contains the user name determined by the domain: bearer, sub-system or service as described in middle tier TS. For LTE this will contain IMSI.	Pre-Release MG 5.0
UWAN-User-Location-Info	SIS-	Oc	The UWAN-User-Location-Info AVP (AVP code 3918) is of type Grouped and contains the UE location in an Untrusted Wireless Access Network (UWAN) which includes the UE local IP address and optionally the UDP source port number (if NAT is detected) as defined in TS 29.274. it may also include WLAN	—

Table 183: PGW Rf attributes (continued)

Attribute name	ACR message support	Presence in ACR	Notes and comments	7750 SR MG release
			location information (SSID and, when available, BSSID of the access point) when provided by the AAA server.	

TDF Rf interface attributes

PGW Rf attributes describes the SSG TDF-specific attributes for the Rf interface.

S6b interface messages

Note that the abbreviations listed in the Presence column in the subsequent tables in this section represent the following characteristics.

- M=Mandatory
- C=Conditional
- O=Optional

Diameter AA-Request

This message is sent from the PGW to the 3GPPP AAA Diameter server for Authorization of a new, updated, and re-authorization of a PDN connection.

Table 184: Diameter AA-Request attributes

Attr no.	Attribute name	Presence	Description	Content type
1	User-Name	M	Contains the IMSI-NAI (default), MSISDN, IMSI, or fixed username.	UTF-8 encoded string

Table 184: Diameter AA-Request attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type
2	User-Password	O	If fixed username and password is configured, the User-Password AVP will be sent un-encrypted in the AAR-I. ¹⁵	UTF-8 encoded string
8	Framed-IP-Address	O	This should be set to the IP address received from the UE. If no IP address was received, this should be set to 0. (RADIUS attribute = 8)	—
9	Framed-IP-Netmask	O	Should be set to 255.255.255.255. (RADIUS attribute = 9)	—
28	Idle-Timeout	O	Contains the maximum number of consecutive seconds of idle connection allowable. Set to the locally configure idle-timeout.	Unsigned 32 bits
97	Framed-IPv6-Prefix	O	This shall be set if an IPv6 address is provided in the PCO ¹⁵ . RADIUS attribute = 97.	—
124	MIP6-Feature-Vector	M	Contains a 64 bit flags field of supported mobility capabilities of the PGW. This is set to "PMIP6_SUPPORTED" flag for S2a sessions (eHRPD), or GTPv2_SUPPORTED for S2b sessions (ePDG).	Unsigned 64 bits
258	Auth-Application-Id	M	Set to 16777272	Unsigned 32 bits
263	Session-Id	M	Session-id is assigned by the PGW to uniquely identify the Diameter session.	UTF-8 encoded string

¹⁵ For information about the supported PCO IE, see *Protocol Configuration Options (PCO) IE*.

Table 184: Diameter AA-Request attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type
264	Origin-Host	M	The PGW host name. It is configured with the origin-host command under config>mobile>pdn>signaling>diameter.	Octet String
274	Auth-Request-Type	M	Set to AUTHORIZE_ONLY	Integer 32 bits
277	Auth-Session-State	O	Using STATE_MAINTAINED	—
283	Destination-Realm	M	Contains the realm for which the AA-Request message is to be routed to.	Octet String
293	Destination-Host	O	Contains the host name of the 3GPP AAA Diameter peer. Only used in case of Re-Authorization.	Octet String
296	Origin-Realm	M	The PGW realm name configured with the origin-realm command under config>mobile>pdn>signaling>diameter.	Octet String
486	MIP6-Agent-Info	O	This is a grouped AVP that contains the PGW identity. Depending on configuration it will contain the IP address or FQDN of the PGW.	Grouped
278	Origin-State-Id	O	Indicates the state of the origin-host generating this message.	Unsigned 32 bits
493	Service-Selection	M	Contains the APN.	UTF-8 encoded string
600	Visited-Network-Identifier	O	Contains the PGW's PLMN ID.	Octet String
628	Supported-Features	O	If present, this AVP contain the list of features supported by the origin host for the lifetime of the Diameter session. The	—

Table 184: Diameter AA-Request attributes (continued)

Attr no.	Attribute name	Presence	Description	Content type
			PGW supports Feature-List-ID 1 feature bit 0 indicating support for P-CSCF Restoration for WLAN, as defined in 3GPP TS 29.273.	
1032	RAT-Type	O	Contains the radio access technology (RAT) that is serving the UE.	Unsigned 32 bits

Table 185: Diameter AA-Request 26/10415 VSAs attributes

Attr. no.	Attribute name	Presence	Description
1	3GPP-IMSI	O	The IMSI can be derived from the NAI format inside the Mobile Node Identifier option.
2	3GPP-Charging-Id	O	The charging ID for the IP-CAN bearer.
3	3GPP-PDP-Type	O	For an EPS bearer this indicates the PDN type (Ipv4, Ipv6 or Ipv4/Ipv6). For a PDP context this indicates PDP type IP.
5	3GPP-GPRS-Negotiated-QoS-Profile	O	For a PDP context, this is the requested QoS-profile as specified in 24.008 Release 7. For an EPS bearer, the format differs for GBR and non-GBR bearers. For a GBR bearer, the QoS-profile shall contain the ARP, QCI, UL/DL MBR and UL/DL GBR. For a non-GBR bearer the QoS-profile shall contain the ARP, QCI and UL/DL APN-AMBR.
6	3GPP-SGSN-Address	O	For a GGSN session, this is the IPv4 address of the SGSN. For a PGW session, this is the IPv4 address of the SGW. For a combined SGW/PGW session, this is the IPv4 address of the MME.

Table 185: Diameter AA-Request 26/10415 VSAs attributes (continued)

Attr. no.	Attribute name	Presence	Description
			For an LMA-eHRPD session, this is the IPv4 address of an HSGW.
7	3GPP-GGSN-Address	O	For GGSN, this is the IP address of the GTP control plane (Gn Ipv4 address.) For PGW, this is the IP address of the GTP control plane (S5 Ipv4 address) or PMIP control plane (S2a Ipv4 address).
8	3GPP-IMSI-MCC-MNC	O	This shall be set to the MCC and MNC extracted from the IMSI.
9	3GPP-GGSN-MCC-MNC	O	This shall be set to the PGW/GGSN MCC and MNC (otherwise known as the PLMN ID.)
10	3GPP-NSAPI	O	For GGSN, this is the NSAPI received via GTP. For PGW, this is the EPS Bearer ID received via GTP.
12	3GPP-Selection-Mode	O	For PGW/GGSN this is the Selection Mode received via GTP.
13	3GPP-Charging-Characteristics	O	For PGW/GGSN this is the Charging-Characteristics value received via GTP or produced internally from selected charging profile.
15	3GPP-SGSN-Ipv6-Address	O	For a GGSN session, this is the IPv6 address of the SGSN. For a PGW session, this is the IPv6 address of the SGW. For a combined SGW/PGW session, this is the IPv6 address of the MME. For an LMA-eHRPD session, this is the IPv6 address of an HSGW.
16	3GPP-GGSN-Ipv6-Address	O	For GGSN, this is the Ipv6 address of the GTP control plane (Gn Ipv6 address.)

Table 185: Diameter AA-Request 26/10415 VSAs attributes (continued)

Attr. no.	Attribute name	Presence	Description
			For PGW, this is the Ipv6 address of the GTP control plane (S5 Ipv6 address) or PMIP control plane (S2a Ipv6 address).
18	3GPP-SGSN-MCC-MNC	O	MCC and MNC extracted from the RAI within the Create PDP Context Request or Update PDP Context Request message.
20	3GPP-IMEISV	O	This is the International Mobile Equipment ID and its Software Version. It is conveyed in GTP messages.
21	3GPP-RAT-Type	O	Indicate which Radio Access Technology is currently serving the UE.
22	3GPP-User-Location-Info	O	This shall be set to the ULI received via GTP.
23	3GPP-MS-TimeZone	O	For GGSN this shall be set to the MS Time Zone received via GTP. For PGW this shall be set to the UE Time Zone received via GTP.
27	3GPP-Allocate-IP-Type	O	Indicates whether the Access-Request is sent for user authentication only and/or for allocation of Ipv4 and/or Ipv6 address.
1536	Origination-Time-Stamp	O	The Origination-TimeStamp AVP is of type Unsigned64. Indicates the time that the message was sent, representing the number of milliseconds since January 1, 1900 00:00 UTC, based on the 64-bit NTP time stamp format defined in RFC 5905, which includes 32-bit unsigned seconds field and 32-bit fraction field. M-bit must be clear for this AVP.
1537	Maximum-Wait-Time	O	The Maximum-Wait-Time AVP is of type Unsigned32. Indicates the number of milliseconds since Origination Timestamp that the originator would wait for response.

Table 185: Diameter AA-Request 26/10415 VSAs attributes (continued)

Attr. no.	Attribute name	Presence	Description
			M-bit must be clear for this AVP.

Diameter AA-Answer

This message is sent from the 3GPP AAA Diameter server to the PGW in response to the AA-Request.

Table 186: Diameter AA-Answer attributes

Attr. no.	Attribute name	Presence	Description	Content type
27	Session-Timeout	C	Contains the absolute timeout value to be enforced for the PDN connection. The value overrides the local configuration. The following values are supported: • value 0 disables the session timeout timer • a value greater than 0 but less than 1800 will be increased to 1800 and used • a value greater than or equal to 1800 up to 4294967295 will be used	Unsigned 32 bits
28	Idle-Timeout	O	Contains the idle-timeout to be applied to the session. The value overrides the local configuration. The following values are supported: • value 0 disables the idle timeout timer • a value greater than 0 but less than 300 will be increased to 300 and used • a value greater than or equal to 300 up to 2592000 will be used	Unsigned 32 bits

Table 186: Diameter AA-Answer attributes (continued)

Attr. no.	Attribute name	Presence	Description	Content type
			a value greater than 2592000 is rounded down to 2592000 and used	
8	Framed-IP-Address	O	Contains IPv4 address assigned by the RADIUS.	—
88	Framed-Pool	O	Contains the name of an assigned address pool that should be used to assign an IP address for the UE	Octet String
97	Framed-IPv6-Prefix	O	The IPv6 prefix allocated by the RADIUS server.	—
100	Framed-IPv6-Pool	—	Name of the name of an IPv6 prefix pool that should be used to assign an address for the user.	—
124	MIP6-Feature-Vector	C	Contains a 64 bit flags field of supported mobility capabilities. If this AVP is returned in the AA-Answer it should be set to "PMIP6_SUPPORTED" flag for S2a sessions (eHRPD), or GTPv2_SUPPORTED for S2b sessions (ePDG).	Unsigned 64 bits
258	Auth-Application-Id	M	Set to 16777272	Unsigned 32 bits
263	Session-Id	M	Contains the Session-id assigned by the PGW to uniquely identify the Diameter session.	UTF-8 encoded string
264	Origin-Host	M	Contains the 3GPP AAA server's host name.	Octet String

Table 186: Diameter AA-Answer attributes (continued)

Attr. no.	Attribute name	Presence	Description	Content type
268	Result-Code	M	Defined in RFC 6733. If the value is anything other than DIAMETER_SUCCESS the PGW will reject the PDN connection.	Unsigned 32 bits
274	Auth-Request-Type	M	Should be set to AUTHORIZE_ONLY. For S2a the PGW will reject the session if set to AUTHORIZE_AUTHENTICATE.	Integer 32 bits
277	Auth-Session-State	O	May be either STATE_MAINTAINED or NO_STATE_MAINTAINED. If STATE_MAINTAINED the PGW will send a Session-Termination-Request	Integer 32 bits
279	Failed-AVP	O	If error was due to a Bad AVP such as an improperly constructed AVP, an unsupported or an unrecognized AVP, the Failed AVP will contain the bad AVP. The Failed-AVP is of type Grouped.	Grouped
281	Error-Message	O	It may accompany a Result-Code AVP in case of an error as a human readable error message. The Error-Message AVP is not intended to be useful in real time.	UTF-8 encoded string
285	Re-Auth-Request-Type	O	Should be set to AUTHORIZE_ONLY. For S2a-based PDN connection, the PGW will reject the session if set to AUTHORIZE_AUTHENTICATE.	Integer 32 bits
291	Authorization-Lifetime	O	Contains the authorization lifetime granted by the 3GPP AAA server.	Unsigned 32 bits
294	Error-Reporting-Host	O	Contains the identity of the Diameter host that sent the Result-Code AVP to a value other than 2001 (Success), only if the host setting the Result-Code is different from	Octet String

Table 186: Diameter AA-Answer attributes (continued)

Attr. no.	Attribute name	Presence	Description	Content type
			the one encoded in the Origin-Host AVP. If received this AVP will be logged with the intended to be used for troubleshooting purposes.	
296	Origin-Realm	M	Contains the 3GPP AAA server's realm name.	Octet String
297	Experimental-Result	O	If present the result code will override the value in Result-Code AVP.	—
444	Subscription-Id	O	Is of type Grouped and indicates the user identity to be used for charging purposes. The Subscription-Id AVP contains one of the following: - Subscription-Id-Type AVP == END_USER_E164 and Subscription-Id-Data AVP == MSISDN included in the Rf messages when Rf-based charging is enabled - Subscription-Id-Type AVP == END_USER_NAI with Subscription-Id-Data AVP == External-Identifier included in the Rf/PCRF messages when Rf-based charging/Gx is enabled The External-Identifier must be in the form of 10-alphanumeric-characters@domain, up to 255 characters	—
626	Supported-Features	O	If present, this AVP contains the list of features supported by the origin host for the lifetime of the Diameter session.	—
848	Served-Party-IP-Address	O	Contains the IP address to be assigned to the UE.	—

Table 187: Diameter AA-Answer 26/10415 VSAs attributes

Attr. no.	Attribute name	Description
5	3GPP-GPRS-Negotiated-QoS-Profile	For an EPS bearer, the format differs for GBR and non-GBR bearers. For a GBR bearer, the QoS-profile shall contain the ARP, QCI, UL/DL MBR and UL/DL GBR. For a non-GBR bearer the QoS-profile shall contain the ARP, QCI and UL/DL APN-AMBR.
17	3GPP-IPv6-DNS-Servers	This shall contain the list of IPv6 DNS server addresses to be provided to the UE.

Table 188: Diameter AA-Answer 26/12951 VSAs attributes

Attr. no.	Attribute name	Description
6115	Self-Activation-Status	The Self-Activation-Status AVP is of type Enumerated and contains the self-activation status of the subscriber. The AVP MAY be included in the AA-Answer to indicate the self-activation status of the subscriber. If the AVP is NOT included the default value of CONTINUE shall be implied. The following values are defined for the Self-Activation-Status AVP: • Continue (0) - The session shall be allowed to continue in a normal manner. • REACTIVATION_DISALLOWED_TO_APN (1) - – For EUTRAN access, this will cause the PGW to set PCO-Action = (5) Self-Activation in the CSResp and disconnect the session after IDLE timeout based on the IDLE time out settings for the subscriber and shall include the disconnect cause code (Reactivation disallowed to APN) over GTP indicating that the UE should not reconnect. – For EHRPD access, the PGW sets the PCO-Action = (5) Self-Activation in the PBA, disconnects the session after an IDLE timeout based on the IDLE time out settings for the subscriber, and includes cause code "apn-denied-no-subscription" over registration revocation to indicate that the UE should not reconnect.

Table 188: Diameter AA-Answer 26/12951 VSAs attributes (continued)

Attr. no.	Attribute name	Description
		The Self-Activation-Status AVP must not be changed mid-session.
6116	Virtual-APN-Name	The Virtual-APN-Name AVP is of type UTF8String and when sent include subscriber enterprise APN in response to the session's first AAR. Virtual-APN-name must not be changed mid-session.
6068	Charging-Gateway-Function-Host	Charging-Gateway-Function-Host AVP is of type Diameter Identity. It contains the FQDN of the CGF, if provided in the response, this value should be included in the RF accounting records. This will be used by the CDF to route the packets to the appropriate CGF. Charging-Gateway- Function-Host must not be changed mid-session.
6069	Charging-Group-ID	Charging-Group-Id AVP is of type UTF8String. This value if provided by the 3GPP AAA will be populated in the Rf accounting records. Charging-Group-ID must not be changed mid-session.

The Self-Activation-Status AVP is sent by the AAA in the AA-Answer if the PDN session is identified as a "Self-Activation".

Table 189: Diameter AA-Answer Self-Activation-Status Attributes

Attr. no.	Attribute name	Description
0	ontinue	The session continues in a normal manner.
1	Reactivation-Disallowed-to-APN	For EUTRAN access, the PGW sets the PCO-Action = (5) Self-Activation in the CS Response, disconnects the session after an IDLE timeout based on the IDLE timeout settings for the subscriber, and includes the cause code "PDN connection to this APN disallowed (09)" over GTP, indicating that the UE should not reconnect. For EHRPD access, the PGW sets the PCO-Action = (5) Self-Activation in the PBA, disconnects the session after an IDLE timeout based on

Table 189: Diameter AA-Answer Self-Activation-Status Attributes (continued)

Attr. no.	Attribute name	Description
		the IDLE time out settings for the subscriber, and includes cause code "apn-denied-no-subscription" over registration revocation to indicate that the UE should not reconnect.

Diameter AS-Request

Table 190: Diameter AS-Request 26/12951 VSAs attributes

Attr. no.	Attribute name	Description
6114	PGW-Relocation-Indication	PGW-Relocation-Indication AVP is of type grouped and contains the PGW relocation indication. The PGW relocation is only included by the 3GPP AAA in the S6b:ASR request sent to the PGW on which the session was anchored prior to relocation. When this AVP is present in the S6b:ASR request, the PGW shall indicate handover to the serving SGW to avoid paging the UE as the session is terminated. The MIP6-Agent-Info AVP may optionally be present and it contains the information on the new PGW.

Diameter Re-Auth-Request

This message is sent from the 3GPP AAA Diameter server to the PGW to request a re-authorization to be made for an existing PDN connection.

Table 191: Diameter Re-Auth-Request attributes

Attr. no.	Attribute name	Presence	Description	Content type
1	User-Name	O	Contains the username for the session.	UTF-8 encoded string

Table 191: Diameter Re-Auth-Request attributes (continued)

Attr. no.	Attribute name	Presence	Description	Content type
258	Auth-Application-Id	M	Set to 16777272	Unsigned 32 bits
263	Session-Id	M	Contains the session ID to be reauthorized.	UTF-8 encoded string
264	Origin-Host	M	Contains the 3GPP AAA server's hostname.	Octet String
274	Auth-Request-Type	M	Should be set to AUTHORIZE_ONLY. For S2a the PGW will reject the session if set to AUTHORIZE_AUTHENTICATE.	Integer 32 bits
278	Origin-State-Id	O	Contains a restart counter maintained by the 3GPP AAA server. If this value changes the PGW will act as if the peer has restarted.	Unsigned 32 bits
283	Destination-Realm	M	Contains the realm name of the PGW.	Octet String
284	Proxy-Info	O	If this AVP is present the PGW will copy it as is and add it to the response.	Grouped
293	Destination-Host	M	Contains the hostname of the PGW.	Octet String
296	Origin-Realm	M	Contains the 3GPP AAA server's realm name.	Octet String
1522	RAR-Flags	O	Contains a bit mask. Bit 1 indicates P-CSCF Restoration Request.	Unsigned 32 bits

Diameter Re-Auth-Answer

This message is sent from the PGW to the 3GPP AAA Diameter server in response to the Re-Auth-Request.

Table 192: Diameter Re-Auth-Answer attributes

Attr no.	Attribute name	Presence	Description	Content type
1	User-Name	O	Contains the username for the session.	UTF-8 encoded string
263	Session-Id	M	Contains the session ID to be reauthorized.	UTF-8 encoded string
264	Origin-Host	M	Contains the PGW hostname.	Octet String
268	Result-Code	M	Result-code set to 2001. In case of errors appropriate result-code will be used as defined in RFC 6733.	Unsigned 32 bits
279	Failed-AVP	O	If error was found due to a Bad AVP such as an improperly constructed AVP, an unsupported or an unrecognized AVP, the Failed AVP will contain the bad AVP". The Failed-AVP is of type Grouped.	Grouped
281	Error-Message	O	It may accompany a Result-Code AVP in case of an error as a human readable error message. The Error-Message AVP is not intended to be useful in real time.	UTF-8 encoded string
284	Proxy-Info	O	If this AVP is present in the request, the PGW will copy it as is in the response.	Grouped
296	Origin-Realm	M	Contains the PGW realm name.	Octet String

Diameter Session-Termination-Request

This message is sent from the PGW to the 3GPP AAA Diameter server to indicate the deletion of a PDN connection.

Table 193: Diameter Session-Termination-Request attributes

Attr. no.	Attribute name	Presence	Description	Content type
1	User-Name	O	Contains the username for the session.	UTF-8 encoded string
258	Auth-Application-Id	M	Set to 16777272	Unsigned 32 bits
263	Session-Id	M	Contains the session ID to be reauthorized.	UTF-8 encoded string
264	Origin-Host	M	Contains the PGW hostname.	Octet String
278	Origin-State-Id	O	This is a restart counter maintained by the PGW.	Unsigned 32 bits
283	Destination-Realm	M	Contains the realm name of the 3GPP AAA server.	Octet String
293	Destination-Host	O	Contains the hostname of the 3GPP AAA server.	Octet String
295	Termination-Cause	M	Contains the appropriate cause value according to the reason the session has terminated.	Integer 32 bits
296	Origin-Realm	M	Contains the PGW realm name.	Octet String

Diameter Session-Termination-Answer

This message is sent from the 3GPP AAA Diameter server to the PGW in response to the Session-Termination-Request.

Table 194: Diameter Session-Termination-Answer attributes

Attr. no.	Attribute name	Presence	Description	Content type
1	User-Name	O	Contains the username for the session.	UTF-8 encoded string
263	Session-Id	M	Contains session-id that applies to this session.	UTF-8 encoded string
264	Origin-Host	M	Contains the 3GPP AAA server hostname.	Octet String
268	Result-Code	M	Result-code set to 2001. In case of errors appropriate result-code should be used as defined in RFC 6733.	Unsigned 32 bits
278	Origin-State-Id	O	Contains a restart counter maintained by the 3GPP AAA server. If this value changes the PGW will act as if the peer has restarted.	Unsigned 32 bits
279	Failed-AVP	O	If error was found due to a Bad AVP such as an improperly constructed AVP, an unsupported or an unrecognized AVP, the Failed AVP will contain the bad AVP". The Failed-AVP is of type Grouped.	Grouped
281	Error-Message	O	It may accompany a Result-Code AVP in case of an error as a human readable error message. The Error-Message AVP is not intended to be useful in real time.	UTF-8 encoded string
283	Destination-Realm	M	Contains the realm name of the 3GPP AAA server.	Octet String
294	Error-Reporting-Host	O	Contains the identity of the Diameter host that sent the Result-Code AVP to a value other than 2001 (Success), only if the host setting the Result-Code is different from the one encoded in the Origin-Host AVP. If received this AVP will be logged with the	Octet String

Table 194: Diameter Session-Termination-Answer attributes (continued)

Attr. no.	Attribute name	Presence	Description	Content type
			intended to be used for troubleshooting purposes.	
295	Termination-Cause	M	Contains the appropriate cause value according to the reason the session has terminated.	Integer 32 bits
296	Origin-Realm	M	Contains the 3GPP AAA server's realm name.	Octet String
297	Experimental-Result	O	Quietly accepted, but not further processed if received in a session termination answer.	—
626	Supported-Features	O	Quietly accepted, but not further processed if received in a session termination answer.	—

Diameter Abort-Session-Request

This message is sent from the Diameter server to the PGW to request that the PGW delete the PDN connection.

Table 195: Diameter Abort-Session-Request attributes

Attr. no.	Attribute name	Presence	Description	Content type
1	User-Name	O	Contains the username for the session.	UTF-8 encoded string
258	Auth-Application-Id	M	Set to 16777272	Unsigned 32 bits
263	Session-Id	M	Contains session-id that applies to this session.	UTF-8 encoded string

Table 195: Diameter Abort-Session-Request attributes (continued)

Attr. no.	Attribute name	Presence	Description	Content type
264	Origin-Host	M	Contains the 3GPP AAA server hostname.	Octet String
277	Auth-Session-State	M	May be either STATE_MAINTAINED or NO_STATE_MAINTAINED. If STATE_MAINTAINED the PGW will send a Session-Termination-Request upon PMIPv6 session termination. If NO_STATE_MAINTAINED no STR require send from the PGW before terminating the PDN session.	Integer 32 bits
278	Origin-State-Id	O	Contains a restart counter maintained by the 3GPP AAA server. If this value changes the PGW will act as if the peer has restarted.	Unsigned 32 bits
283	Destination-Realm	M	Contains the realm name of the PGW.	Octet String
284	Proxy-Info	O	If this AVP is present the PGW will copy it as is and add it to the response.	Grouped
293	Destination-Host	M	Contains the hostname of the PGW.	Octet String
296	Origin-Realm	M	Contains the 3GPP AAA server's realm name.	Octet String
626	Supported-Features	O	Quietly accepted, but not further processed if received in a session termination answer.	—

Diameter Abort-Session-Answer

This message is sent from the PGW to the 3GPP AAA Diameter server in response to the Abort-Session-Request.

Table 196: Diameter Abort-Session-Answer attributes

Attr. no.	Attribute name	Presence	Description	Content type
1	User-Name	O	Contains the username for the session.	UTF-8 encoded string
263	Session-Id	M	Contains session-id that applies to this session.	UTF-8 encoded string
264	Origin-Host	M	Contains the PGW hostname.	Octet String
268	Result-Code	M	Defined in RFC 6733.	Unsigned 32 bits
278	Origin-State-Id	O	This is a restart counter maintained by the PGW.	Unsigned 32 bits
279	Failed-AVP	O	If the error was due to a Bad AVP such as an improperly constructed AVP, an unsupported or an unrecognized AVP, the Failed AVP will contain the bad AVP. The Failed-AVP is of type Grouped.	Grouped
281	Error-Message	O	It may accompany a Result-Code AVP in case of an error as a human readable error message. The Error-Message AVP is not intended to be useful debugging not in real time.	UTF-8 encoded string
284	Proxy-Info	O	If this AVP is present in the request, the PGW will copy it as is in the response.	Grouped
296	Origin-Realm	M	Contains the PGW realm name.	Octet String

SWm interface messages

Note that the abbreviations listed in the Presence column in the subsequent tables in this section represent the following characteristics.

- M=Mandatory
- C=Conditional
- O=Optional

Diameter EAP-Request

The Authentication and Authorization Request uses the Diameter EAP-Request. This is part of the initial authentication and authorization that is performed when the IKEv2 SA is being established on the ePDG. The Diameter EAP-Request is sent from the ePDG to the AAA server.

Table 197: Diameter EAP-Request SWm attributes

Attribute name	Presence	Description
Session-Id	M	Session-id is assigned by the ePDG to uniquely identify the Diameter session.
Auth-Application-Id	M	Set to 5.
Origin-Host	M	The ePDG hostname. It is configured with the origin-host command under config>mobile-gateway>pdn>signaling>diameter.
Origin-Realm	M	The ePDG realm name. It is configured with the origin-realm command under config>mobile>pdn-signaling>diameter.
Destination-Realm	M	This is the realm name of the Diameter peer.
Auth-Request-Type	M	Set to AUTHORIZE_AUTHENTICATE.
EAP-Payload	M	This contains the EAP-Response/Identity or EAP-Response/AKA-Challenge. The EAP-Response/Identity must contain the client identity received from IKEv2.

Table 197: Diameter EAP-Request SWm attributes (continued)

Attribute name	Presence	Description
User-Name	O	Contains the IKEv2 identity.
RAT-Type	O	Set by the ePDG if the access type is known or to a default.
Service-Selection	O	The APN presented by the UE via IKEv2, if available.
MIP6-Feature-Vector	O	This is set to "GTPv2_SUPPORTED".
QoS-Capability	O	—
Visited-Network-Identifier	O	Set to the ePDG's PLMN ID in case the PGW is in another operator's network (roaming case). The format is "mnc<MNC>.mcc<MCC>.3gppnetwork.org".
AAA-Failure-Indication	O	If present, this indicates that the request is sent after the ePDG has determined that a previously assigned 3GPP AAA Server is unavailable.
Supported-Features	O	—
UE-Local-IP-Address	O	The ePDG shall include this IE based on local policy for Fixed Broadband access network interworking as specified in 3GPP TS 23.139. If present, it contains the source IPv4 or IPv6 address of the IKE_SA_AUTH message from the UE.

Diameter EAP-Answer

The Authentication and Authorization Request uses the Diameter EAP-Answer. This is part of the initial authentication and authorization that is performed when the IKEv2 SA is being established on the ePDG. The Diameter EAP-Answer is sent from the AAA server to the ePDG.

Table 198: Diameter EAP-Answer SWm attributes

Attribute name	Presence	Description
Session-Id	M	The session ID from the corresponding request.
Auth-Application-Id	M	Set to 5.
Auth-Request-Type	M	Set to AUTHORIZE_AUTHENTICATE.
Result-Code	M	Must be DIAMETER-SUCCESS.
Origin-Host	M	This is the AAA hostname.
Origin-Realm	M	This is the AAA realm name.
EAP-Payload	M	This contains the EAP-Response/AKA-Challenge or EAP-Success.
EAP-Master-Session-Key	O	Sent if EAP-Success.
APN-OI-Replacement	O	This is to replace the OI used by the ePDG to select the PGW via DNS.
APN-Configuration	O	A single APN-Configuration is received if the Result-Code = DIAMETER-SUCCESS.
MIP6-Feature-Vector	O	This is set to GTPv2_SUPPORTED. If not, discard or flag as error.
Mobile-Node-Identifier	O	If the IKEv2 identity was not a permanent one (IMSI) this AVP provides a way for the AAA to provide the IMSI to the ePDG. Should be sent as the IMSI over S2b.
Subscription-ID	O	Should contain the MSISDN to be sent over S2b.
Session-Timeout	O	This is the authorization lifetime.

Table 198: Diameter EAP-Answer SWm attributes (continued)

Attribute name	Presence	Description
MIP6-Agent-Info	O	Ignored.
3GPP-Charging-Characteristics	O	Charging-characteristics sent over GTPv2
Redirect-Host	O	We need to be able to redirect in case we have selected the wrong AAA.
Supported-Features	O	—
Authorization-Lifetime	O	This contains the authorization lifetime granted by the 3GPP AAA server.

Diameter Re-Auth-Request

SWm defines a reauthorization procedure whereby the AAA can request the ePDG to reauthorize its session. The AAA triggers reauthorization using the ReAuth-Request message. The ePDG responds by using AA-Request to reauthorize its session. The Diameter Re-Auth-Request is sent from the AAA server to the ePDG.

Table 199: Diameter Re-Auth-Request SWm attributes

Attribute name	Presence	Description
Session-Id	M	The session ID to be reauthorized.
Origin-Host	M	This is the Diameter server's hostname.
Origin-Realm	M	This is the Diameter server's realm name.
Destination-Realm	M	This is the realm name of the ePDG. This request is ignored if the Destination-Realm does not match the ePDG Origin-Realm.

Table 199: Diameter Re-Auth-Request SWm attributes (continued)

Attribute name	Presence	Description
Destination-Host	M	This is the hostname of the ePDG. This request is ignored if the Destination-Host does not match the ePDG Origin-Host.
Auth-Application-Id	M	Set to 1.
Re-Auth-Request-Type	M	Should be set to AUTHORIZE_ONLY. For S2b the PGW rejects the session if set to AUTHORIZE_AUTHENTICATE.
Origin-State-Id	O	This is a restart counter maintained by the Diameter server.
Proxy-Info	O	The ePDG should ignore this AVP.
Route-Record	O	The ePDG should ignore this AVP.
User-Name	O	This contains the username for the session.

Diameter Re-Auth-Answer

SWm defines a reauthorization procedure whereby the AAA can request the ePDG to reauthorize its session. The AAA triggers reauthorization using the ReAuth-Request message. The ePDG responds by using AA-Request to reauthorize its session. The Diameter Re-Auth-Answer is sent from the ePDG to the AAA server.

Table 200: Diameter Re-Auth-Answer SWm attributes

Attribute name	Presence	Description
Session-Id	M	The session-id that applies to this session.
Result-Code	M	Defined in RFC 6733.

Table 200: Diameter Re-Auth-Answer SWm attributes (continued)

Attribute name	Presence	Description
Origin-Host	M	This is the ePDG hostname. It is configured with the origin-host command under config>mobile>pdn>signaling>diameter.
Origin-Realm	M	This is the ePDG realm name. It is configured with the origin-realm command under config>mobile>pdn>signaling>diameter.
User-Name	O	This contains the username for the session.

Diameter AA-Request

The Diameter AA-Request is sent from the ePDG to the AAA server.

Table 201: Diameter AA-Request SWm attributes

Attribute name	Presence	Description
Session-Id	M	Session-id is assigned by the ePDG to uniquely identify the Diameter session.
Auth-Application-Id	M	Set to 1.
Origin-Host	M	This is the PGW hostname. It is configured with the origin-host command under config>mobile>pdn>signaling>diameter.
Origin-Realm	M	This is the PGW realm name. It is configured with the origin-realm command under config>mobile>pdn>signaling>diameter.
Destination-Realm	M	This is the realm name of the Diameter peer.

Table 201: Diameter AA-Request SWm attributes (continued)

Attribute name	Presence	Description
Auth-Request-Type	M	Set to AUTHORIZE_ONLY.
User-Name	M	This contains the username for the session.

Diameter AA-Answer

The Diameter AA-Answer is sent from the AAA server to the ePDG.

Table 202: Diameter AA-Answer SWm attributes

Attribute Name	Presence	Description
Session-Id	M	Session-id is assigned by the ePDG to uniquely identify the Diameter session.
Auth-Application-Id	M	Set to 1.
Auth-Request-Type	M	Set to AUTHORIZE_ONLY
Result-Code	M	—
Origin-Host	M	This is the PGW hostname. It is configured with the origin-host command under config>mobile>pdn>signaling>diameter.
Origin-Realm	M	This is the PGW realm name. It is configured with the origin-realm command under config>mobile>pdn>signaling>diameter.
User-Name	O	This contains the username for the session.
APN-OI-Replacement	O	Replaces the OI used by the ePDG to select the PGW via DNS.

Table 202: Diameter AA-Answer SWm attributes (continued)

Attribute Name	Presence	Description
APN-Configuration	O	Can receive multiple APN-Configuration AVPs.
Subscription-ID	O	Contains the MSISDN.
3GPP-Charging-Characteristics	O	Charging-characteristics sent over GTPv2
Session-Timeout	O	This is the authorization lifetime.

Diameter Session-Termination-Request

This message is sent from the ePDG to the AAA upon deletion of a PDN connection.

Table 203: Diameter Session-Termination-Request SWm attributes

Attribute Name	Presence	Description
Session-Id	M	The session-id that applies to this session.
Auth-Application-Id	M	Set to 1.
Origin-Host	M	This is the ePDG hostname. It is configured with the origin-host command under config>mobile>pdn>signaling>diameter.
Origin-Realm	M	This is the ePDG realm name. It is configured with the origin-realm command under config>mobile>pdn>signaling>diameter.
Destination-Realm	M	This is the realm name of the Diameter peer

Table 203: Diameter Session-Termination-Request SWm attributes (continued)

Attribute Name	Presence	Description
Termination-Cause	M	The ePDG sends the appropriate cause value according to the reason the session has terminated.
User-Name	O	This contains the username for the session.

Diameter Session-Termination-Answer

This message is sent from the Diameter server to the ePDG, in response to the Session-Termination-Request.

Table 204: Diameter Session-Termination-Answer SWm attributes

Attribute name	Presence	Description
Session-Id	M	The session-id that applies to this session.
Result-Code	M	Defined in RFC 6733.
Origin-Host	M	This is the ePDG hostname. It is configured with the origin-host command under config>mobile>pdn>signaling>diameter.
Origin-Realm	M	This is the ePDG realm name. It is configured with the origin-realm command under config>mobile>pdn>signaling>diameter.
User-Name	O	This contains the username for the session.

Diameter Abort-Session-Request

This message is sent from the Diameter server to the ePDG, to request the ePDG to delete a specific session.

Table 205: Diameter Abort-Session-Request SWm attributes

Attribute Name	Presence	Description
Session-Id	M	The session ID to be terminated.
Origin-Host	M	This is the PGW hostname. It is configured with the origin-host command under config>mobile>pdn>signaling>diameter.
Origin-Realm	M	This is the PGW realm name. It is configured with the origin-realm command under config>mobile>pdn>signaling>diameter.
Destination-Realm	M	This is the realm name of the Diameter peer.
Destination-Host	M	This is the hostname of the ePDG. This request is ignored if the Destination-Host does not match the ePDG Origin-Host.
Auth-Application-Id	O	Set to 1.
Auth-Session-State	O	May be either STATE_MAINTAINED or NO_STATE_MAINTAINED.
User-Name	O	The username for the session.
Proxy-Info	O	This field contains information of the host.
Route-Record	O	—
Supported-Features	O	—

Diameter Abort-Session-Answer

This message is sent from the ePDG to the Diameter server in response to the Abort-Session-Request.

Table 206: Diameter Abort-Session-Answer SWm attributes

Attribute Name	Presence	Description
Session-Id	M	The session ID to be terminated.
Result-Code	M	Defined in RFC 6733.
Origin-Host	M	This is the PGW hostname. It is configured with the origin-host command under config>mobile>pdn>signaling>diameter.
Origin-Realm	M	This is the PGW realm name. It is configured with the origin-realm command under config>mobile>pdn>signaling>diameter.

IKEv2 Notify messages

This section describes custom operator-supported IKEv2 Notify Message error types for the ePDG to the UE and for the UE to the ePDG.

Table 207: ePDG to UE Notify message types

Notify message type	Value	Description
UNSUPPORTED_CRITICAL_PAYLOAD	1	Sent if the ePDG finds an unknown payload that has a Critical bit set in its payload header.
INVALID_MAJOR_VERSION	5	Sent as response if the ePDG receives a message with invalid major version.
INVALID_SYNTAX	7	Sent if the ePDG receives a message that misses required payloads or has the incorrect format. This error is not sent in IKE_SA_INIT response.

Table 207: ePDG to UE Notify message types (continued)

Notify message type	Value	Description
NO_PROPOSAL_CHOSEN	14	Sent if the ePDG fails to choose a proposal from the MS proposed list.
INVALID_KE_PAYLOAD	17	Sent if the KE payload from the UE is invalid.
AUTHENTICATION_FAILED	24	Sent if there is an error in the authentication process. The ePDG terminates the call immediately following this notification.
NO_ADDITIONAL_SAS	35	Sent if the ePDG is not willing to accept any more CHILD_SA on the IKE_SA or if it does not accept CHILD_SA re-key requests.
INTERNAL_ADDRESS_FAILURE	36	Sent if the ePDG experiences a failure in address assignment.
FAILED_CP_REQUIRED	37	Sent if CP (CFG_REQUEST) is expected but not received.
TS_UNACCEPTABLE	38	Sent if multiple TSi or TSr are received.

Table 208: UE to ePDG Notify message types

Notify message type	Value	Description
UNSUPPORTED_CRITICAL_PAYLOAD	1	The ePDG ignores this error code.
INVALID_SYNTAX	7	The ePDG may record the error and initiate IKEv2 Delete exchange to terminate the call.
INVALID_SPI	11	The ePDG will delete the ESP identified by the SPI sent in this Notify payload.

Table 208: UE to ePDGNotify message types (continued)

Notify message type	Value	Description
INVALID_KEY_PAYLOAD	17	The ePDG expects to receive this error if it initiates a re-key and the DH group does not match the MS configuration. In this case, the ePDG sends a new proposal to match the DH group. For other cases, the ePDG ignores the notification.
NO_ADDITIONAL_SAS	35	The ePDG ignores the notification and the CHILD_SA re-key fails upon receiving the notification.
INVALID_SELECTORS	39	The ePDG ignores this error code.

When the ePDG receives an SWm error code, the ePDG responds to the UE with an AUTHENTICATION_FAILURE notification by implementing the error code mapping specified in *SWm error mapping to UE IKEv2 error codes*.

Table 209: SWm error mapping to UE IKEv2 error codes

Code	AVP name	AVP type	IKE response
2001	DIAMETER_SUCCESS	Success	Not an error, IKE SA is accepted
2002	DIAMETER_LIMITED_SUCCESS	Success	NOTIFY (ID 24: AUTHENTICATION_FAILURE)
3002	DIAMETER_UNABLE_TO_DELIVER	Protocol_Error	The ePDG attempts to contact two AAAs (the primary and the secondary) and will only attempt to select the secondary AAA. If the secondary AAA peer sends this error code, the ePDG will send NOTIFY (ID 24: AUTHENTICATION_FAILURE).
3003	DIAMETER_REALM_NOT_SERVED	Protocol_Error	The ePDG attempts to contact two AAAs (the primary and the secondary) and will only attempt to select the secondary AAA. If the secondary AAA

Table 209: SWm error mapping to UE IKEv2 error codes (continued)

Code	AVP name	AVP type	IKE response
			peer sends this error code, the ePDG will send NOTIFY (ID 24: AUTHENTICATION_FAILURE).
3004	DIAMETER_TOO_BUSY	Protocol_Error	The ePDG attempts to contact two AAAs (the primary and the secondary) and will only attempt to select the secondary AAA. If the secondary AAA peer sends this error code, the ePDG will send NOTIFY (ID 24: AUTHENTICATION_FAILURE).
3006	DIAMETER_REDIRECT_INDICATION	Protocol_Error	The ePDG attempts to contact the redirect AAA. The ePDG will only attempt to redirect once. If the redirect AAA peer sends this error code or if the redirect AAA does not respond at all, the ePDG will send NOTIFY (ID 24: AUTHENTICATION_FAILURE).
4001	DIAMETER_AUTHENTICATION_REJECTED	Transient	NOTIFY (ID 24: AUTHENTICATION_FAILURE)
5002	DIAMETER_UNKNOWN_SESSION_ID	Permanent	NOTIFY (ID 24: AUTHENTICATION_FAILURE)
5003	DIAMETER_AUTHORIZATION_REJECTED	Permanent	NOTIFY (ID 24: AUTHENTICATION_FAILURE)
5004	DIAMETER_INVALID_AVP_VALUE	Permanent	NOTIFY (ID 24: AUTHENTICATION_FAILURE)
5005	DIAMETER_MISSING_AVP	Permanent	NOTIFY (ID 24: AUTHENTICATION_FAILURE)
5006	DIAMETER_RESOURCES_EXCEEDED	Permanent	NOTIFY (ID 24: AUTHENTICATION_FAILURE)
5008	DIAMETER_AVP_NOT_ALLOWED	Permanent	NOTIFY (ID 24: AUTHENTICATION_FAILURE)
5012	DIAMETER_UNABLE_TO_COMPLY	Permanent	NOTIFY (ID 24: AUTHENTICATION_FAILURE)

Table 209: SWm error mapping to UE IKEv2 error codes (continued)

Code	AVP name	AVP type	IKE response
3xxx	Any other 3xxx cause code not specified above	Protocol_ Error	NOTIFY (ID 24: AUTHENTICATION_FAILURE)
4xxx	Any other 4xxx cause code not specified above	Transient	NOTIFY (ID 24: AUTHENTICATION_FAILURE)
5xxx	Any other 5xxx cause code not specified above	Permanent	NOTIFY (ID 24: AUTHENTICATION_FAILURE)
Experimental Error Codes (TS 29.272)			
5001	DIAMETER_ERROR_USER_UNKNOWN	Permanent	NOTIFY (ID 24: AUTHENTICATION_FAILURE)
5003	DIAMETER_ERROR_IDENTITY_NOT_REGISTERED	Permanent	NOTIFY (ID 24: AUTHENTICATION_FAILURE)
5004	DIAMETER_ERROR_ROAMING_NOT_ALLOWED	Permanent	NOTIFY (ID 24: AUTHENTICATION_FAILURE)
5005	DIAMETER_ERROR_IDENTITY_ALREADY_REGISTERED	Permanent	NOTIFY (ID 24: AUTHENTICATION_FAILURE)
5450	DIAMETER_ERROR_USER_NO_NON_3GPP_SUBSCRIPTION	Permanent	NOTIFY (ID 24: AUTHENTICATION_FAILURE)
5451	DIAMETER_ERROR_USER_NO_APN_SUBSCRIPTION	Permanent	NOTIFY (ID 24: AUTHENTICATION_FAILURE)
5452	DIAMETER_ERROR_RAT_TYPE_NOT_ALLOWED	Permanent	NOTIFY (ID 24: AUTHENTICATION_FAILURE)
5xxx	Any other 5xxx cause code not specified above	Permanent	NOTIFY (ID 24: AUTHENTICATION_FAILURE)

When the ePDG receives an S2b cause code, the ePDG responds to the UE with as INTERNAL_ADDRESS_FAILURE by implementing the error code mapping specified in *S2b error mapping to UE IKE error code*.

Table 210: S2b error mapping to UE IKE error code

Message type	GTP-C cause value	Description	IKE response
Request/initial message	1	Reserved	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	2	Local detach	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	3	Complete detach	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	4	RAT changed from 3GPP to non-3GPP	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	5	ISR deactivation	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	6	Error indication received from RNC/eNodeB/S4-SGSN	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	7	IMSI detach only	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	8	Reactivation requested	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	9	PDN reconnection to this APN disallowed	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	10	Access changed from non-3GPP to 3GPP	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)

Table 210: S2b error mapping to UE IKE error code (continued)

Message type	GTP-C cause value	Description	IKE response
	11	PDN connection inactivity timer expired	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	12	PGW not responding	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	13	Network failure	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	14	QoS parameter mismatch	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	15	Spare. This value range can be used by cause values in an initial/request message.	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
Acceptance in a response/triggered message	16	Request accepted	IKE SA establishment proceeds
	17	Request accepted partially	IKE SA establishment proceeds
	18	New PDN type due to network preference	IKE SA establishment proceeds
	19	New PDN type due to single address bearer only	IKE SA establishment proceeds
	20 to 63	Spare. This value range can be used by cause values in an acceptance response/triggered message.	IKE SA establishment proceeds

Table 210: S2b error mapping to UE IKE error code (continued)

Message type	GTP-C cause value	Description	IKE response
Rejection in a response/triggered message	64	Context Not Found	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	65	Invalid Message Format	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	66	Version not supported by next peer	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	67	Invalid length	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	68	Service not supported	The ePDG should select another PGW. The ePDG will attempt to establish a PDN connection on up to three PGWs. If none of these PGWs respond, the ePDG will send a NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE) message.
	69	Mandatory IE incorrect	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	70	Mandatory IE missing	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	71	Not used	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	72	System failure	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	73	No resources available	The ePDG should select another PGW. The ePDG will attempt to establish a PDN connection on up to three

Table 210: S2b error mapping to UE IKE error code (continued)

Message type	GTP-C cause value	Description	IKE response
			PGWs. If none of these PGWs respond, the ePDG will send a NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE) message.
	74	Semantic error in the TFT operation	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	75	Syntactic error in the TFT operation	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	76	Semantic errors in packet filters	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	77	Syntactic errors in packet filters	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	78	Missing or unknown APN	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	79	Not used	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	80	GRE key not found	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	81	Relocation failure	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	82	Denied in RAT	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	83	Preferred PDN type not supported	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)

Table 210: S2b error mapping to UE IKE error code (continued)

Message type	GTP-C cause value	Description	IKE response
	84	All dynamic addresses are occupied	The ePDG should select another PGW. The ePDG will attempt to establish a PDN connection on up to three PGWs. If none of these PGWs respond, the ePDG will send a NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE) message.
	85	UE context without TFT already activated	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	86	Protocol type not supported	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	87	UE not responding	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	88	UE refuses	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	89	Service denied	The ePDG should select another PGW. The ePDG will attempt to establish a PDN connection on up to three PGWs. If none of these PGWs respond, the ePDG will send a NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE) message.
	90	Unable to page UE	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	91	No memory available	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)

Table 210: S2b error mapping to UE IKE error code (continued)

Message type	GTP-C cause value	Description	IKE response
	92	User authentication failed	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	93	APN access denied - no subscription	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	94	Request rejected (reason not specified)	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	95	P-TMSI signature mismatch	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	96	IMSI/IMEI not known	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	97	Semantic error in the TAD operation	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	98	Syntactic error in the TAD operation	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	99	Not used	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	100	Remote peer not responding	The ePDG should select another PGW. The ePDG will attempt to establish a PDN connection on up to three PGWs. If none of these PGWs respond, the ePDG will send a NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE) message.
	101	Collision with network initiated request	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)

Table 210: S2b error mapping to UE IKE error code (continued)

Message type	GTP-C cause value	Description	IKE response
	102	Unable to page UE due to suspension	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	103	Conditional IE missing	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	104	APN Restriction type Incompatible with currently active PDN connection	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	105	Invalid overall length of the triggered response message and a piggybacked initial message	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	106	Data forwarding not supported	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	107	Invalid reply from remote peer	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	108	Fallback to GTPv1	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	109	Invalid peer	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	110	Fallback to GTPv1	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	111	Modifications not limited to S1-U bearers	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)

Table 210: S2b error mapping to UE IKE error code (continued)

Message type	GTP-C cause value	Description	IKE response
	112	Request rejected for a PMIPv6 reason (see 3GPP TS 29.275)	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	113	APN Congestion	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	114	Bearer handling not supported	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	115	UE already re-attached	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	116	Multiple PDN connections for a given APN not allowed	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	117	Target access restricted for the subscriber	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	118	Not used	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	119	MME/SGSN refuses due to VPLMN Policy	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	120	GTP-C Entity Congestion	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	121	Late Overlapping Request	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	122	Timed out Request	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)

Table 210: S2b error mapping to UE IKE error code (continued)

Message type	GTP-C cause value	Description	IKE response
	123	UE is temporarily not reachable due to power saving	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	124	Relocation failure due to NAS message redirection	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
	125 to 239	Spare. For future use in a triggered/response message.	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)
Request/initial message	240 to 255	Spare. For future use in an initial/request message.	NOTIFY (ID 36: INTERNAL_ADDRESS_FAILURE)

Private Notify message - error types describes the error types that indicate an error during the IKEv2 SA negotiation for the PDN setup. The errors can relate either to GTPv2 or DIAMETER errors.

Table 211: Private Notify message - error types

Notify message type	Value	Description
PDN_CONNECTION_REJECTION	8192	With an IP address information in Notification Data field: The PDN connection corresponding to the IP address information has been rejected. Without Notification Data field: The PDN connection corresponding to the requested APN has been rejected. No additional PDN connections to the given APN can be established. If the rejected PDN connection is the first PDN connection for the given APN, this APN is not allowed for the UE.

Table 211: Private Notify message - error types (continued)

Notify message type	Value	Description
MAX_CONNECTION_REACHED	8193	The PDN connection has been rejected. No additional PDN connections can be established for the UE due to the network policies or capabilities. The maximum number of PDN connections per UE allowed to be established simultaneously is 11 due to a limitation in the network mobility procedures.
NON_3GPP_ACCESS_TO_EPC_NOT_ALLOWED	9000	Corresponds to: DIAMETER_ERROR_USER_NO_NON_3GPP_SUBSCRIPTION Result code IE as specified in 3GPP TS 29.273; or Other scenarios when the UE is not allowed to use non-3GPP access to EPC.
USER_UNKNOWN	9001	Corresponds to: DIAMETER_ERROR_USER_UNKNOWN Result code IE as specified in 3GPP TS 29.273; or Other scenarios when the user identified by the IMSI is unknown.
NO_APN_SUBSCRIPTION	9002	Corresponds to: DIAMETER_ERROR_USER_NO_APN_SUBSCRIPTION Result code IE as specified in 3GPP TS 29.273; or Other scenarios when the requested APN is not included in the user's profile, and therefore is not authorized for that user.
AUTHORIZATION_REJECTED	9003	Corresponds to: DIAMETER_AUTHORIZATION_REJECTED Result code IE as specified in 3GPP TS 29.273; or

Table 211: Private Notify message - error types (continued)

Notify message type	Value	Description
		Other scenarios when the user is barred from using the non-3GPP access or the subscribed APN.
NETWORK_FAILURE	10500	Corresponds to: DIAMETER_ERROR_UNABLE_TO_COMPLY Result code IE as specified in 3GPP TS 29.273; or Other scenarios when the network has determined that the requested procedure cannot be completed successfully due to network failure.
RAT_TYPE_NOT_ALLOWED	11001	Corresponds to: DIAMETER_RAT_TYPE_NOT_ALLOWED Result code IE as specified in 3GPP TS 29.273; or Other scenarios when the access type is restricted to the user.
PLMN_NOT_ALLOWED	11011	Corresponds to: DIAMETER_ERROR_ROAMING_NOT_ALLOWED Result code IE as specified in 3GPP TS 29.273; or Other scenarios when the UE requests service in a PLMN where the UE is not allowed to operate.

Protocol Configuration Options (PCO) IE

The PCO Information Elements (IE) are used to transfer information between the UE and the network (PGW). The PCO message types and values used by the UE and network are described in 3GPP TS 24.008. These values may be sent in the PCO, the Additional Protocol Configuration Options (APCO), or the Extended Protocol Configuration Options (EPCO).

The PCO/APCO has a maximum length of 253 octets. The SGW can transparently pass the EPCO with a maximum length of 2560 octets and PGW (including combined SGW/PGW) can support the EPCO with a maximum length of 500 bytes.

PCO IE consists of configuration protocol options list and additional parameter list, which both contain a variable number of logical units. Each unit is of variable length and consists of the following:

- protocol identifier (2 octets)
- length of the protocol identifier contents of the unit (1 octet)
- protocol identifier contents itself (n octets)

Table 212: Protocol identifiers in configuration protocol options list of PCO IE

Identifier	Description
C023H	Used to pass PAP username and password from the UE. For more information, refer to RFC 1334.
C223H	Used to pass CHAP challenge and response from the UE. For more information, refer to RFC 1334.
8021H	Used to negotiate IPCP parameters with the UE. For more information, refer to RFC 1332.

Table 213: Containers of additional parameter list, UE to Network direction

Value	Message type	Description
0001H	P-CSCF IPv6 Address Request	The container is used to request P-CSCF IPv6 addresses when the PDN connection is activated. The container contents are empty.
0002H	IM CN Subsystem Signaling Flag	The container is used to indicate that default bearer is to be used only for IMS signaling. The container contents are empty.

Table 213: Containers of additional parameter list, UE to Network direction (continued)

Value	Message type	Description
0003H	DNS Server IPv6 Address Request	The container is used to request DNS server IPv6 addresses when the PDN connection is activated. The container contents are empty.
0005H	MS Support of Network Requested Bearer Control indicator	The container is used to indicate that UE supports Bearer Control Mode (BCM) "MS/NW". This means that dedicated bearers may be created both by MS and network. The container contents are empty.
000BH	IPv4 address allocation via DHCPv4	The container is used to request DHCP IPv4 addresses when the PDN connection is activated. The container contents are empty.
000CH	P-CSCF IPv4 Address Request	The container is used to request P-CSCF IPv4 addresses when the PDN connection is activated. The container contents are empty.
000DH	DNS Server IPv4 Address Request	The container is used to request DNS IPv4 addresses when the PDN connection is activated. The container contents are empty.
000EH	MSISDN Request	The container is used to request MSISDN value. The container contents are empty.
0010H	IPv4 Link MTU Request	The container is used to request the link MTU for IPv4. The container contents are empty.
0012H	P-CSCF Re-selection support	This PCO parameter may be present only if a container with PCSCF IPv4 Address Request or P-CSCF IPv6 Address Request is present. This information indicates that the UE supports P-CSCF re-selection based on procedures specified in 3GPP TS 24.229 subclauses B.2.2.1C, L.2.2.1C and R.2.2.1C

Table 213: Containers of additional parameter list, UE to Network direction (continued)

Value	Message type	Description
0015H	Non-IP Link MTU Request	The container is used by the MS to request the link MTU for the non-IP PDN connection. The container contents are empty.
0016H	APN rate control support indicator	UE supporting APN rate control will include this container.
0019H	Additional APN rate control for exception data support indicator	UE supporting additional APN rate control for exception data will include this container.
FF00H to FFFFH	This message type is reserved for operator specific use	—

Table 214: Containers of additional parameter list, Network to UE direction

Value	Message Type	Description
0001H	P-CSCF IPv6 Address	The container identifier contents field contains one IPv6 address corresponding to a P-CSCF address (see 3GPP TS 24.229) encoded as a 128-bit address (RFC 3513). Additional logical units with container identifier indicating P-CSCF address are used as needed to include multiple P-CSCF addresses.
0002H	IM CN Subsystem Signaling Flag	The container is used to indicate that bearer can be used only for IMS signaling. The container contents are empty.
0003H	DNS Server IPv6 Address	When the container identifier indicates DNS Server IPv6 Address, the container identifier contents field contains one IPv6 DNS server address (see 3GPP TS 27.060). This IPv6 address is encoded as a 128-bit address according to IETF RFC 4291.

Table 214: Containers of additional parameter list, Network to UE direction (continued)

Value	Message Type	Description
		When there is a need to include more than one DNS Server IPv6 address, then more logical units with the container identifier indicating DNS Server IPv6 Address are used
0005H	Selected Bearer Control Mode	The container identifier contents field contains the selected bearer control mode. 01H indicates that MS-only mode has been selected. 02H indicates that 'MS/NW' mode has been selected.
000CH	P-CSCF IPv4 Address	The container identifier contents field contains one IPv4 address corresponding to the P-CSCF address to be used. When there is a need to include more than one P-CSCF IPv4 address, then more logical units with the container identifier indicating P-CSCF IPv4 Address are used. If more than 3 instances of the P-CSCF IPv4 Address logical unit are received by the MS then the MS may ignore all but the first 3 instances of the P-CSCF IPv4 Address logical unit received.
000DH	DNS Server IPv4 Address	When the container identifier indicates DNS Server IPv4 Address, the container identifier contents field contains one IPv4 address corresponding to the DNS server address to be used. When there is a need to include more than one DNS Server IPv4 address, then more logical units with the container identifier indicating DNS Server IPv4 Address are used.
000EH	MSISDN	When the container identifier indicates MSISDN, the container identifier contents field contains the MSISDN (see 3GPP TS 23.003) assigned to the MS.
0010H	IPv4 Link MTU	The length of container identifier contents indicates a length equal to two. The container identifier contents field contains the binary coded representation of the IPv4 link MTU size in octets. Bit 8 of the first octet of the container identifier contents field contains the most significant bit and bit 1 of the second octet of the container identifier contents

Table 214: Containers of additional parameter list, Network to UE direction (continued)

Value	Message Type	Description
		field contains the least significant bit. If the length of the container identifier contents is different from two octets, then it is ignored by the receiver.
0015H	Non-IP Link MTU	The length of container identifier contents indicates a length equal to two. The container identifier contents field contains the binary coded representation of the link MTU size for the non-IP PDN connection in octets, which is at least 128 octets. Bit 8 of the first octet of the container identifier contents field contains the most significant bit and bit 1 of the second octet of the container identifier contents field contains the least significant bit.
0016H	APN rate control parameters	The container contains the APN rate control parameters: maximum uplink rate for the defined time unit, and the indication whether additional exceptional reports are allowed to exceed the uplink rate limit.
0019H	Additional APN rate control for exception data parameters	The container contains the additional APN rate control parameters for exception data: maximum uplink rate for the defined time unit.
FF00H to FFFFH	Reserved for operator specific use	—

Gx interface Supported Features of Feature-List-ID 1

Table 215: Gx interface Supported Features of Feature-List-ID 1

Feature bit	Feature	Presence	Description
0	Rel8	M	This feature indicates the support of base 3GPP Rel-8 Gx functionality, including the AVPs and corresponding procedures

Table 215: Gx interface Supported Features of Feature-List-ID 1 (continued)

Feature bit	Feature	Presence	Description
			supported by the base 3GPP Rel-7 Gx standard, but excluding those features represented by separate feature bits.
1	Rel9	M	This feature indicates the support of base 3GPP Rel-9 Gx functionality, including the AVPs and corresponding procedures supported by the Rel8 feature bit, but excluding those features represented by separate feature bits.
3	Rel10	M	This feature indicates the support of base 3GPP Rel-10 Gx functionality, including the AVPs and corresponding procedures supported by the Rel8 and Rel9 feature bit, but excluding those features represented by separate feature bits.
4	Sponsored Connectivity	O	This feature indicates support for sponsored data connectivity feature. If the PCEF supports this feature, the PCRF may authorize sponsored data connectivity to the subscriber. To enable this feature use the following command: <pre>config>mobile>profile>gx>supported-features> sponsored-connectivity</pre>
6	ADC	O	This feature indicates support for the Application Detection and Control feature. To enable this feature use the following command: <pre>config>mobile>profile>gx>supported-features>adc</pre>
10	NetLoc	O	This feature indicates the support of the Access Network Information Reporting for GPRS and EPS. To enable this feature use the following command: <pre>config>mobile>profile>gx>supported-features> net loc</pre>
13	Trusted-WLAN	O	This feature indicates the support for the Trusted WLAN access as defined in 3GPP TS 23.402. To enable this feature use the following command:

Table 215: Gx interface Supported Features of Feature-List-ID 1 (continued)

Feature bit	Feature	Presence	Description
			config>mobile>profile>gx>supported-features>trusted-wlan
14	SGW-Rest	O	This feature indicates the support of SGW Restoration procedures as defined in 3GPP TS 23.007. To enable this feature use the following command: <pre>config>mobile>profile>gx>supported-features>sgw-rest</pre>
16	Pending Transaction	O	This feature indicates support for the race condition handling as defined in 3GPP TS 29.213. To enable this feature use the following command: <pre>config>mobile>profile>gx>supported-features>pending-transaction</pre>
21	Conditional APNPolicyInfo	O	This feature indicates support for APN related policy information with condition as defined in 3GPP TS 23.203 subclause A.4.3.3. To enable this feature use the following command: <pre>config>mobile>profile>gx>supported-features>conditional-policy-info</pre>
22	RAN-NAS-Cause	O	This feature indicates the support for the detailed release cause code information (NOTE 1) from the access network. To enable this feature use the following command: <pre>config>mobile>profile>gx>supported-features>ran-nas-cause</pre>
23	CNO-ULI	O	This feature indicates support for Presence Reporting Area Information reporting. To enable this feature use the following command: <pre>config>mobile>profile>gx>supported-features>cno-uli</pre>

Table 215: Gx interface Supported Features of Feature-List-ID 1 (continued)

Feature bit	Feature	Presence	Description
24	PCSCF-Restoration-Enhancement	O	This feature indicates support of P-CSCF Restoration Enhancement. It is used for the PCEF to indicate if it supports P-CSCF Restoration Enhancement.
25	MissionCritical QCI	O	This feature indicates support for the Mission Critical QCI values 65, 69 and 70, and the Non Mission Critical QCI value 66 within the QoS-Class-Identifier AVP. To enable this feature use the following command: <pre>config>mobile>profile>gx>supported-features>mission-critical-qcis</pre>
30	NetLoc-Untrusted-WLAN	O	This feature indicates support of the Access Network Information Reporting for Untrusted WLAN access as defined in 3GPP TS 23.203. If supported, the PCEF will behave as described in annex D.4. It requires that NetLoc feature is also supported.

Gx interface Supported Features of Feature-List-ID 2

Table 216: Gx interface Supported Features of Feature-List-ID 2

Feature bit	Feature	Presence	Description
3	Multiple-PRA	O	This feature indicates support for Multiple Presence Reporting Area Information reporting. If the PCEF supports this feature, the PCRF shall behave as described in 29.212 Annex B.3.17.
14	User-Equipment-Info-Extension	O	This feature indicates support of the User-Equipment-Info-Extension AVP as defined in IETF RFC 8506 [66].

S6b Interface Supported-Features AVP

Table 217: S6b interface Supported-Features AVP

Feature bit	Feature	Presence	Description
0	P-CSCF Restoration for WLAN	O	This feature is applicable to the AAR/AAA command pair over the S6b interface, when the PGW supports the execution of the P-CSCF restoration procedures for WLAN for the related IMS PDN connection as described in 3GPP TS 23.380. If the PGW does not indicate support of this feature in a former AAR command, the 3GPP AAA Server shall not send a RAR command requesting the execution of the HSS-based P-CSCF restoration procedures for WLAN. To enable this feature use the following command: <pre>config>mobile>profile>s6b-profile>wlan-pcscf-restoration</pre>

Appendix B: Standards and protocol support

**NOTE:**

The information presented is subject to change without notice.
Nokia assumes no responsibility for inaccuracies contained herein.

Application Assurance (AA)

3GPP Release 12 (ADC rules over Gx interfaces)

RFC 3507, *Internet Content Adaptation Protocol (ICAP)*

BGP

RFC 1772, *Application of the Border Gateway Protocol in the Internet*

RFC 1997, *BGP Communities Attribute*

RFC 2385, *Protection of BGP Sessions via the TCP MD5 Signature Option*

RFC 2439, *BGP Route Flap Damping*

RFC 2545, *Use of BGP-4 Multiprotocol Extensions for IPv6 Inter-Domain Routing*

RFC 2858, *Multiprotocol Extensions for BGP-4*

RFC 2918, *Route Refresh Capability for BGP-4*

RFC 3107, *Carrying Label Information in BGP-4*

RFC 3392, *Capabilities Advertisement with BGP-4*

RFC 4271, *A Border Gateway Protocol 4 (BGP-4)*

RFC 4360, *BGP Extended Communities Attribute*

RFC 4364, *BGP/MPLS IP Virtual Private Networks (VPNs)*

RFC 4456, *BGP Route Reflection: An Alternative to Full Mesh Internal BGP (IBGP)*

RFC 4486, *Subcodes for BGP Cease Notification Message*

RFC 4659, *BGP/MPLS IP Virtual Private Network (VPN) Extension for IPv6 VPN*

RFC 4684, *Constrained Route Distribution for Border Gateway Protocol/MultiProtocol Label Switching (BGP/MPLS) Internet Protocol (IP) Virtual Private Networks (VPNs)*

RFC 4724, *Graceful Restart Mechanism for BGP (Helper Mode)*

RFC 4760, *Multiprotocol Extensions for BGP-4*

RFC 4798, *Connecting IPv6 Islands over IPv4 MPLS Using IPv6 Provider Edge Routers (6PE)*

RFC 4893, *BGP Support for Four-octet AS Number Space*

RFC 5004, *Avoid BGP Best Path Transitions from One External to Another*

RFC 5065, *Autonomous System Confederations for BGP*

RFC 5291, *Outbound Route Filtering Capability for BGP-4*

RFC 5575, *Dissemination of Flow Specification Rules*

RFC 5668, *4-Octet AS Specific BGP Extended Community*

draft-ietf-idr-add-paths-04, *Advertisement of Multiple Paths in BGP*

draft-ietf-idr-best-external-03, *Advertisement of the best external route in BGP*

Evolved Packet Data Gateway (ePDG)

RFC2104, *HMAC: Keyed-Hashing for Message Authentication*

RFC2404, *the Use of HMAC-SHA-1-96 within ESP and AH*

RFC2451, *the ESP CBC-Mode Cipher Algorithm*

RFC3526, *More Modular Exponential (MODP) Diffie-Hellman groups for Internet Key Exchange (IKE)*

RFC3566, *the AES-XCBC-MAC-96 Algorithm and Its Use with IPSec*

RFC3602, *the AES-CBC Cipher Algorithm and Its Use with IPSec*

RFC3664, *the AES-XCBC-PRF-128 Algorithm for the Internet Key Exchange Protocol (IKE)*

RFC4303, *IP Encapsulating Security Payload (ESP)*

RFC4307, *Cryptographic Algorithms for Use in the Internet Key Exchange Version 2 (IKEv2)*

RFC4308, *Cryptographic Suites for IPSec*

RFC4835, *Cryptographic Algorithm Implementation Requirements for Encapsulating Security Payload (ESP) and Authentication Header (AH).*

RFC 7321, *(Cryptographic Algorithm Implementation Requirements and Usage Guidance for Encapsulating Security Payload (ESP) and Authentication Header (AH))*

RFC 3947, *(Negotiation of NAT-Traversal in the IKE)*

RFC 3948, *(UDP Encapsulation of IPSec ESP Packets)*

RFC 4555, *(IKEv2 Mobility and Multihoming Protocol (MOBIKE))*

RFC 3748, *regarding EAP authenticator functionalities (Extensible Authentication Protocol (EAP))*

RFC 5247, *(Extensible Authentication Protocol (EAP) Key Management Framework)*

RFC 7296, *Internet Key Exchange Protocol Version 2 (IKEv2)*

RFC 7383, *Internet Key Exchange Protocol Version 2 (IKEv2) Message Fragmentation*

RFC 4555, *IKEv2 Mobility and Multihoming Protocol (MOBIKE)*

RFC 4301, *Security Architecture for the Internet Protocol*

RFC 4303, *IP Encapsulating Security Payload (ESP)*

RFC 5998, *An Extension for EAP-Only Authentication in IKEv2*

RFC 7651, *3GPP IP Multimedia Subsystems (IMS) Option for the Internet Key Exchange Protocol Version 2 (IKEv2)*

RFC 6733, *Diameter Base Protocol*

RFC 4945, *The Internet IP Security PKI Profile of IKEv1/ISAKMP, IKEv2, and PKIX*

Ethernet

IEEE 802.1AB, *Station and Media Access Control Connectivity Discovery*

IEEE 802.1ad, *Provider Bridges*

IEEE 802.1ag, *Connectivity Fault Management*

IEEE 802.1ah, *Provider Backbone Bridges*

IEEE 802.1ak, *Multiple Registration Protocol*

IEEE 802.1aq, *Shortest Path Bridging*

IEEE 802.1ax, *Link Aggregation*

IEEE 802.1D, *MAC Bridges*

IEEE 802.1p, *Traffic Class Expediting*

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